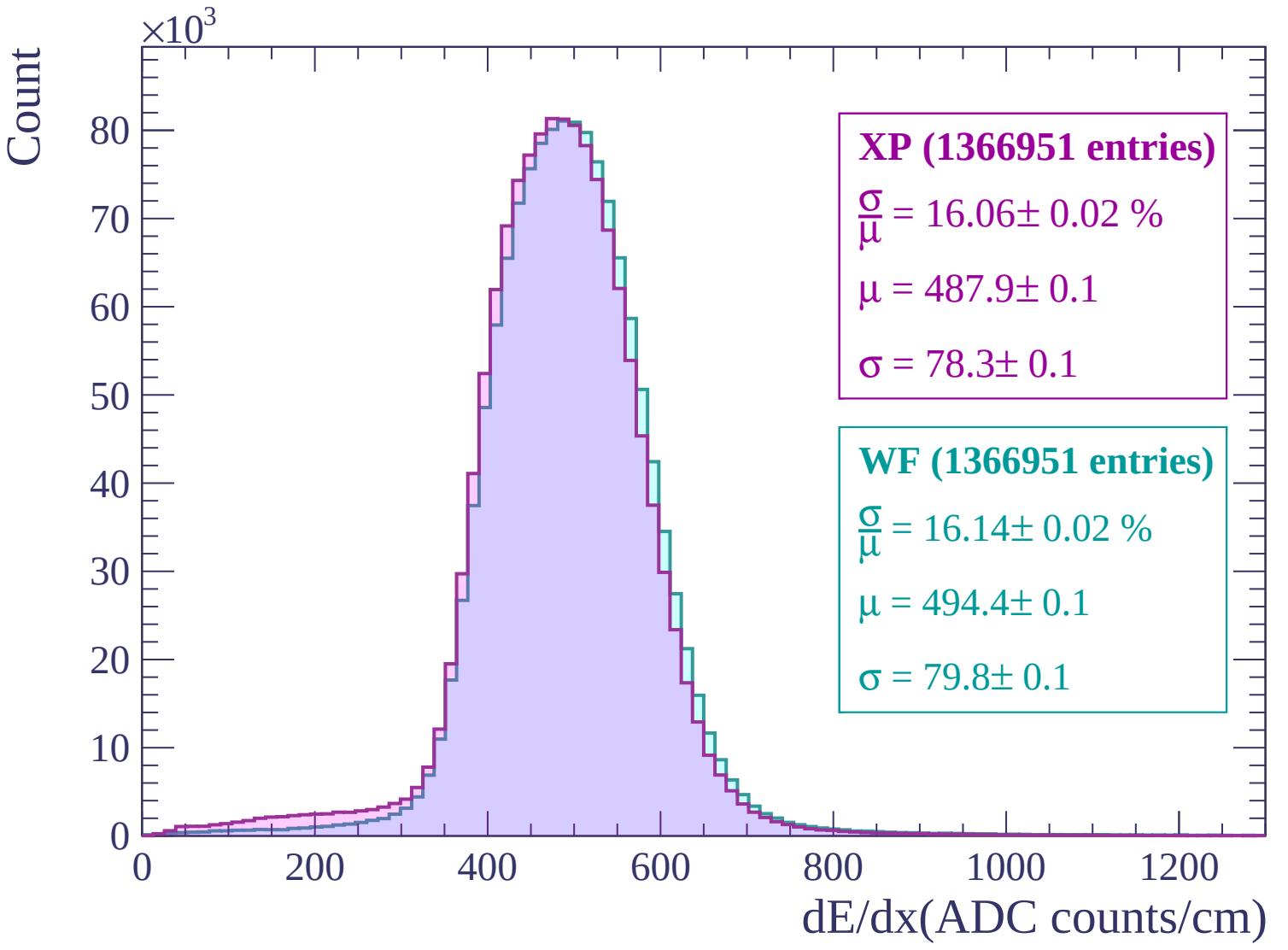
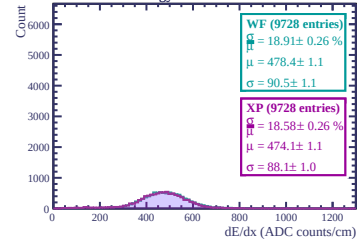
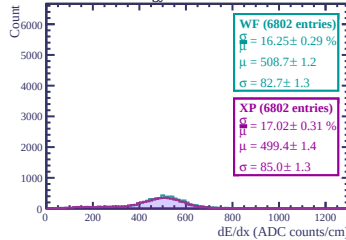




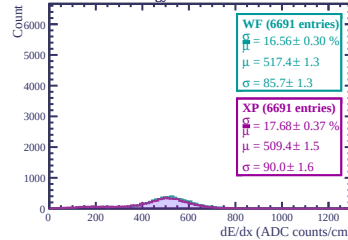
Energy loss in ERAM 14



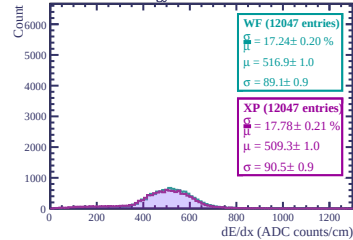
Energy loss in ERAM 30



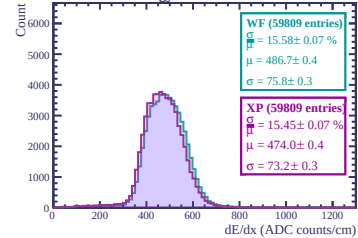
Energy loss in ERAM 28



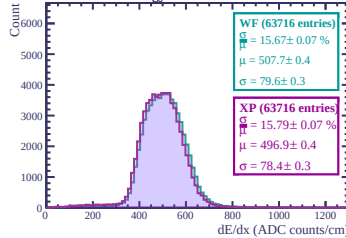
Energy loss in ERAM 19



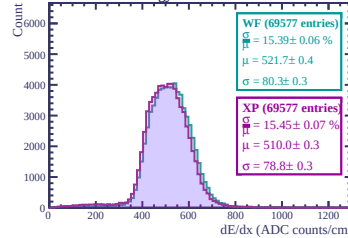
Energy loss in ERAM 21



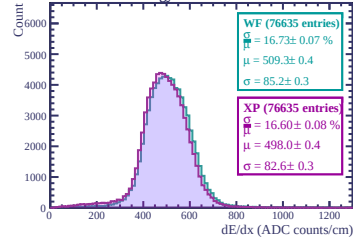
Energy loss in ERAM 13



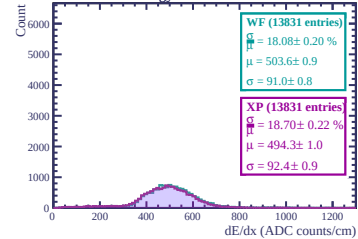
Energy loss in ERAM 9



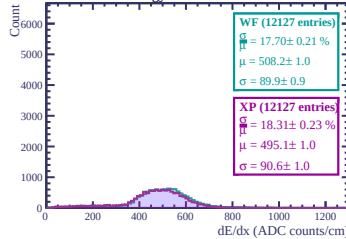
Energy loss in ERAM 2



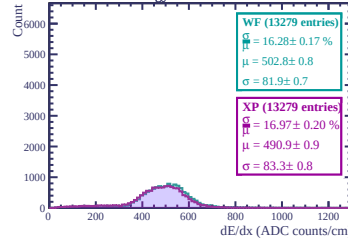
Energy loss in ERAM 26



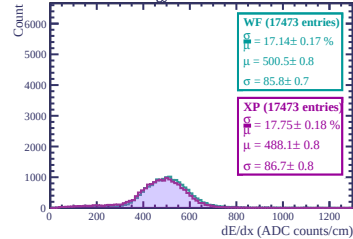
Energy loss in ERAM 17



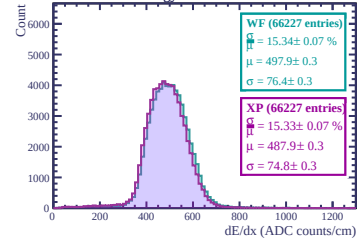
Energy loss in ERAM 23



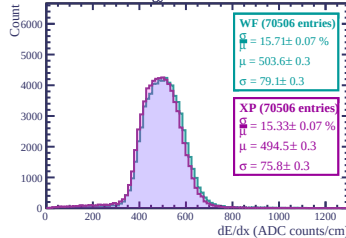
Energy loss in ERAM 29



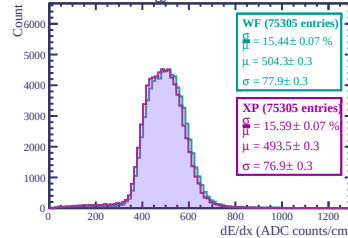
Energy loss in ERAM 1



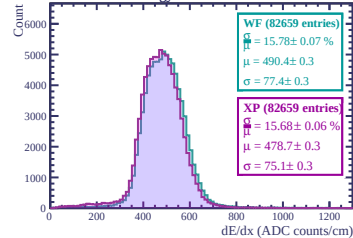
Energy loss in ERAM 10



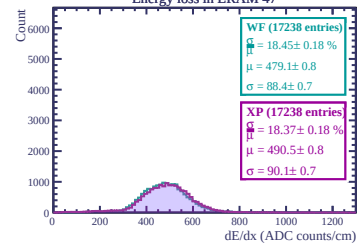
Energy loss in ERAM 11



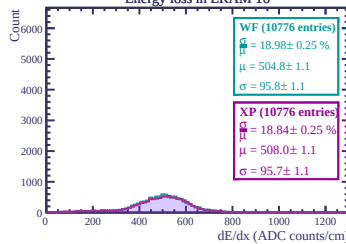
Energy loss in ERAM 3



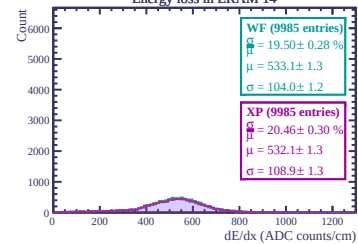
Energy loss in ERAM 47



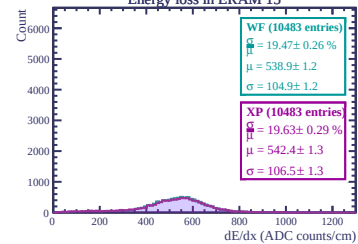
Energy loss in ERAM 16



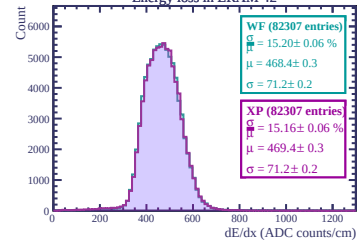
Energy loss in ERAM 14



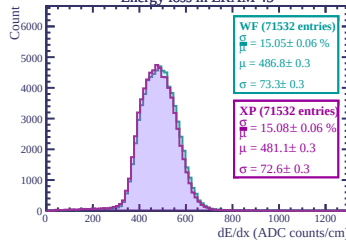
Energy loss in ERAM 15



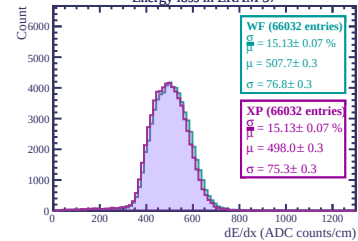
Energy loss in ERAM 42



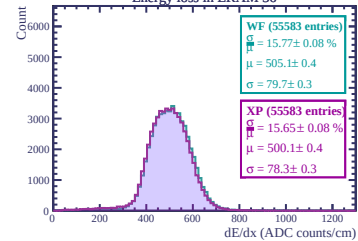
Energy loss in ERAM 45



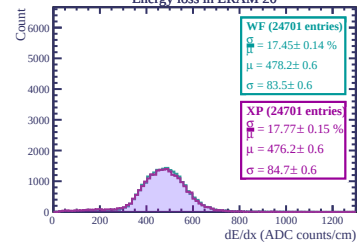
Energy loss in ERAM 37



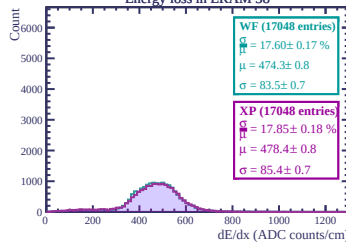
Energy loss in ERAM 36



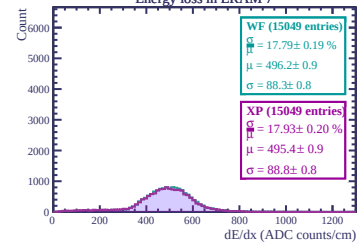
Energy loss in ERAM 20



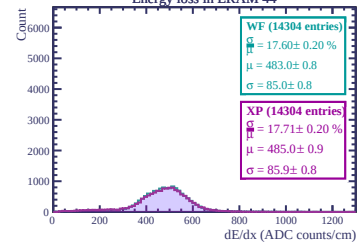
Energy loss in ERAM 38



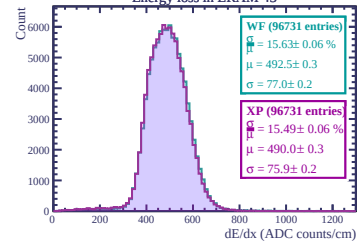
Energy loss in ERAM 7



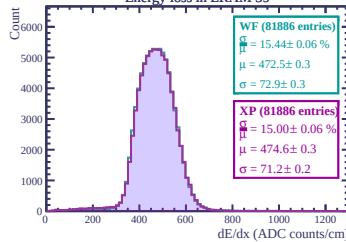
Energy loss in ERAM 44



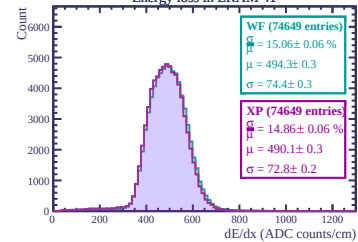
Energy loss in ERAM 43



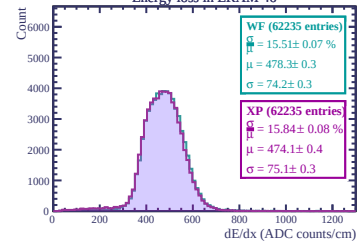
Energy loss in ERAM 39

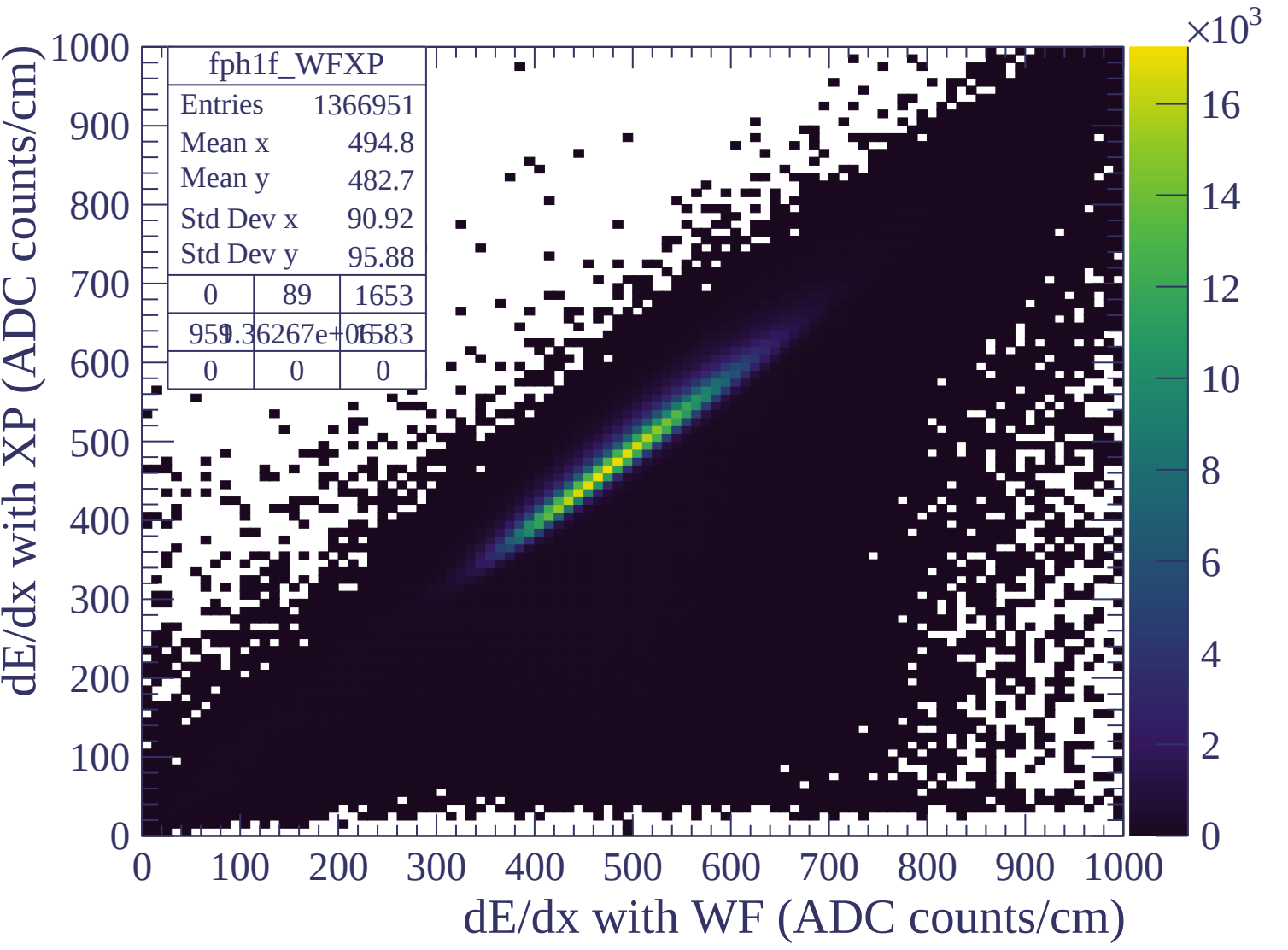


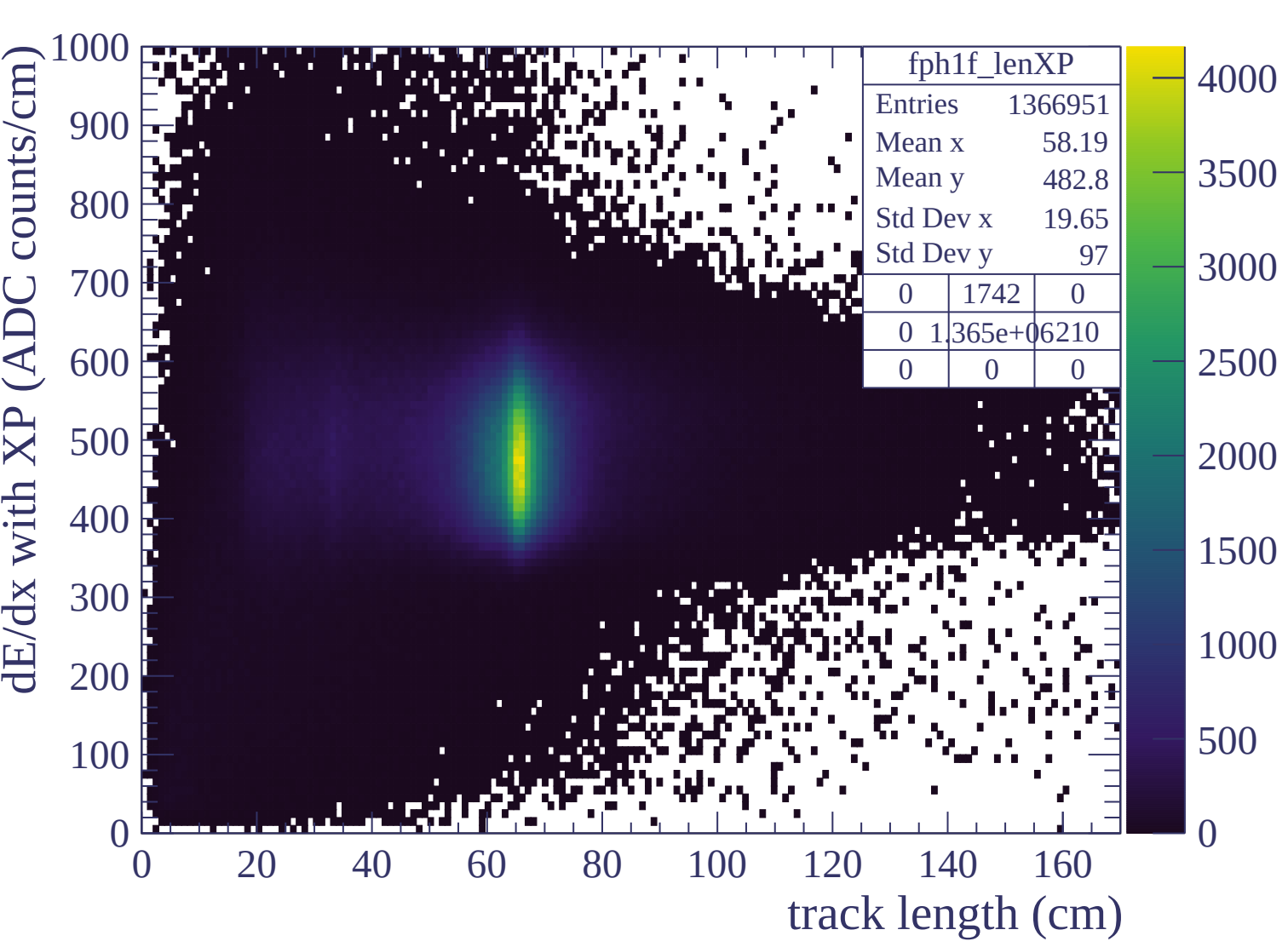
Energy loss in ERAM 41

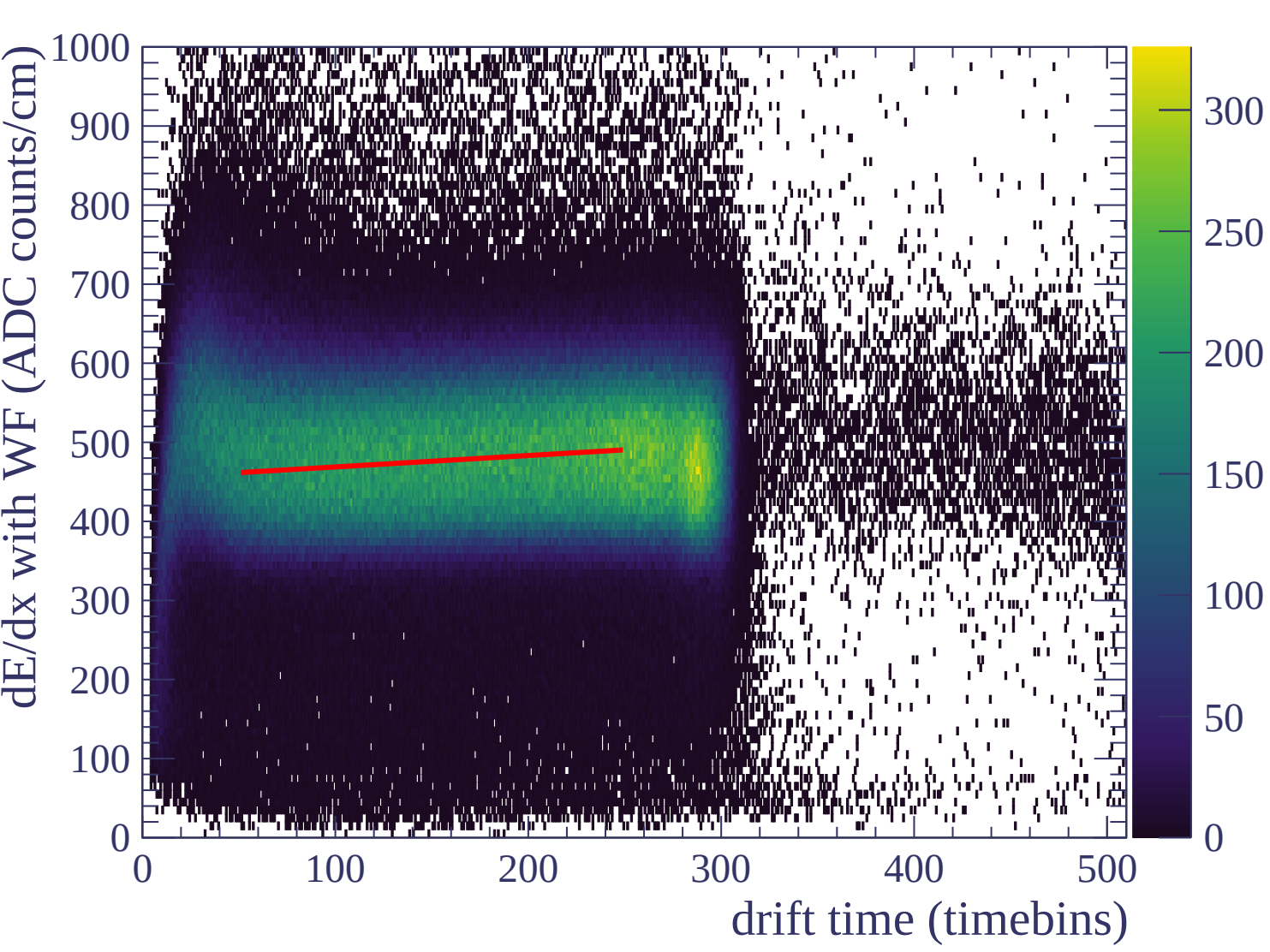


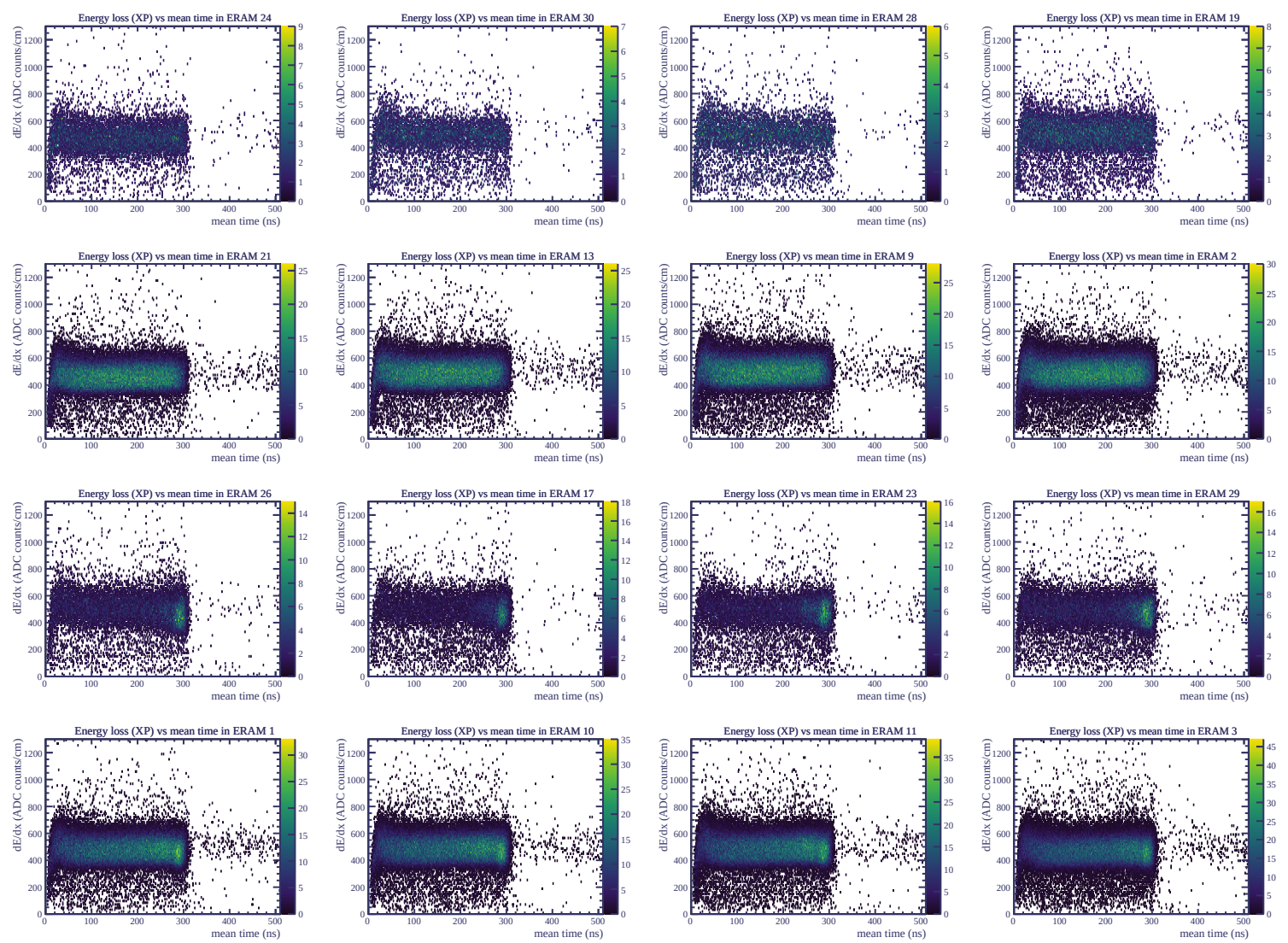
Energy loss in ERAM 46

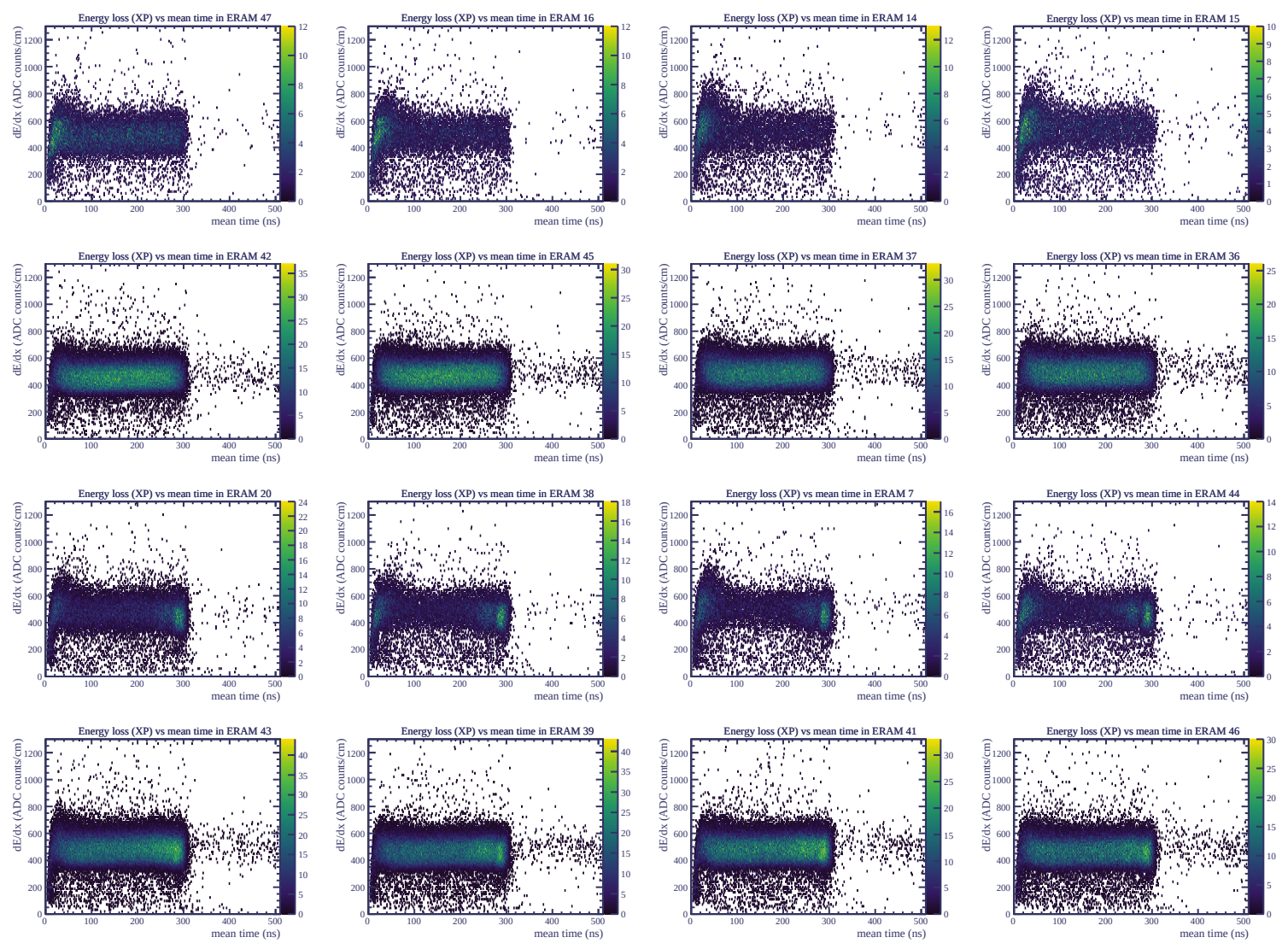




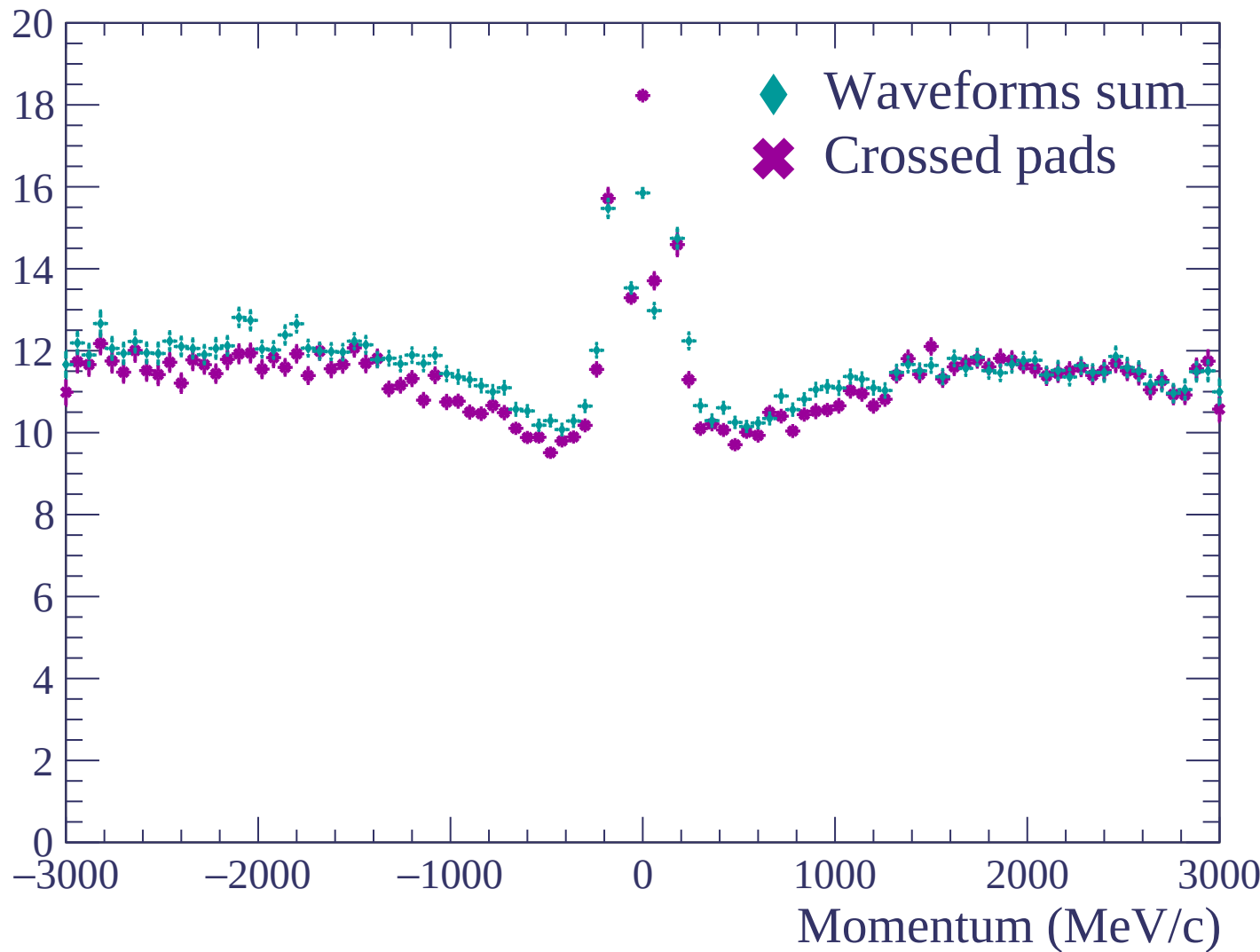


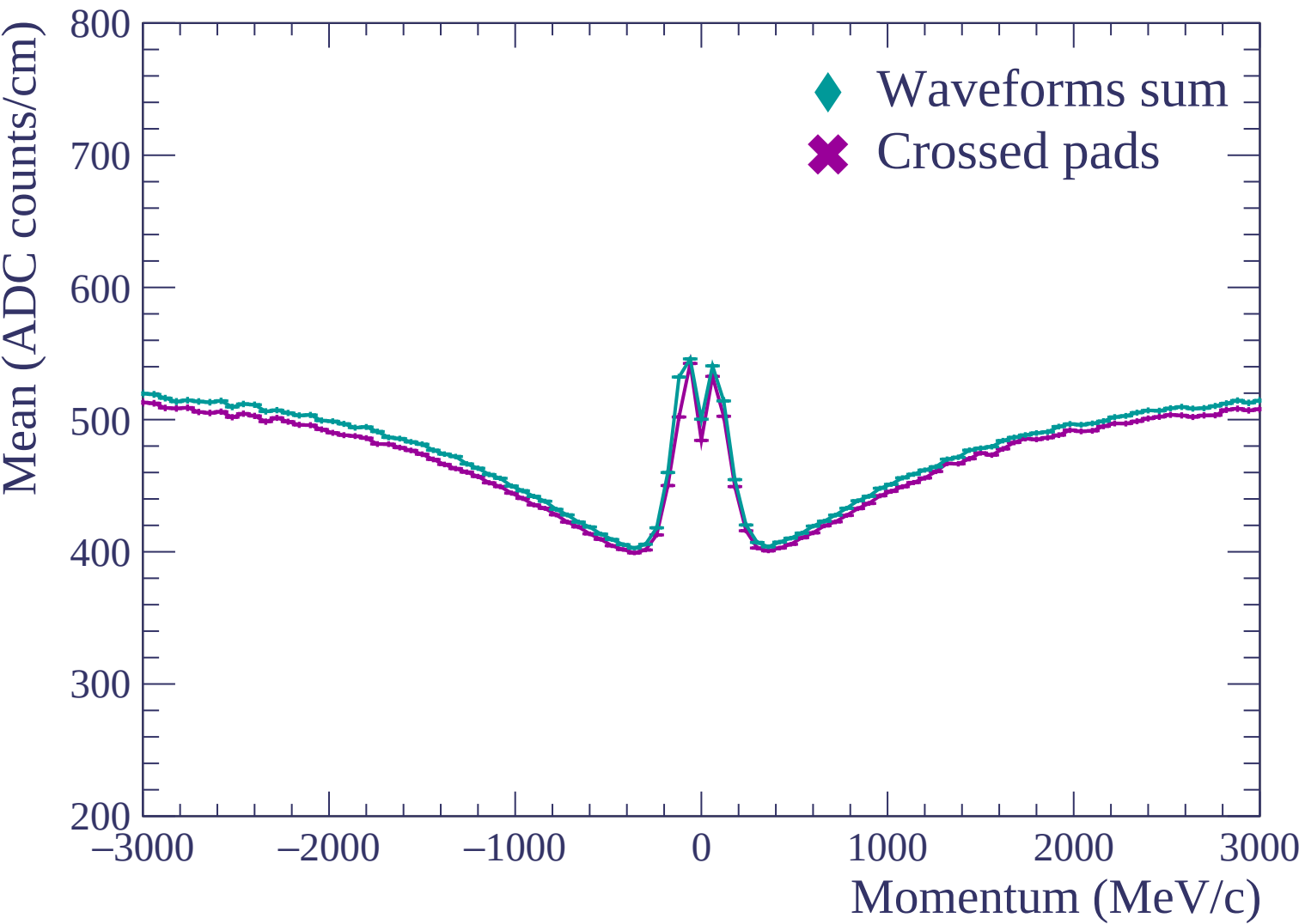


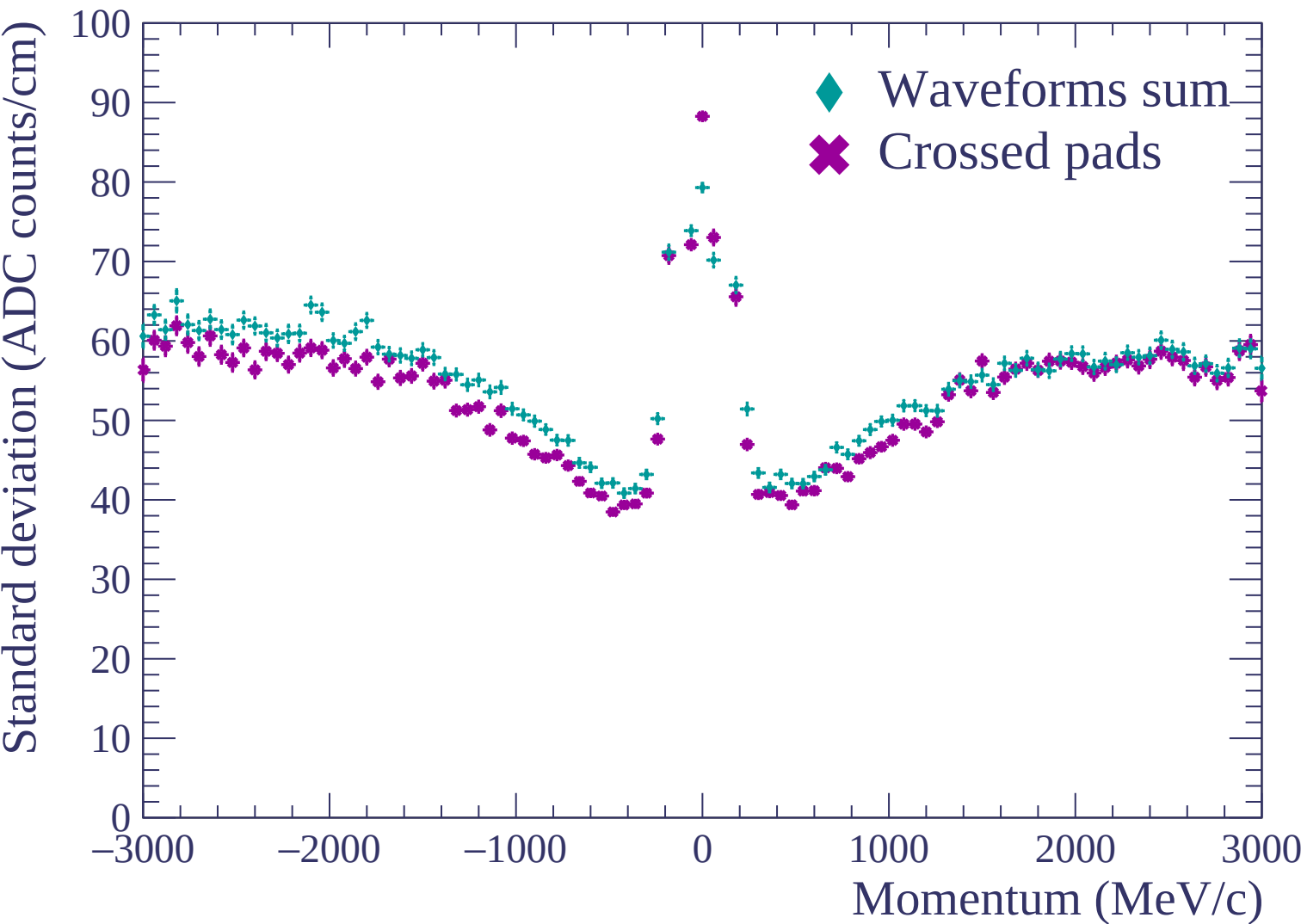


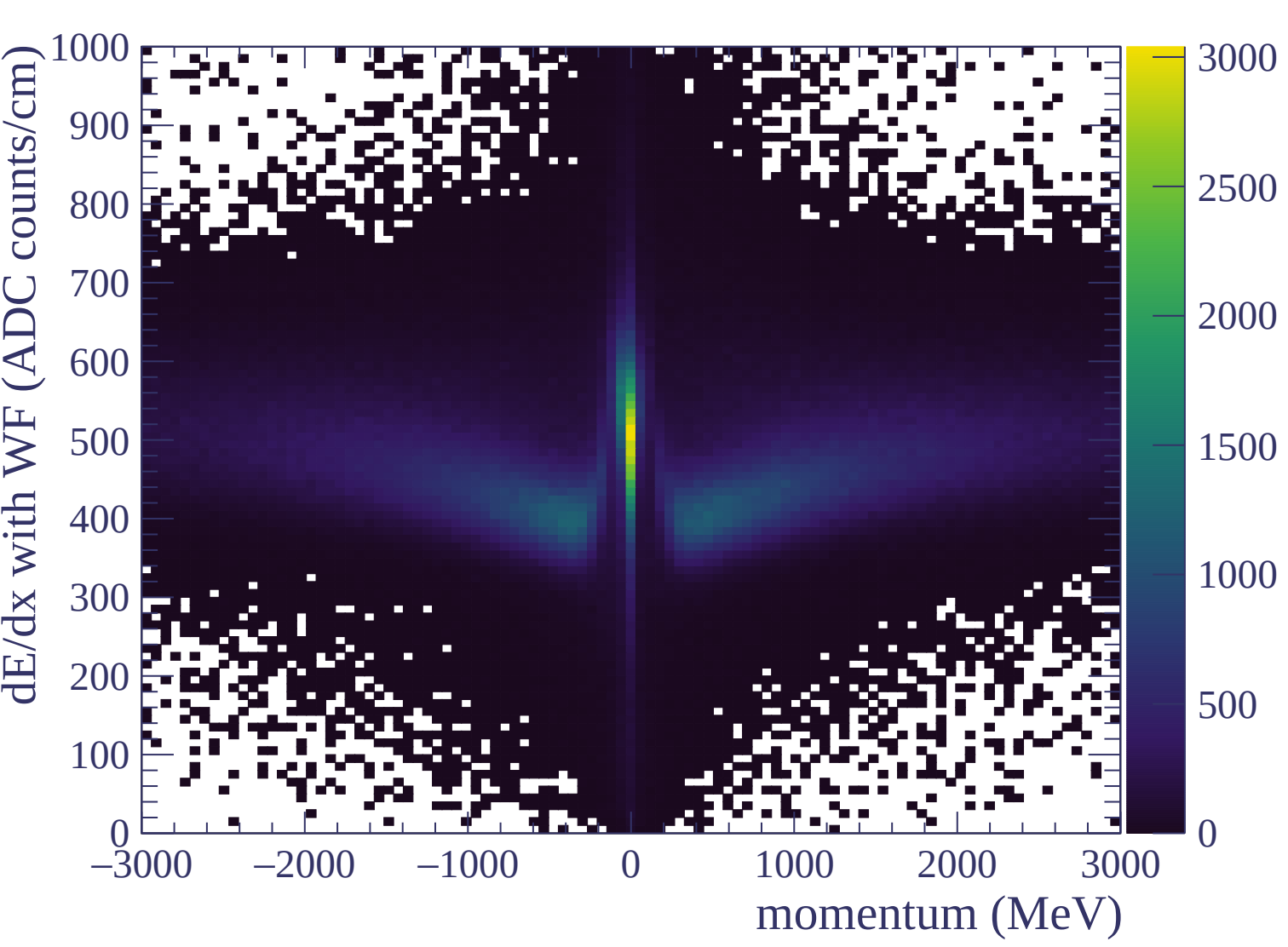


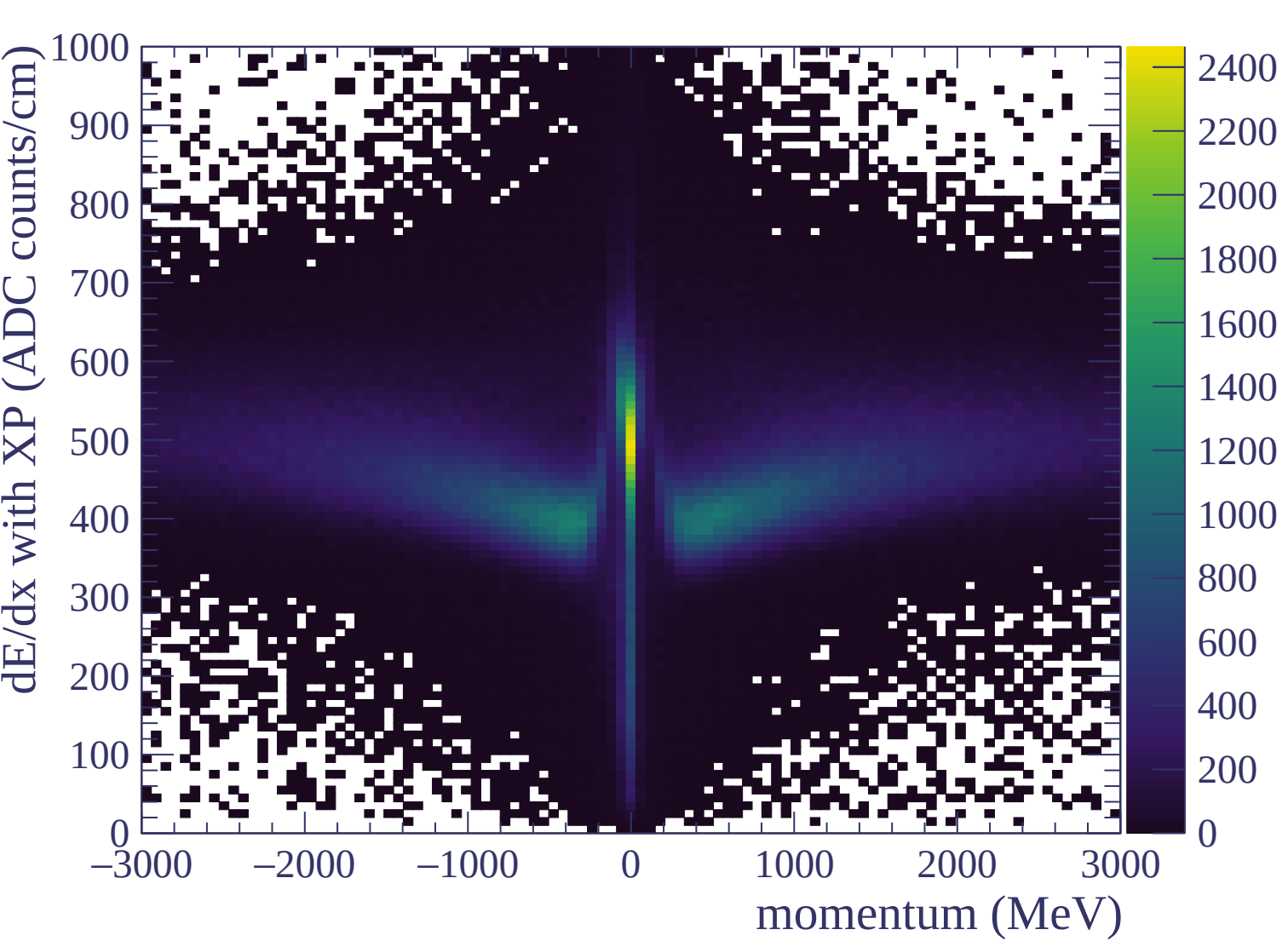
Resolution (%)

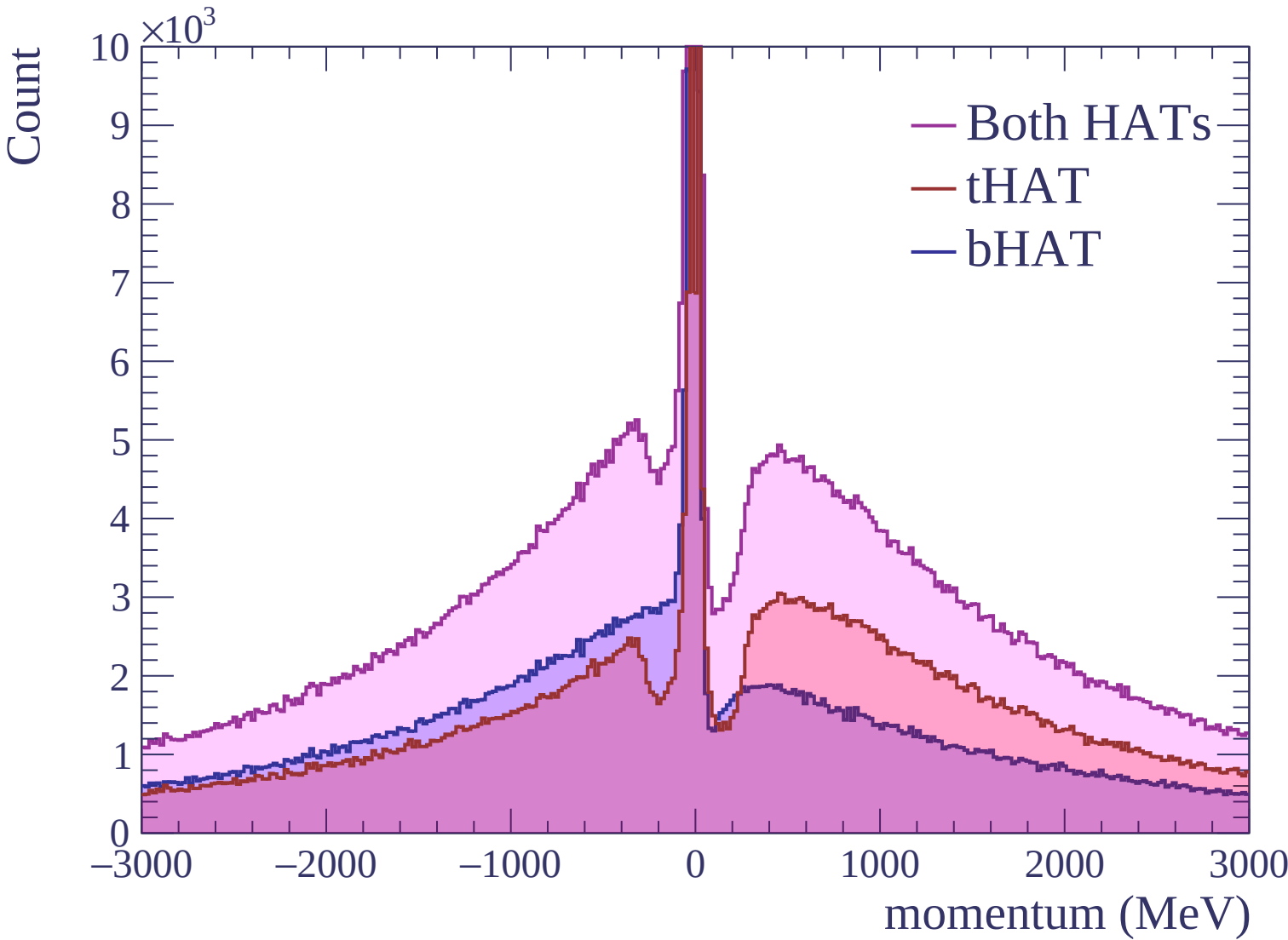


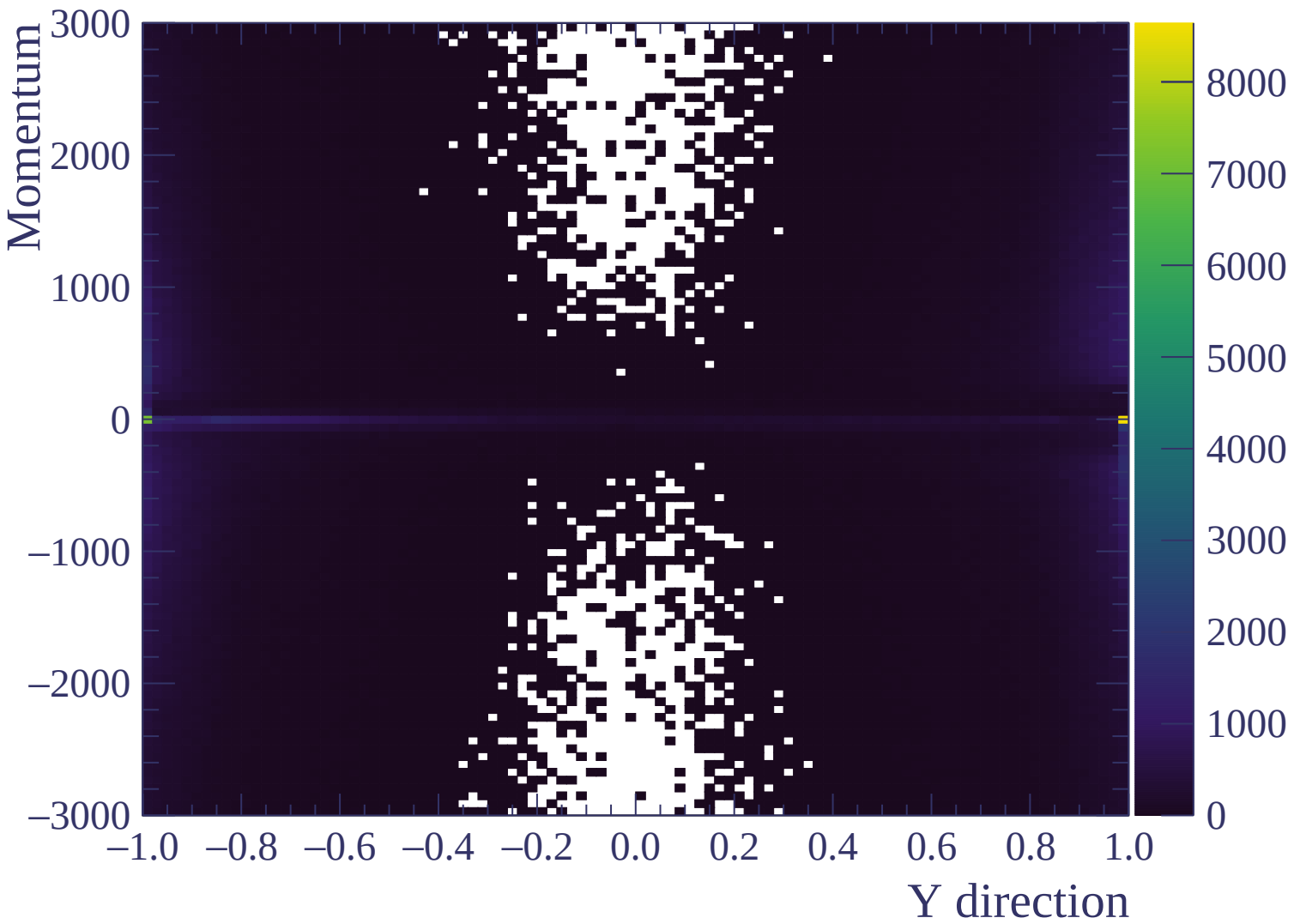


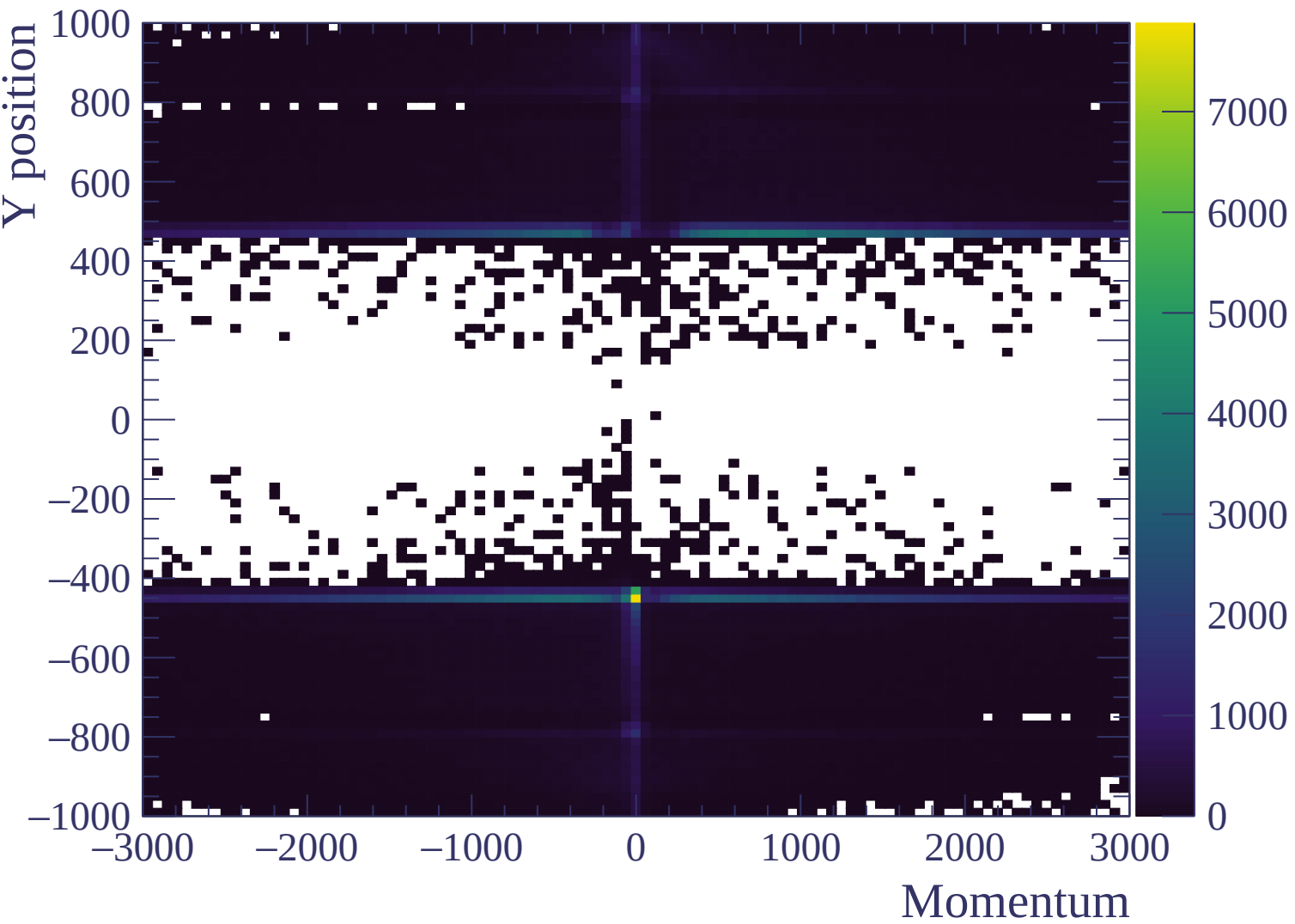


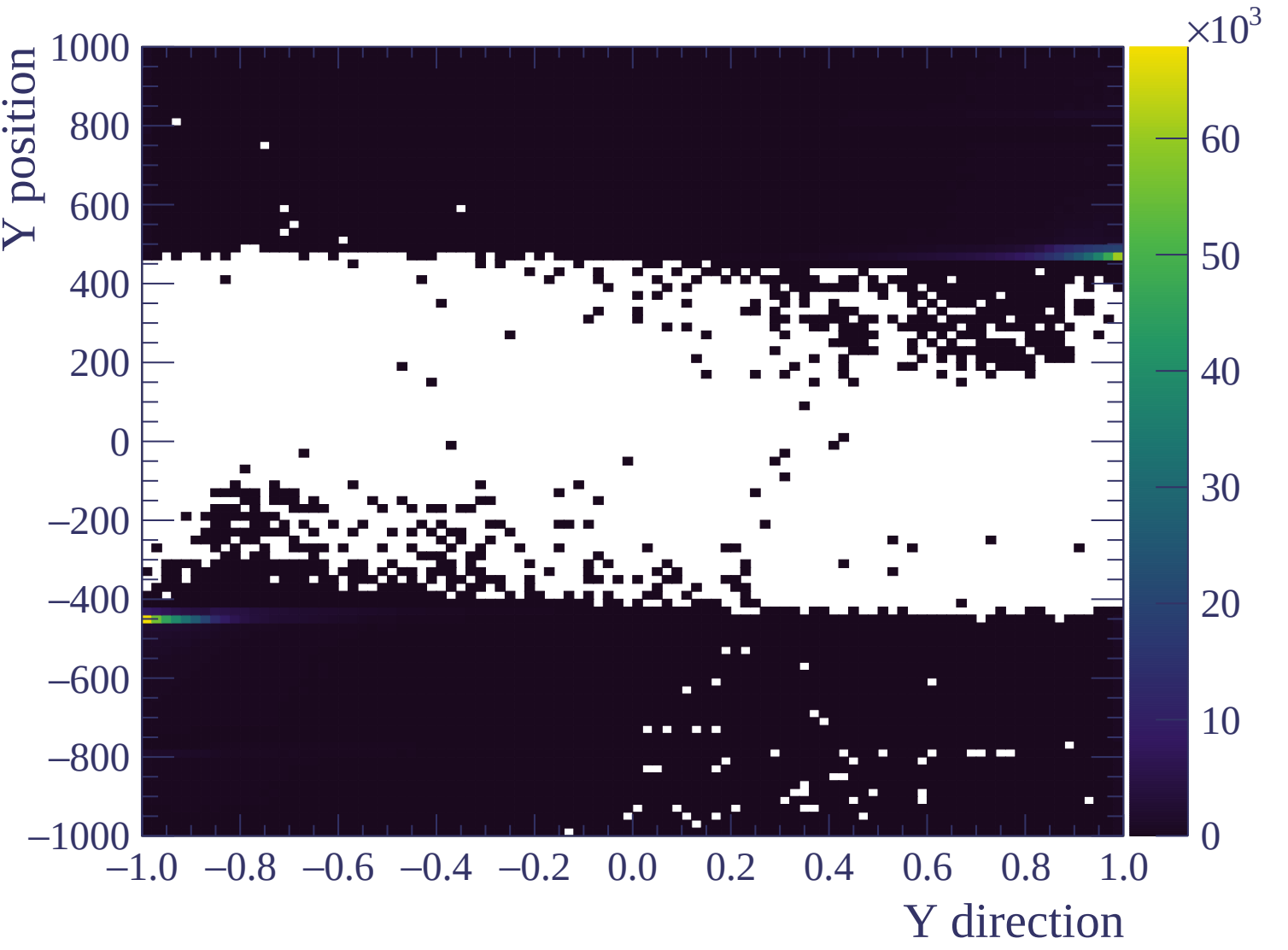






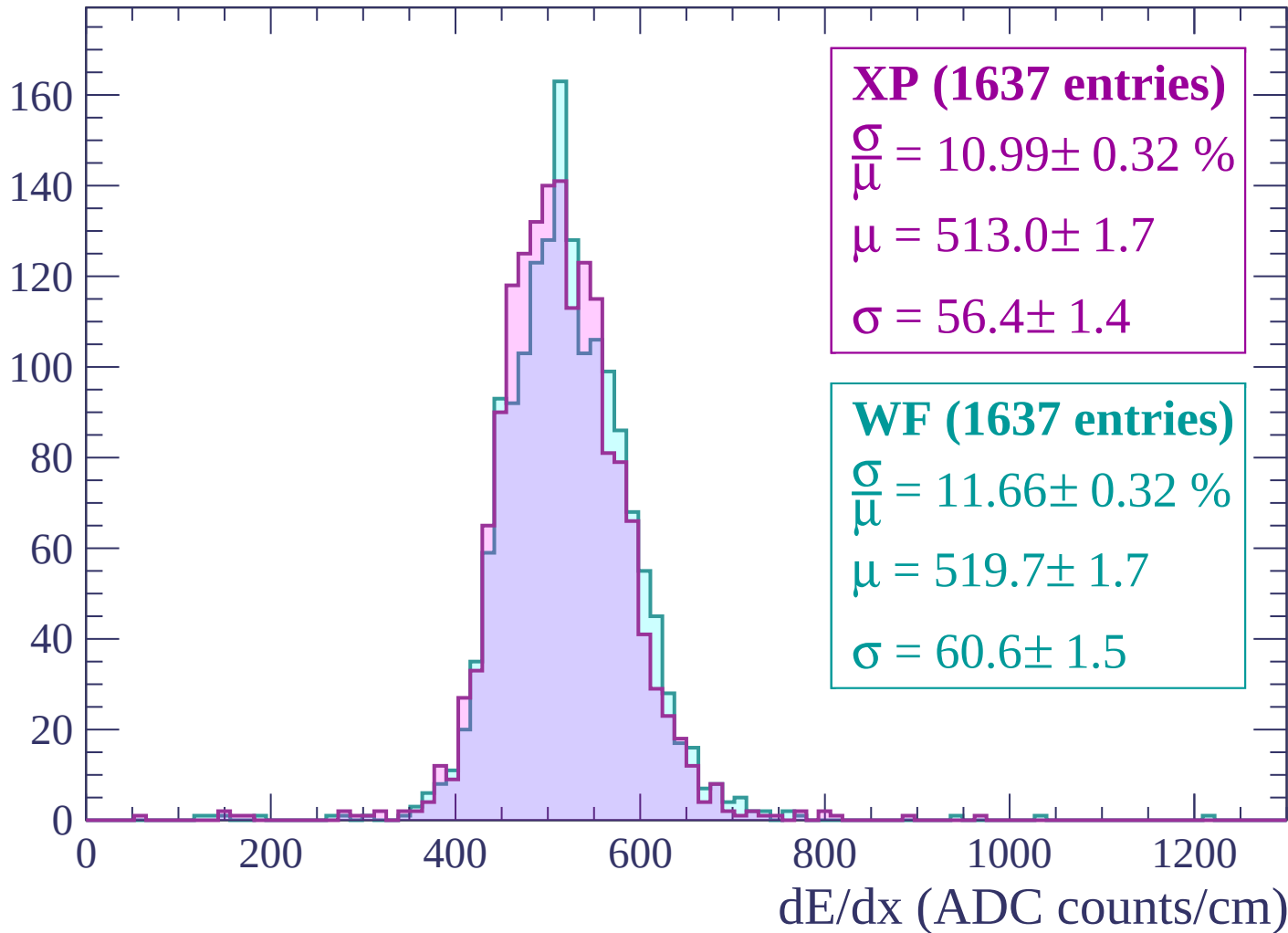






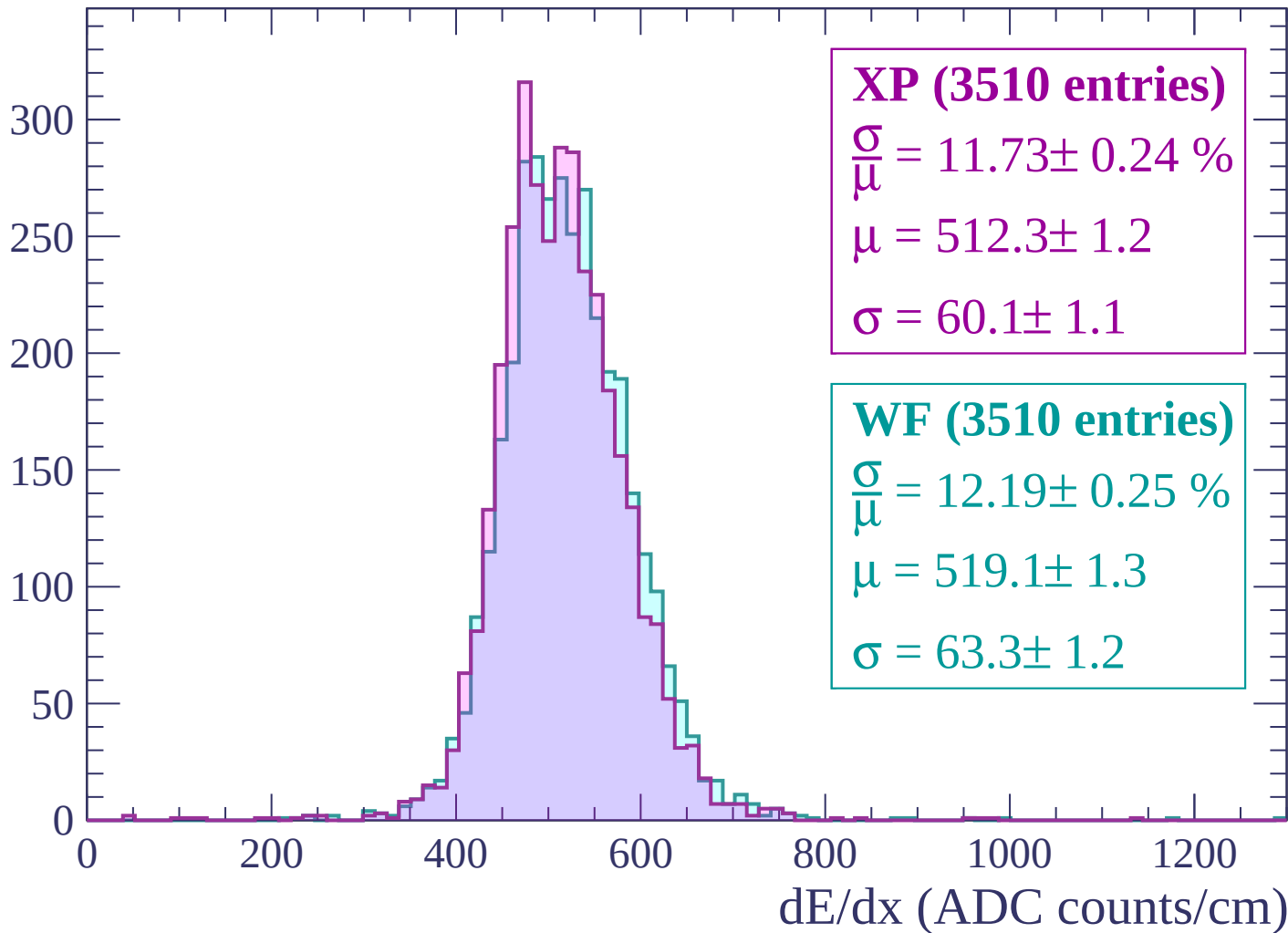
Energy loss | $-3000 < p < -2940$

Count



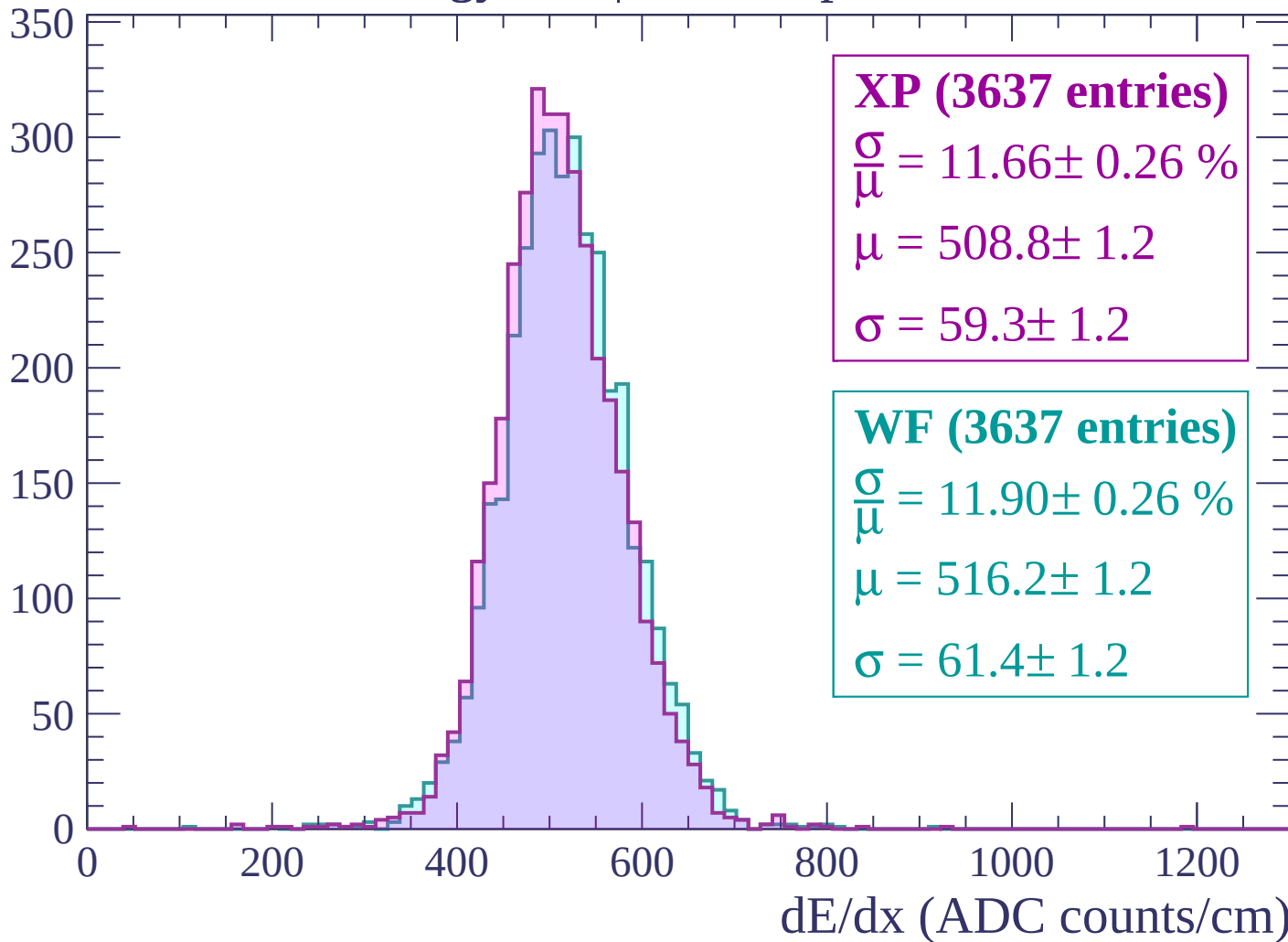
Energy loss | $-2940 < p < -2880$

Count



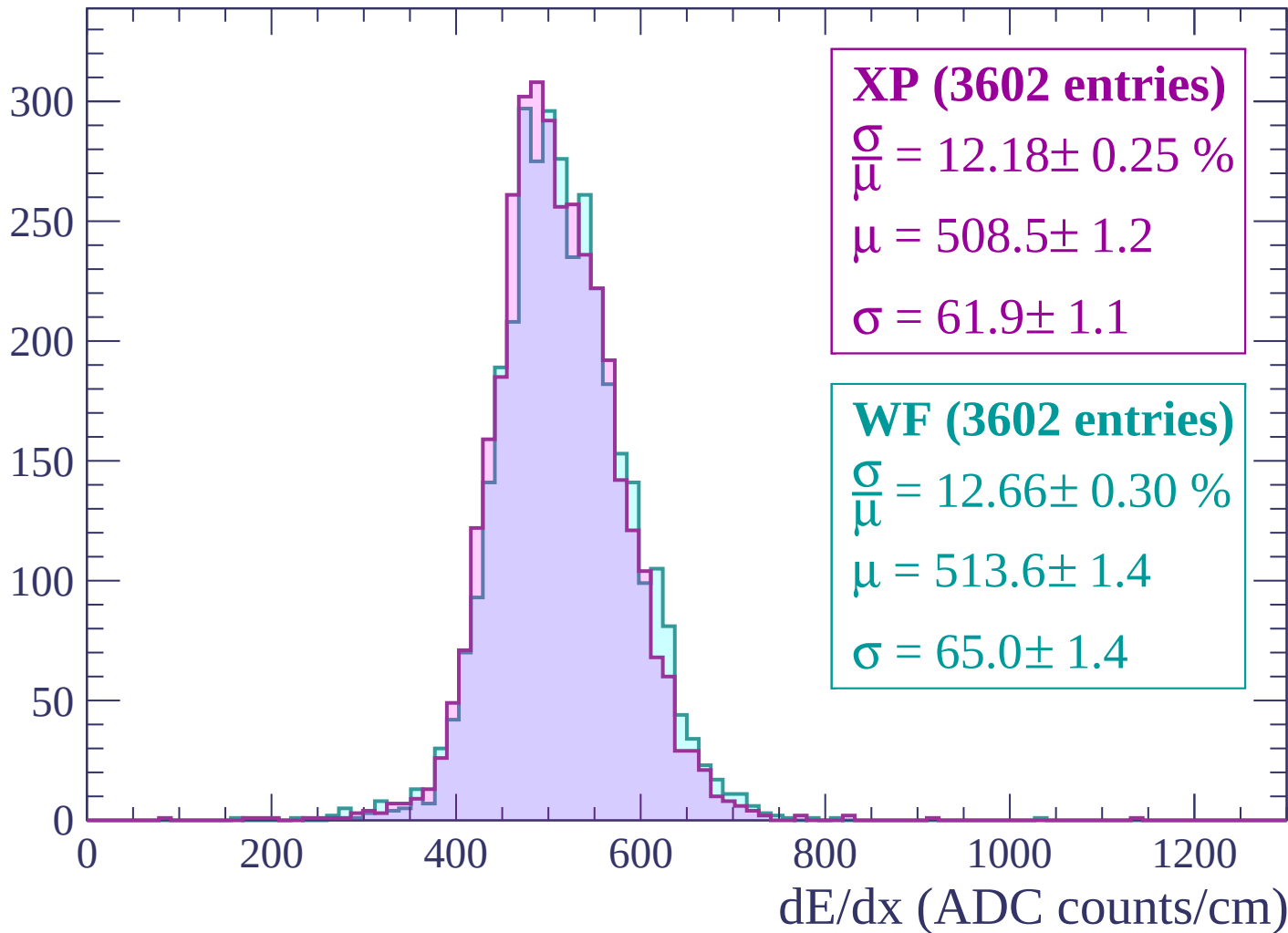
Energy loss | $-2880 < p < -2820$

Count



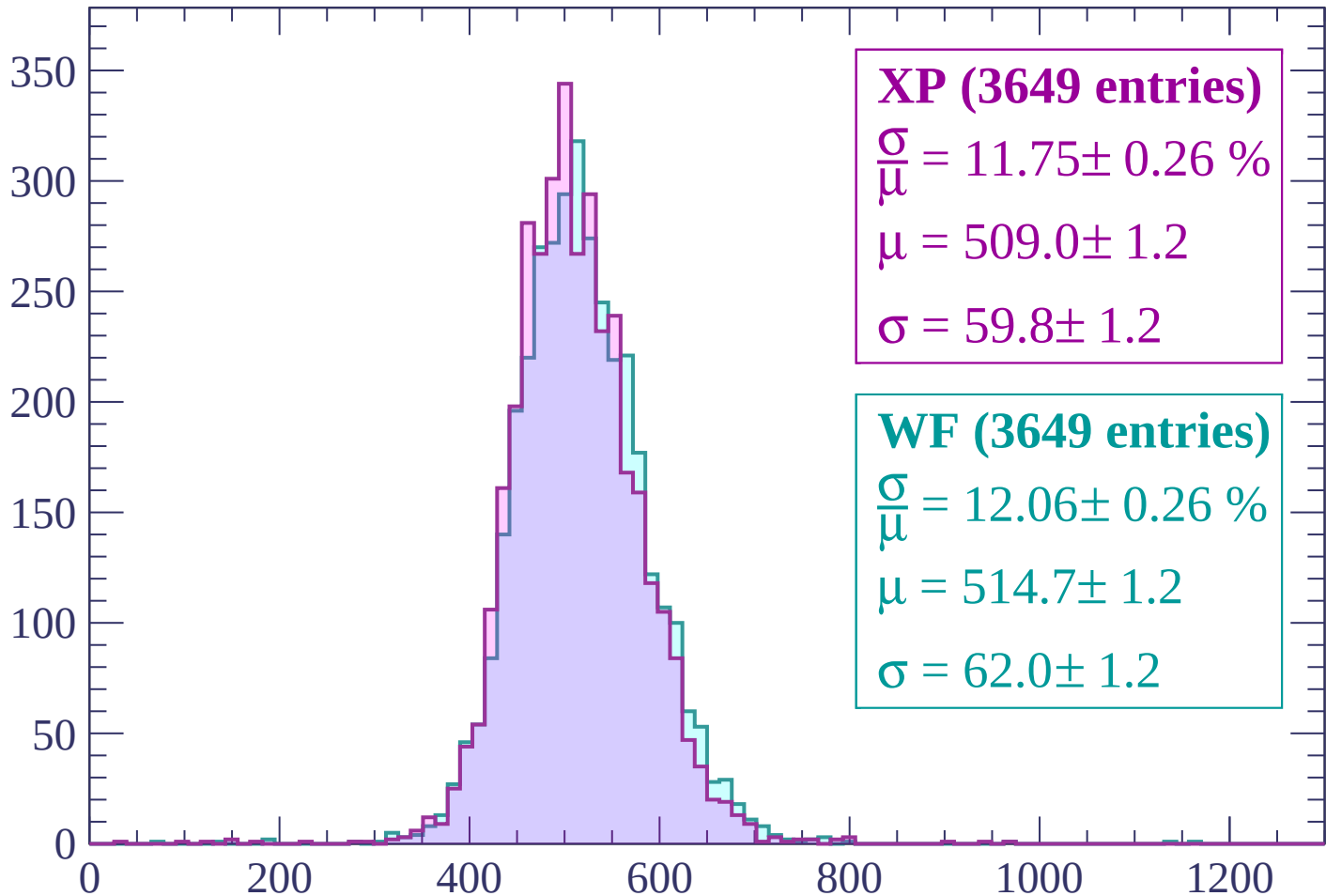
Energy loss | -2820 < p < -2760

Count



Energy loss | -2760 < p < -2700

Count



XP (3649 entries)

$$\frac{\sigma}{\mu} = 11.75 \pm 0.26 \%$$

$$\mu = 509.0 \pm 1.2$$

$$\sigma = 59.8 \pm 1.2$$

WF (3649 entries)

$$\frac{\sigma}{\mu} = 12.06 \pm 0.26 \%$$

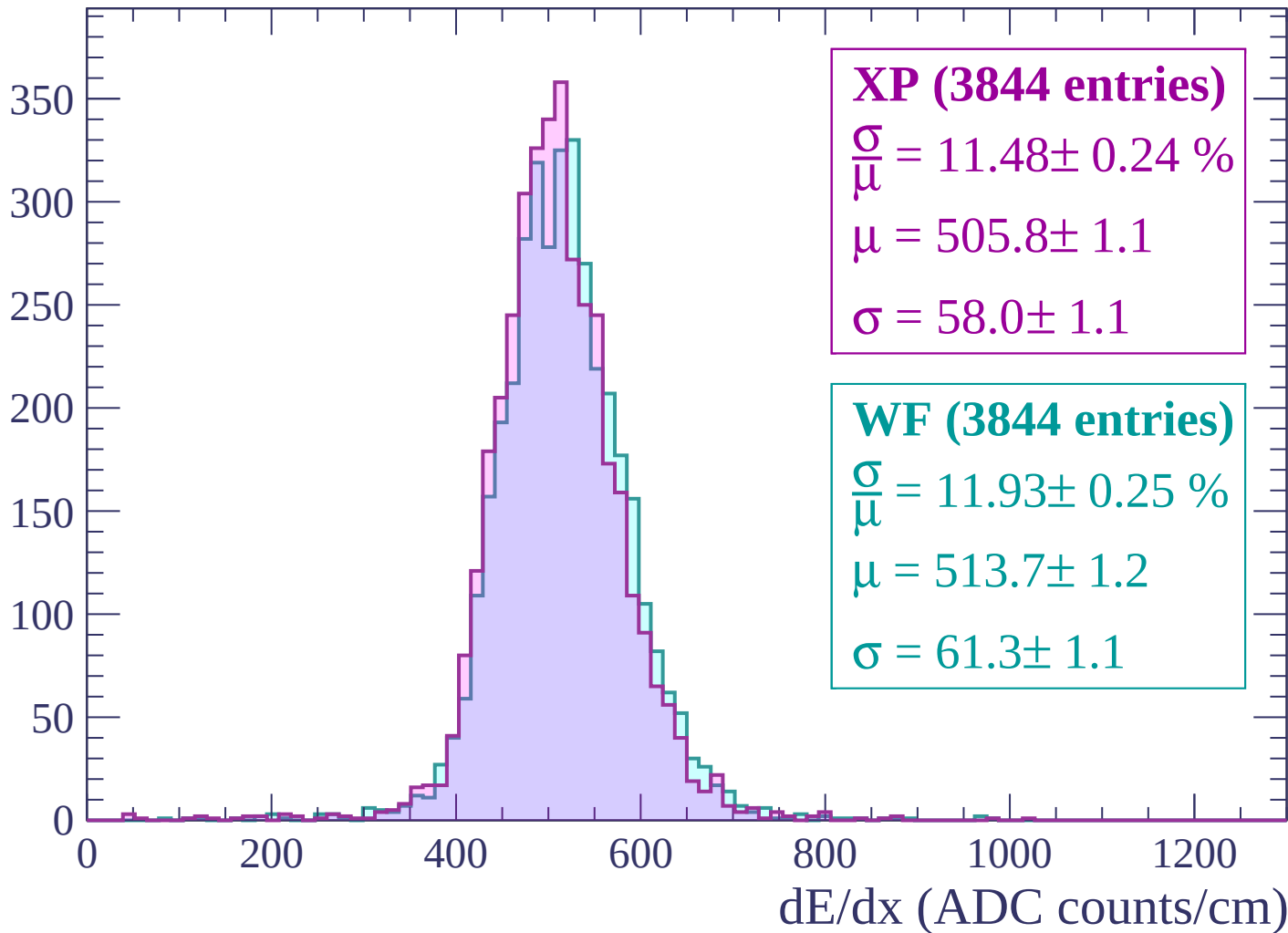
$$\mu = 514.7 \pm 1.2$$

$$\sigma = 62.0 \pm 1.2$$

dE/dx (ADC counts/cm)

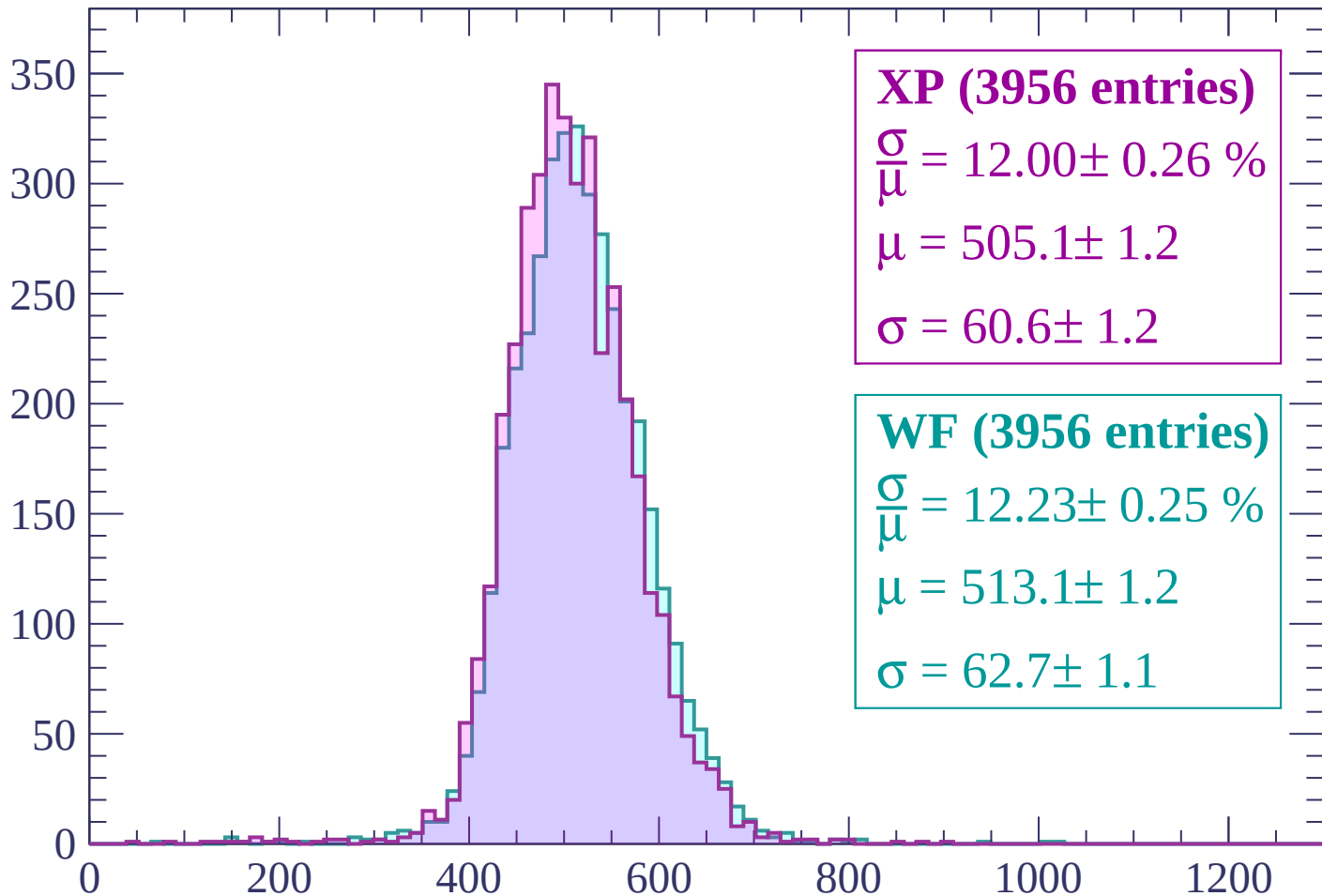
Energy loss | -2700 < p < -2640

Count



Energy loss | $-2640 < p < -2580$

Count



XP (3956 entries)

$$\frac{\sigma}{\mu} = 12.00 \pm 0.26 \%$$

$$\mu = 505.1 \pm 1.2$$

$$\sigma = 60.6 \pm 1.2$$

WF (3956 entries)

$$\frac{\sigma}{\mu} = 12.23 \pm 0.25 \%$$

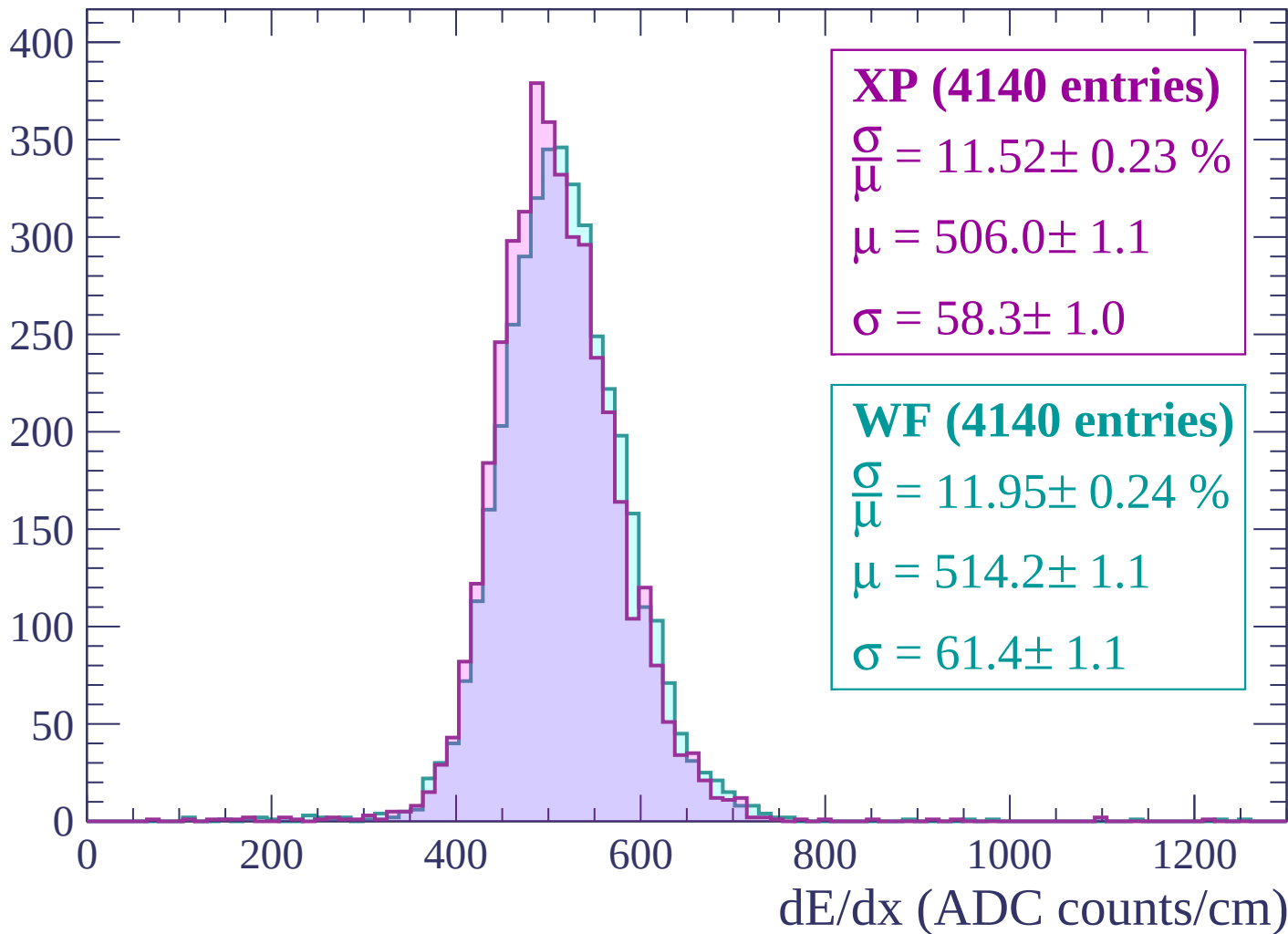
$$\mu = 513.1 \pm 1.2$$

$$\sigma = 62.7 \pm 1.1$$

dE/dx (ADC counts/cm)

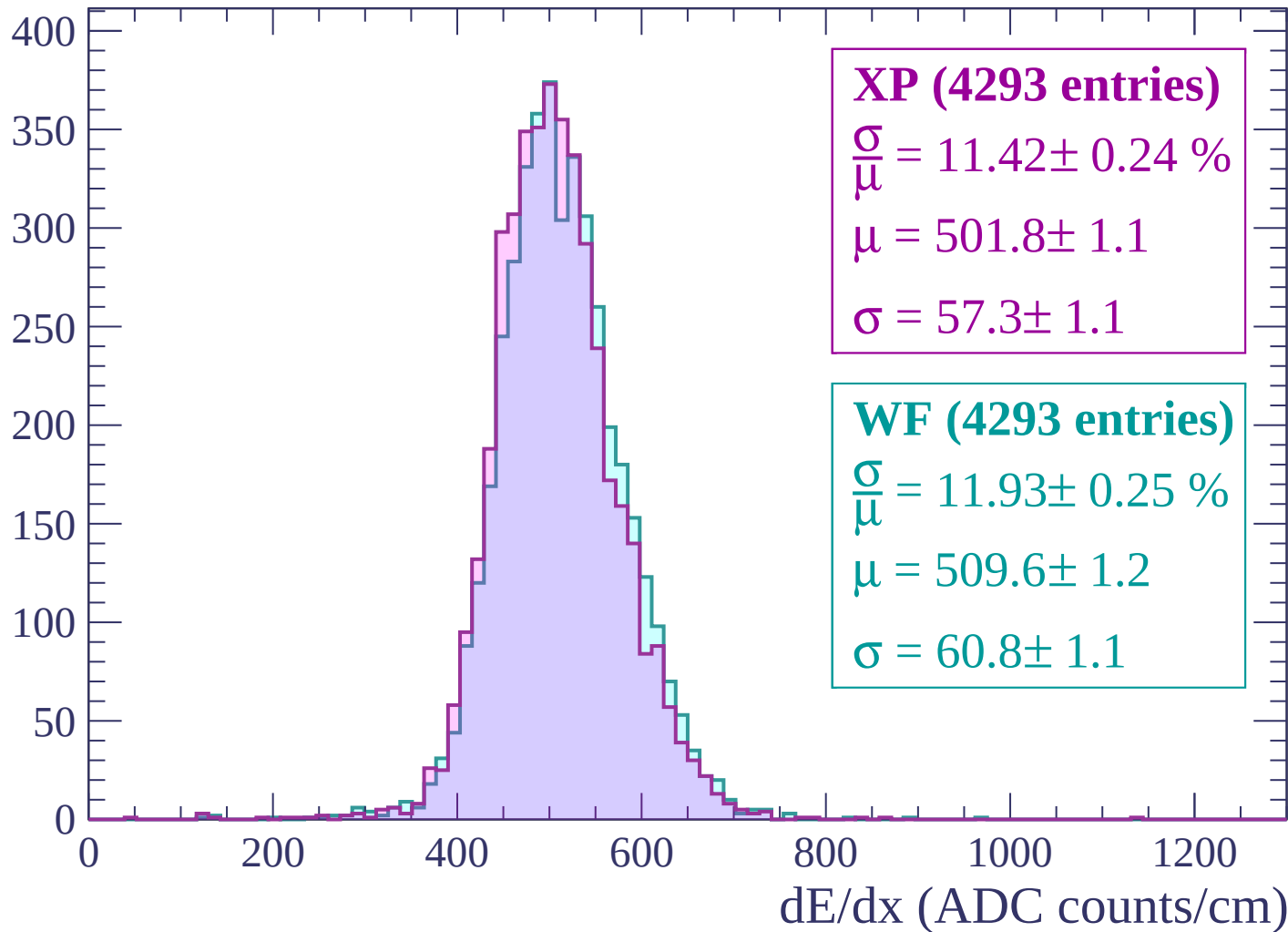
Energy loss | -2580 < p < -2520

Count



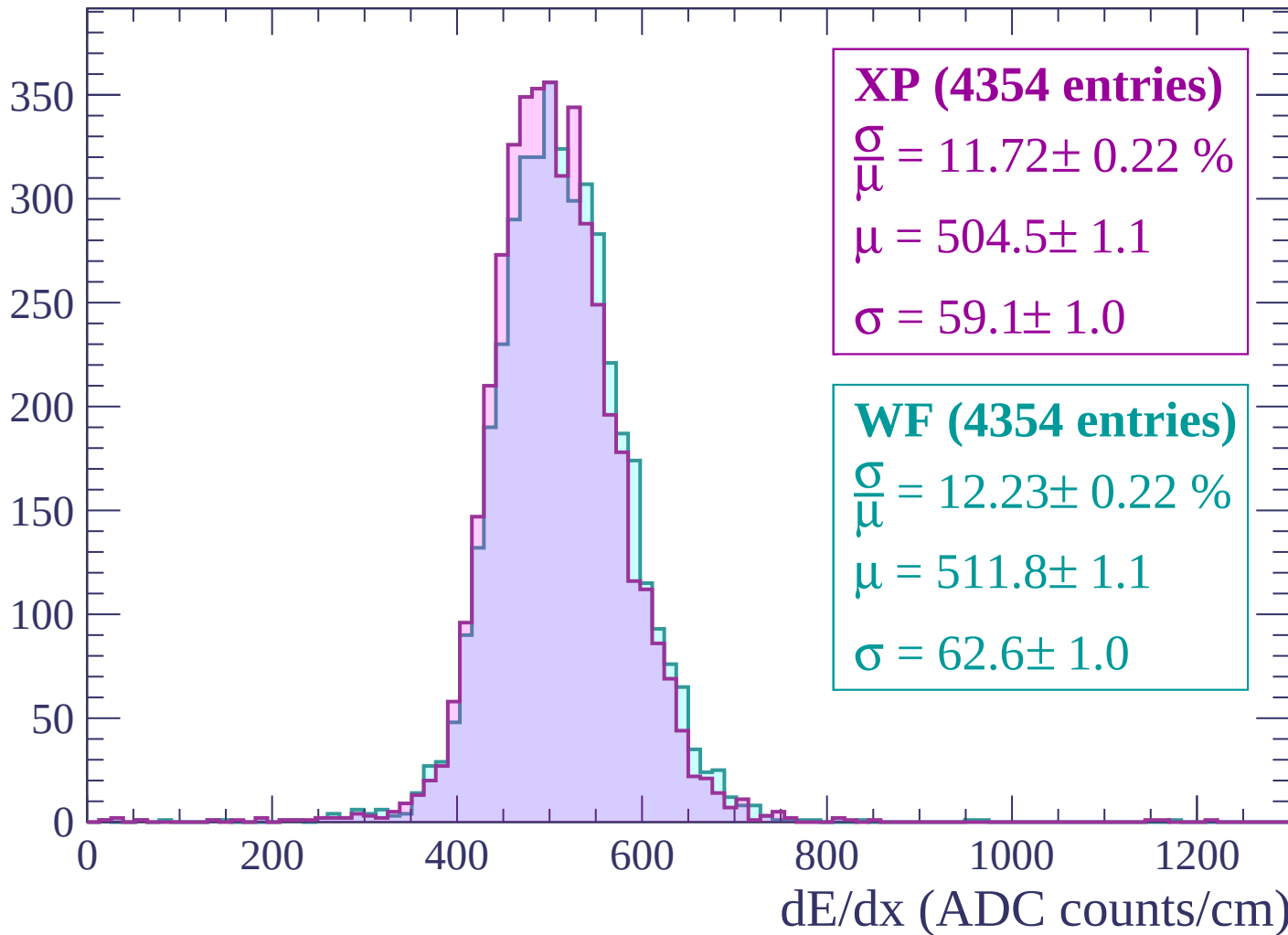
Energy loss | -2520 < p < -2460

Count



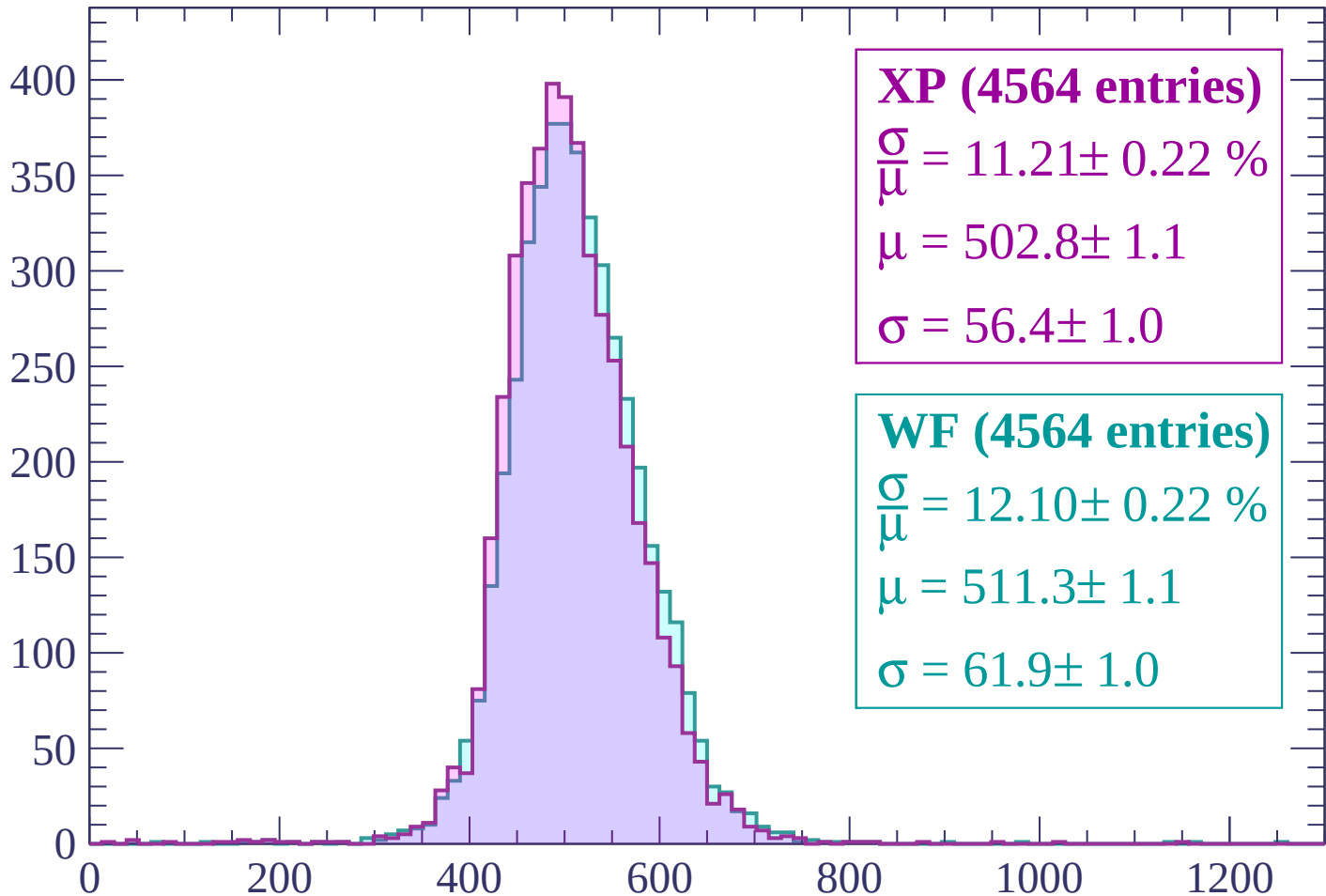
Energy loss | $-2460 < p < -2400$

Count



Energy loss | $-2400 < p < -2340$

Count



XP (4564 entries)

$\frac{\sigma}{\mu} = 11.21 \pm 0.22 \%$

$\mu = 502.8 \pm 1.1$

$\sigma = 56.4 \pm 1.0$

WF (4564 entries)

$\frac{\sigma}{\mu} = 12.10 \pm 0.22 \%$

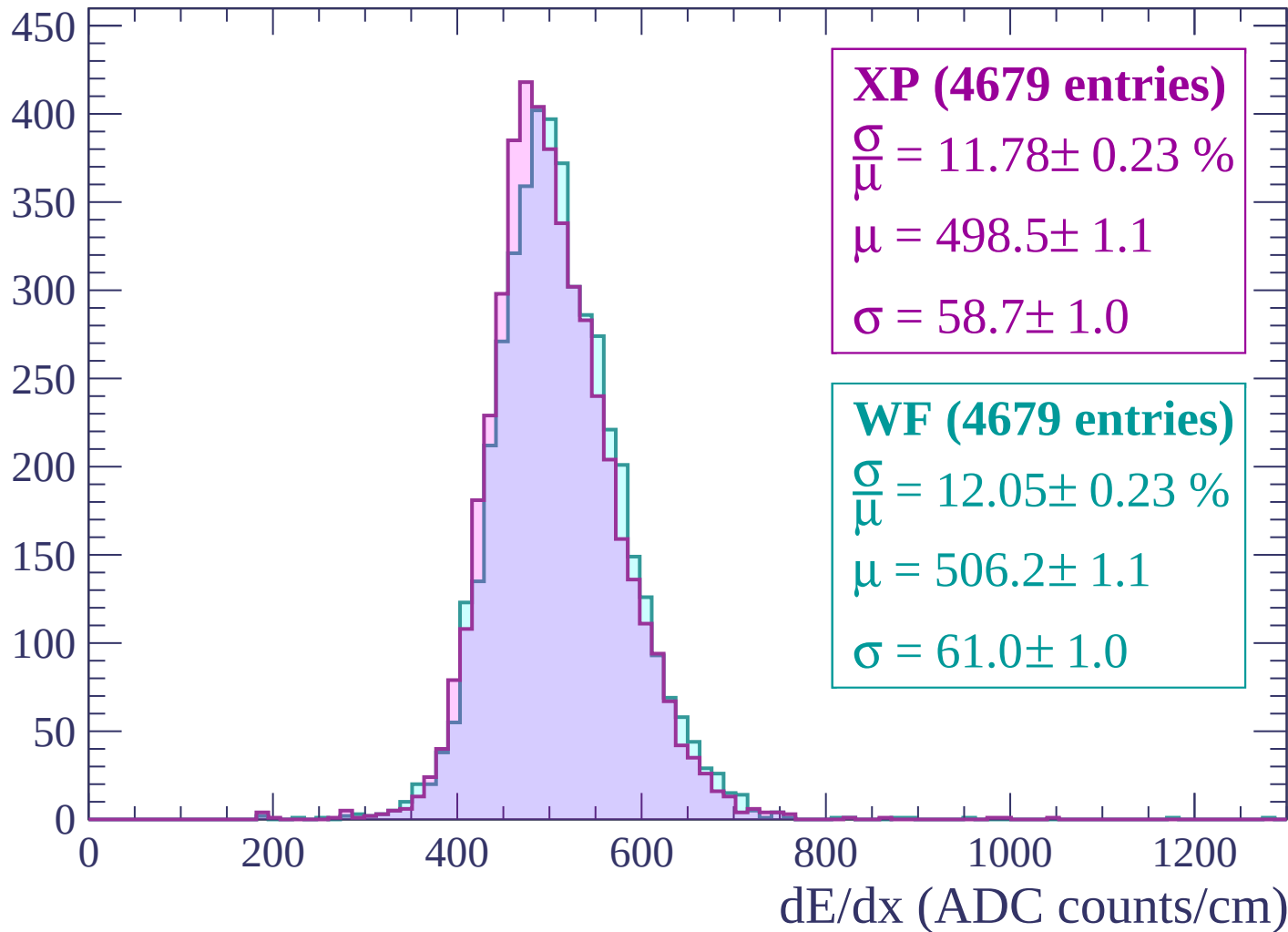
$\mu = 511.3 \pm 1.1$

$\sigma = 61.9 \pm 1.0$

dE/dx (ADC counts/cm)

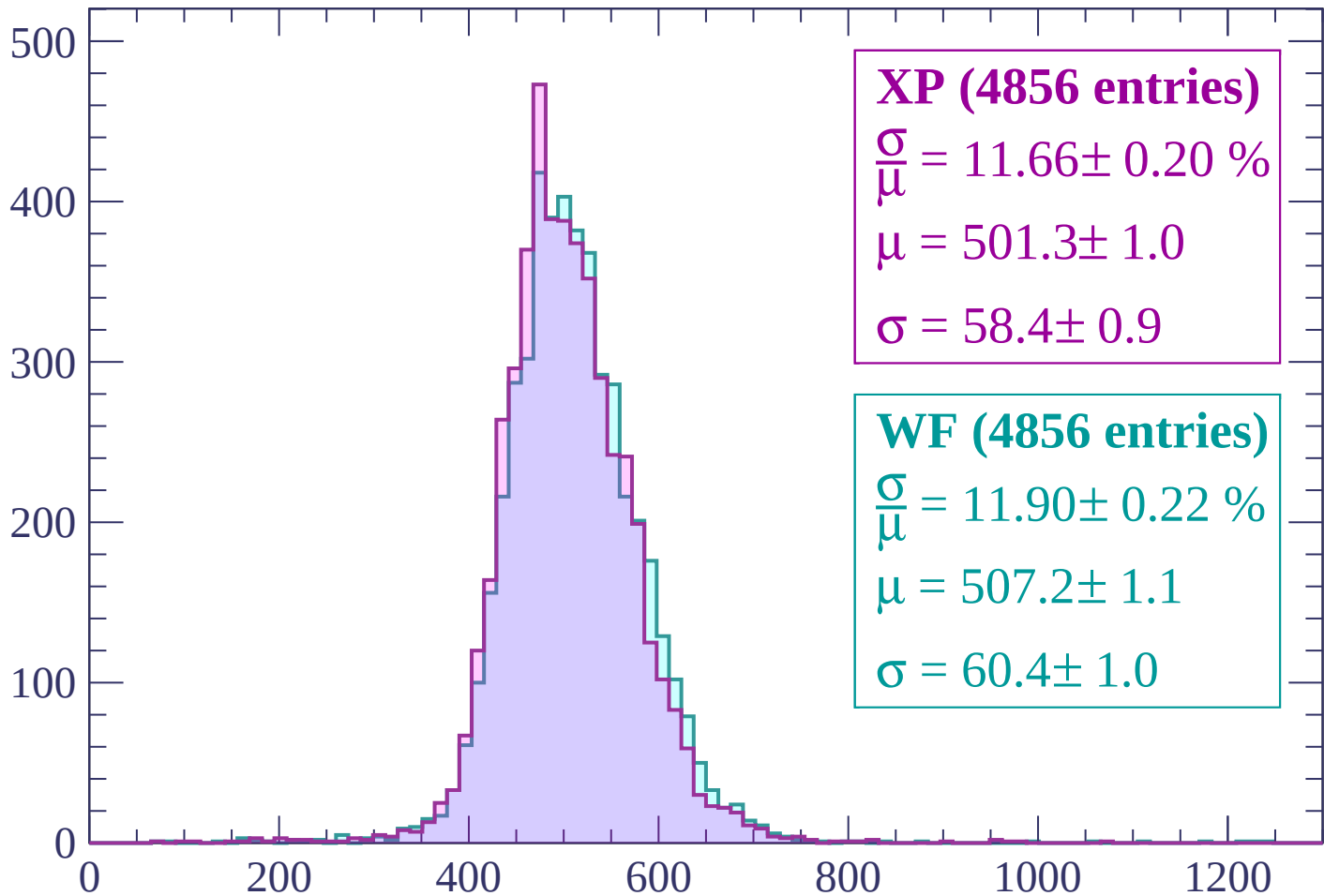
Energy loss | $-2340 < p < -2280$

Count



Energy loss | $-2280 < p < -2220$

Count



XP (4856 entries)

$\frac{\sigma}{\mu} = 11.66 \pm 0.20 \%$

$\mu = 501.3 \pm 1.0$

$\sigma = 58.4 \pm 0.9$

WF (4856 entries)

$\frac{\sigma}{\mu} = 11.90 \pm 0.22 \%$

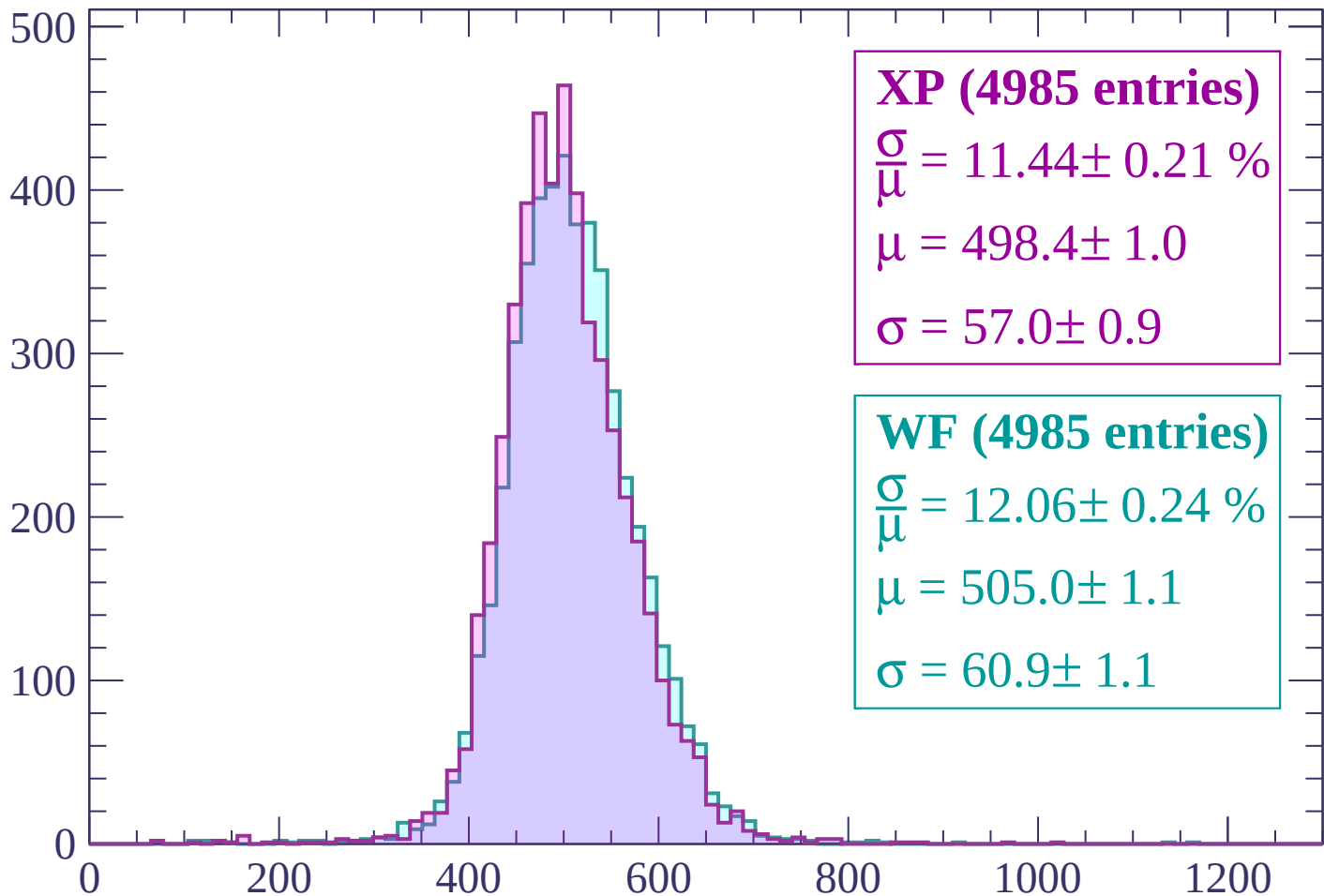
$\mu = 507.2 \pm 1.1$

$\sigma = 60.4 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160

Count



XP (4985 entries)

$$\frac{\sigma}{\mu} = 11.44 \pm 0.21 \%$$

$$\mu = 498.4 \pm 1.0$$

$$\sigma = 57.0 \pm 0.9$$

WF (4985 entries)

$$\frac{\sigma}{\mu} = 12.06 \pm 0.24 \%$$

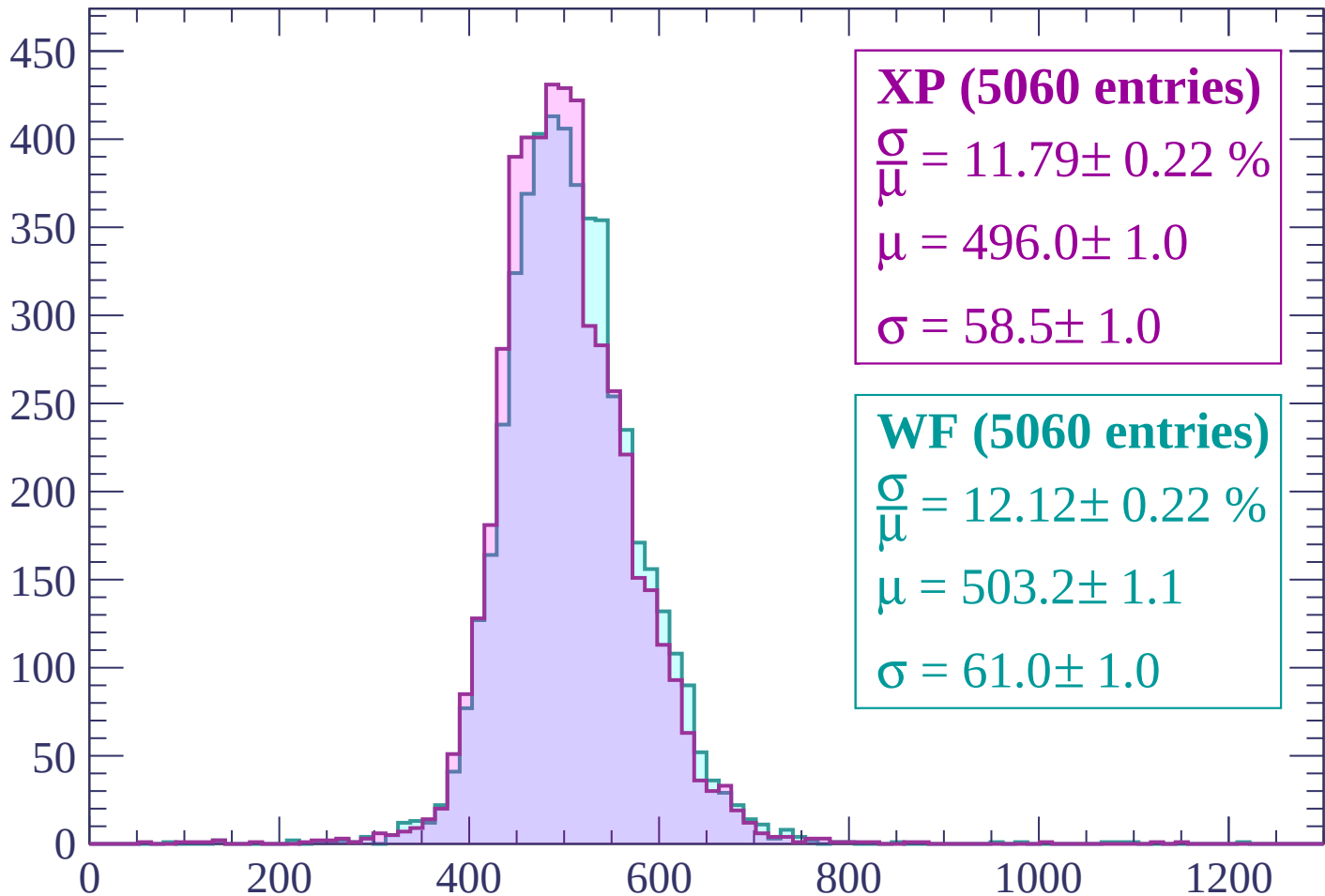
$$\mu = 505.0 \pm 1.1$$

$$\sigma = 60.9 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -2160 < p < -2100

Count



XP (5060 entries)

$$\frac{\sigma}{\mu} = 11.79 \pm 0.22 \%$$

$$\mu = 496.0 \pm 1.0$$

$$\sigma = 58.5 \pm 1.0$$

WF (5060 entries)

$$\frac{\sigma}{\mu} = 12.12 \pm 0.22 \%$$

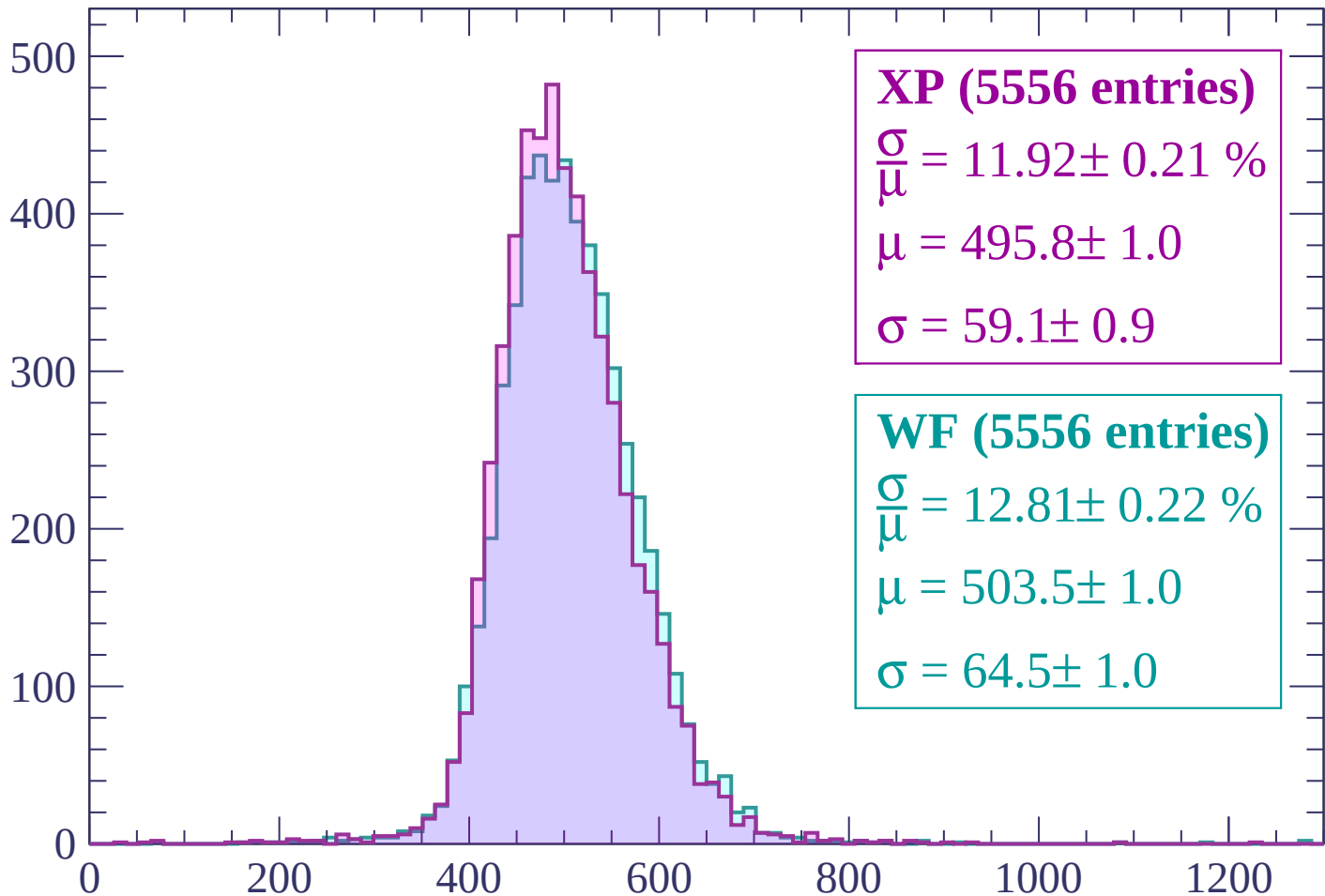
$$\mu = 503.2 \pm 1.1$$

$$\sigma = 61.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count



XP (5556 entries)

$$\frac{\sigma}{\mu} = 11.92 \pm 0.21 \%$$

$$\mu = 495.8 \pm 1.0$$

$$\sigma = 59.1 \pm 0.9$$

WF (5556 entries)

$$\frac{\sigma}{\mu} = 12.81 \pm 0.22 \%$$

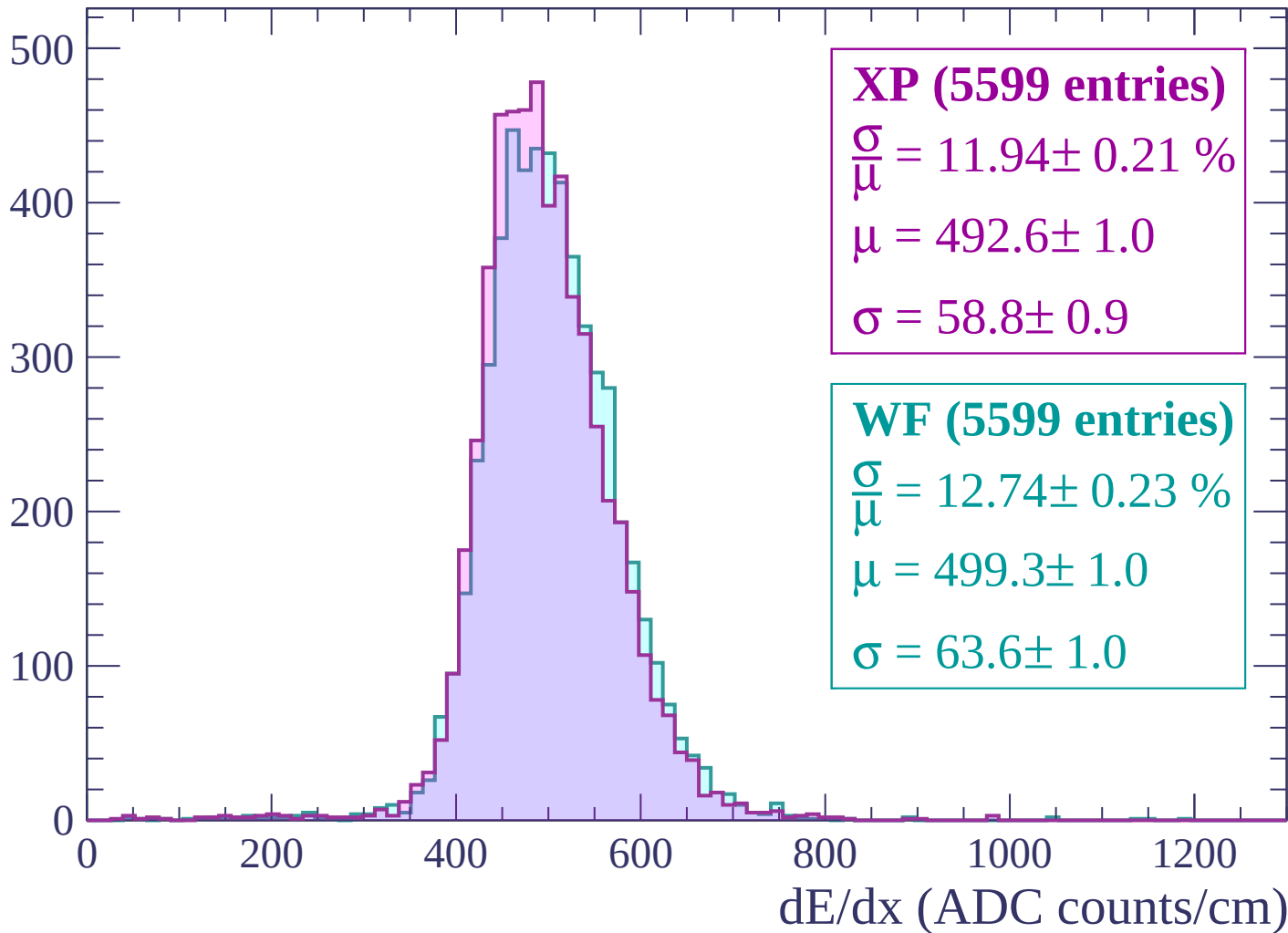
$$\mu = 503.5 \pm 1.0$$

$$\sigma = 64.5 \pm 1.0$$

dE/dx (ADC counts/cm)

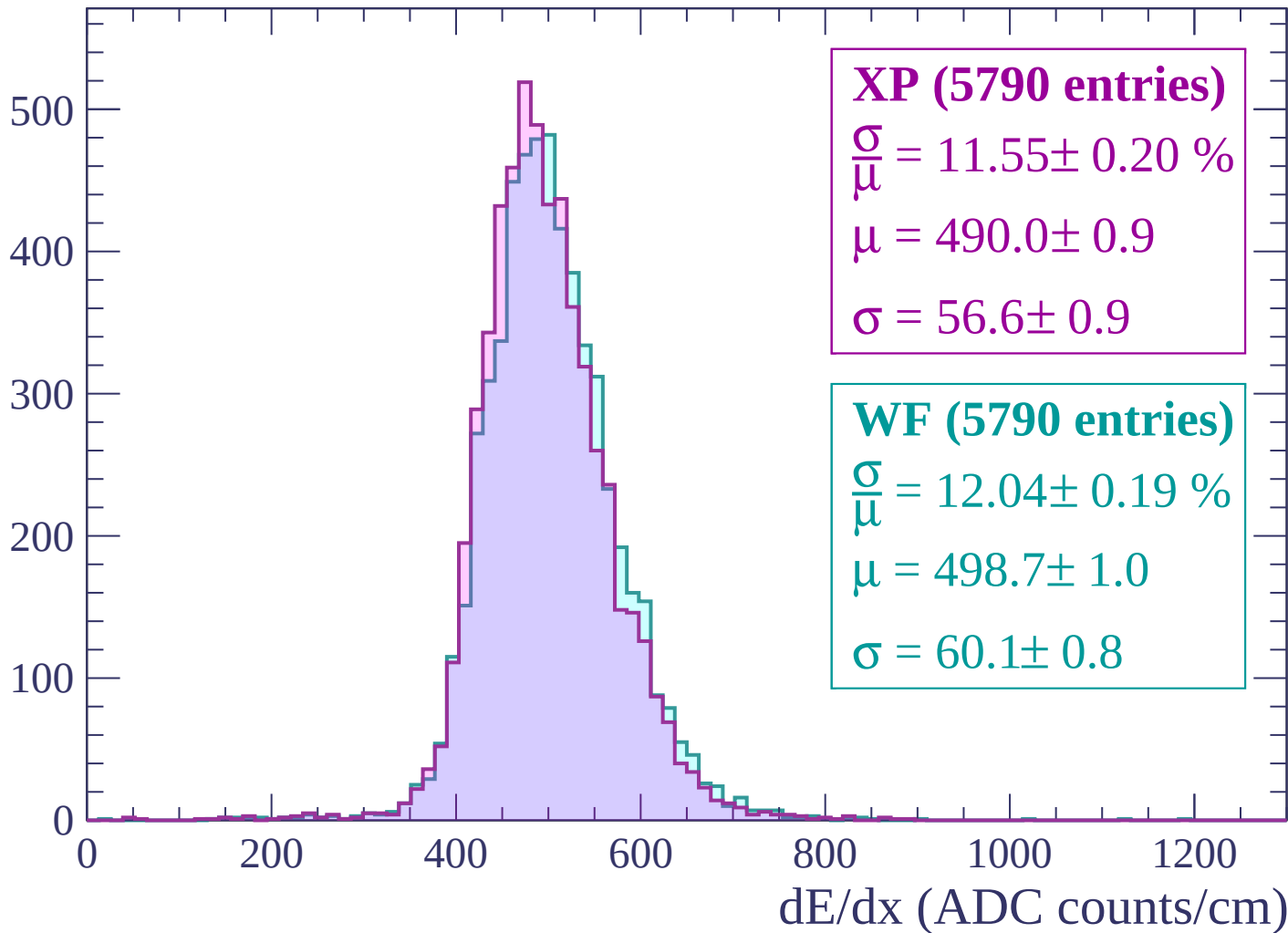
Energy loss | -2040 < p < -1980

Count



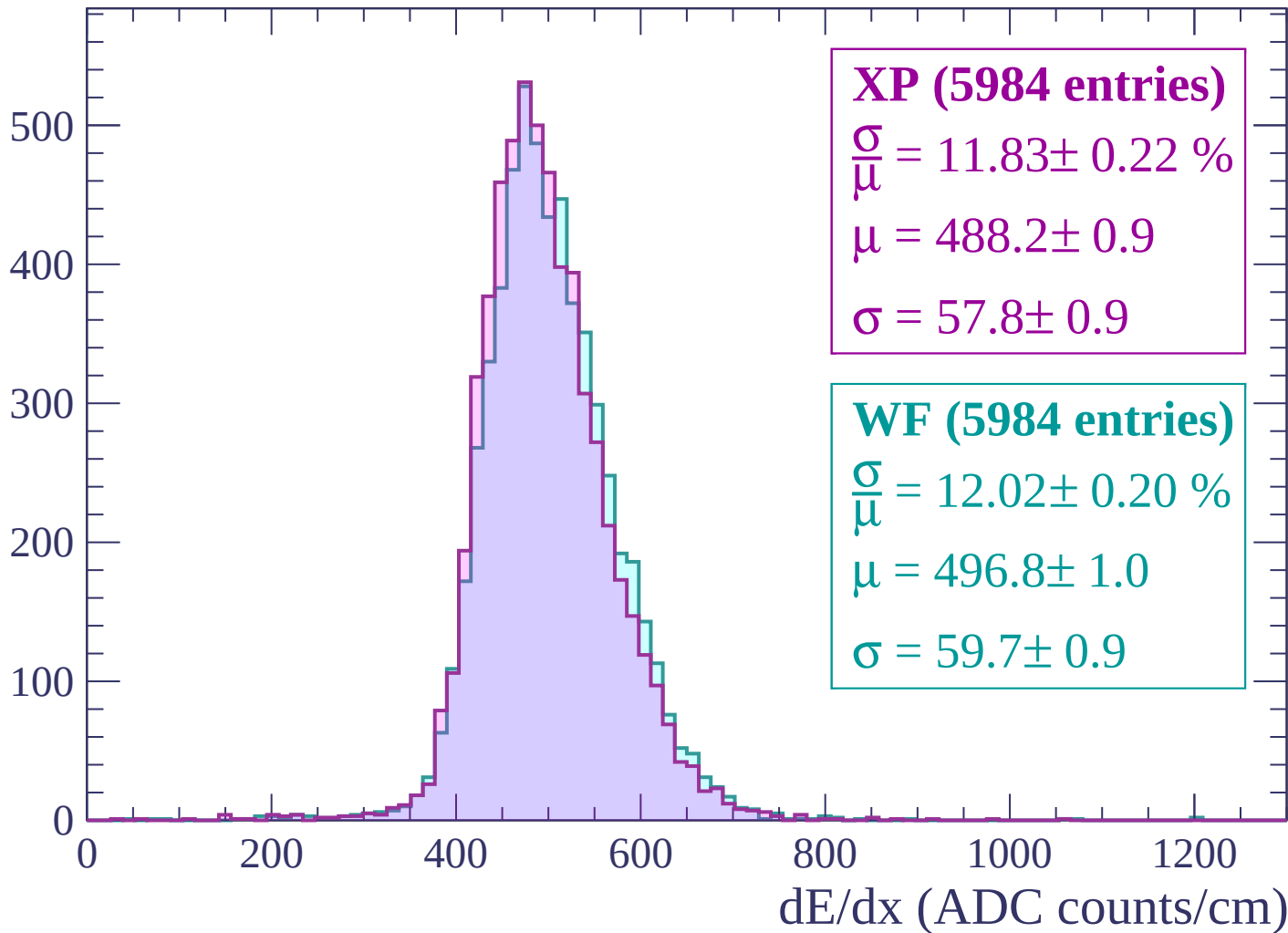
Energy loss | $-1980 < p < -1920$

Count



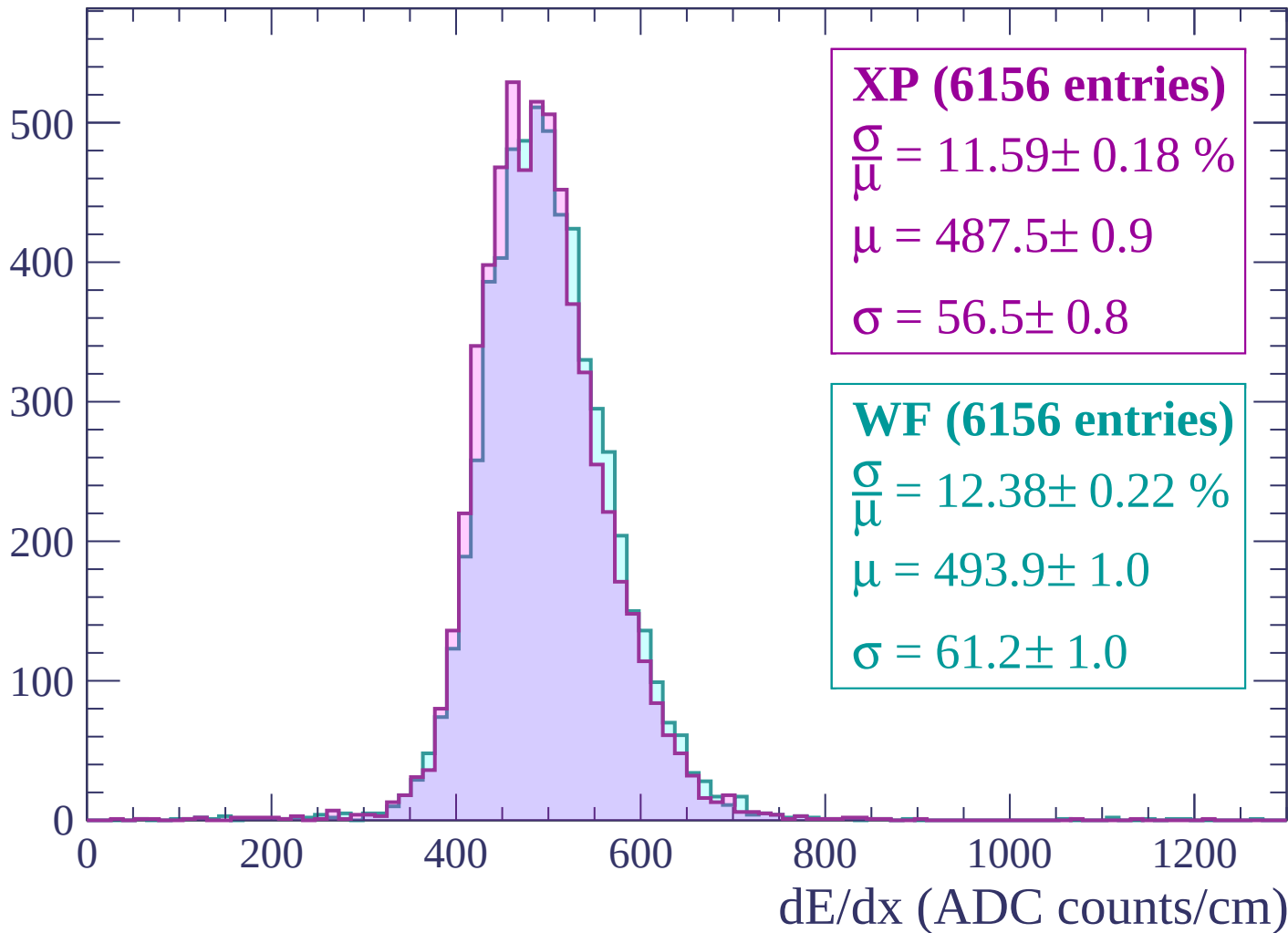
Energy loss | $-1920 < p < -1860$

Count

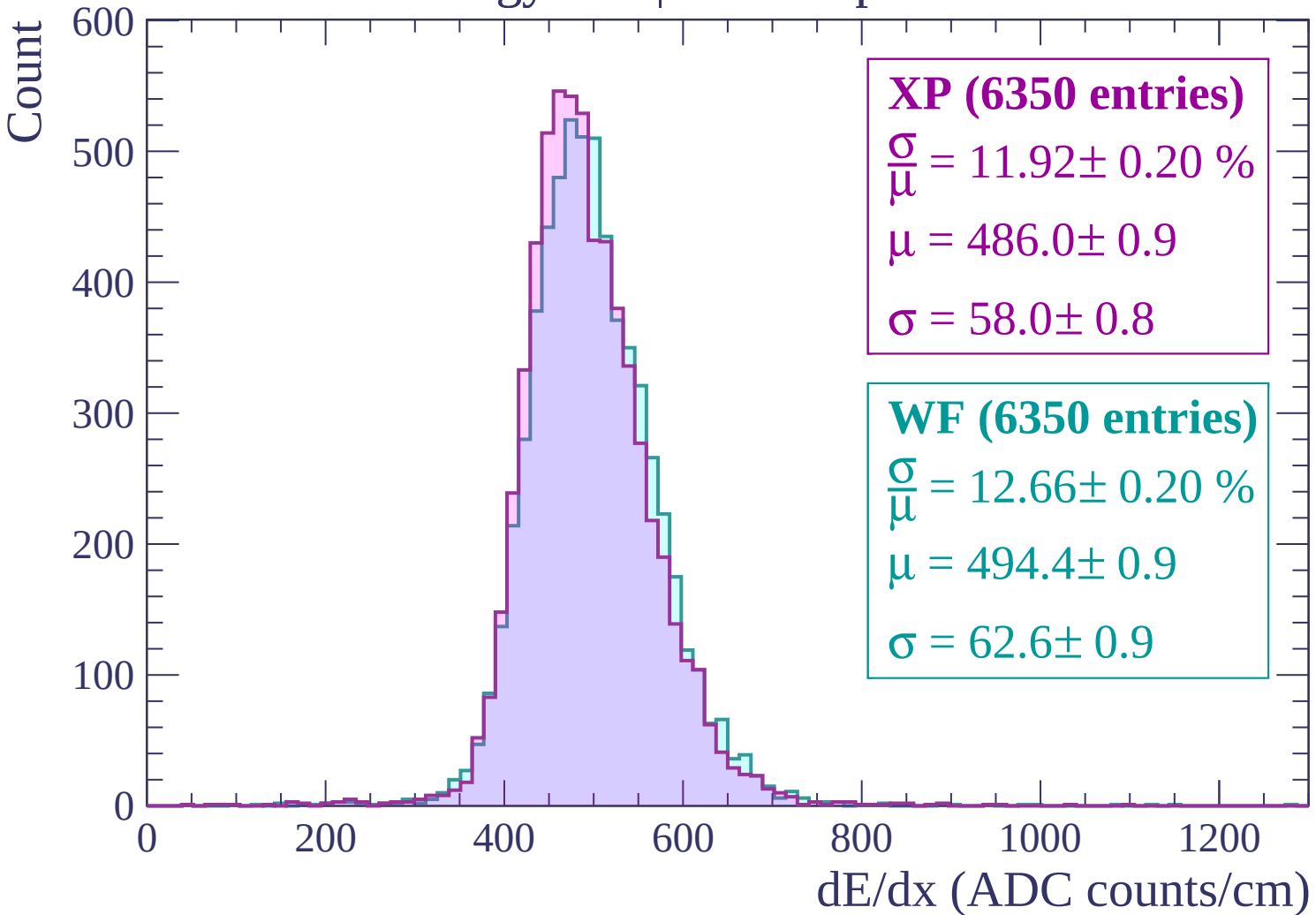


Energy loss | $-1860 < p < -1800$

Count

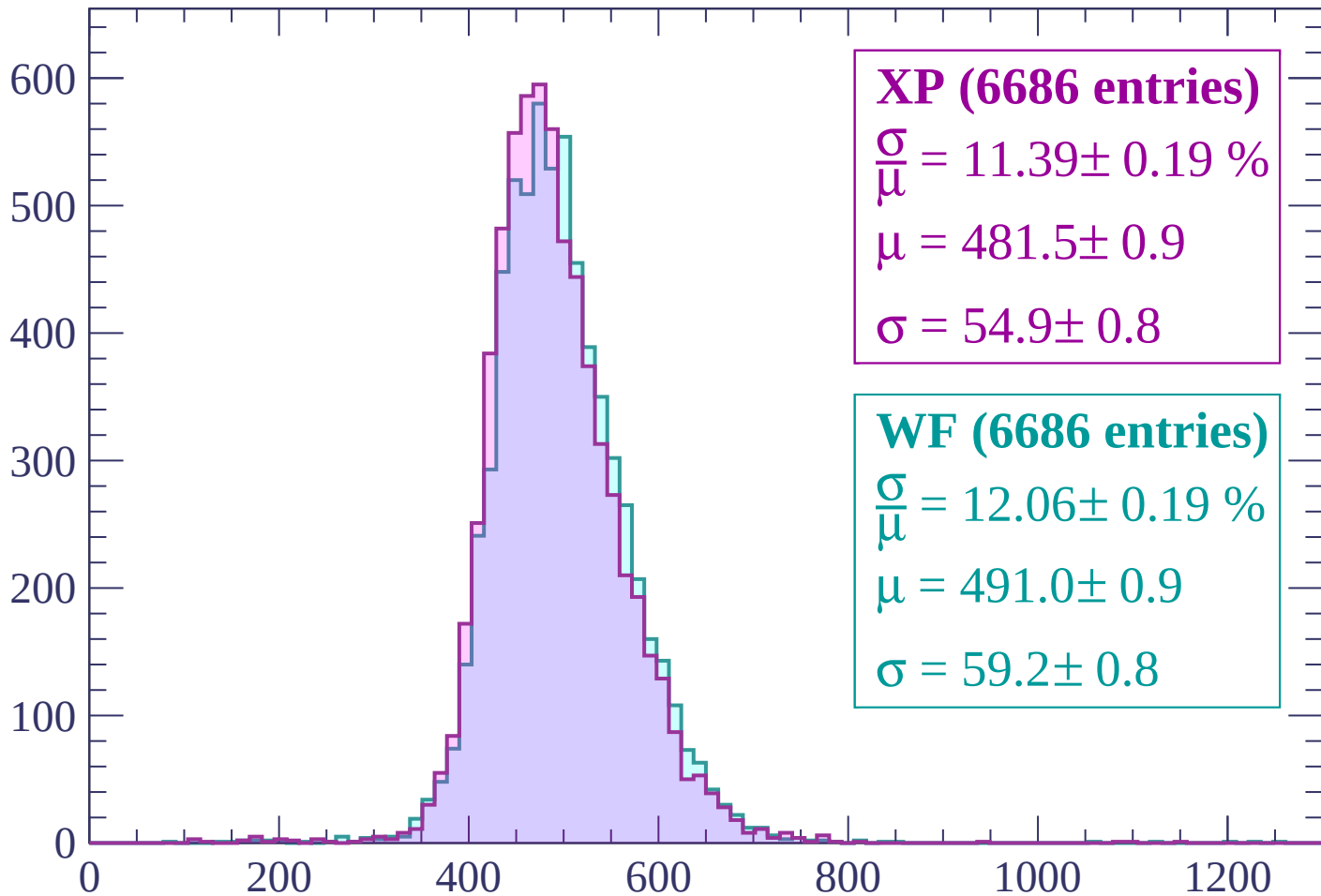


Energy loss | $-1800 < p < -1740$



Energy loss | $-1740 < p < -1680$

Count



XP (6686 entries)

$$\frac{\sigma}{\mu} = 11.39 \pm 0.19 \%$$

$$\mu = 481.5 \pm 0.9$$

$$\sigma = 54.9 \pm 0.8$$

WF (6686 entries)

$$\frac{\sigma}{\mu} = 12.06 \pm 0.19 \%$$

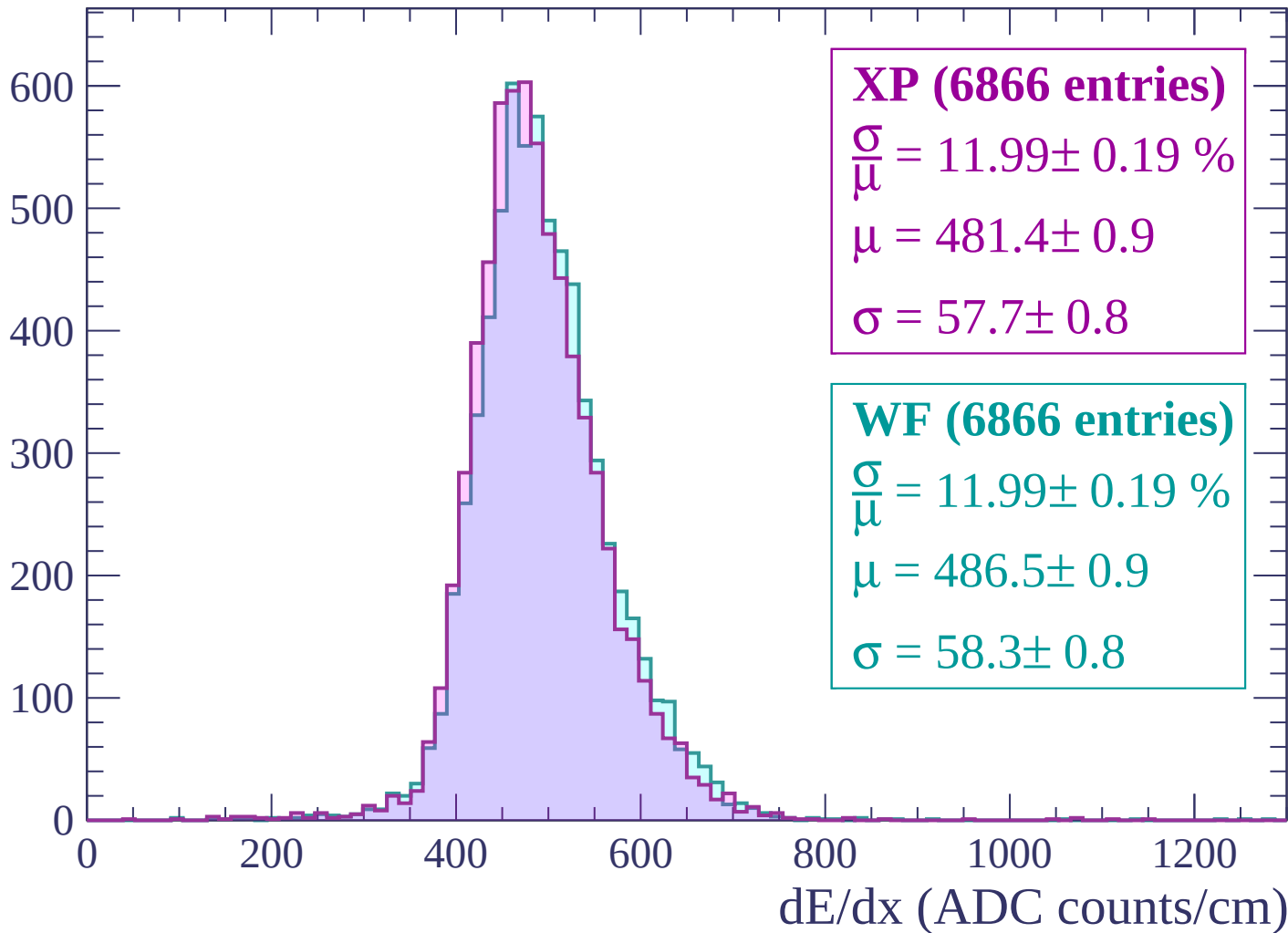
$$\mu = 491.0 \pm 0.9$$

$$\sigma = 59.2 \pm 0.8$$

dE/dx (ADC counts/cm)

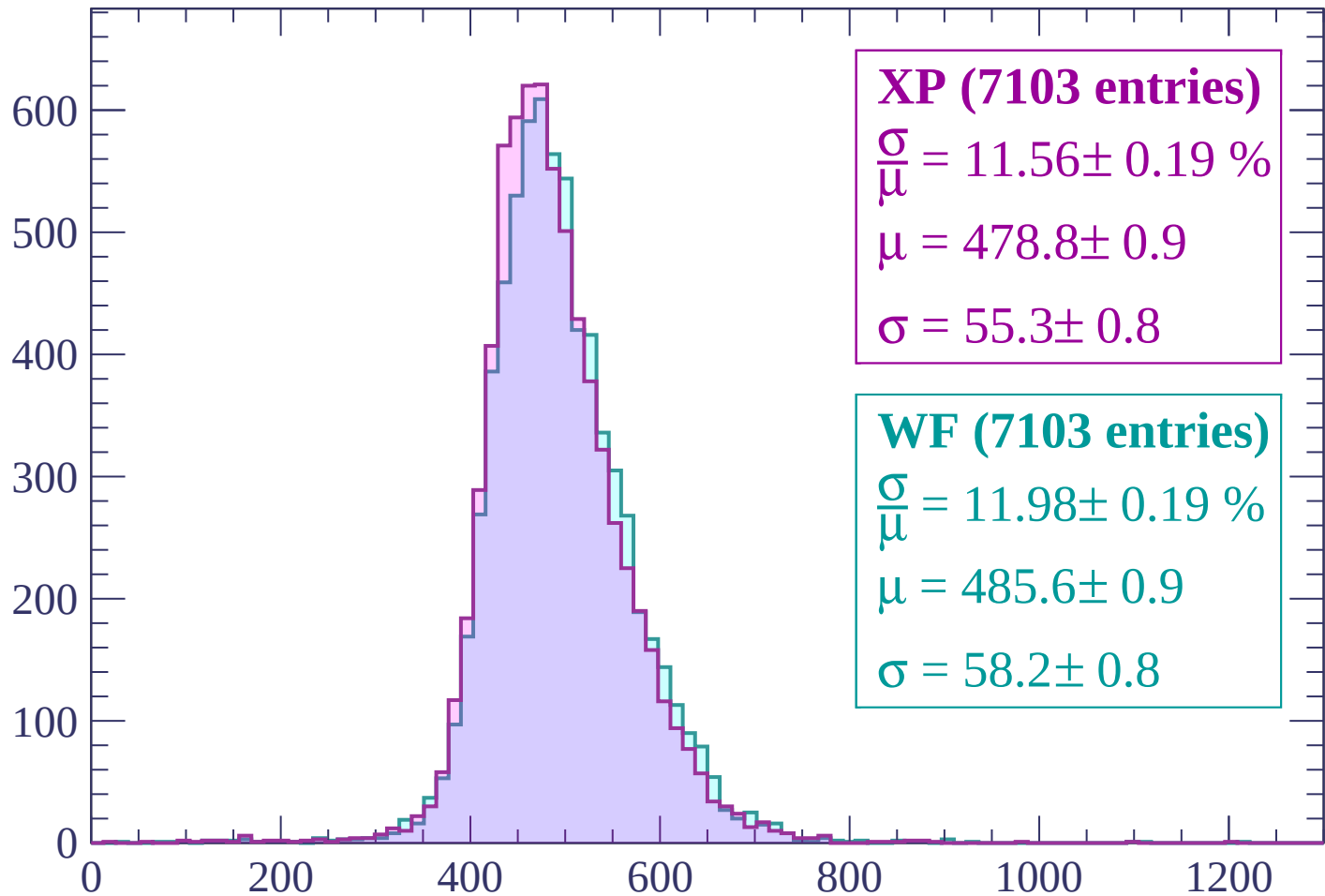
Energy loss | $-1680 < p < -1620$

Count



Energy loss | $-1620 < p < -1560$

Count



XP (7103 entries)

$$\frac{\sigma}{\mu} = 11.56 \pm 0.19 \%$$

$$\mu = 478.8 \pm 0.9$$

$$\sigma = 55.3 \pm 0.8$$

WF (7103 entries)

$$\frac{\sigma}{\mu} = 11.98 \pm 0.19 \%$$

$$\mu = 485.6 \pm 0.9$$

$$\sigma = 58.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count

700

600

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (7357 entries)

$$\frac{\sigma}{\mu} = 11.66 \pm 0.18 \%$$

$$\mu = 476.7 \pm 0.8$$

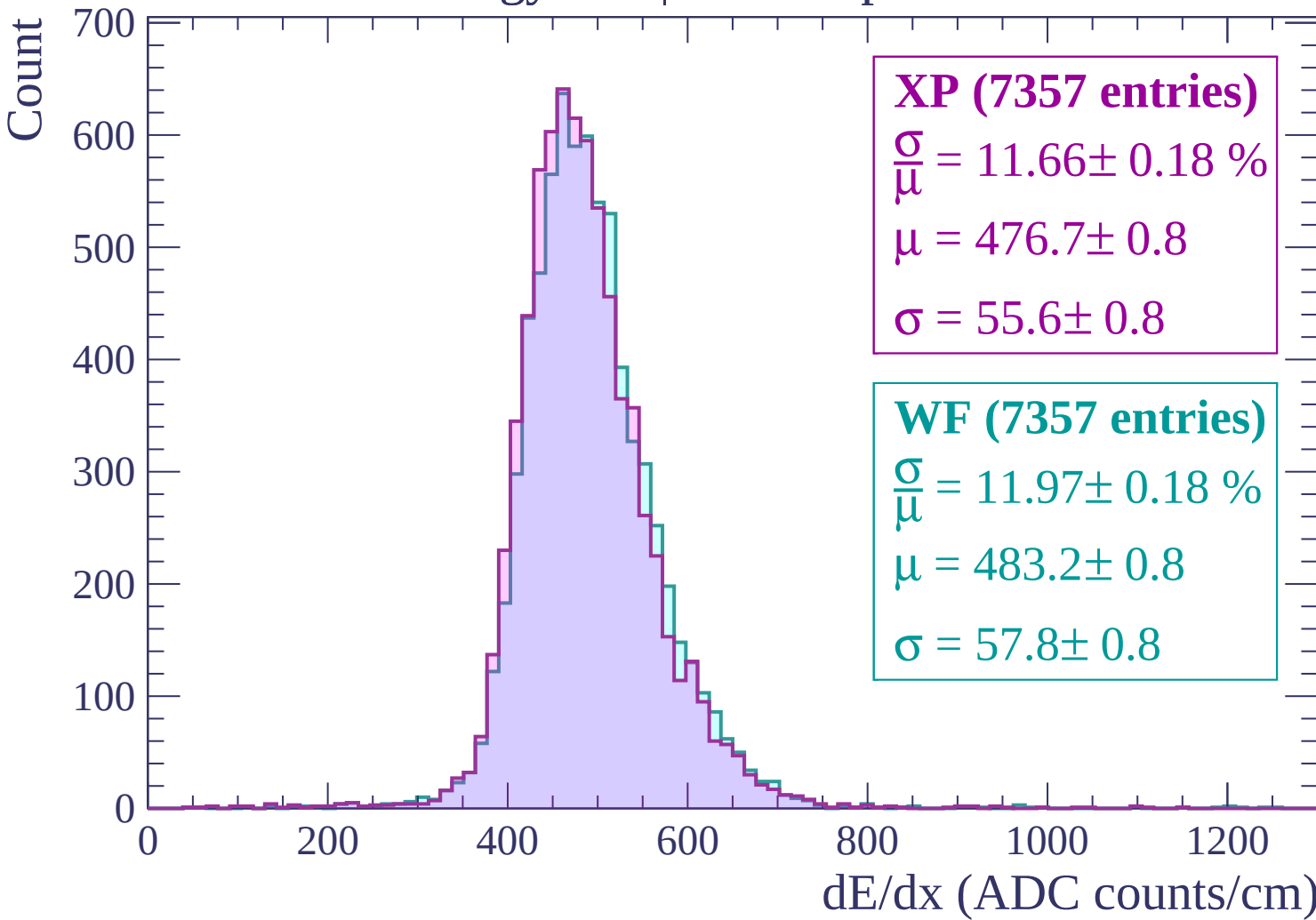
$$\sigma = 55.6 \pm 0.8$$

WF (7357 entries)

$$\frac{\sigma}{\mu} = 11.97 \pm 0.18 \%$$

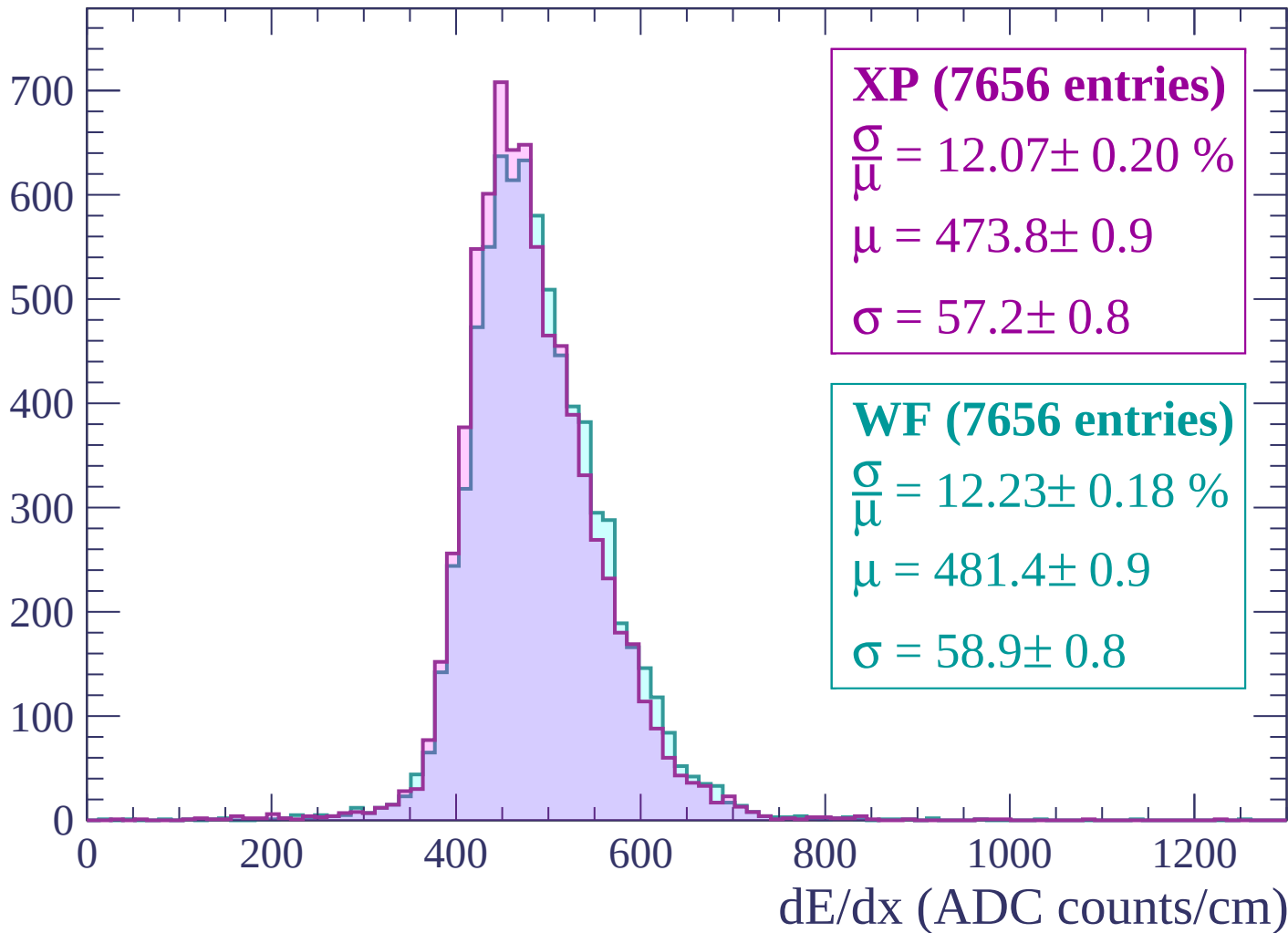
$$\mu = 483.2 \pm 0.8$$

$$\sigma = 57.8 \pm 0.8$$



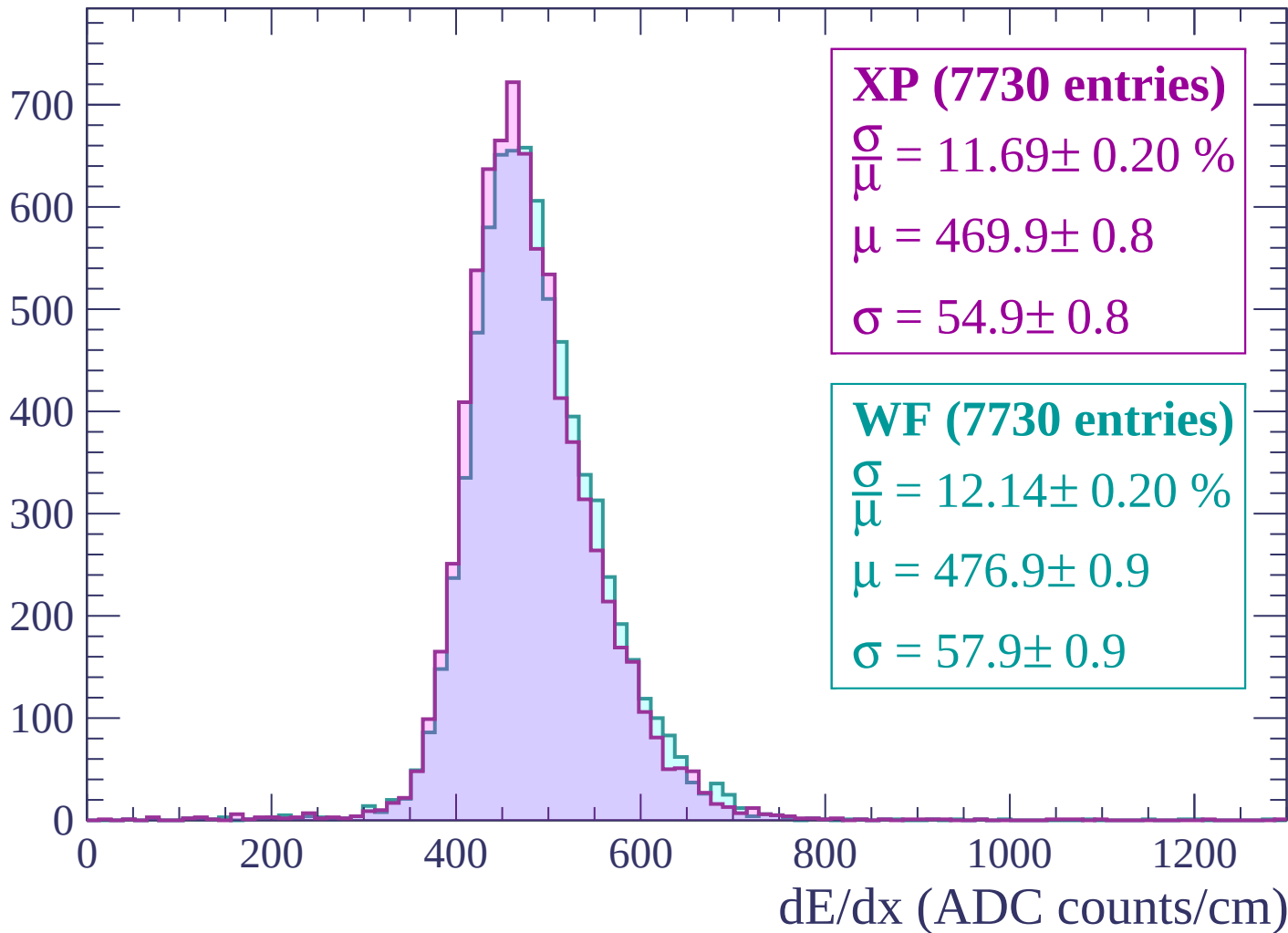
Energy loss | $-1500 < p < -1440$

Count



Energy loss | $-1440 < p < -1380$

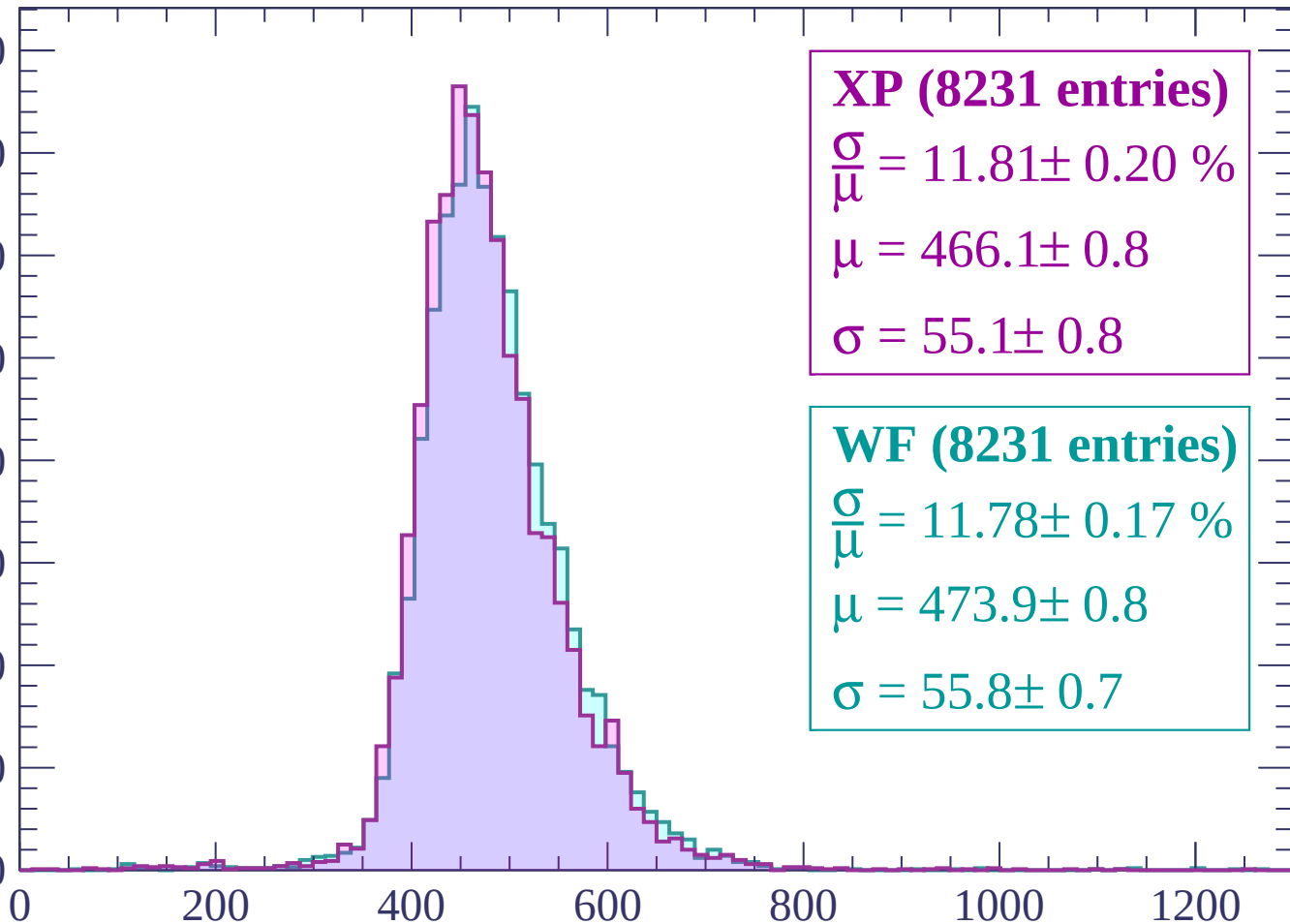
Count



Energy loss | $-1380 < p < -1320$

Count

800
700
600
500
400
300
200
100
0



XP (8231 entries)

$\frac{\sigma}{\mu} = 11.81 \pm 0.20 \%$

$\mu = 466.1 \pm 0.8$

$\sigma = 55.1 \pm 0.8$

WF (8231 entries)

$\frac{\sigma}{\mu} = 11.78 \pm 0.17 \%$

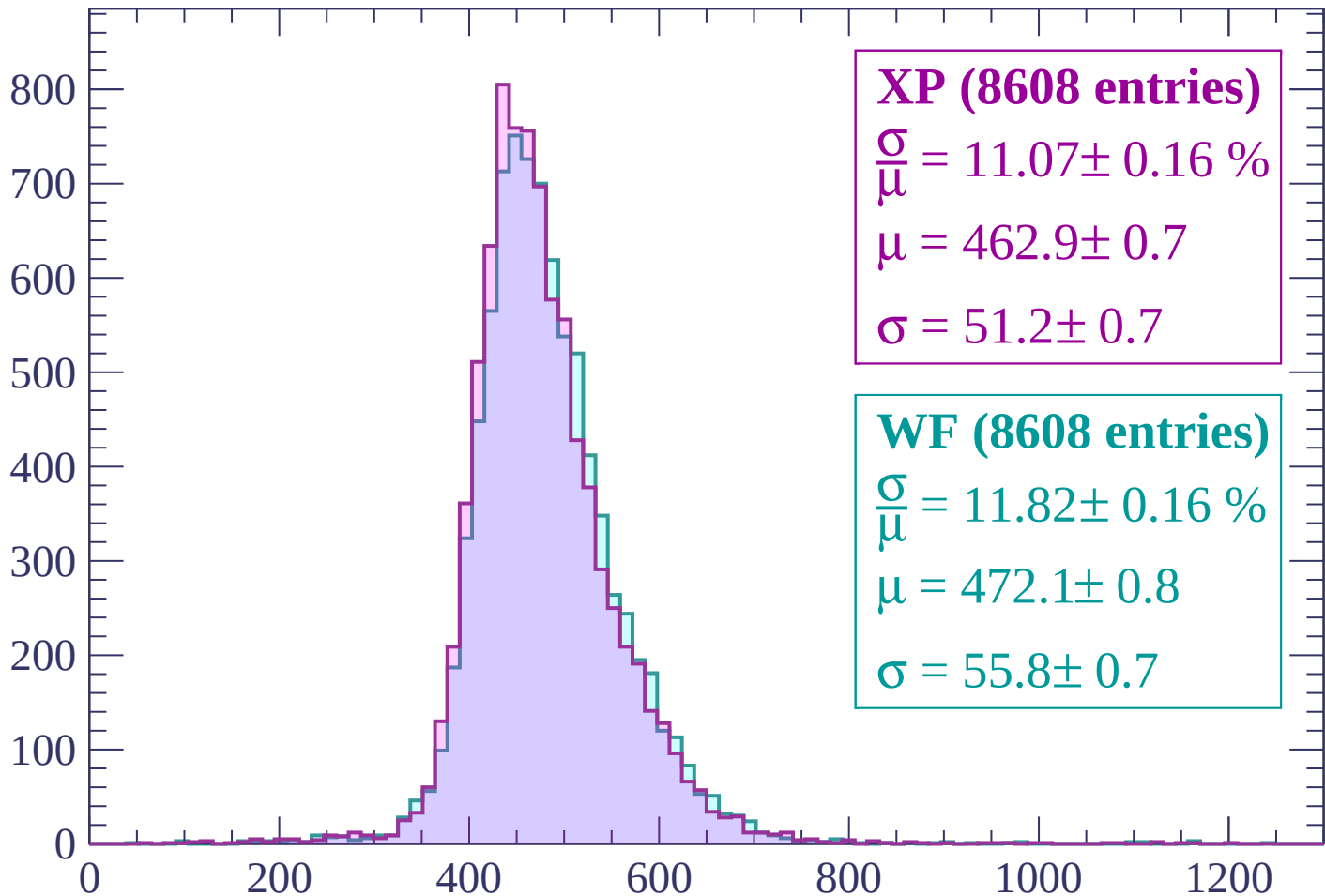
$\mu = 473.9 \pm 0.8$

$\sigma = 55.8 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1260$

Count



XP (8608 entries)

$$\frac{\sigma}{\mu} = 11.07 \pm 0.16 \%$$

$$\mu = 462.9 \pm 0.7$$

$$\sigma = 51.2 \pm 0.7$$

WF (8608 entries)

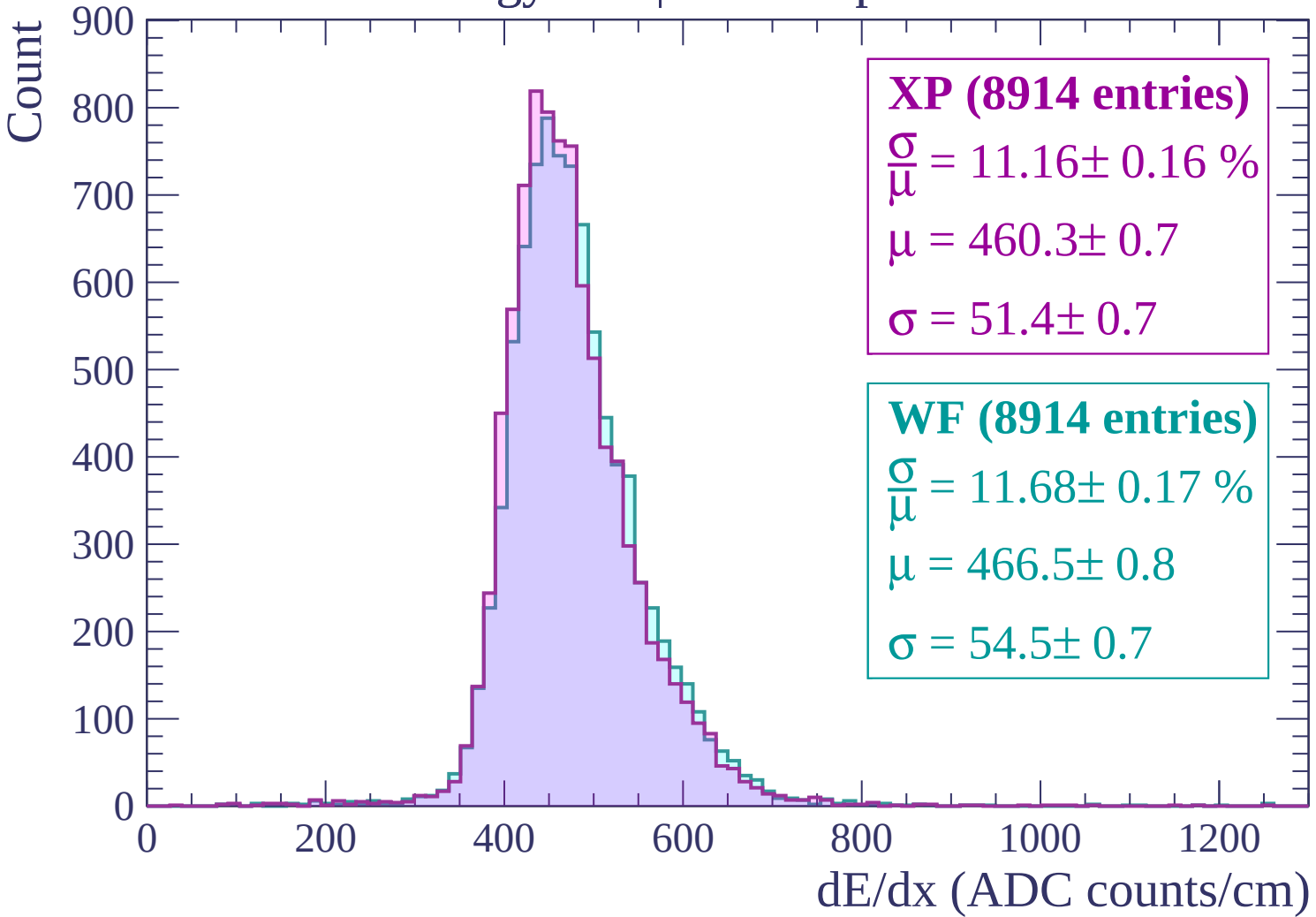
$$\frac{\sigma}{\mu} = 11.82 \pm 0.16 \%$$

$$\mu = 472.1 \pm 0.8$$

$$\sigma = 55.8 \pm 0.7$$

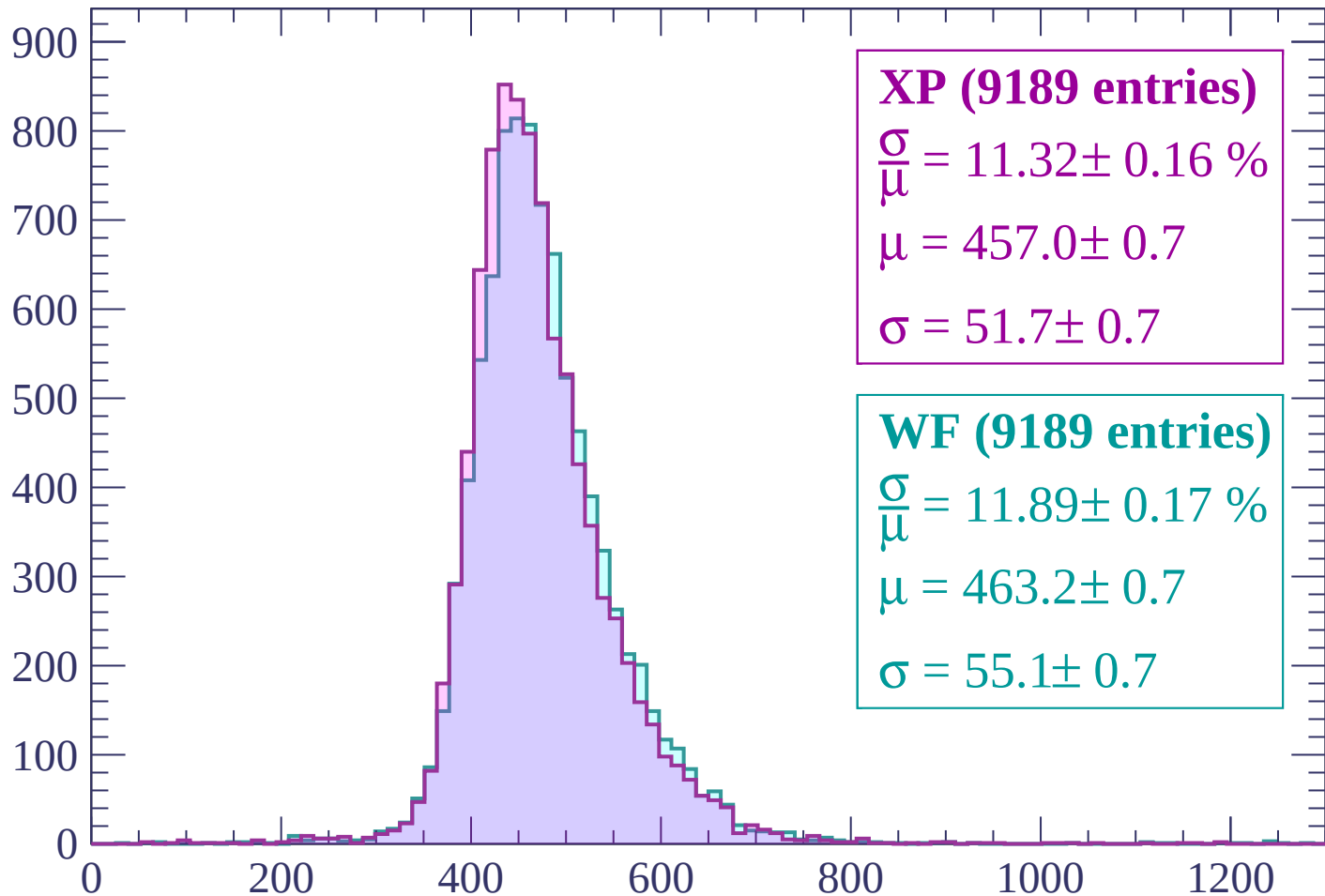
dE/dx (ADC counts/cm)

Energy loss | $-1260 < p < -1200$



Energy loss | -1200 < p < -1140

Count



XP (9189 entries)

$$\frac{\sigma}{\mu} = 11.32 \pm 0.16 \%$$

$$\mu = 457.0 \pm 0.7$$

$$\sigma = 51.7 \pm 0.7$$

WF (9189 entries)

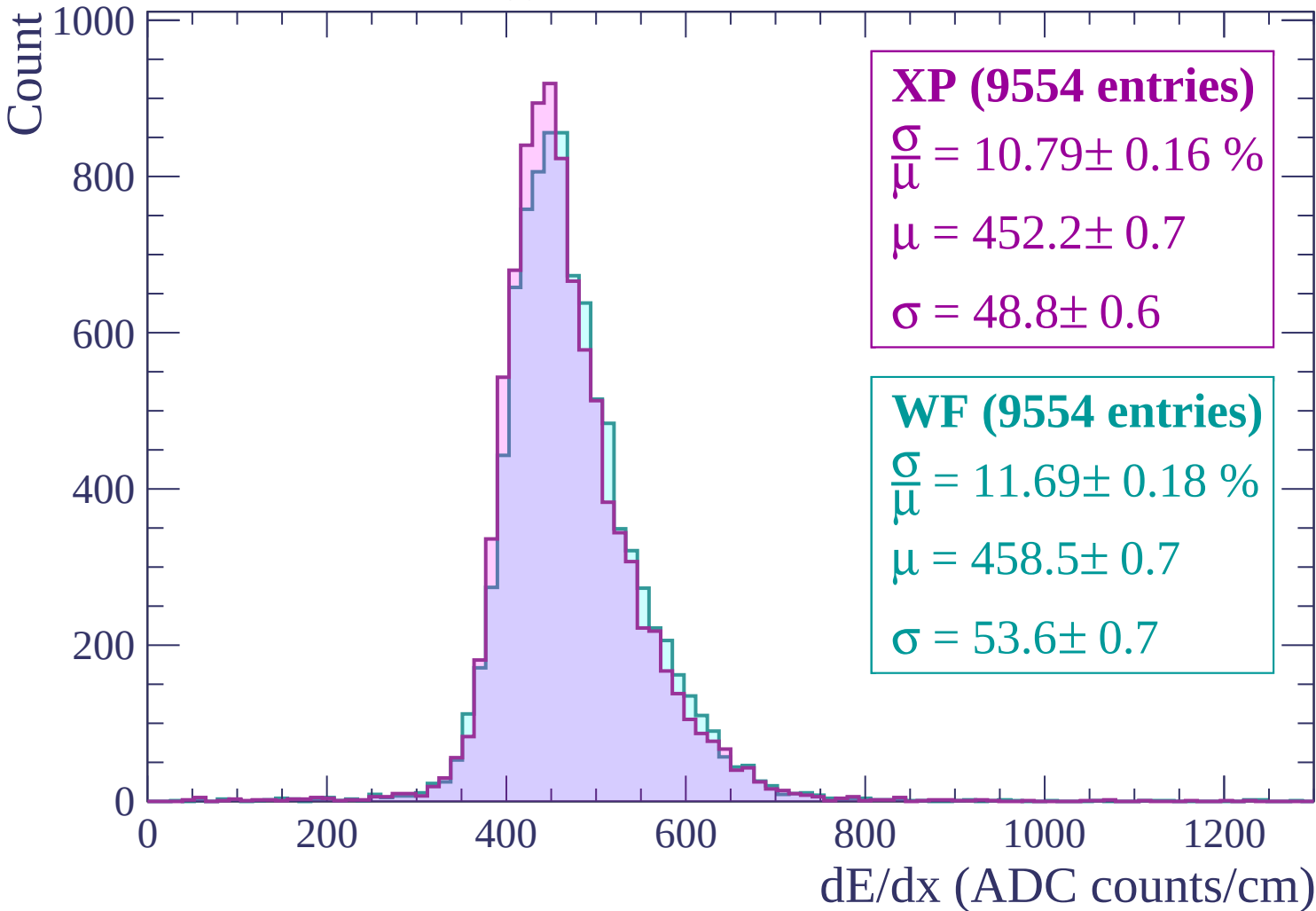
$$\frac{\sigma}{\mu} = 11.89 \pm 0.17 \%$$

$$\mu = 463.2 \pm 0.7$$

$$\sigma = 55.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1140 < p < -1080$



Energy loss | $-1080 < p < -1020$

Count

1000

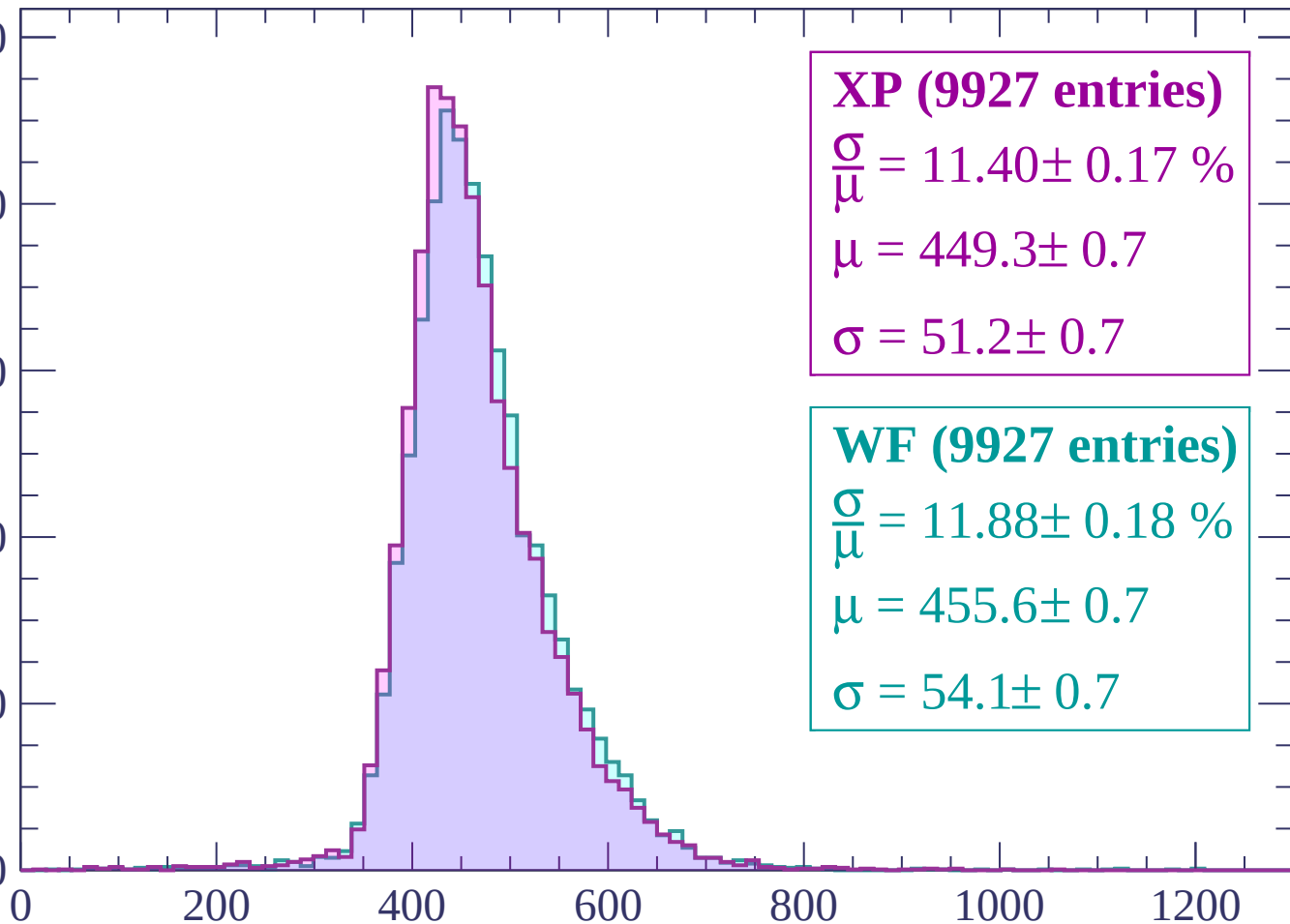
800

600

400

200

0



XP (9927 entries)

$\frac{\sigma}{\mu} = 11.40 \pm 0.17 \%$

$\mu = 449.3 \pm 0.7$

$\sigma = 51.2 \pm 0.7$

WF (9927 entries)

$\frac{\sigma}{\mu} = 11.88 \pm 0.18 \%$

$\mu = 455.6 \pm 0.7$

$\sigma = 54.1 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count

1000

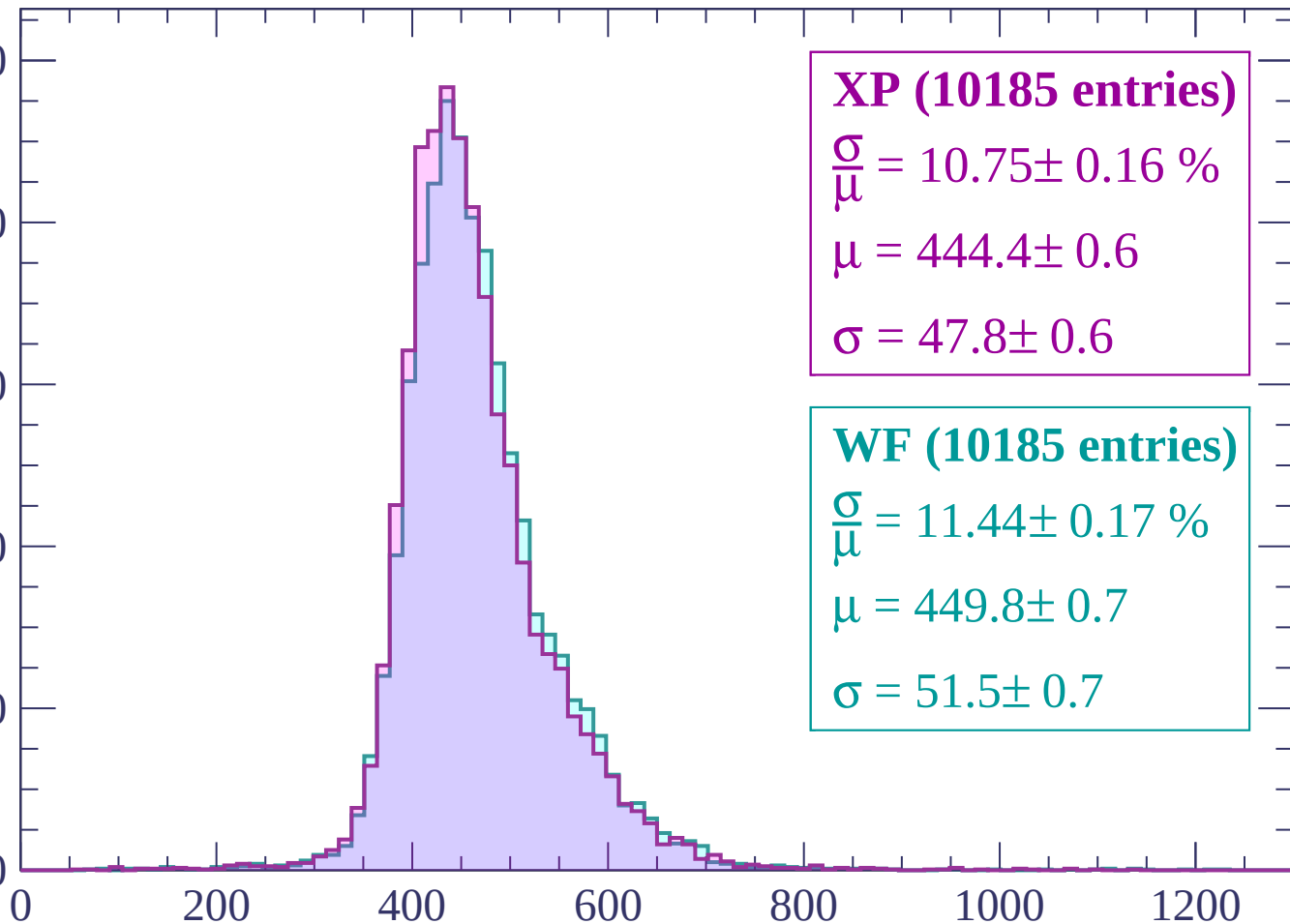
800

600

400

200

0



XP (10185 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.16 \%$

$\mu = 444.4 \pm 0.6$

$\sigma = 47.8 \pm 0.6$

WF (10185 entries)

$\frac{\sigma}{\mu} = 11.44 \pm 0.17 \%$

$\mu = 449.8 \pm 0.7$

$\sigma = 51.5 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count

1000

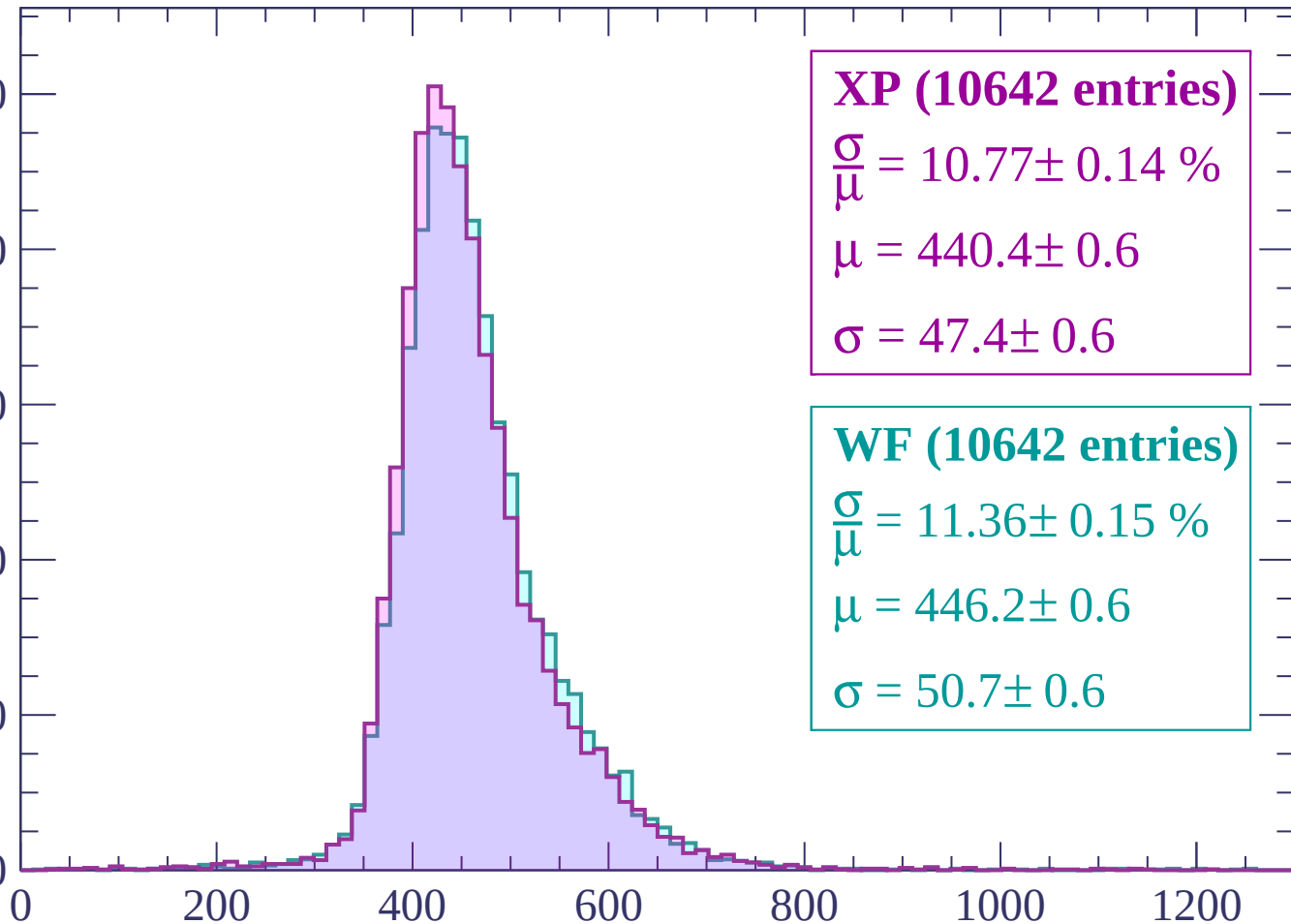
800

600

400

200

0



XP (10642 entries)

$\frac{\sigma}{\mu} = 10.77 \pm 0.14 \%$

$\mu = 440.4 \pm 0.6$

$\sigma = 47.4 \pm 0.6$

WF (10642 entries)

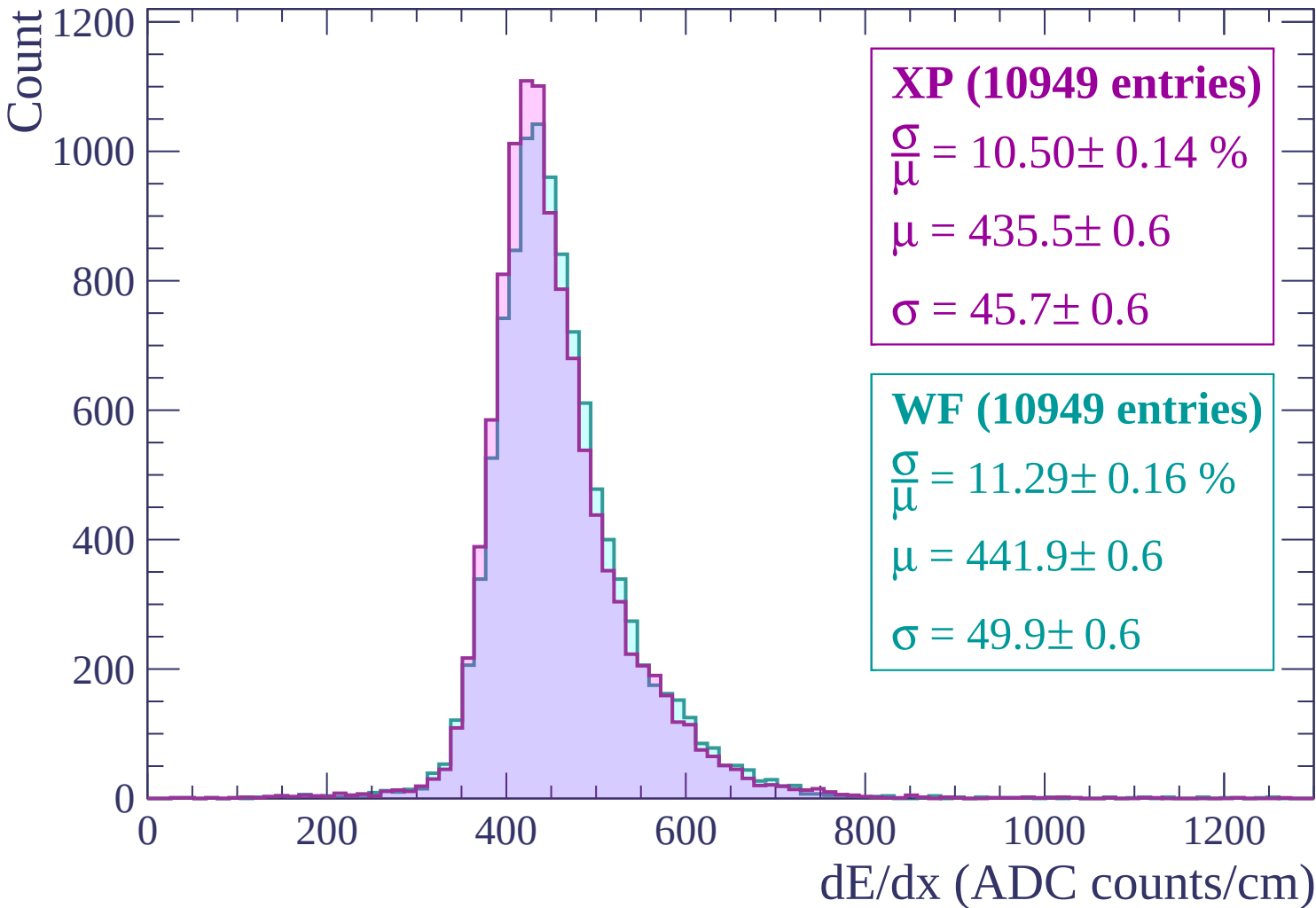
$\frac{\sigma}{\mu} = 11.36 \pm 0.15 \%$

$\mu = 446.2 \pm 0.6$

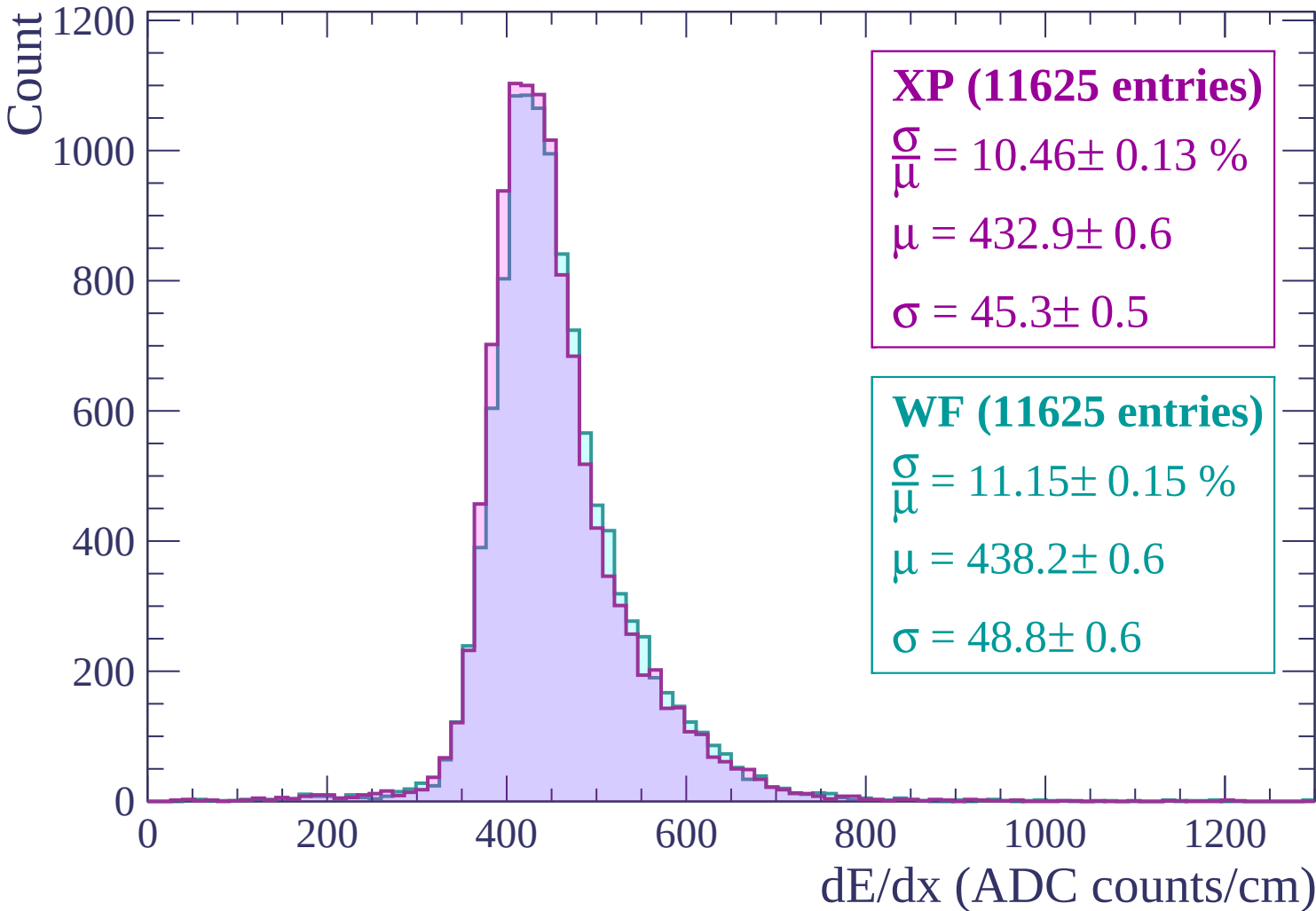
$\sigma = 50.7 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$



Energy loss | $-840 < p < -780$



Energy loss | $-780 < p < -720$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (11910 entries)

$$\frac{\sigma}{\mu} = 10.66 \pm 0.14 \%$$

$$\mu = 428.0 \pm 0.6$$

$$\sigma = 45.6 \pm 0.6$$

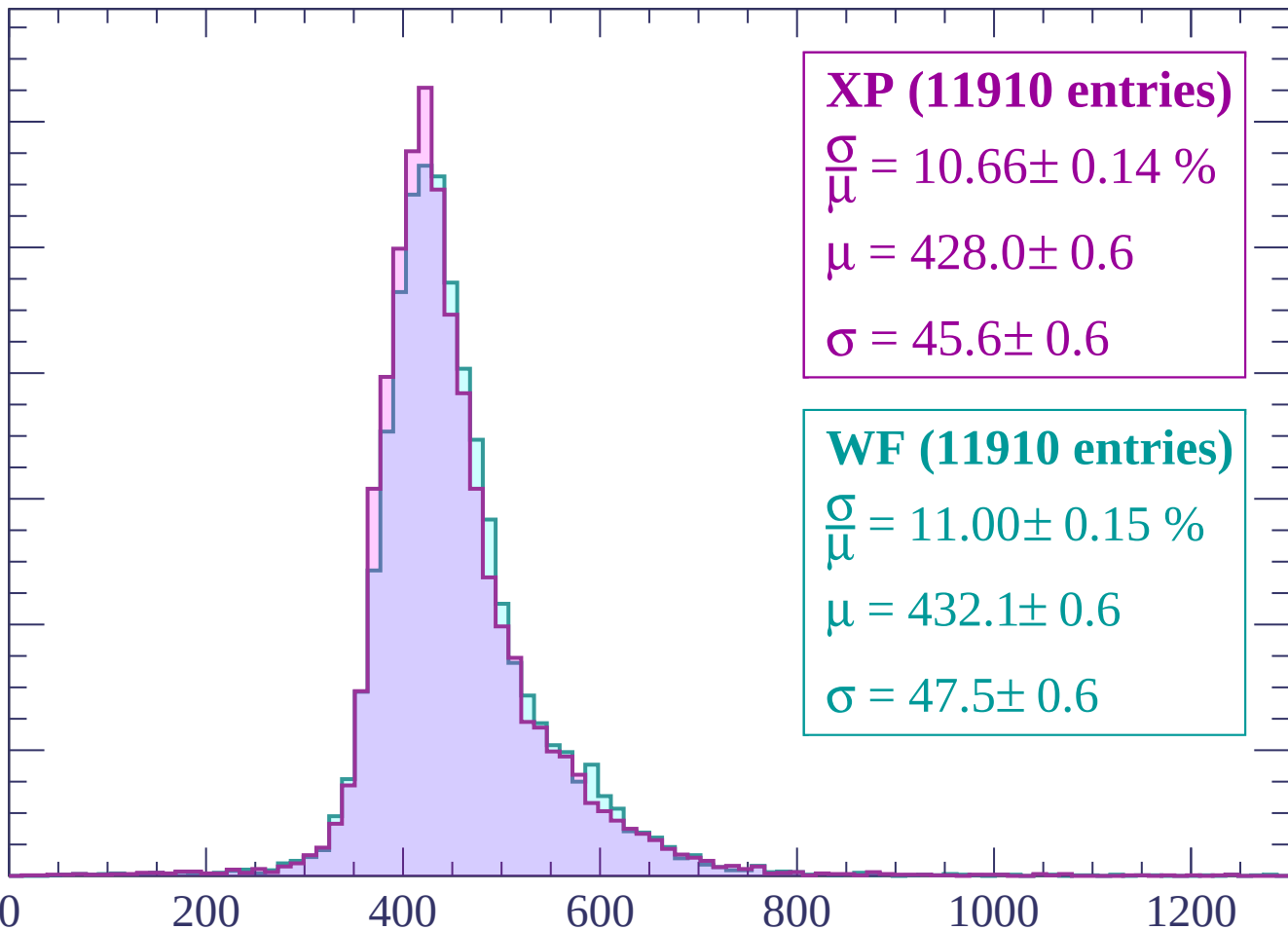
WF (11910 entries)

$$\frac{\sigma}{\mu} = 11.00 \pm 0.15 \%$$

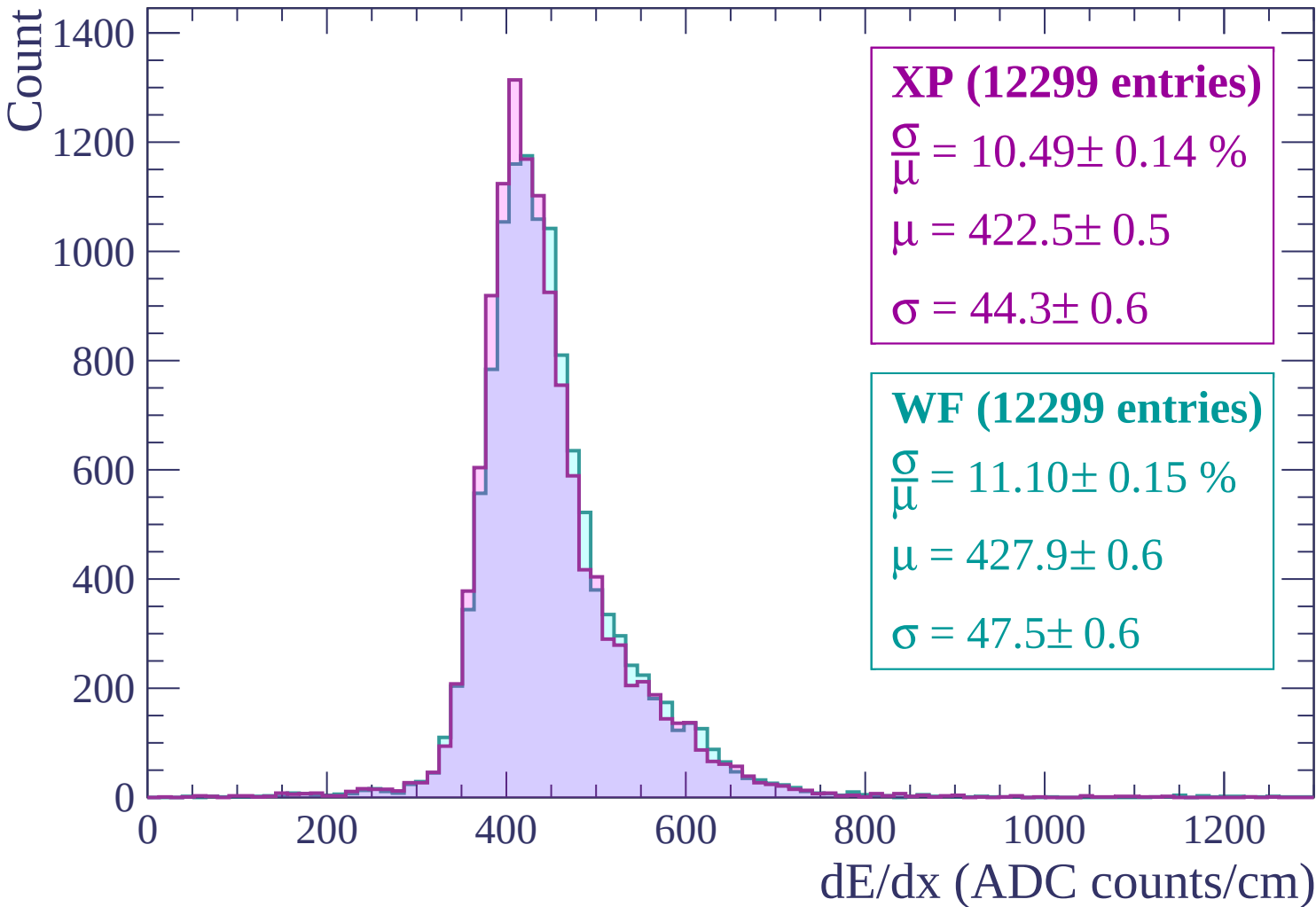
$$\mu = 432.1 \pm 0.6$$

$$\sigma = 47.5 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $-720 < p < -660$



Energy loss | $-660 < p < -600$

Count

1400
1200
1000
800
600
400
200
0

XP (13001 entries)

$$\frac{\sigma}{\mu} = 10.11 \pm 0.12 \%$$

$$\mu = 418.9 \pm 0.5$$

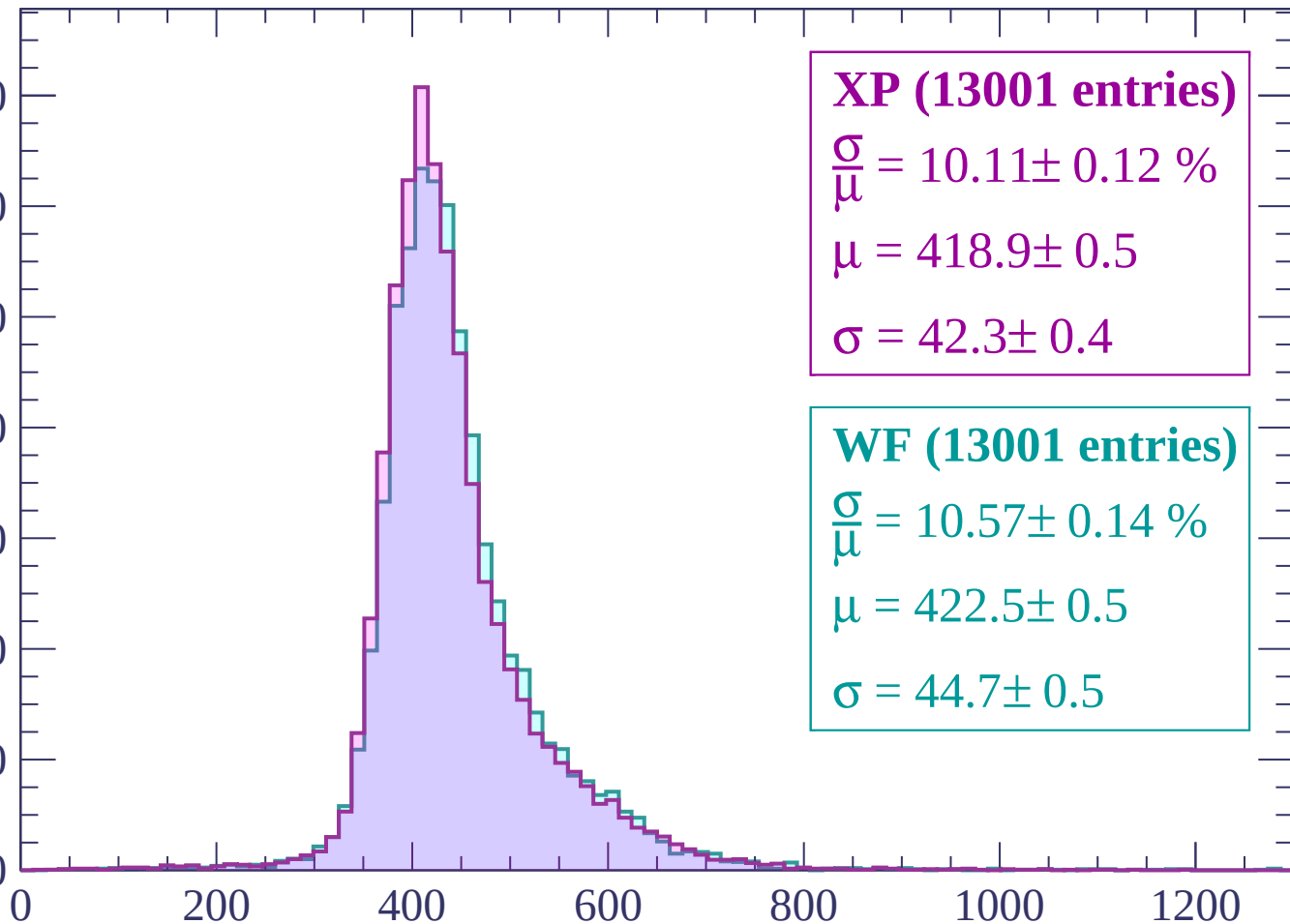
$$\sigma = 42.3 \pm 0.4$$

WF (13001 entries)

$$\frac{\sigma}{\mu} = 10.57 \pm 0.14 \%$$

$$\mu = 422.5 \pm 0.5$$

$$\sigma = 44.7 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count

1400
1200
1000
800
600
400
200
0

XP (13305 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.12 \%$$

$$\mu = 413.6 \pm 0.5$$

$$\sigma = 40.9 \pm 0.4$$

WF (13305 entries)

$$\frac{\sigma}{\mu} = 10.53 \pm 0.13 \%$$

$$\mu = 418.8 \pm 0.5$$

$$\sigma = 44.1 \pm 0.5$$

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14029 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.11 \%$$

$$\mu = 409.4 \pm 0.4$$

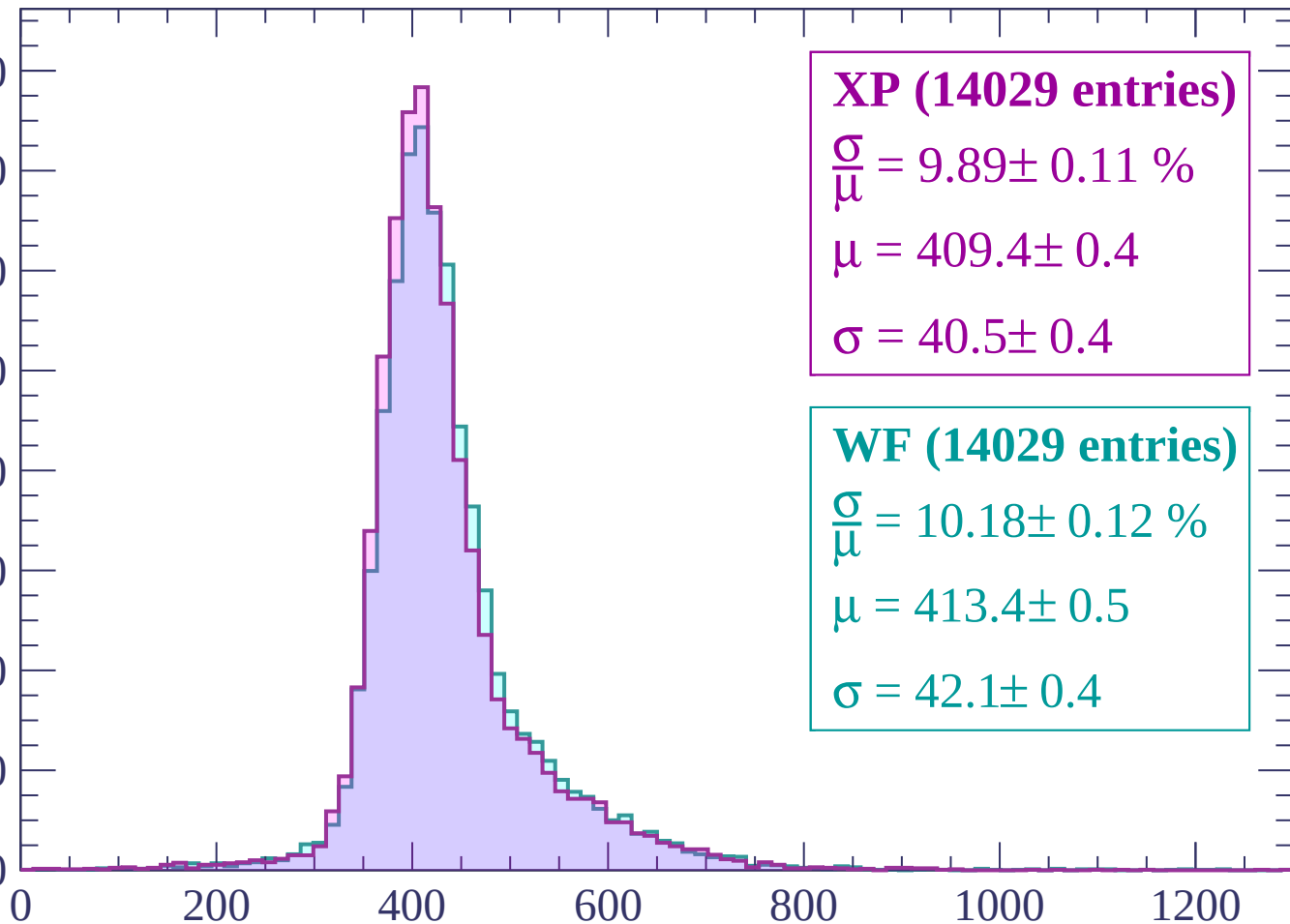
$$\sigma = 40.5 \pm 0.4$$

WF (14029 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.12 \%$$

$$\mu = 413.4 \pm 0.5$$

$$\sigma = 42.1 \pm 0.4$$



dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14467 entries)

$$\frac{\sigma}{\mu} = 9.51 \pm 0.11 \%$$

$$\mu = 404.6 \pm 0.4$$

$$\sigma = 38.5 \pm 0.4$$

WF (14467 entries)

$$\frac{\sigma}{\mu} = 10.29 \pm 0.12 \%$$

$$\mu = 409.3 \pm 0.5$$

$$\sigma = 42.1 \pm 0.5$$

0

200

400

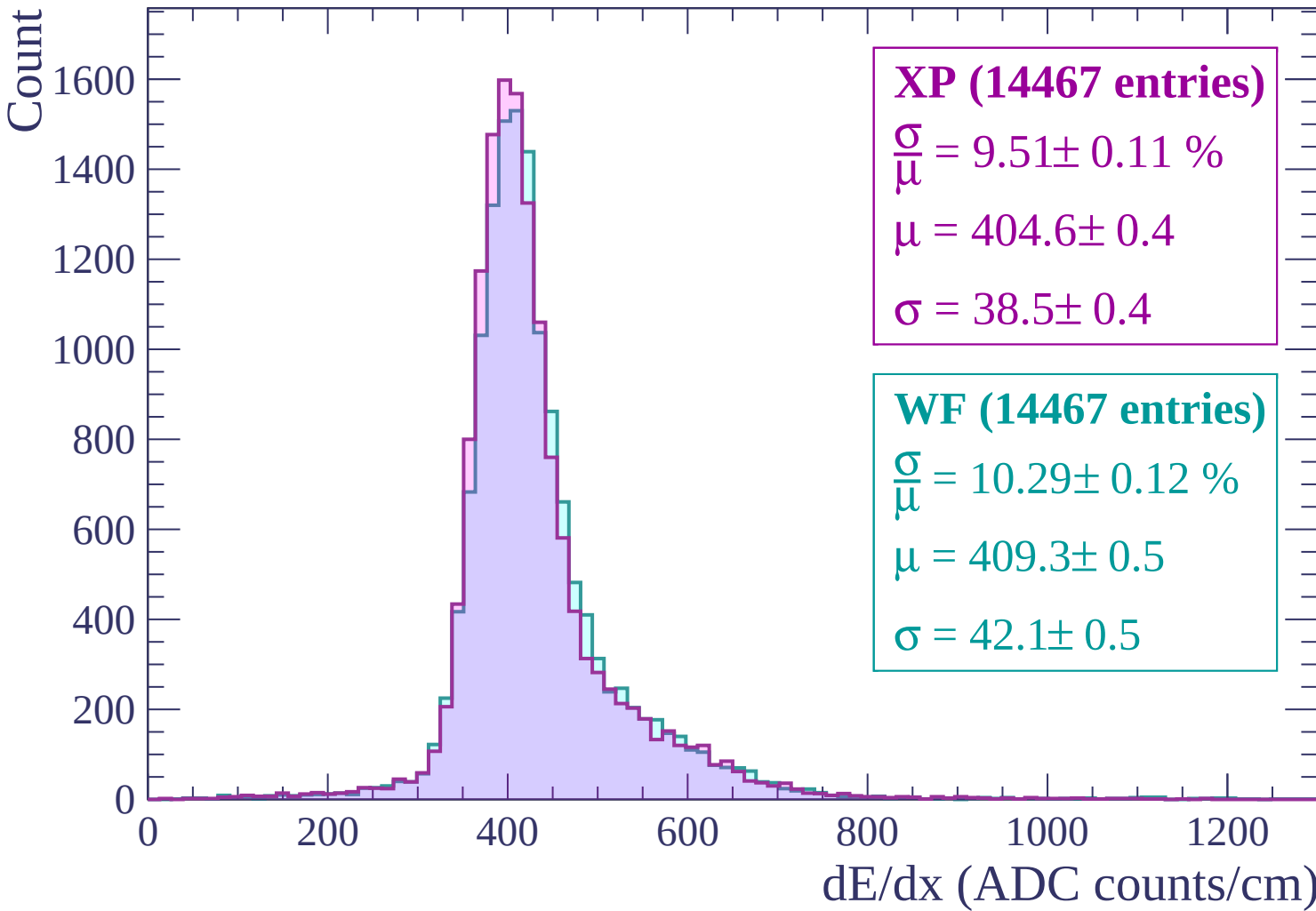
600

800

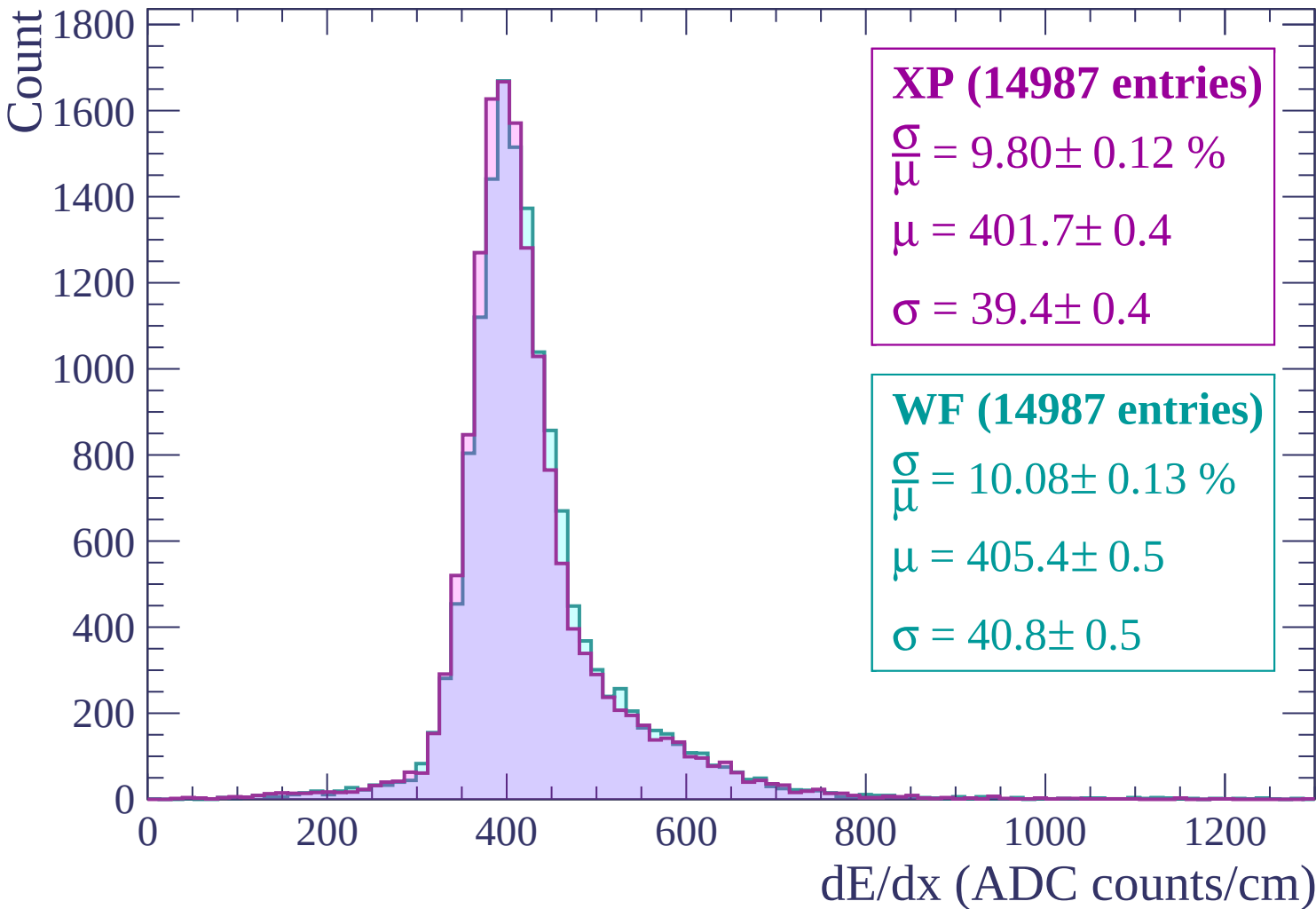
1000

1200

dE/dx (ADC counts/cm)



Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count

1800
1600
1400
1200
1000
800
600
400
200
0

XP (15606 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.12 \%$$

$$\mu = 399.1 \pm 0.4$$

$$\sigma = 39.5 \pm 0.4$$

WF (15606 entries)

$$\frac{\sigma}{\mu} = 10.28 \pm 0.13 \%$$

$$\mu = 402.8 \pm 0.4$$

$$\sigma = 41.4 \pm 0.5$$

0

200

400

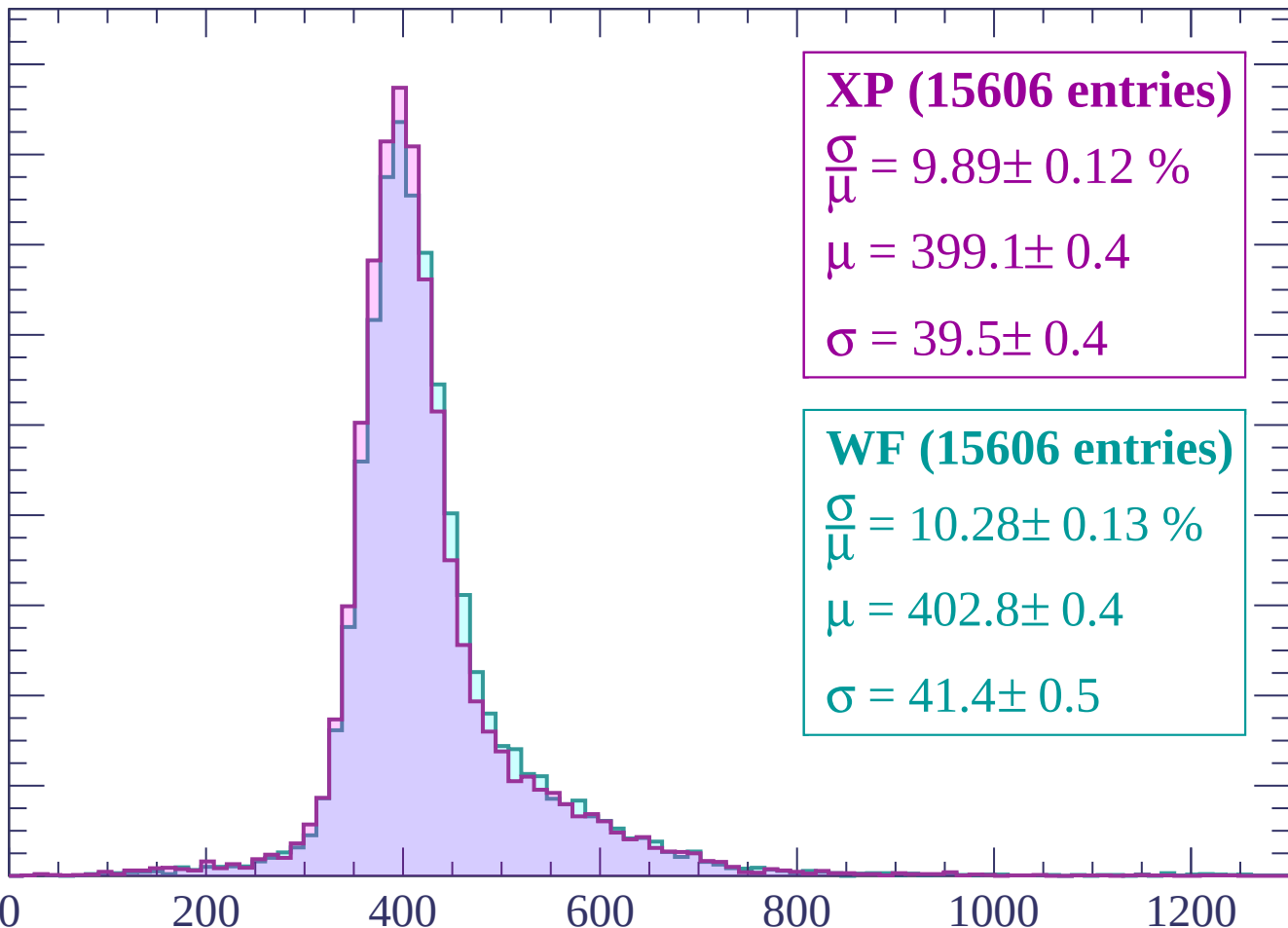
600

800

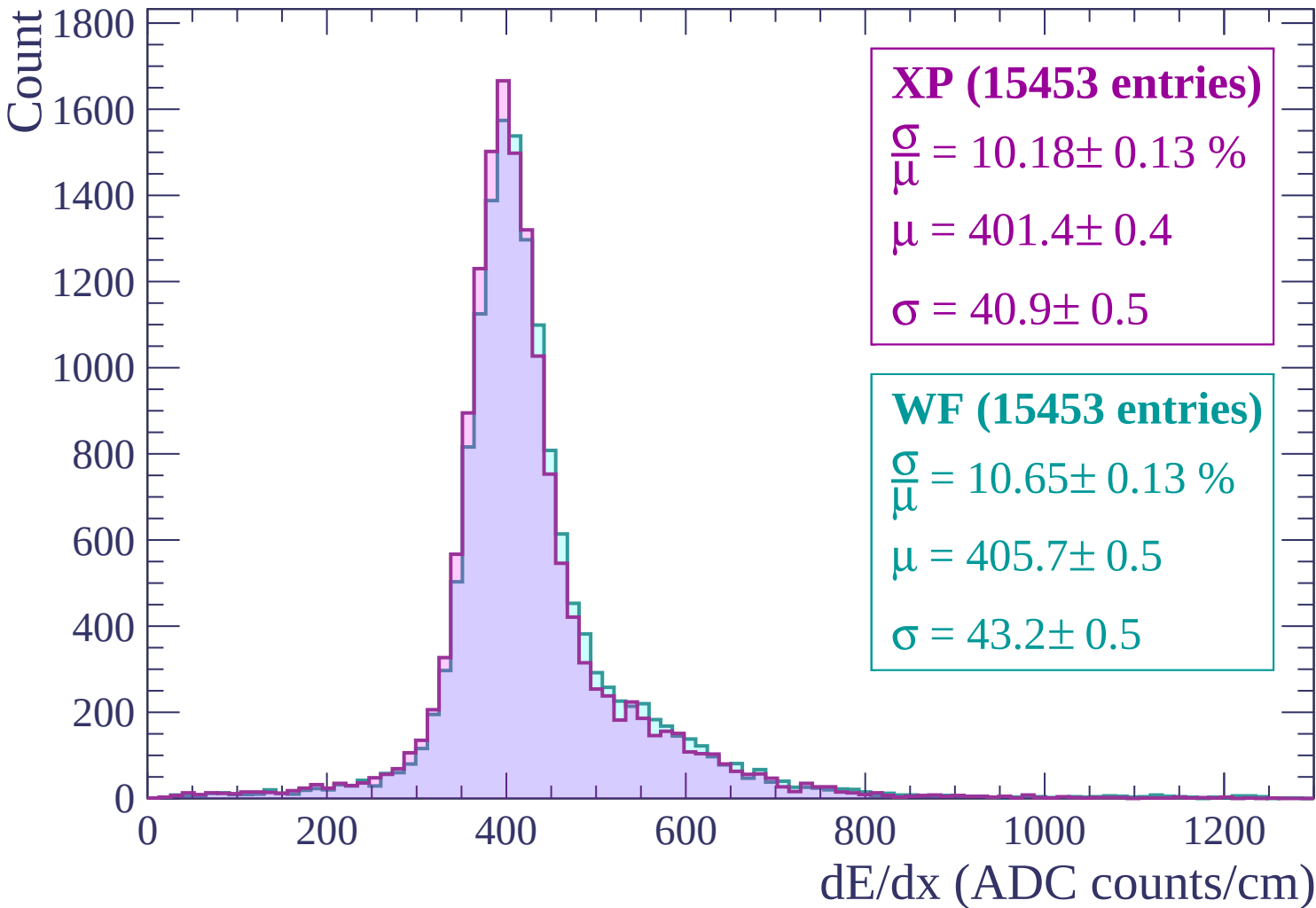
1000

1200

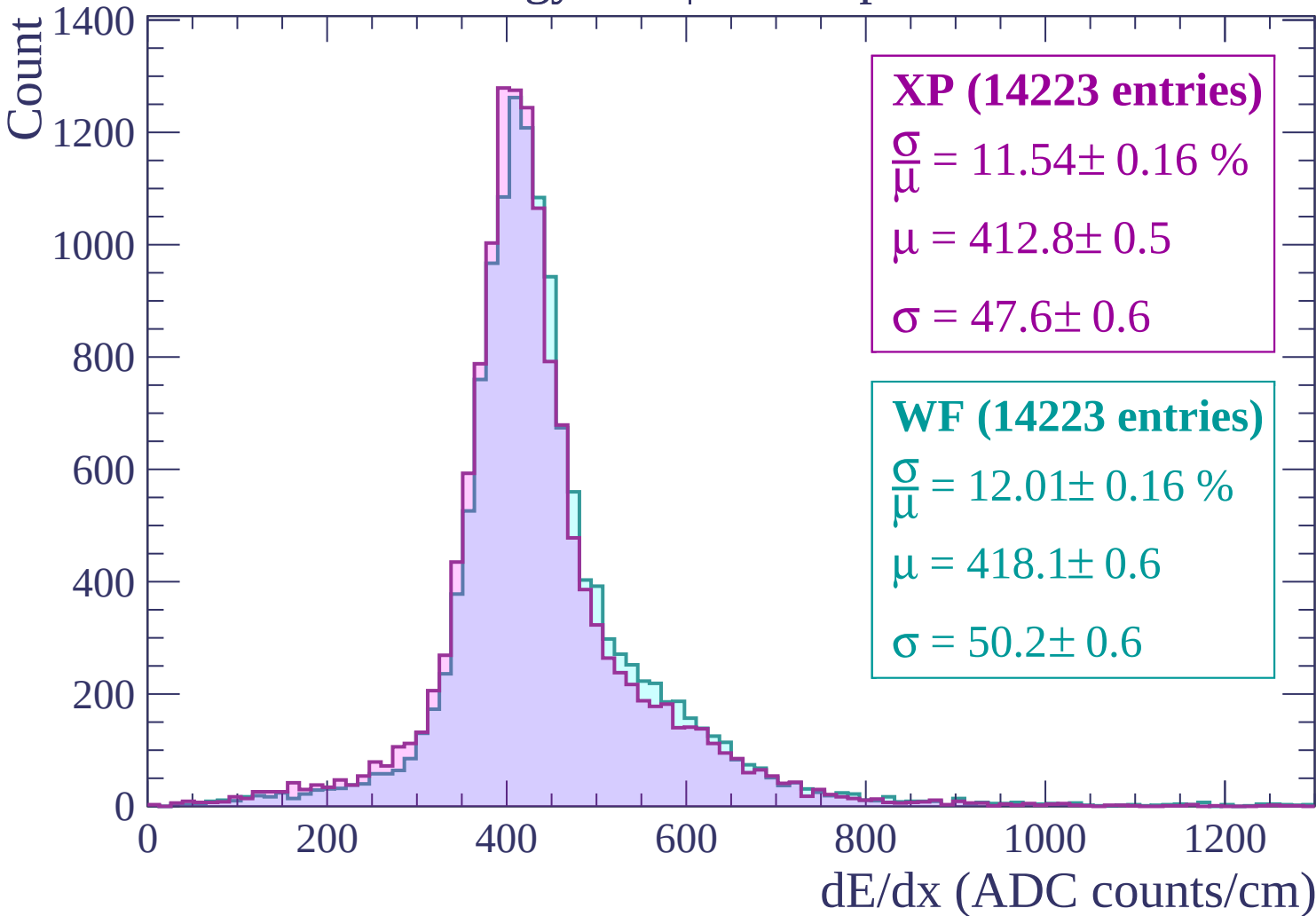
dE/dx (ADC counts/cm)



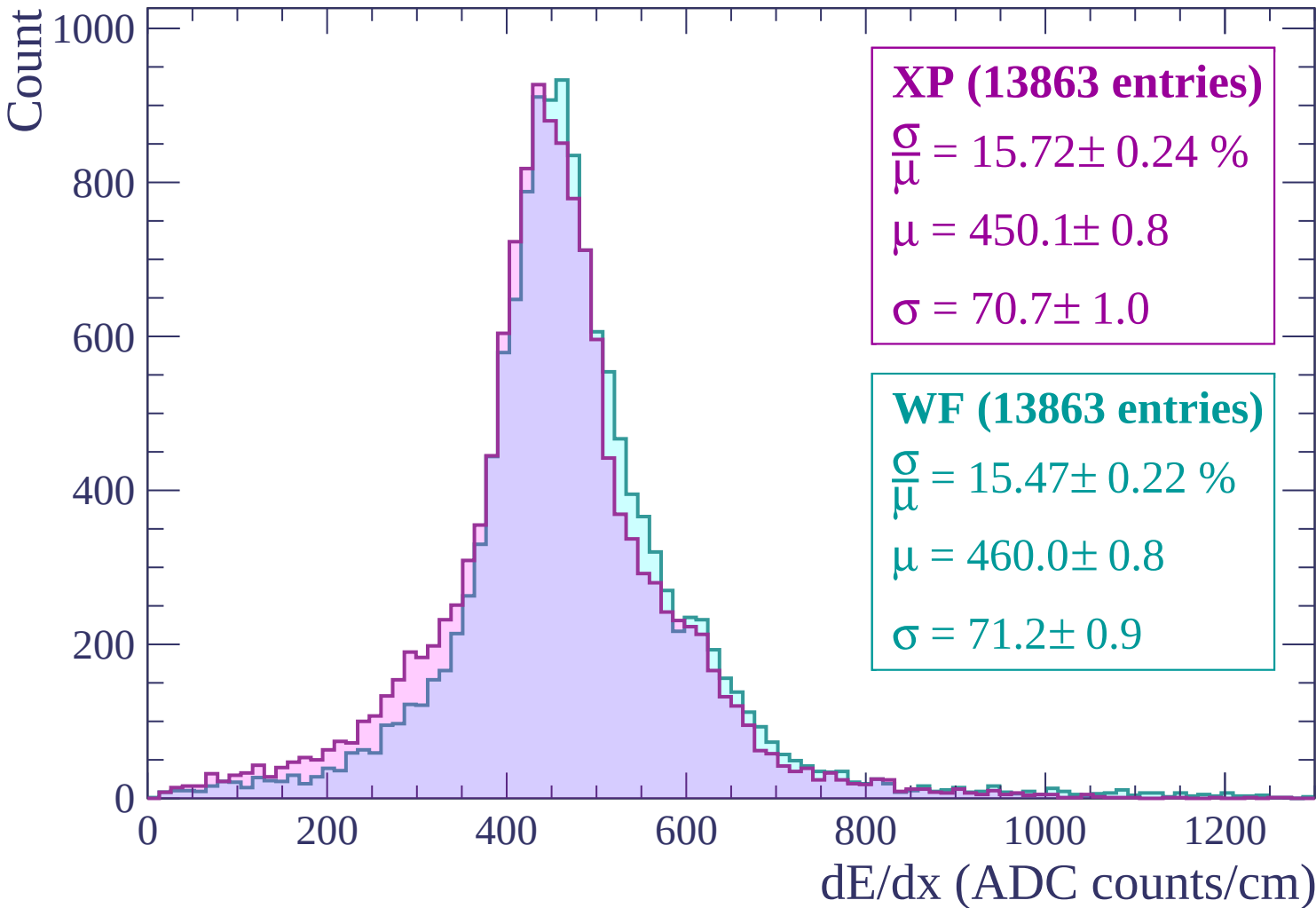
Energy loss | $-300 < p < -240$



Energy loss | $-240 < p < -180$

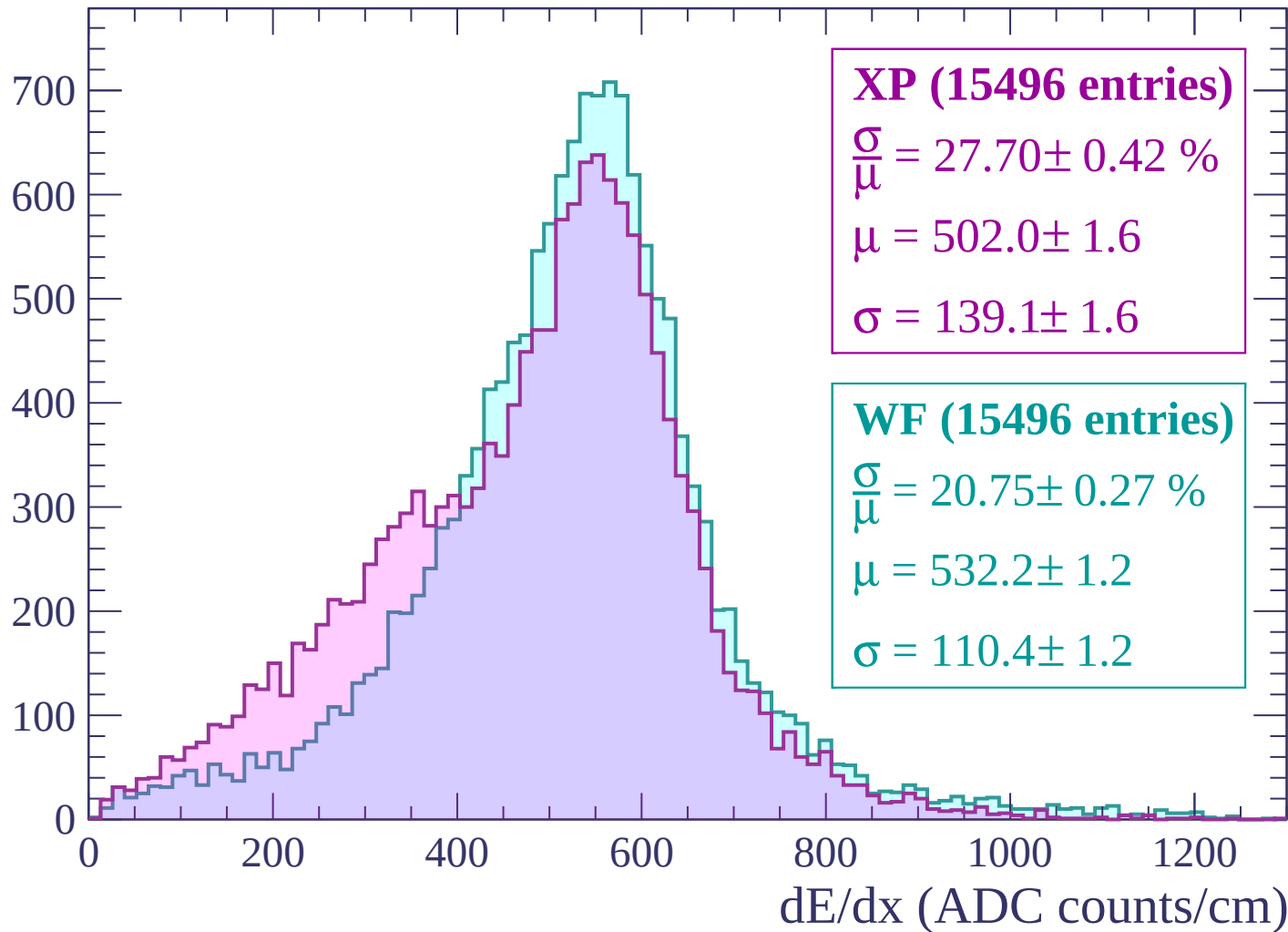


Energy loss | -180 < p < -120

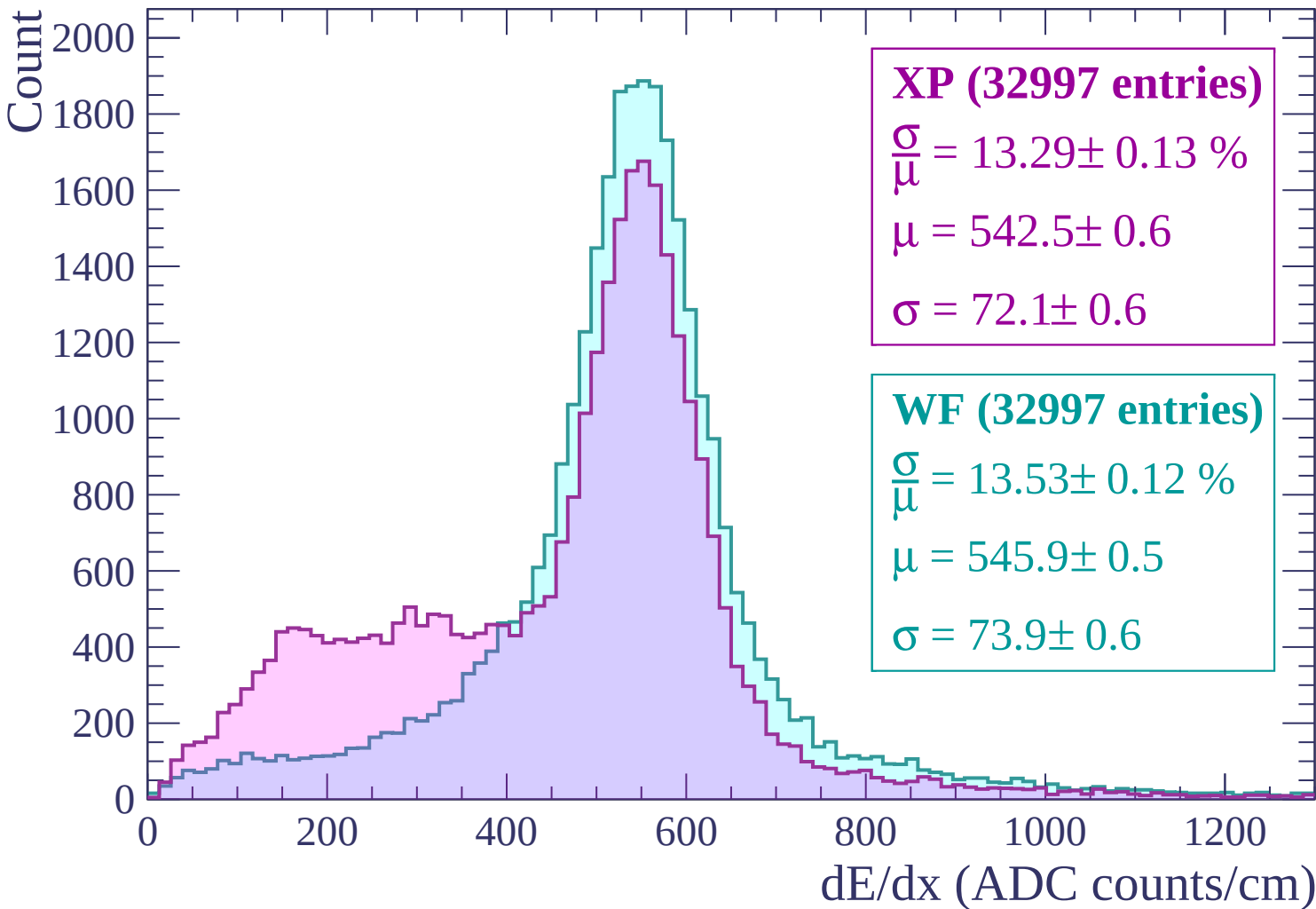


Energy loss | $-120 < p < -60$

Count

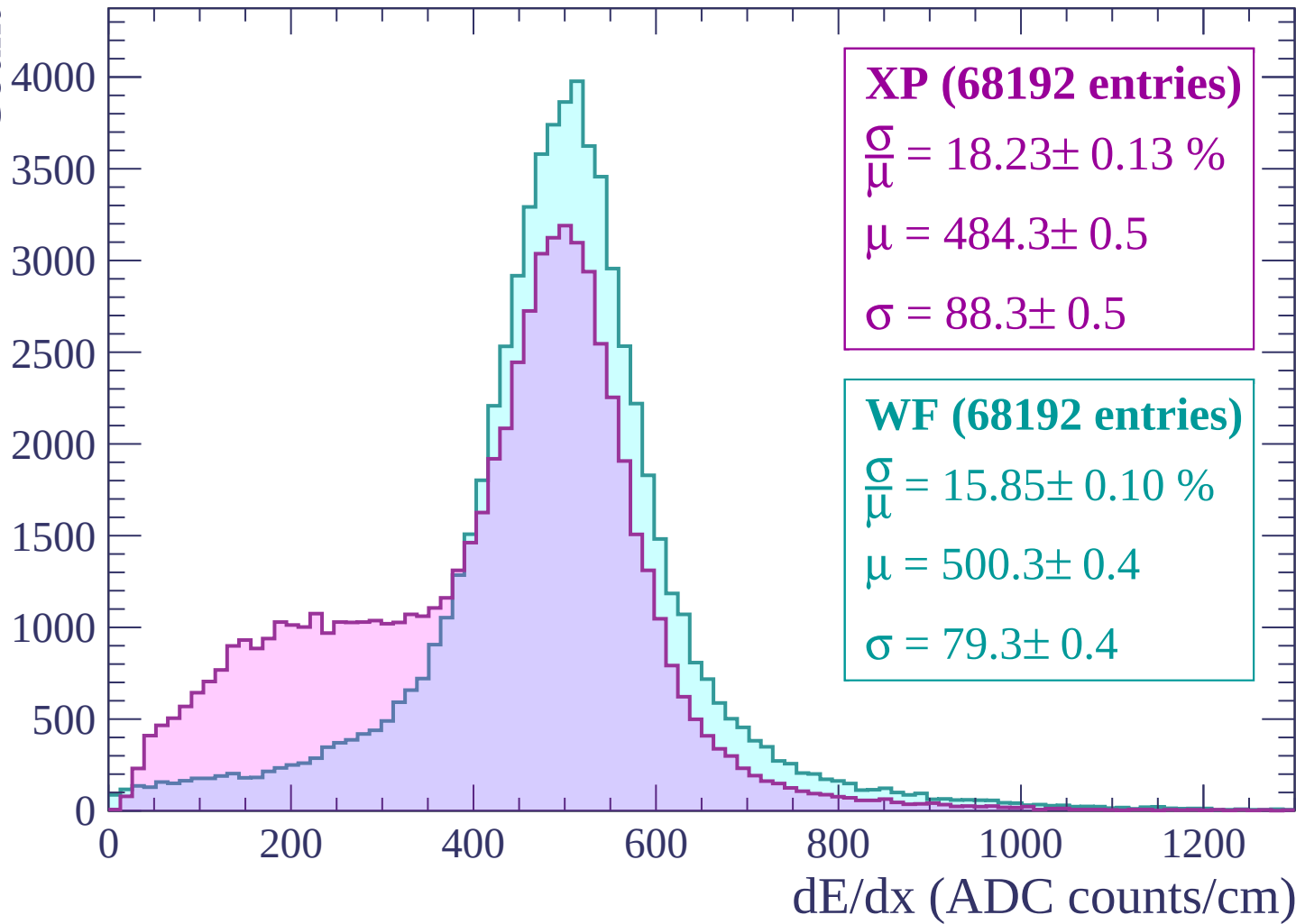


Energy loss | $-60 < p < 0$



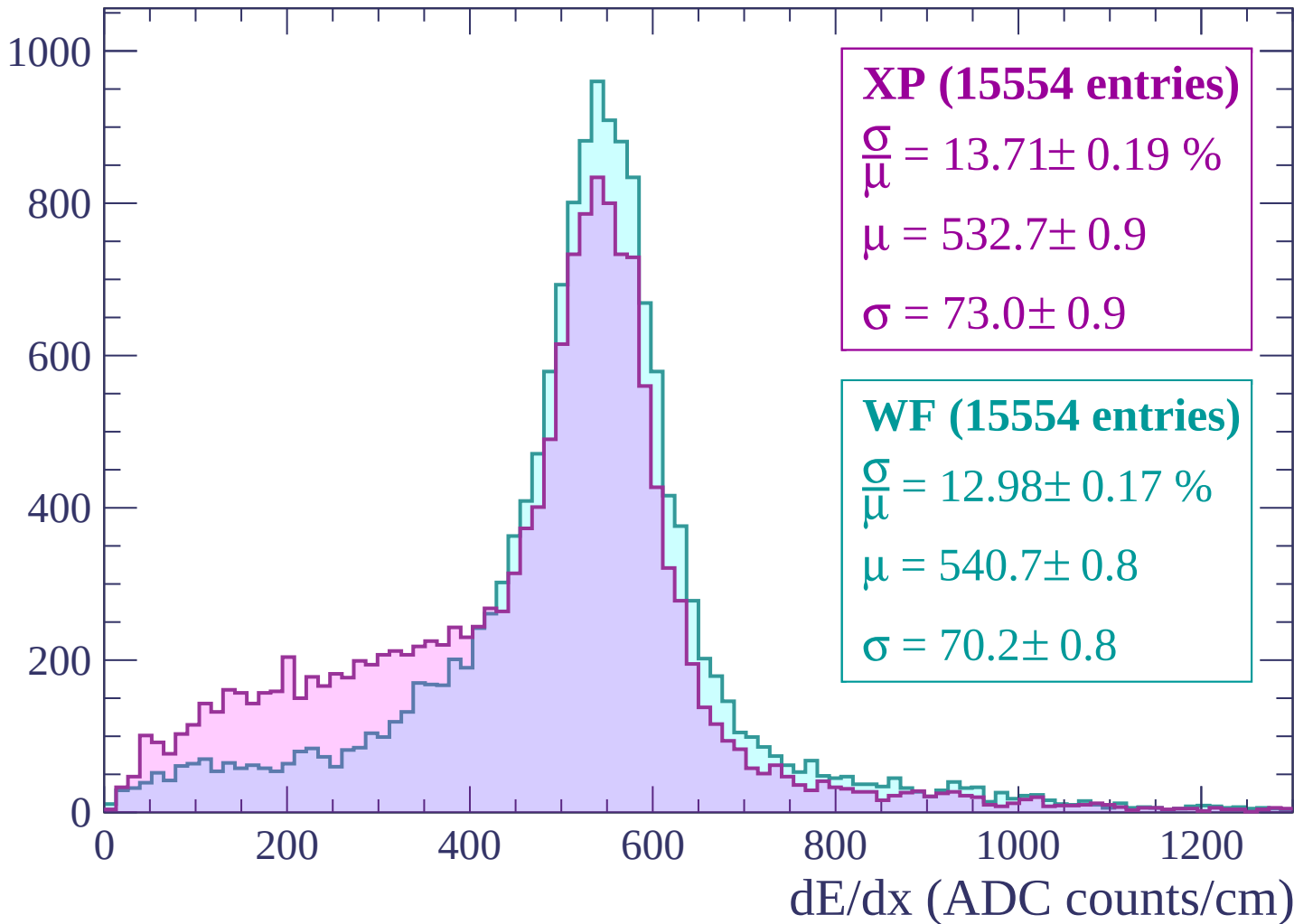
Energy loss | $0 < p < 60$

Count



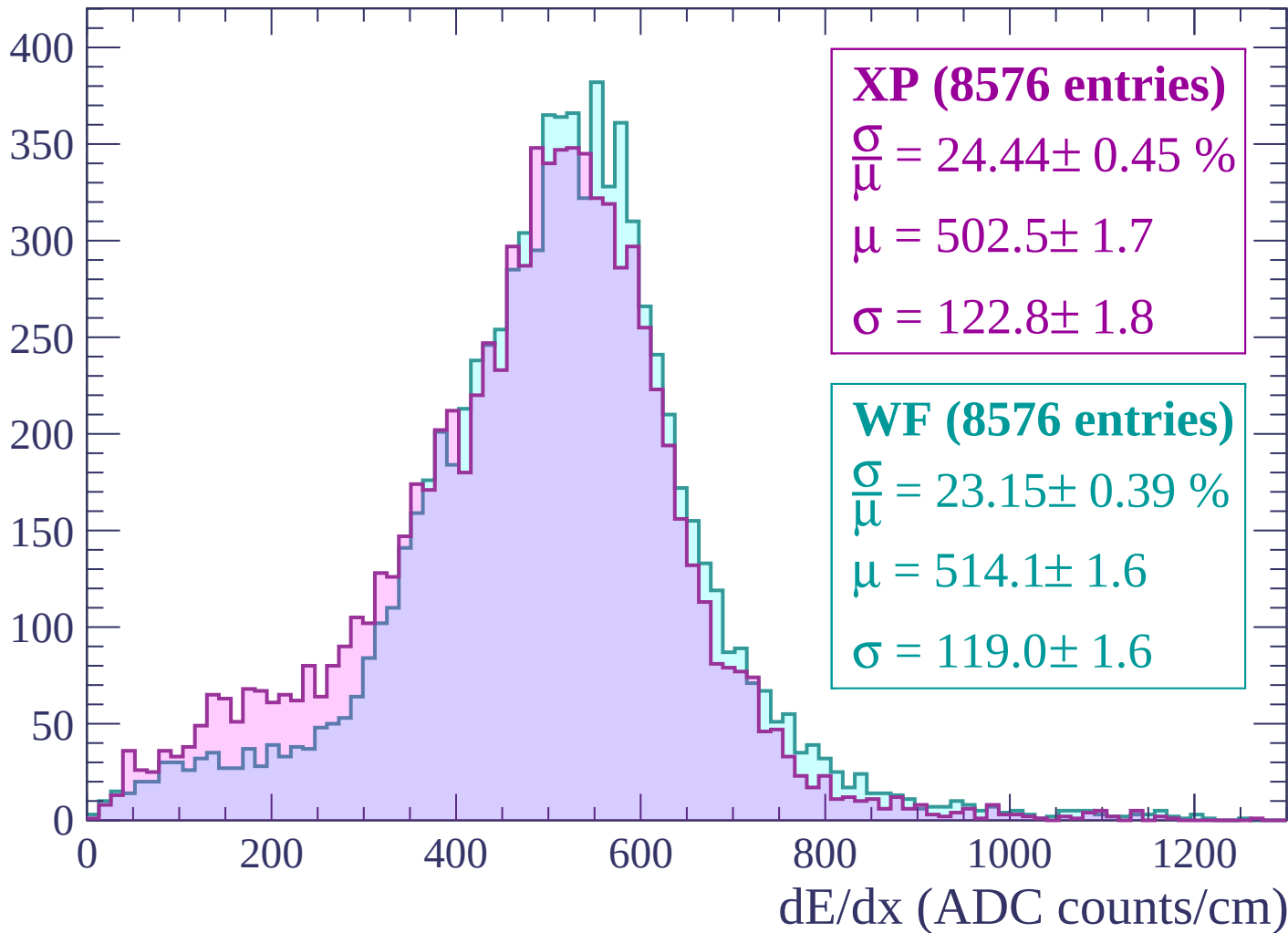
Energy loss | $60 < p < 120$

Count



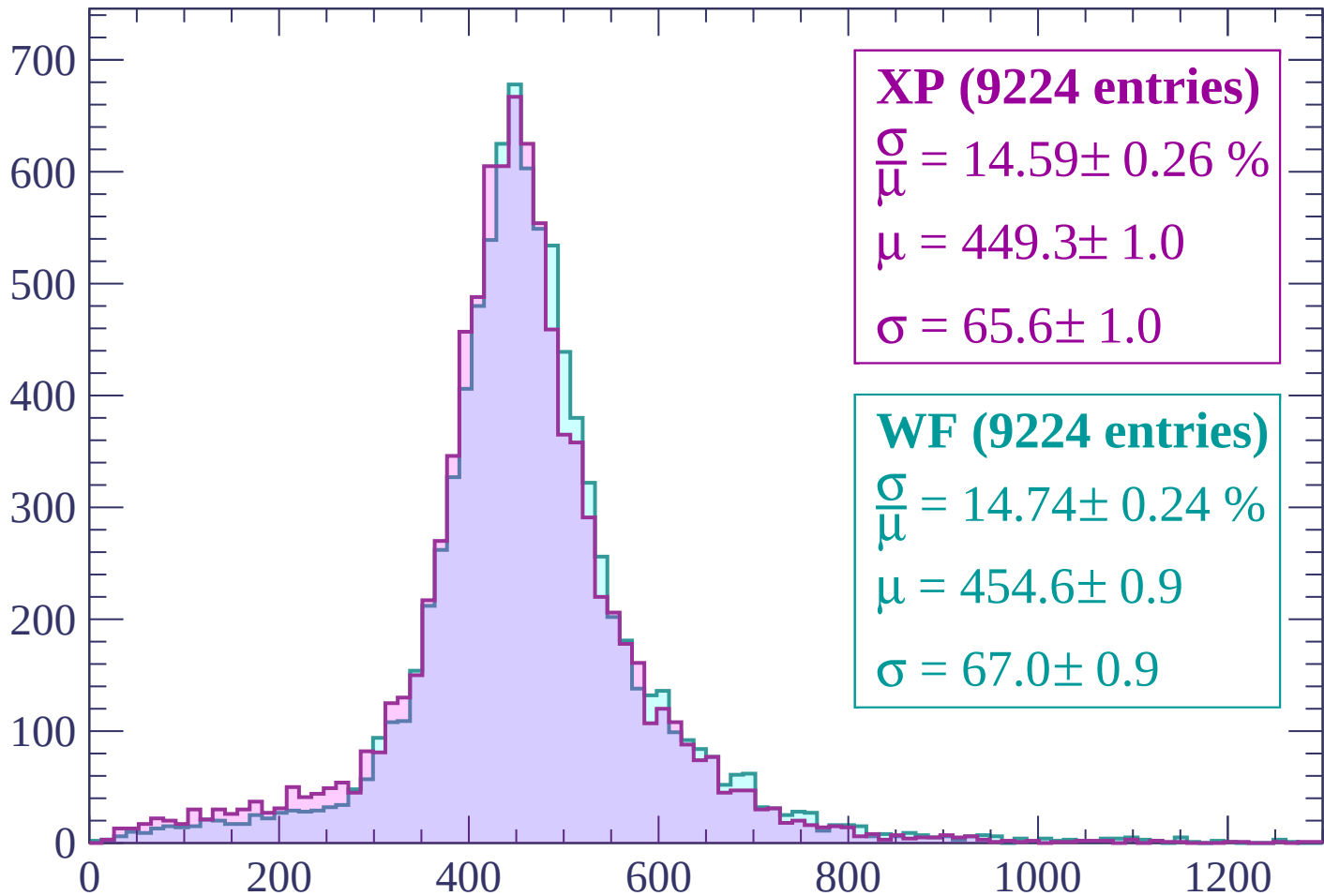
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (9224 entries)

$\frac{\sigma}{\mu} = 14.59 \pm 0.26 \%$

$\mu = 449.3 \pm 1.0$

$\sigma = 65.6 \pm 1.0$

WF (9224 entries)

$\frac{\sigma}{\mu} = 14.74 \pm 0.24 \%$

$\mu = 454.6 \pm 0.9$

$\sigma = 67.0 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count

1000

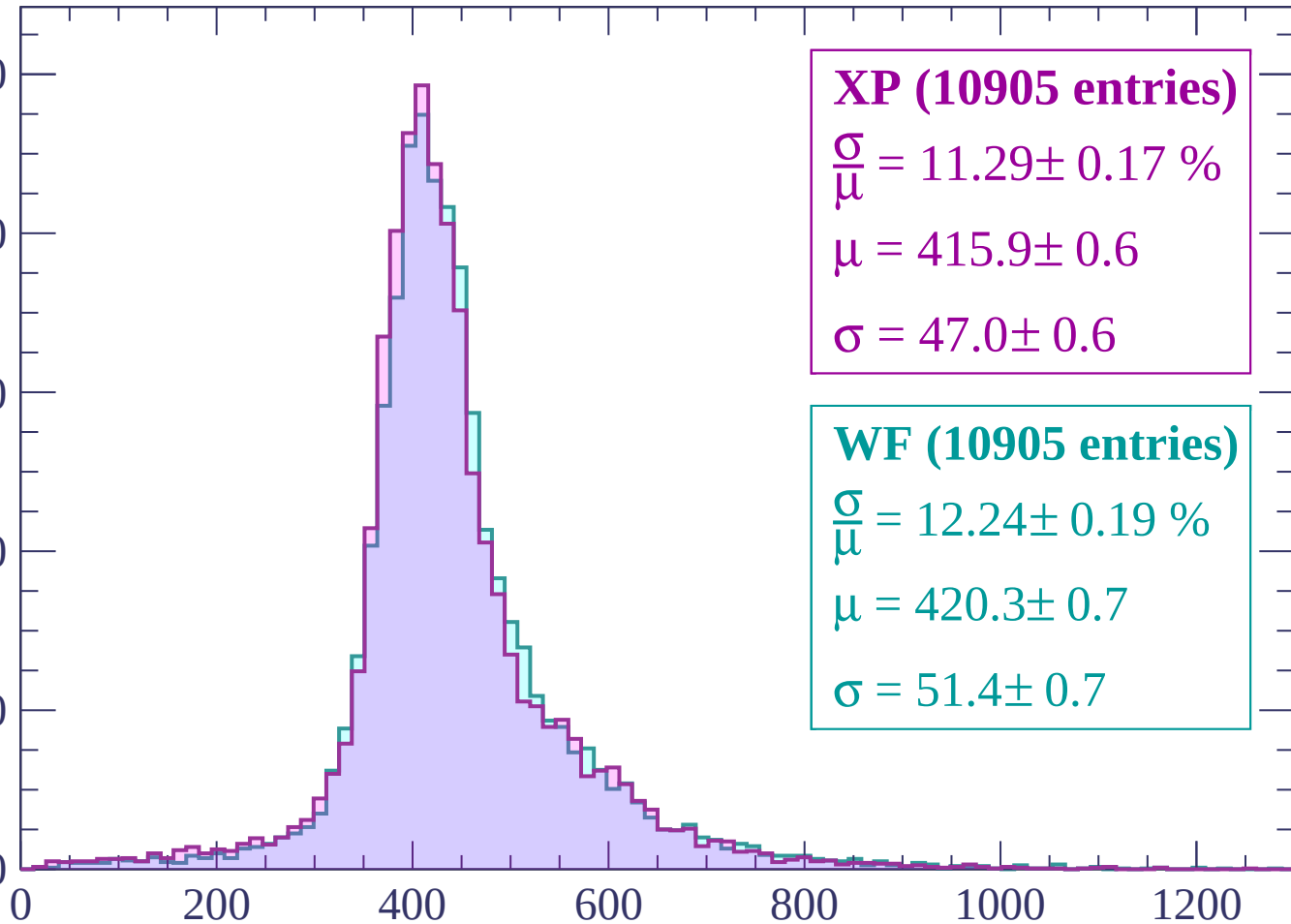
800

600

400

200

0



XP (10905 entries)

$\frac{\sigma}{\mu} = 11.29 \pm 0.17 \%$

$\mu = 415.9 \pm 0.6$

$\sigma = 47.0 \pm 0.6$

WF (10905 entries)

$\frac{\sigma}{\mu} = 12.24 \pm 0.19 \%$

$\mu = 420.3 \pm 0.7$

$\sigma = 51.4 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count

1400
1200
1000
800
600
400
200
0

XP (13444 entries)

$$\frac{\sigma}{\mu} = 10.10 \pm 0.13 \%$$

$$\mu = 402.9 \pm 0.5$$

$$\sigma = 40.7 \pm 0.5$$

WF (13444 entries)

$$\frac{\sigma}{\mu} = 10.66 \pm 0.14 \%$$

$$\mu = 407.1 \pm 0.5$$

$$\sigma = 43.4 \pm 0.5$$

0

200

400

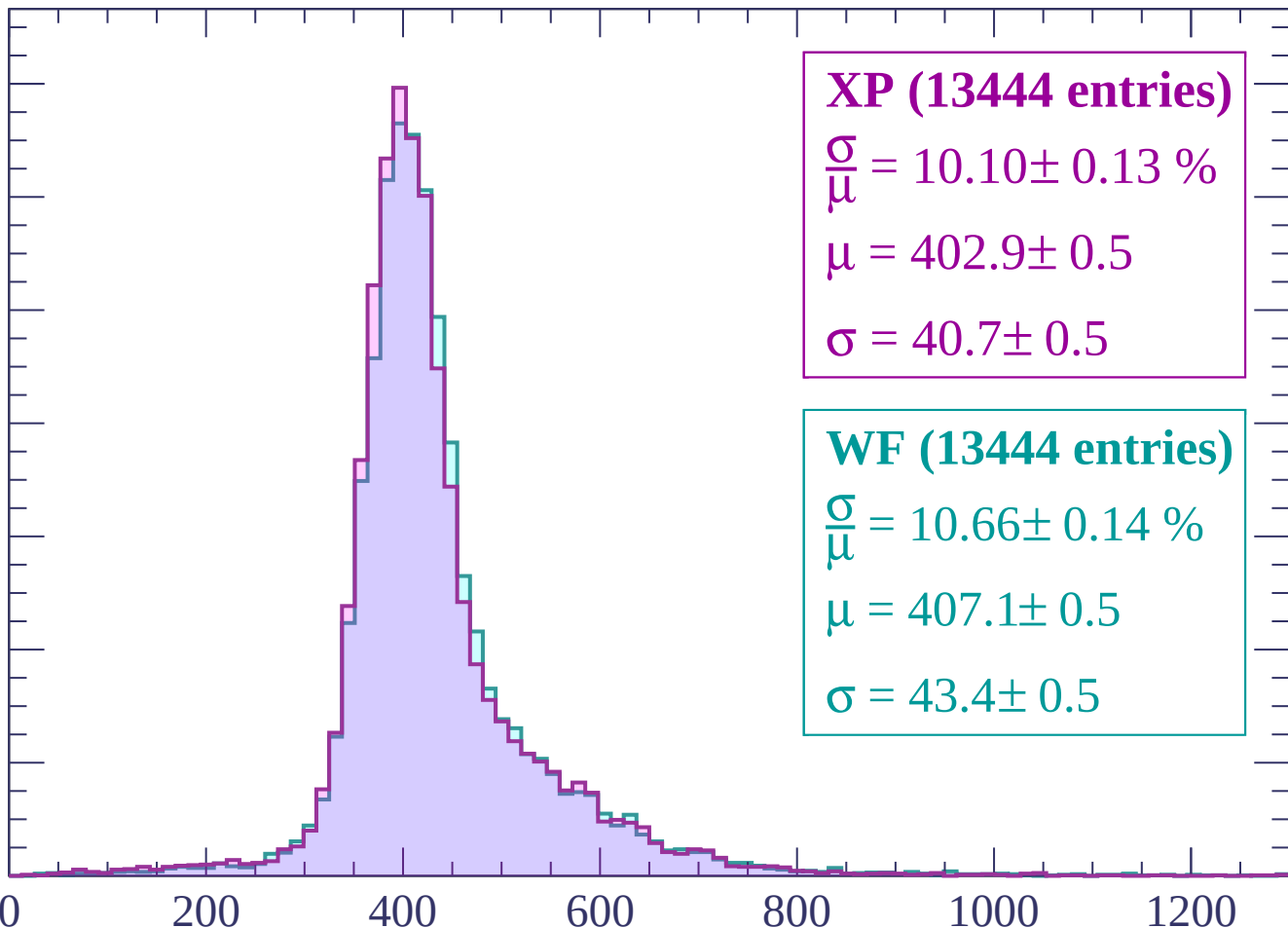
600

800

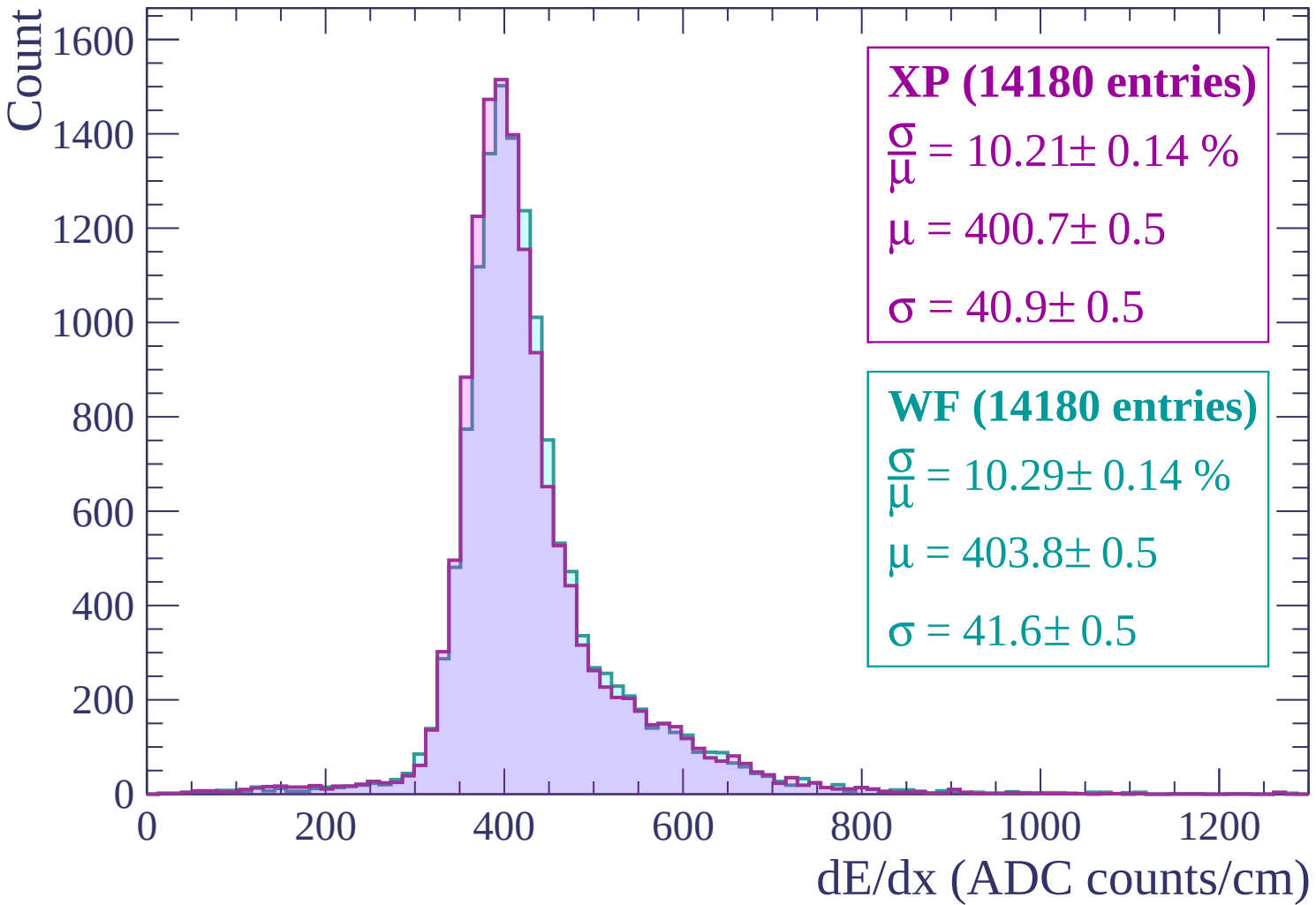
1000

1200

dE/dx (ADC counts/cm)



Energy loss | $360 < p < 420$



Energy loss | $420 < p < 480$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14589 entries)

$\frac{\sigma}{\mu} = 10.07 \pm 0.13 \%$

$\mu = 402.9 \pm 0.5$

$\sigma = 40.5 \pm 0.5$

WF (14589 entries)

$\frac{\sigma}{\mu} = 10.61 \pm 0.13 \%$

$\mu = 407.3 \pm 0.5$

$\sigma = 43.2 \pm 0.5$

0

200

400

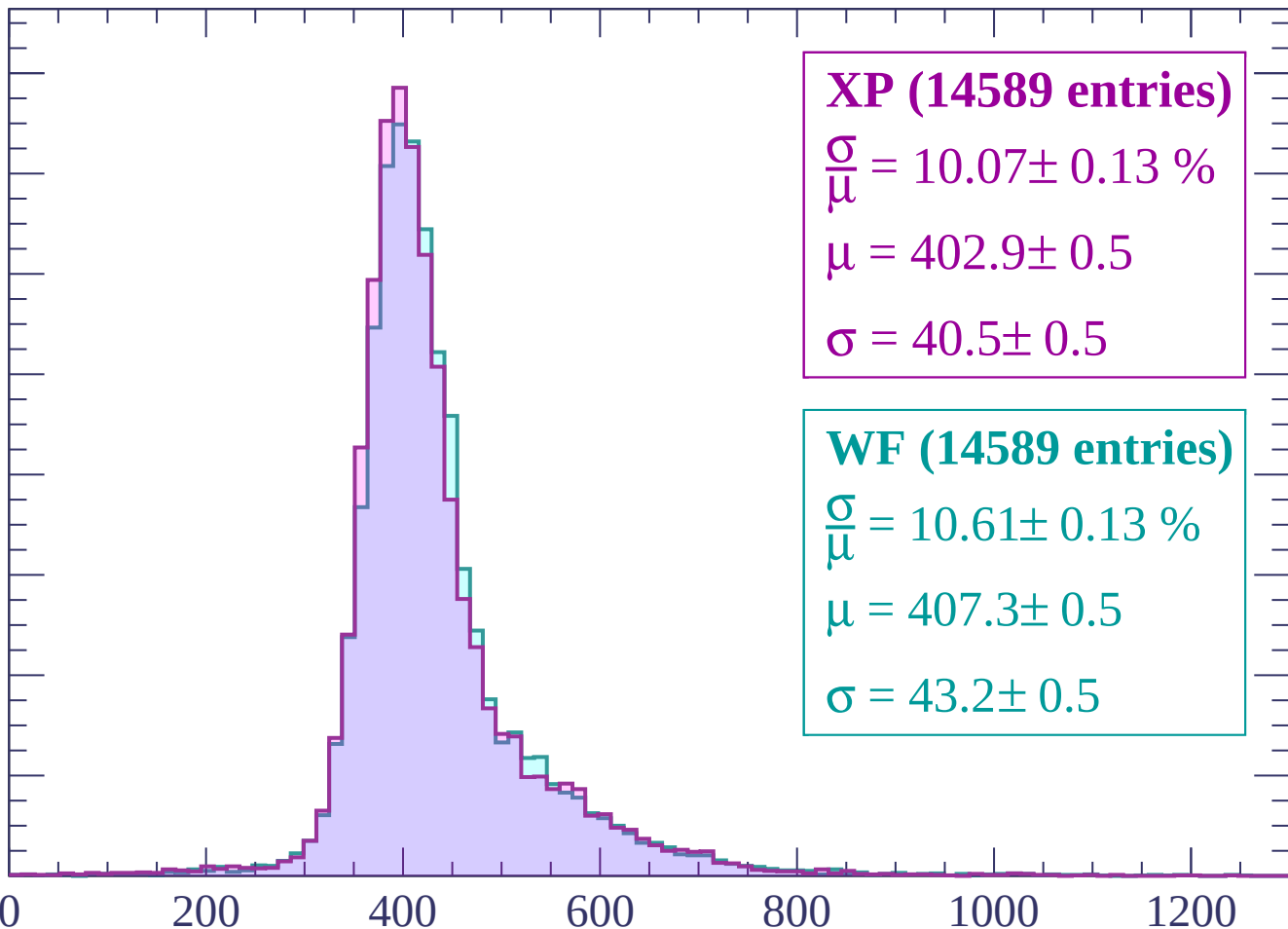
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $480 < p < 540$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14553 entries)

$$\frac{\sigma}{\mu} = 9.71 \pm 0.12 \%$$

$$\mu = 405.7 \pm 0.4$$

$$\sigma = 39.4 \pm 0.4$$

WF (14553 entries)

$$\frac{\sigma}{\mu} = 10.25 \pm 0.12 \%$$

$$\mu = 410.3 \pm 0.5$$

$$\sigma = 42.1 \pm 0.5$$

0

200

400

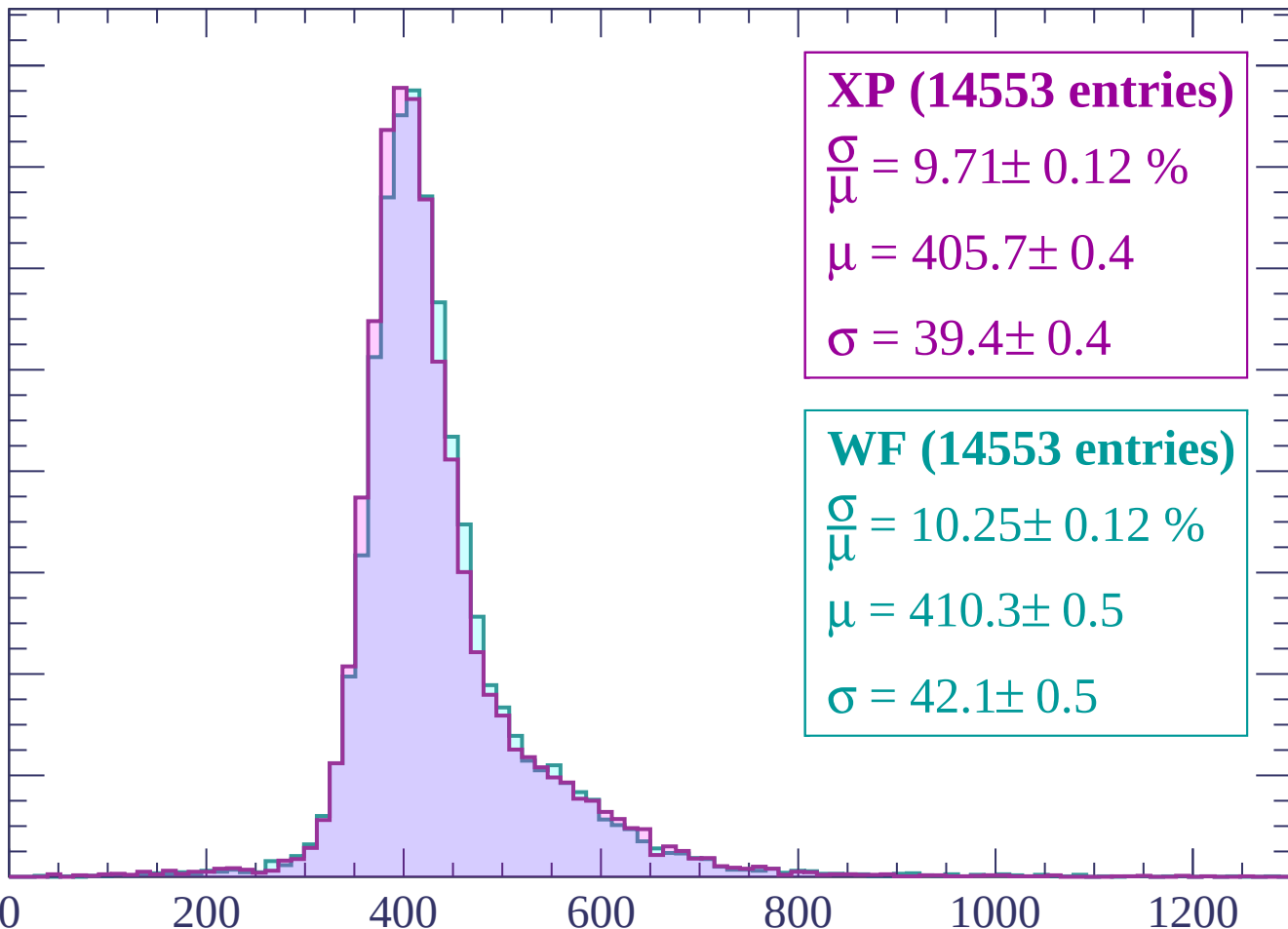
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $540 < p < 600$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14376 entries)

$$\frac{\sigma}{\mu} = 10.01 \pm 0.12 \%$$

$$\mu = 410.6 \pm 0.4$$

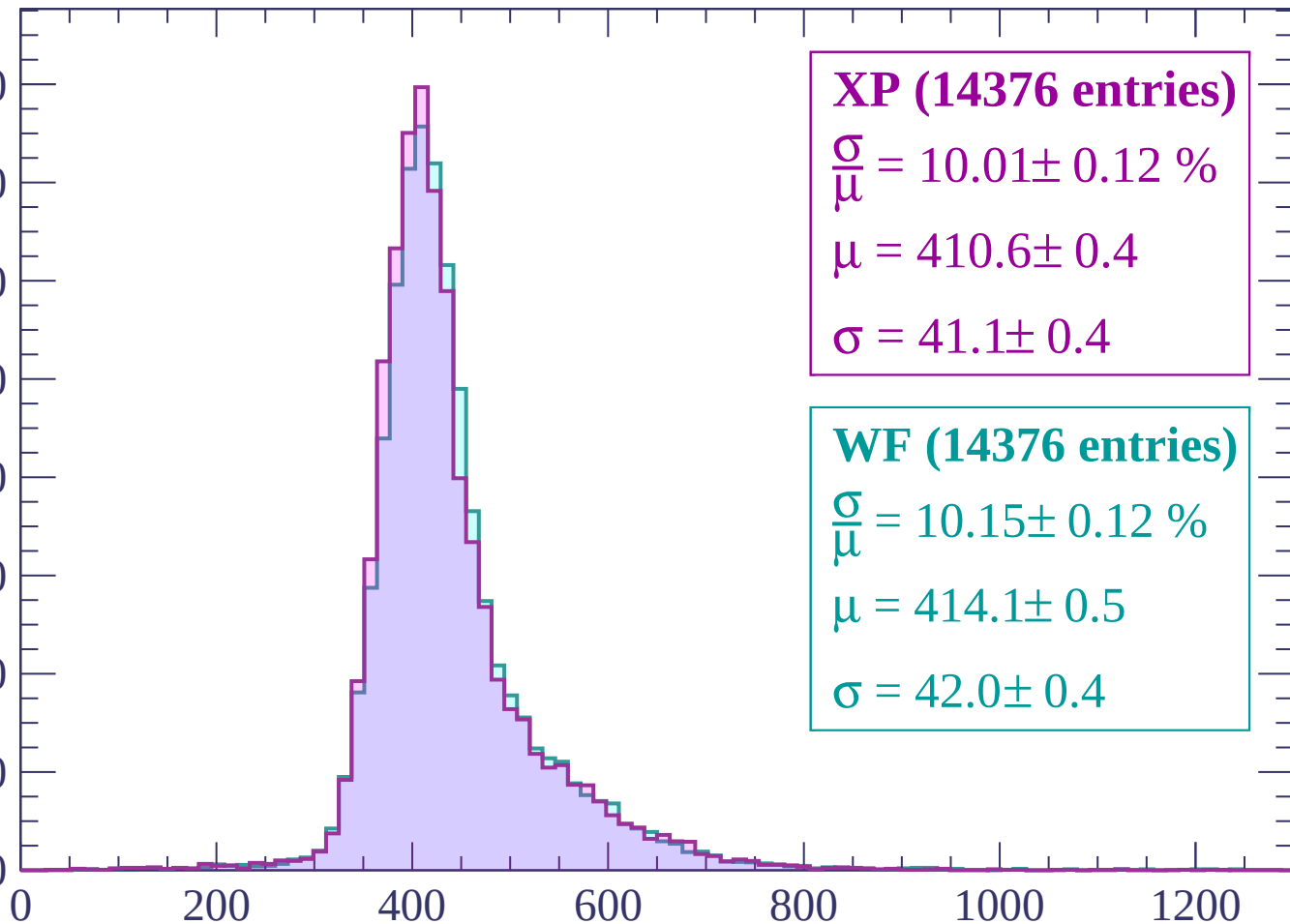
$$\sigma = 41.1 \pm 0.4$$

WF (14376 entries)

$$\frac{\sigma}{\mu} = 10.15 \pm 0.12 \%$$

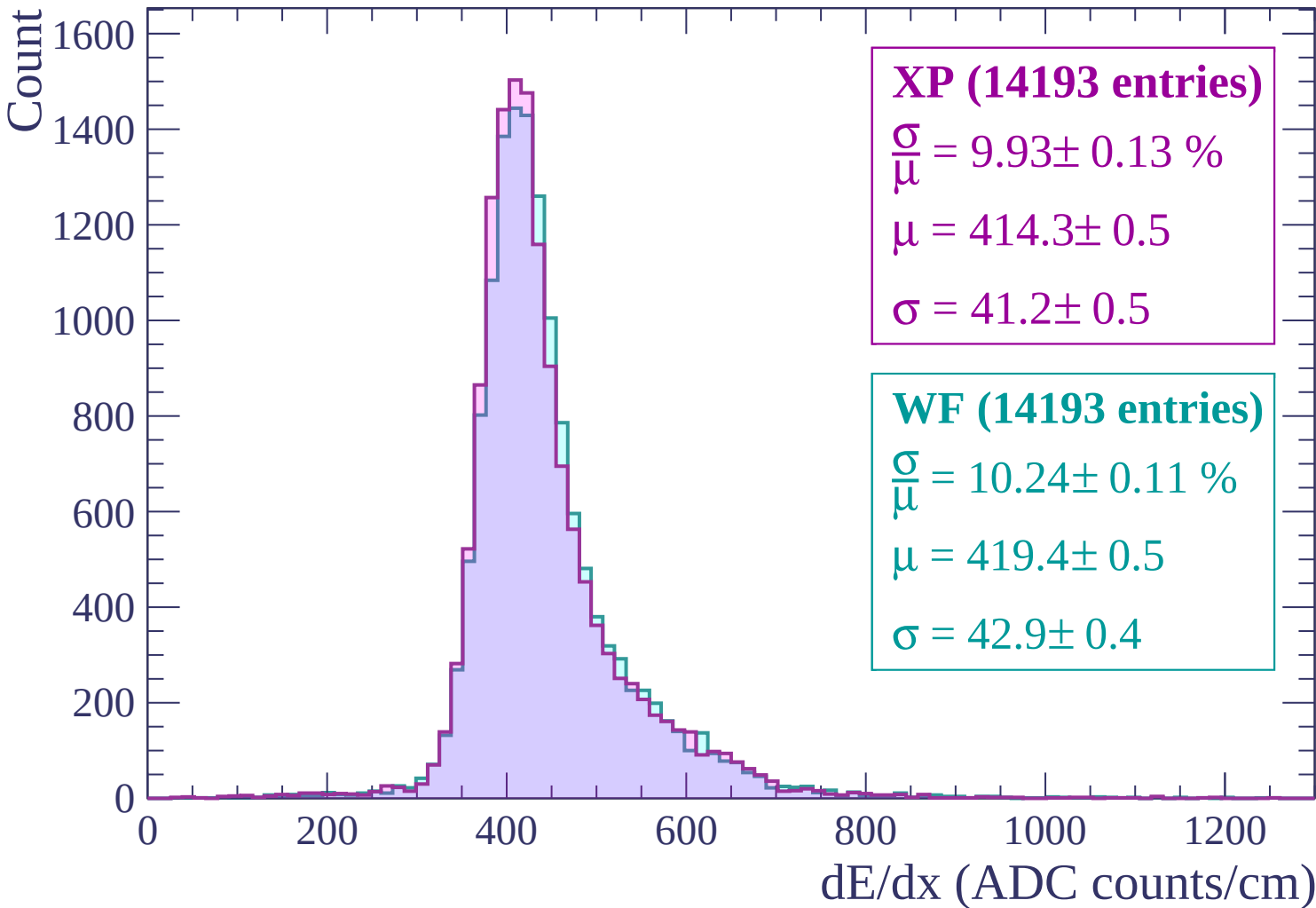
$$\mu = 414.1 \pm 0.5$$

$$\sigma = 42.0 \pm 0.4$$



dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$



Energy loss | $660 < p < 720$

Count

1400
1200
1000
800
600
400
200
0

XP (13767 entries)

$$\frac{\sigma}{\mu} = 10.50 \pm 0.14 \%$$

$$\mu = 419.6 \pm 0.5$$

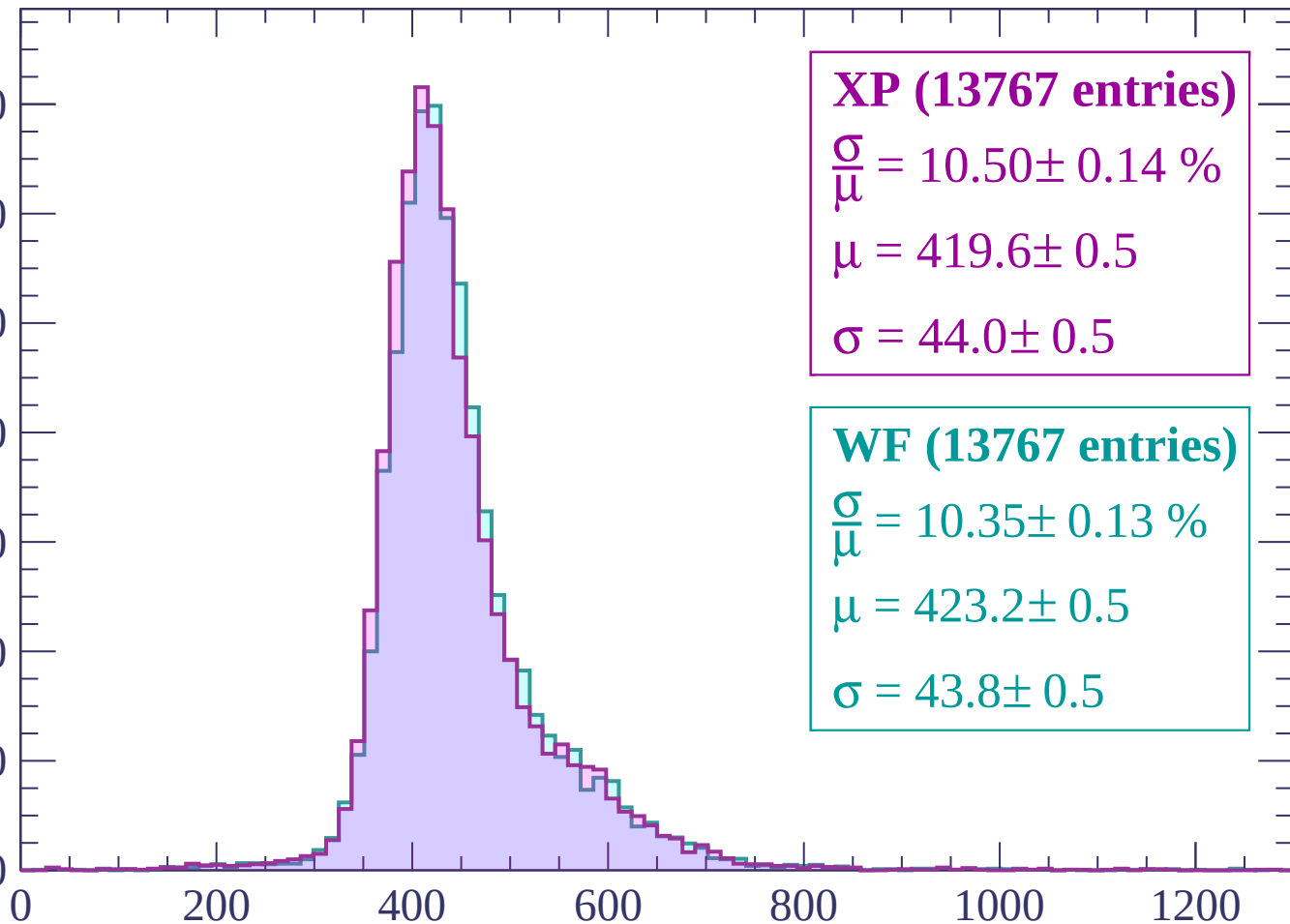
$$\sigma = 44.0 \pm 0.5$$

WF (13767 entries)

$$\frac{\sigma}{\mu} = 10.35 \pm 0.13 \%$$

$$\mu = 423.2 \pm 0.5$$

$$\sigma = 43.8 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count

1400
1200
1000
800
600
400
200
0

XP (13509 entries)

$$\frac{\sigma}{\mu} = 10.40 \pm 0.14 \%$$

$$\mu = 422.7 \pm 0.5$$

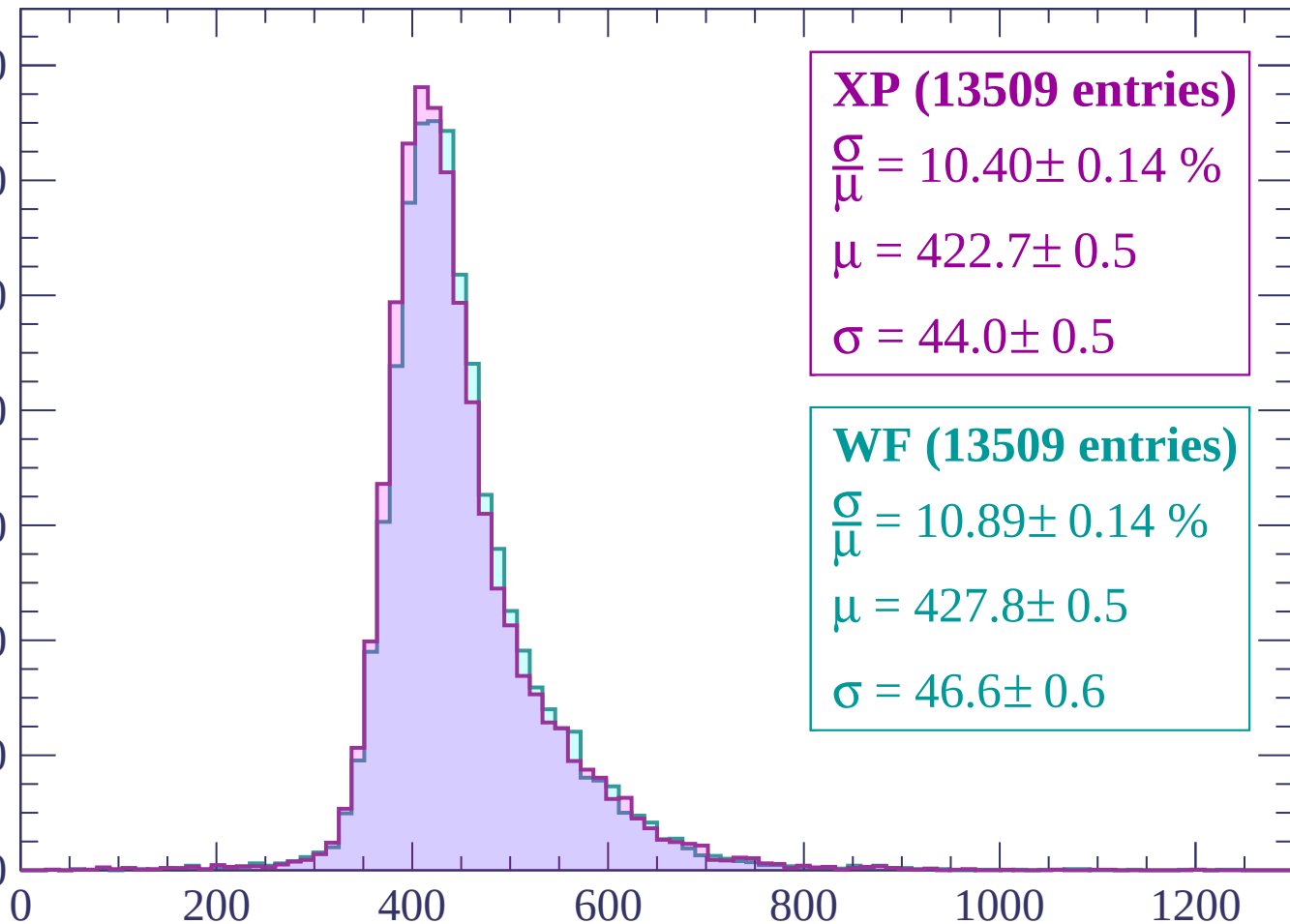
$$\sigma = 44.0 \pm 0.5$$

WF (13509 entries)

$$\frac{\sigma}{\mu} = 10.89 \pm 0.14 \%$$

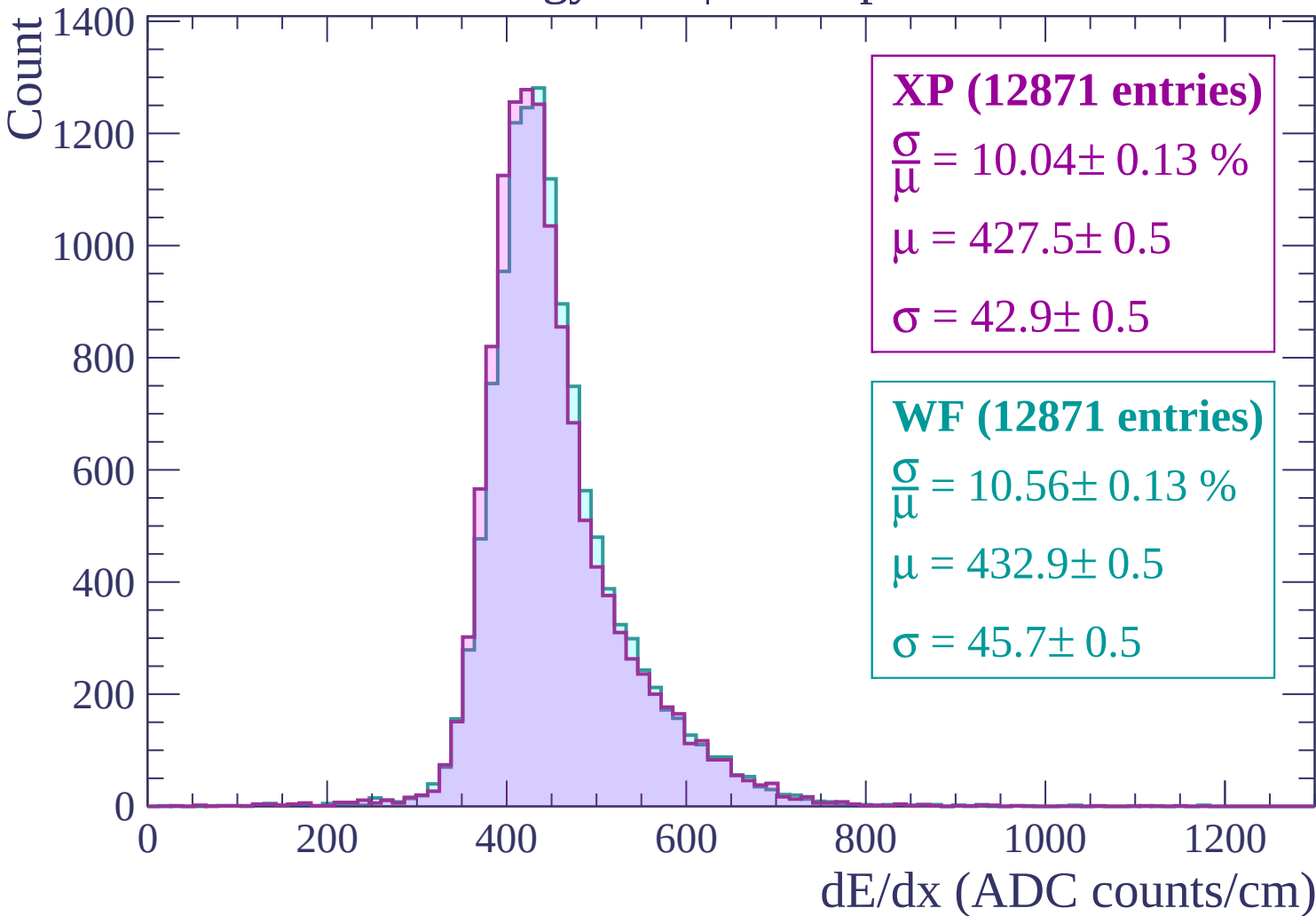
$$\mu = 427.8 \pm 0.5$$

$$\sigma = 46.6 \pm 0.6$$



dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$



Energy loss | $840 < p < 900$

Count

1200

1000

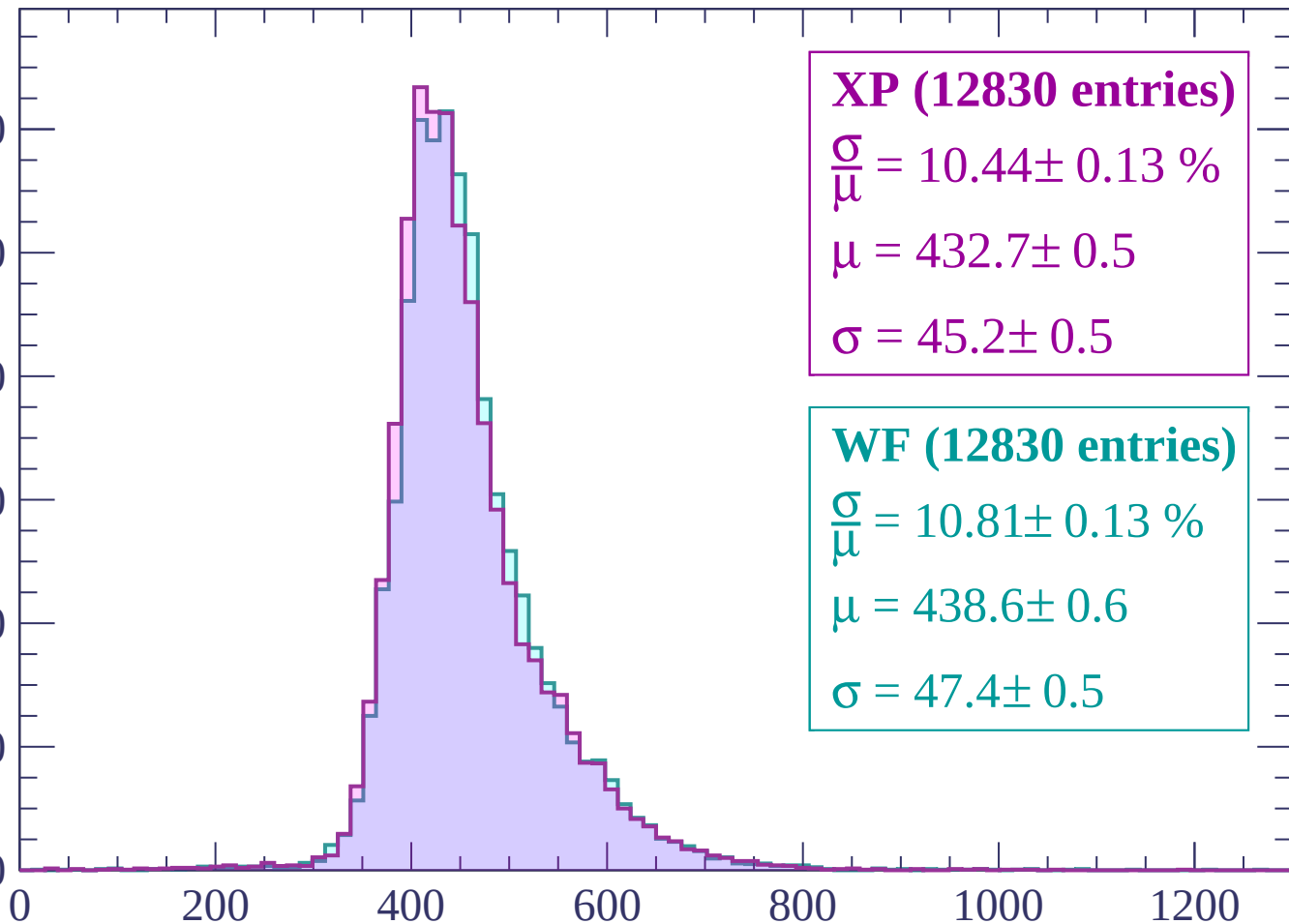
800

600

400

200

0



XP (12830 entries)

$\frac{\sigma}{\mu} = 10.44 \pm 0.13 \%$

$\mu = 432.7 \pm 0.5$

$\sigma = 45.2 \pm 0.5$

WF (12830 entries)

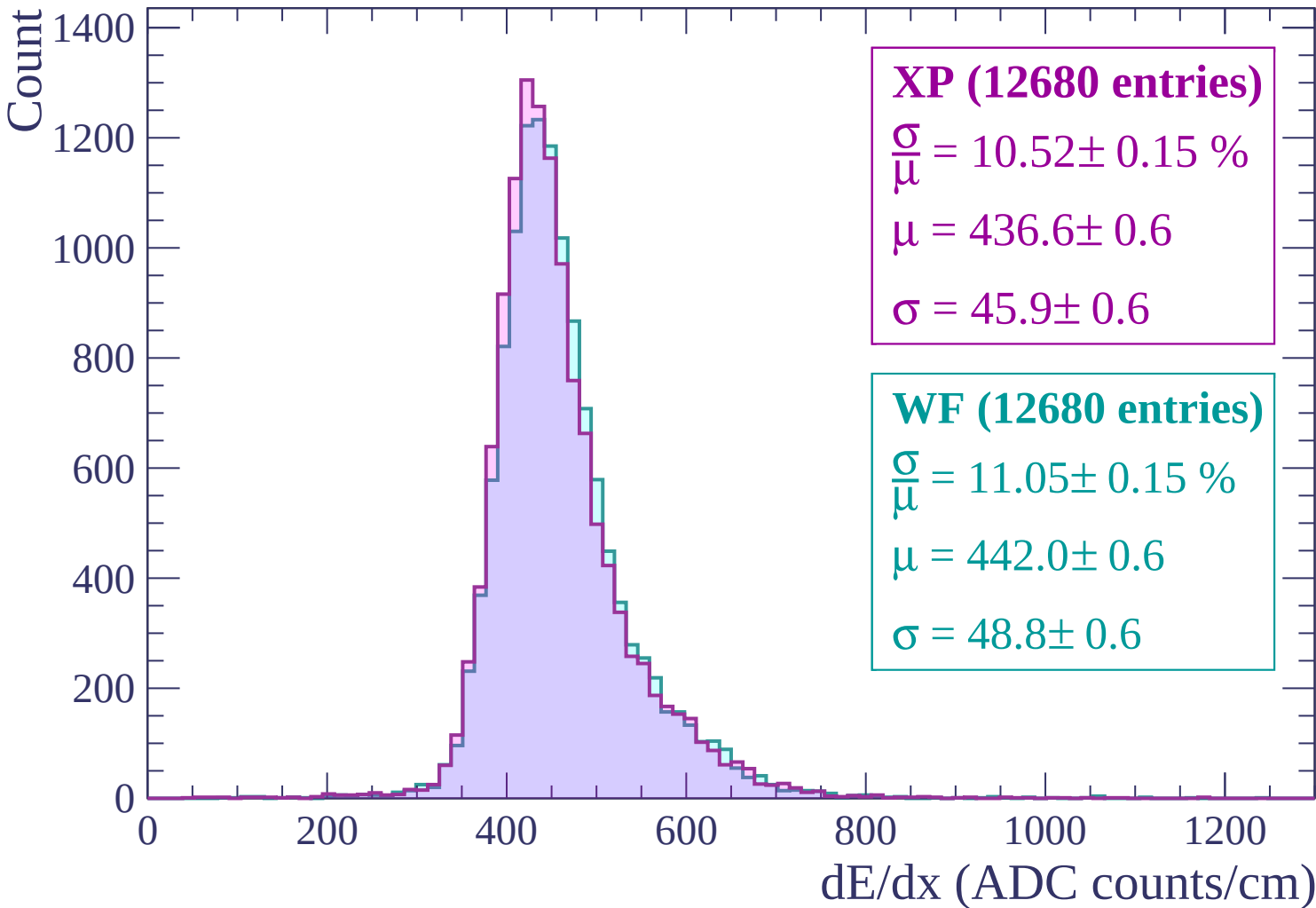
$\frac{\sigma}{\mu} = 10.81 \pm 0.13 \%$

$\mu = 438.6 \pm 0.6$

$\sigma = 47.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$



Energy loss | $960 < p < 1020$

Count

1200

1000

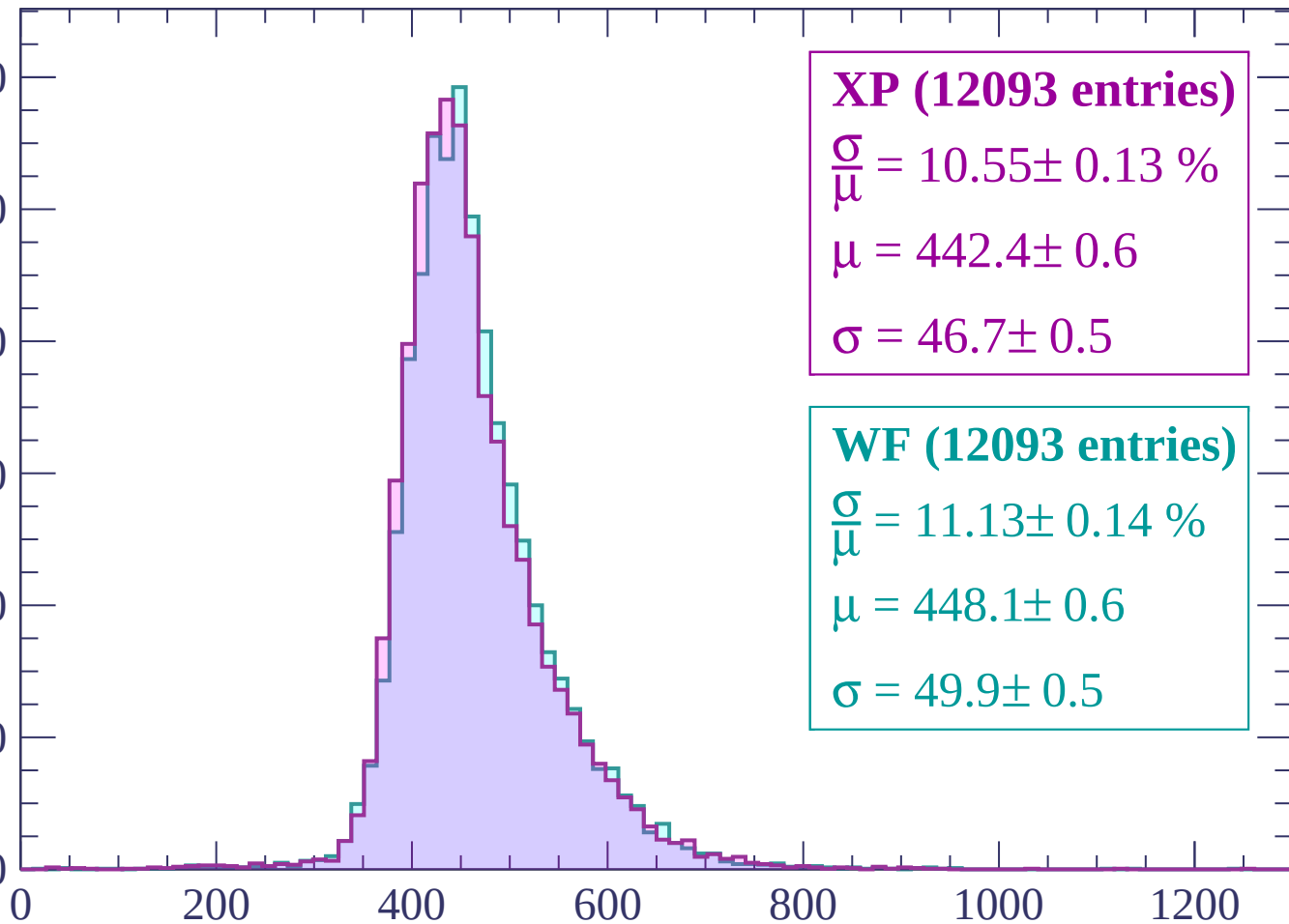
800

600

400

200

0



XP (12093 entries)

$\frac{\sigma}{\mu} = 10.55 \pm 0.13 \%$

$\mu = 442.4 \pm 0.6$

$\sigma = 46.7 \pm 0.5$

WF (12093 entries)

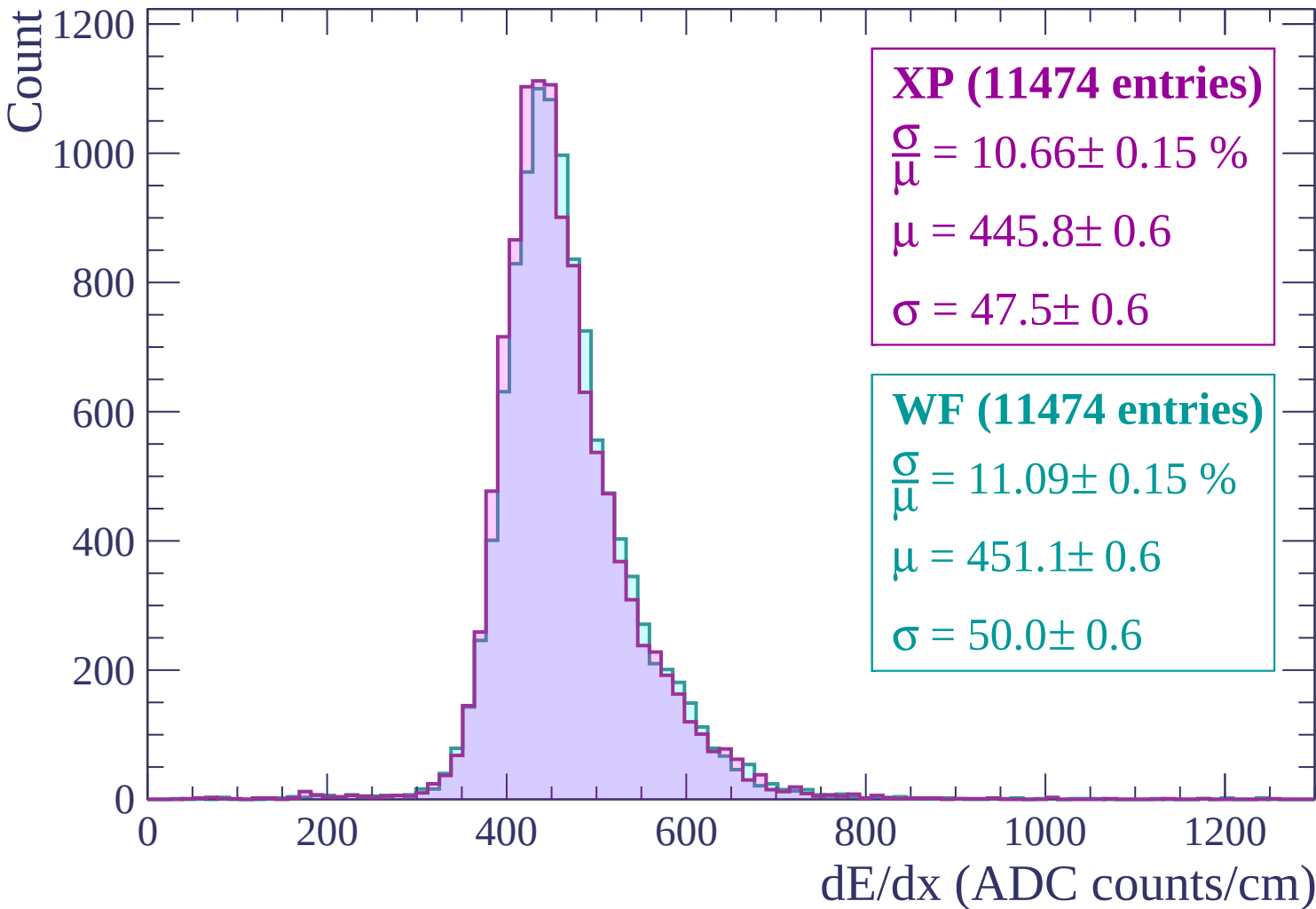
$\frac{\sigma}{\mu} = 11.13 \pm 0.14 \%$

$\mu = 448.1 \pm 0.6$

$\sigma = 49.9 \pm 0.5$

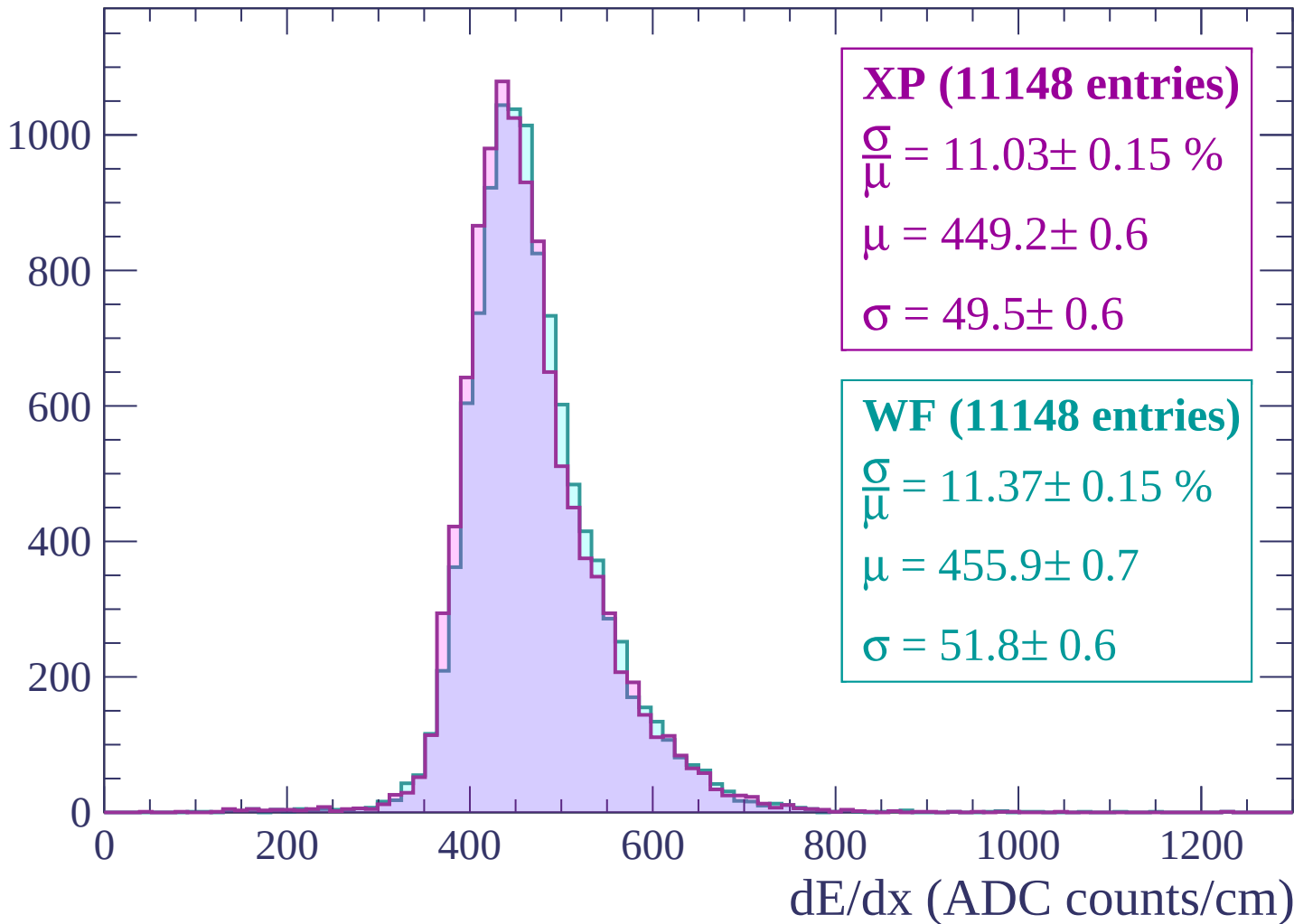
dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$



Energy loss | $1080 < p < 1140$

Count



Energy loss | $1140 < p < 1200$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (10820 entries)

$\frac{\sigma}{\mu} = 10.95 \pm 0.15 \%$

$\mu = 452.4 \pm 0.6$

$\sigma = 49.5 \pm 0.6$

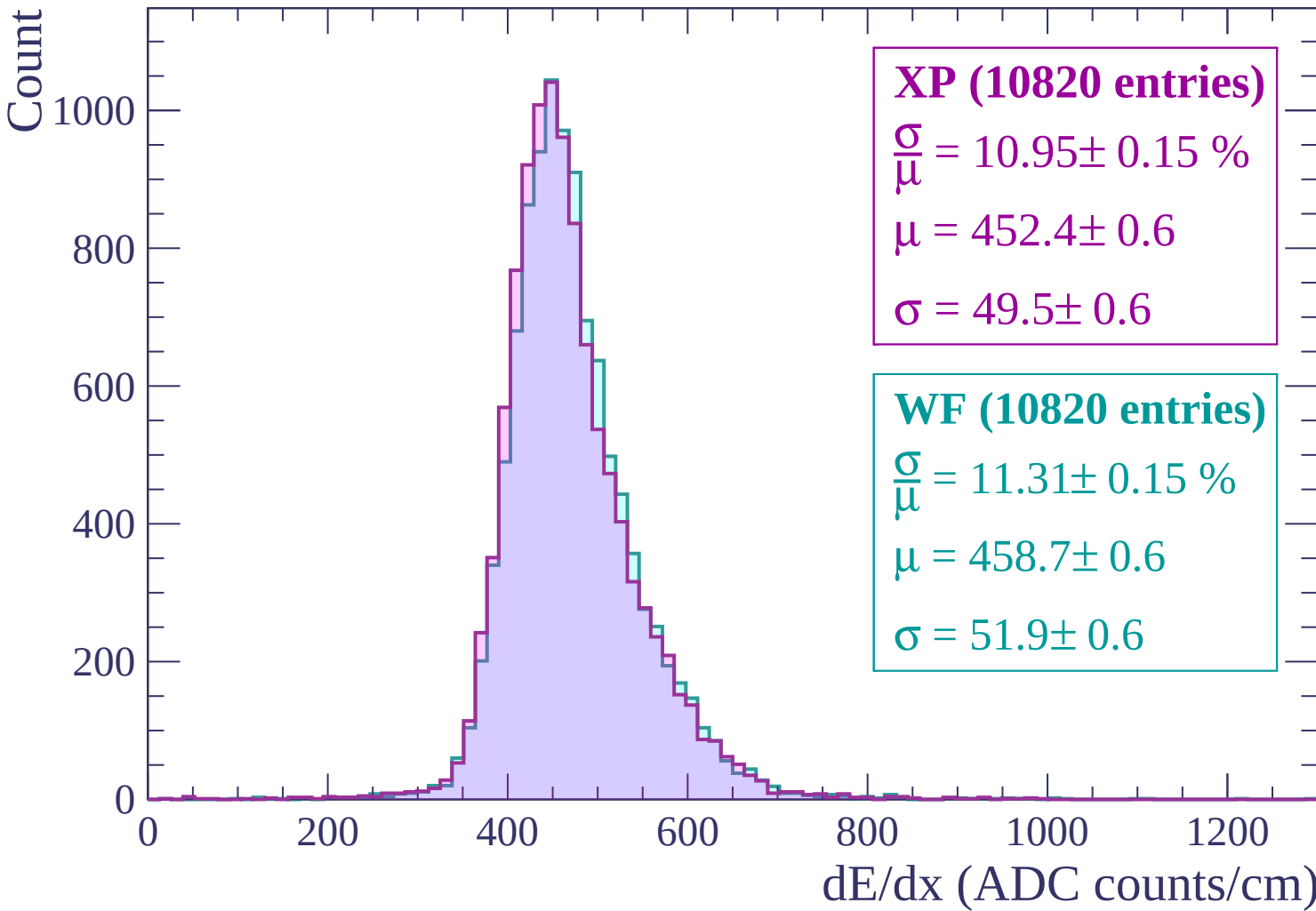
WF (10820 entries)

$\frac{\sigma}{\mu} = 11.31 \pm 0.15 \%$

$\mu = 458.7 \pm 0.6$

$\sigma = 51.9 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $1200 < p < 1260$

Count

1000

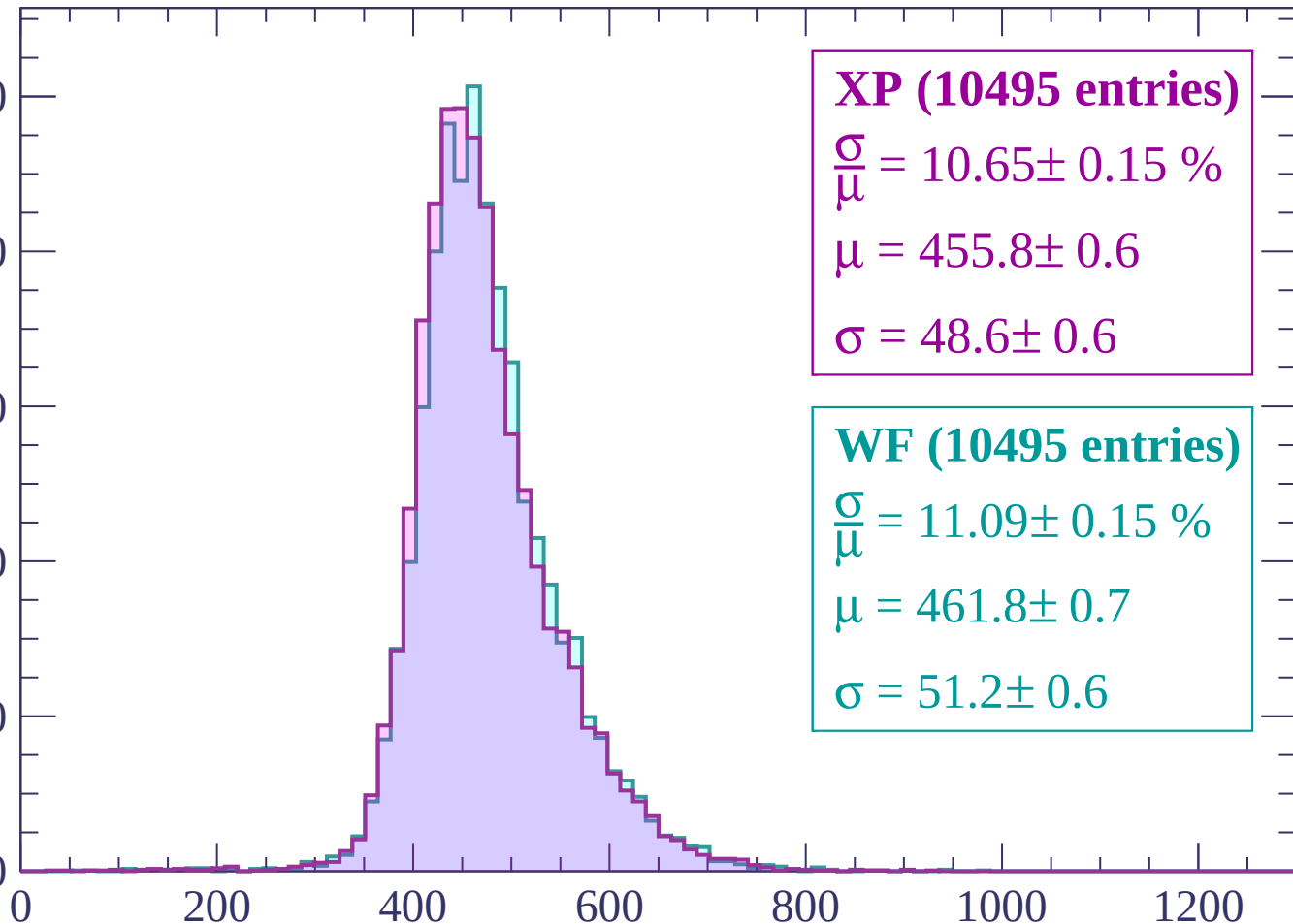
800

600

400

200

0



XP (10495 entries)

$\frac{\sigma}{\mu} = 10.65 \pm 0.15 \%$

$\mu = 455.8 \pm 0.6$

$\sigma = 48.6 \pm 0.6$

WF (10495 entries)

$\frac{\sigma}{\mu} = 11.09 \pm 0.15 \%$

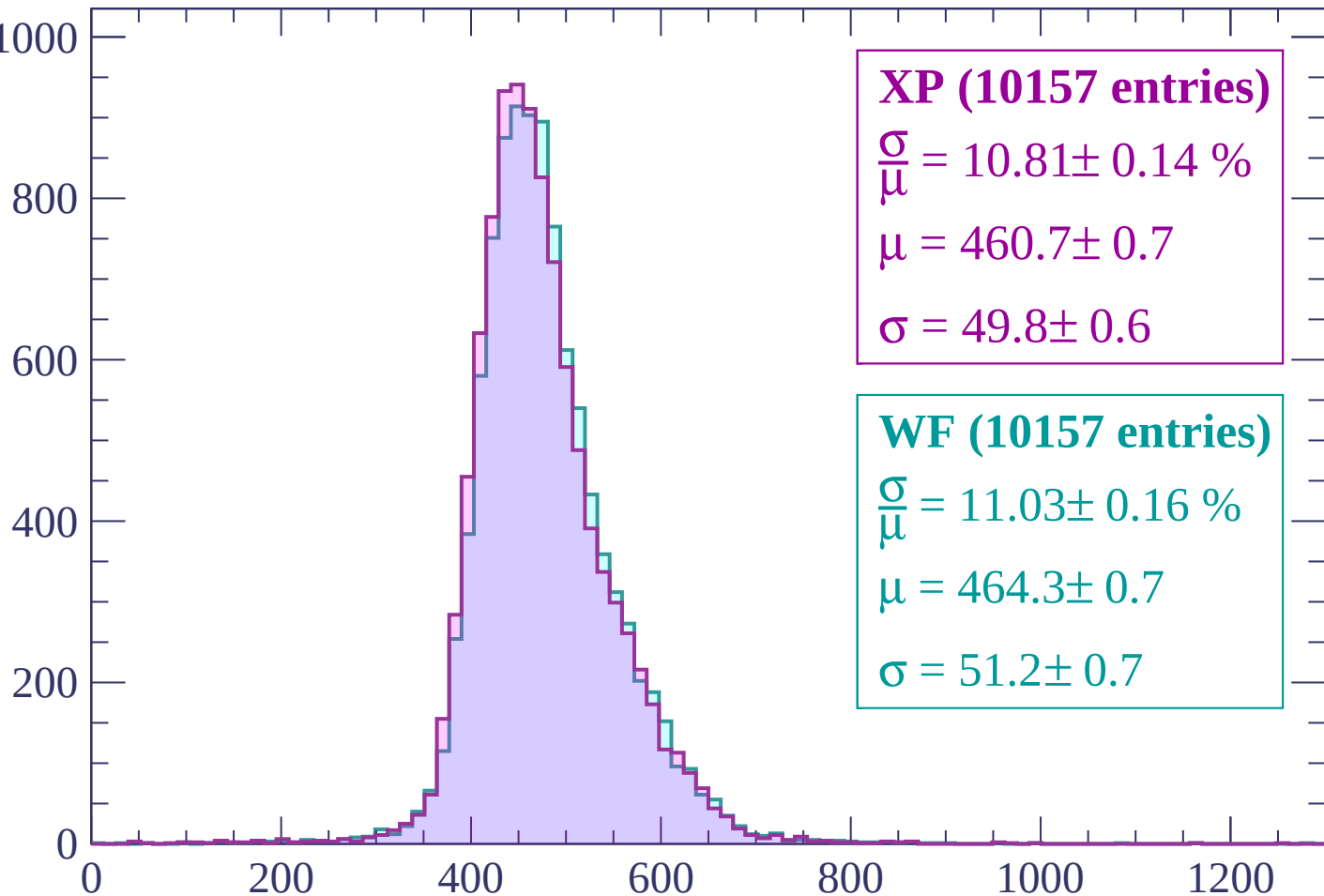
$\mu = 461.8 \pm 0.7$

$\sigma = 51.2 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (10157 entries)

$$\frac{\sigma}{\mu} = 10.81 \pm 0.14 \%$$

$$\mu = 460.7 \pm 0.7$$

$$\sigma = 49.8 \pm 0.6$$

WF (10157 entries)

$$\frac{\sigma}{\mu} = 11.03 \pm 0.16 \%$$

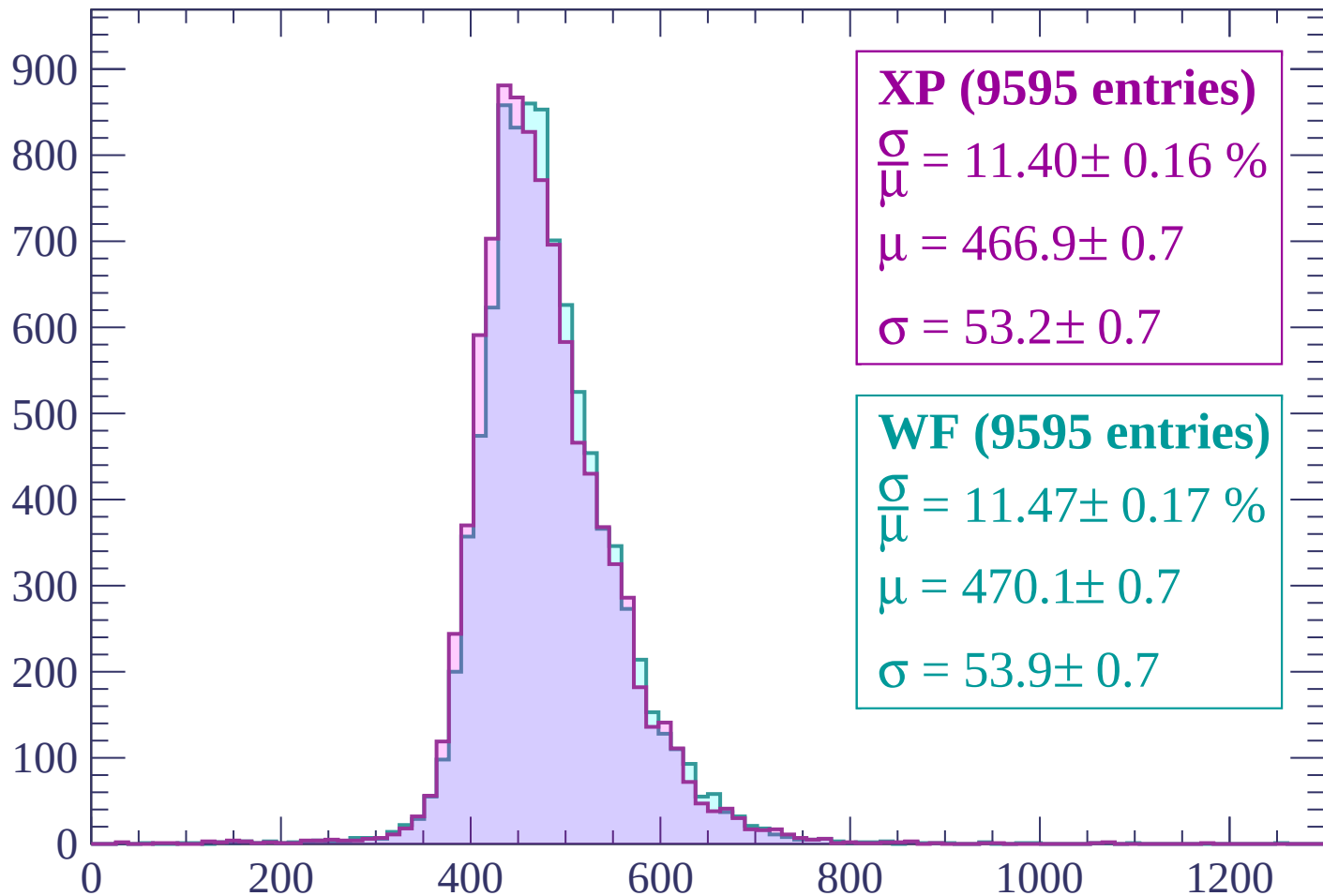
$$\mu = 464.3 \pm 0.7$$

$$\sigma = 51.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (9595 entries)

$$\frac{\sigma}{\mu} = 11.40 \pm 0.16 \%$$

$$\mu = 466.9 \pm 0.7$$

$$\sigma = 53.2 \pm 0.7$$

WF (9595 entries)

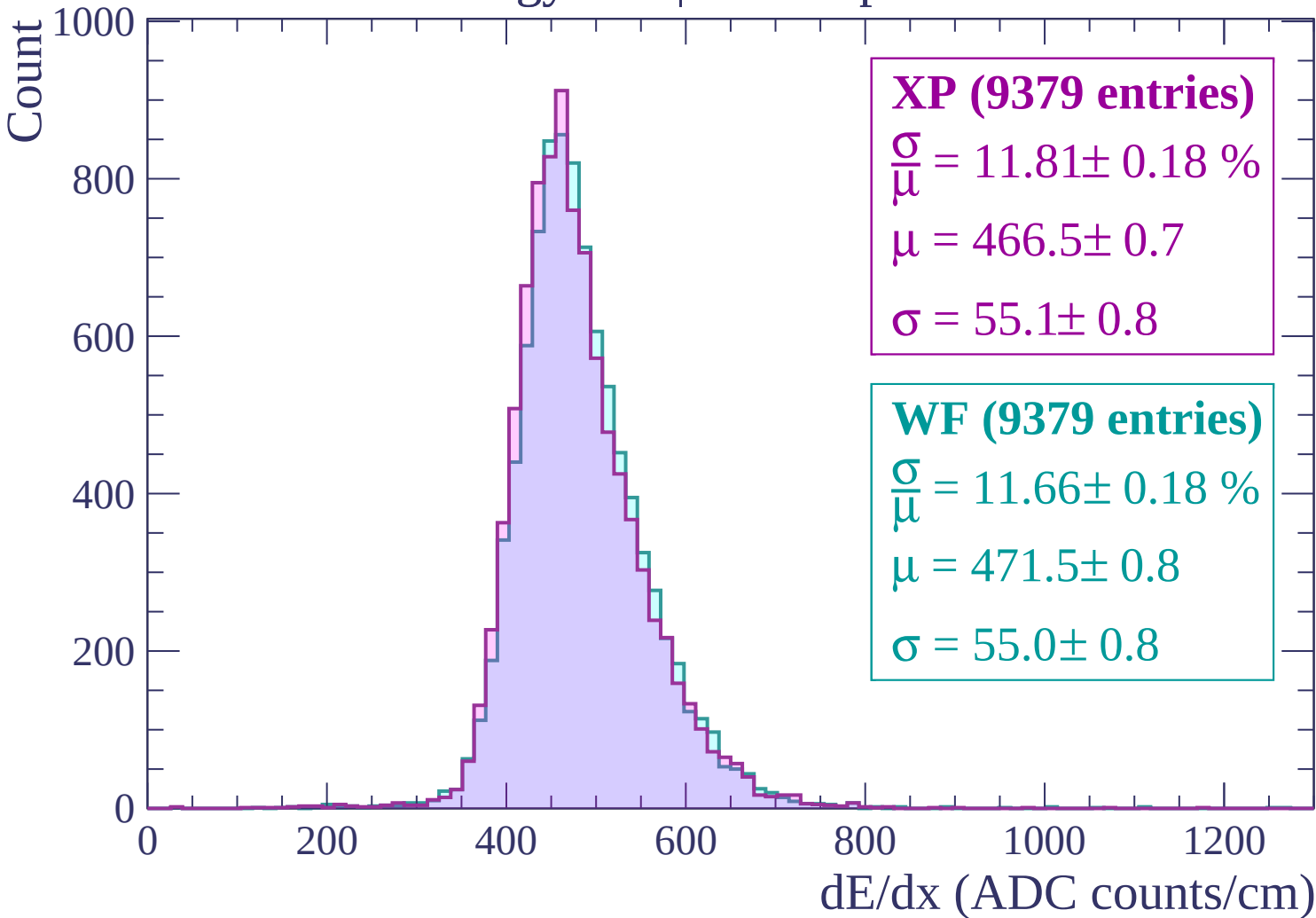
$$\frac{\sigma}{\mu} = 11.47 \pm 0.17 \%$$

$$\mu = 470.1 \pm 0.7$$

$$\sigma = 53.9 \pm 0.7$$

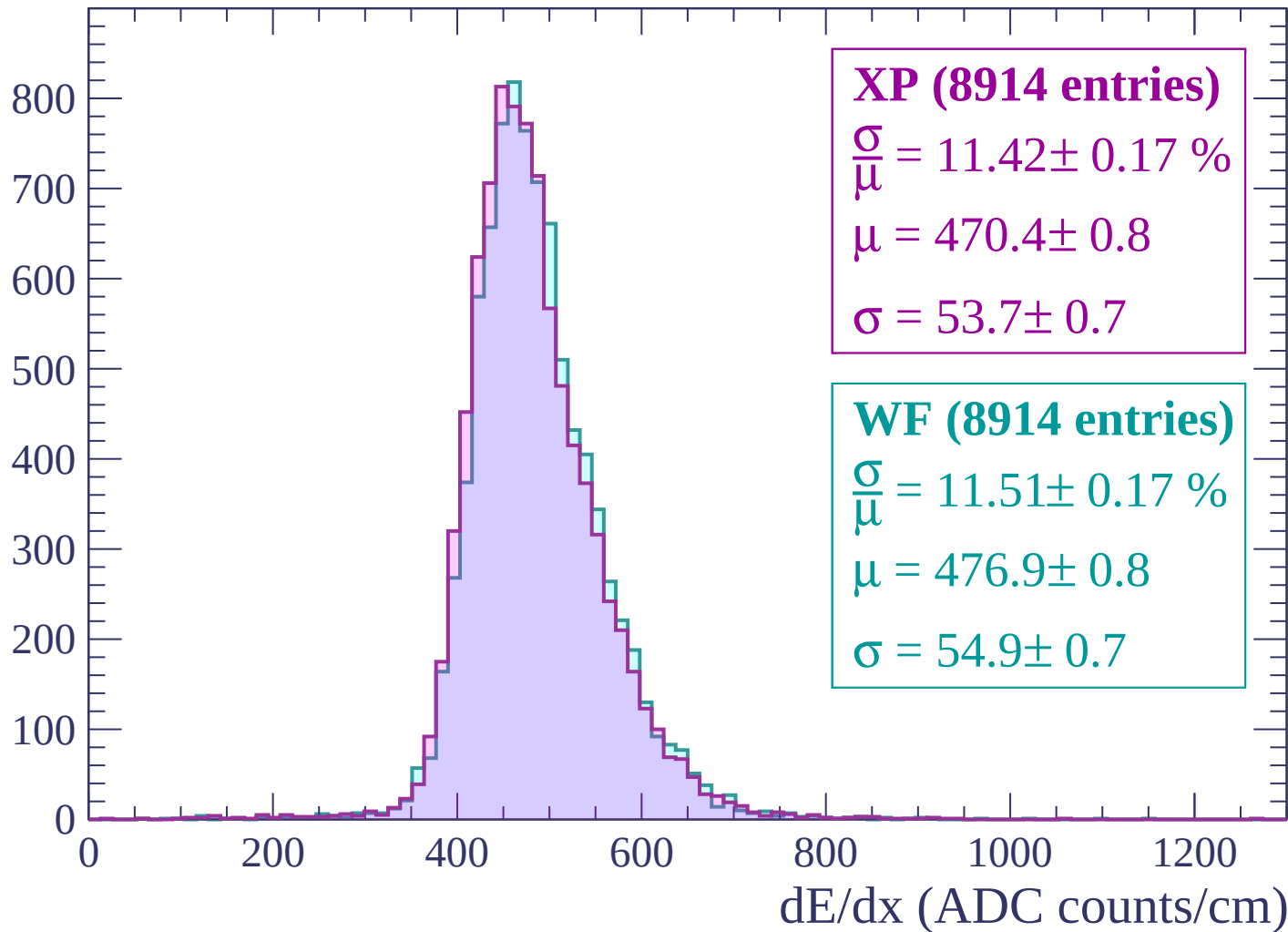
dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$



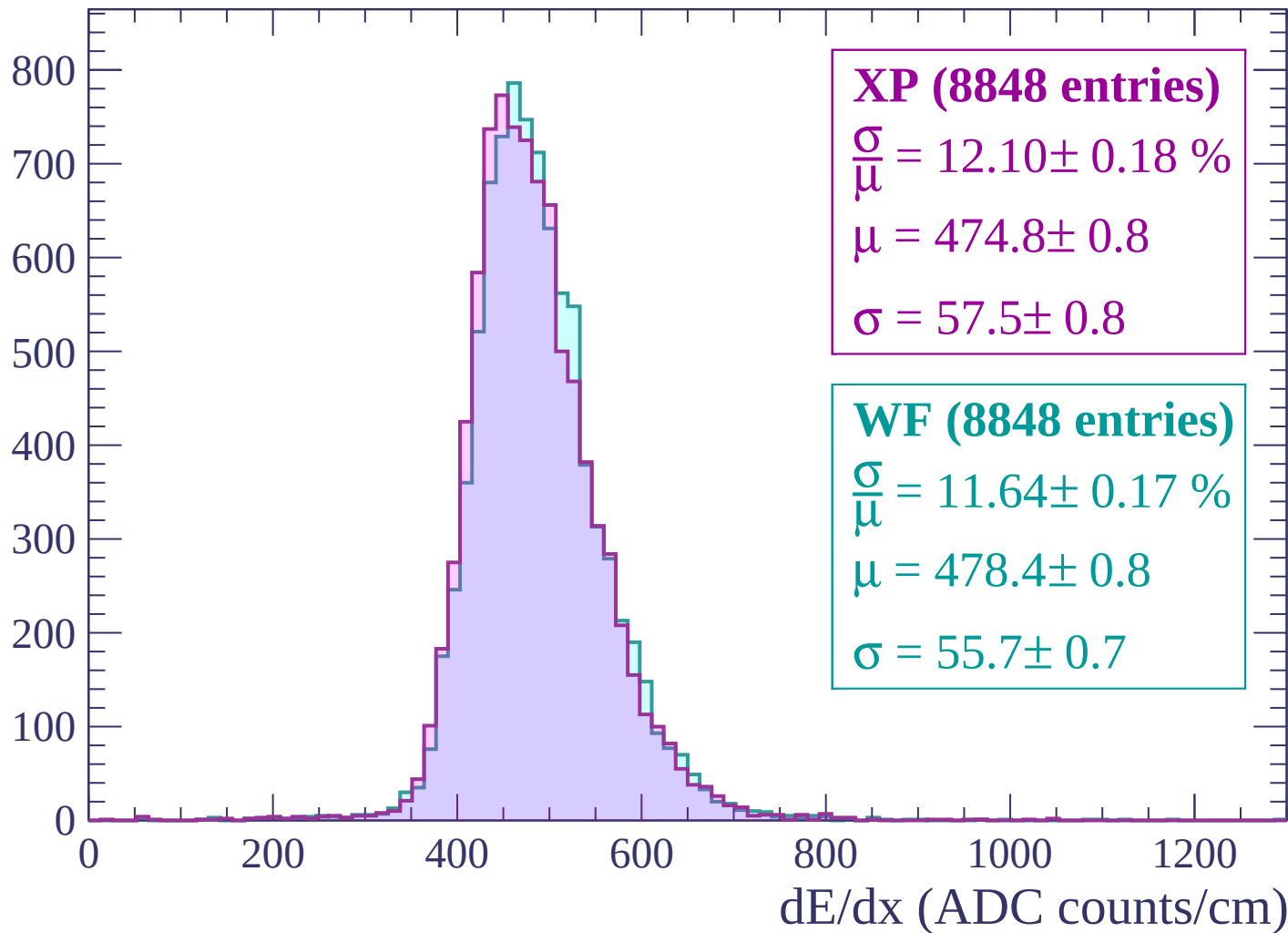
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count

800
700
600
500
400
300
200
100
0

XP (8305 entries)

$$\frac{\sigma}{\mu} = 11.31 \pm 0.17 \%$$

$$\mu = 473.2 \pm 0.8$$

$$\sigma = 53.5 \pm 0.7$$

WF (8305 entries)

$$\frac{\sigma}{\mu} = 11.37 \pm 0.17 \%$$

$$\mu = 479.5 \pm 0.8$$

$$\sigma = 54.5 \pm 0.7$$

0

200

400

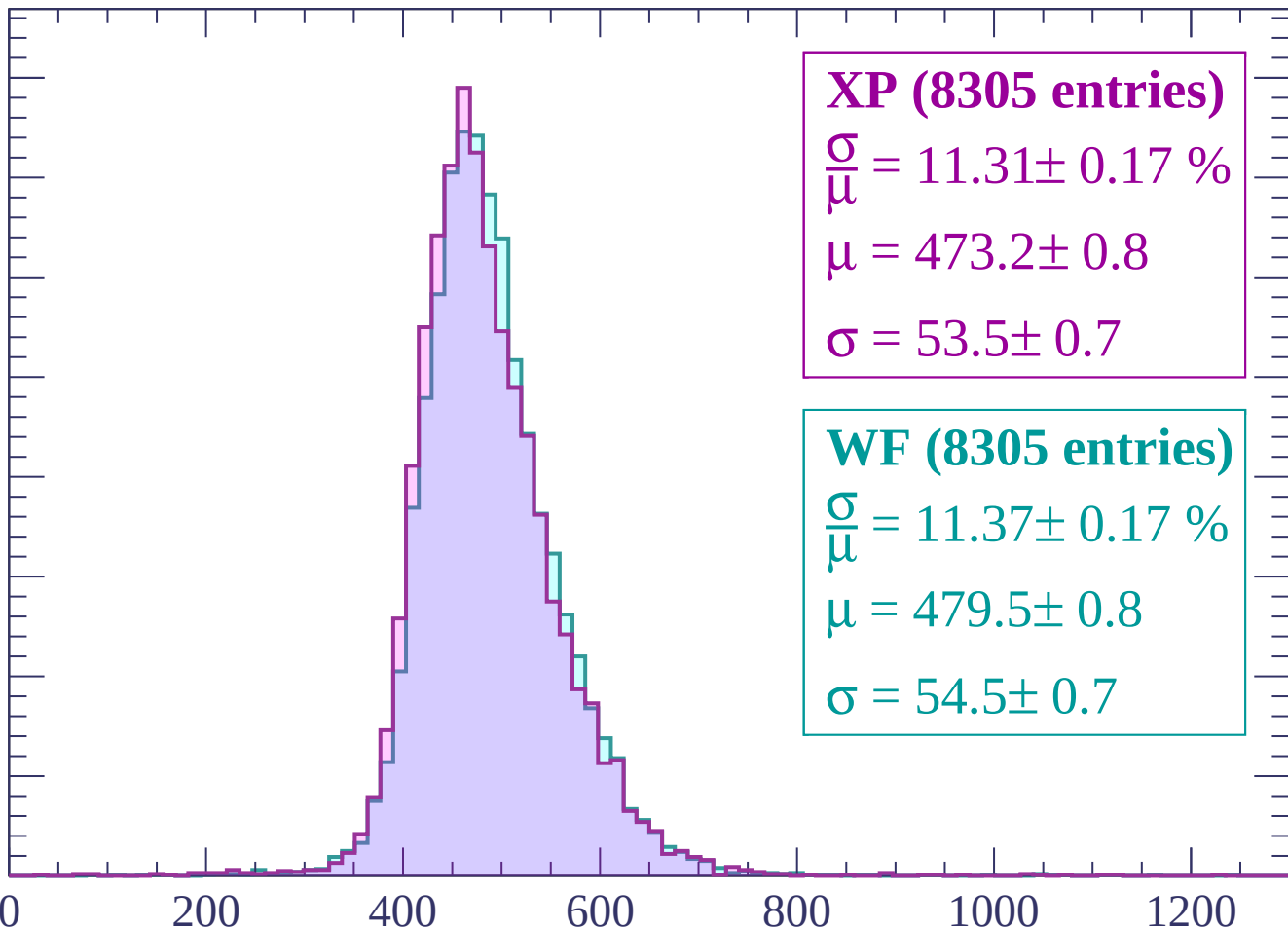
600

800

1000

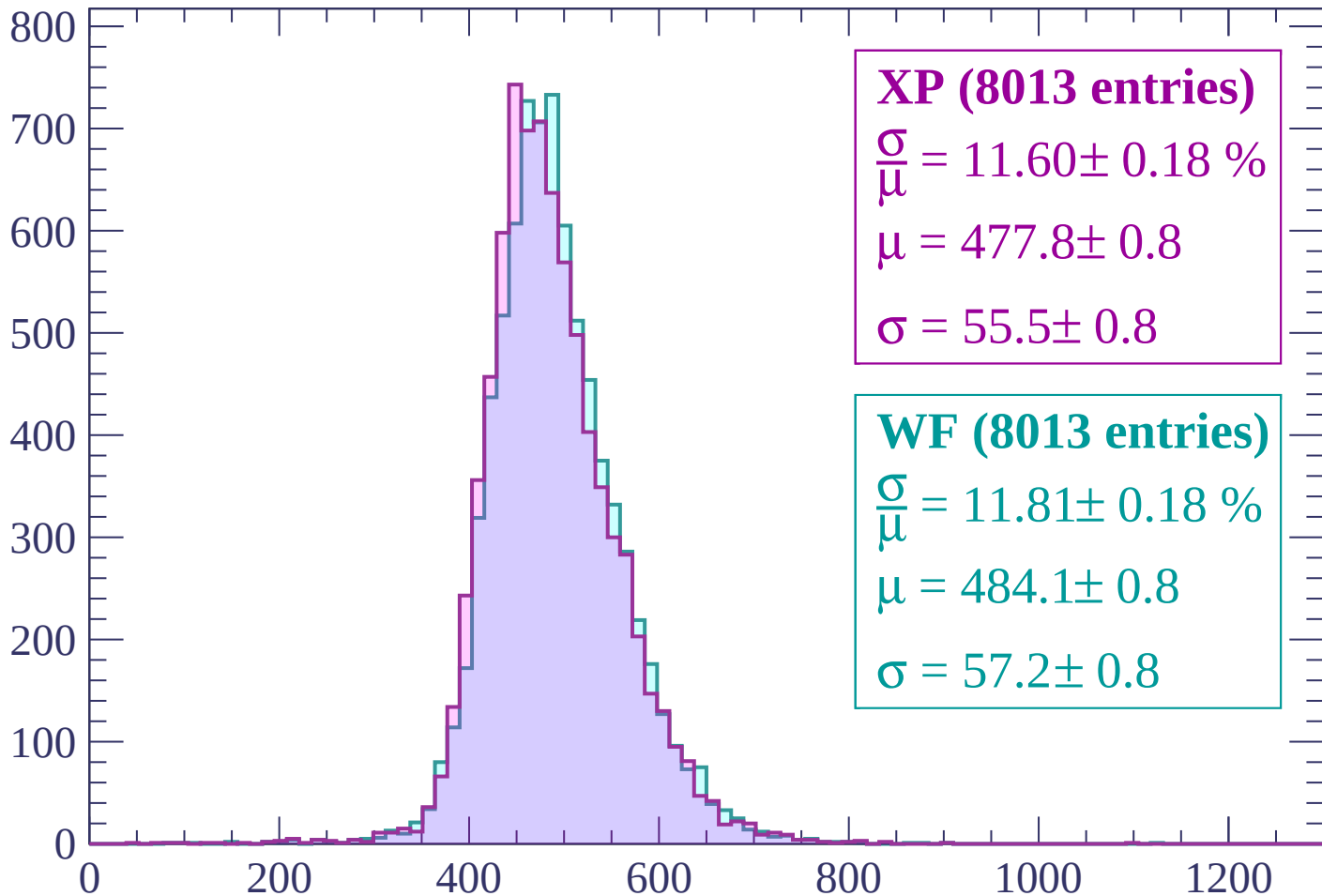
1200

dE/dx (ADC counts/cm)



Energy loss | $1620 < p < 1680$

Count



XP (8013 entries)

$$\frac{\sigma}{\mu} = 11.60 \pm 0.18 \%$$

$$\mu = 477.8 \pm 0.8$$

$$\sigma = 55.5 \pm 0.8$$

WF (8013 entries)

$$\frac{\sigma}{\mu} = 11.81 \pm 0.18 \%$$

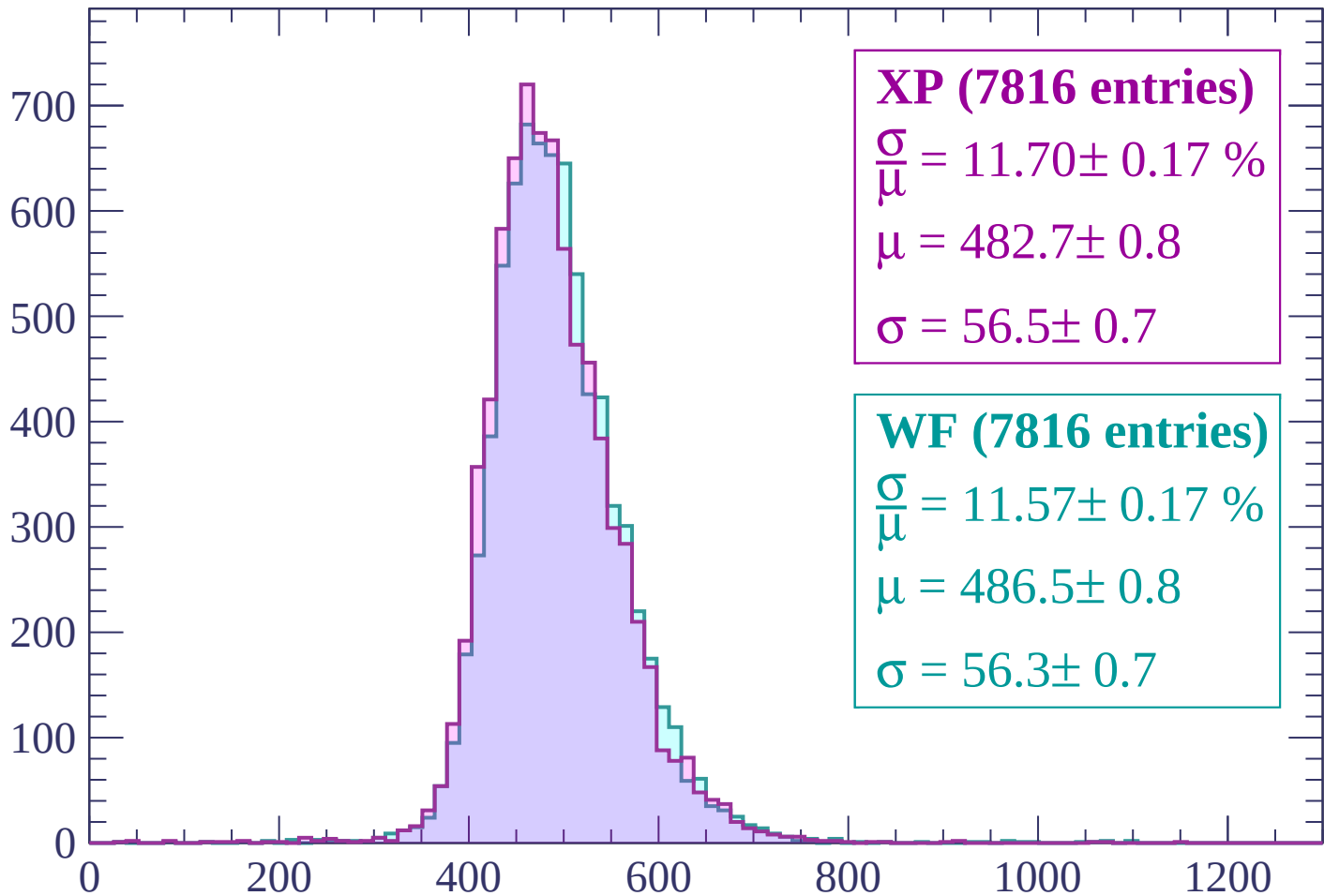
$$\mu = 484.1 \pm 0.8$$

$$\sigma = 57.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (7816 entries)

$$\frac{\sigma}{\mu} = 11.70 \pm 0.17 \%$$

$$\mu = 482.7 \pm 0.8$$

$$\sigma = 56.5 \pm 0.7$$

WF (7816 entries)

$$\frac{\sigma}{\mu} = 11.57 \pm 0.17 \%$$

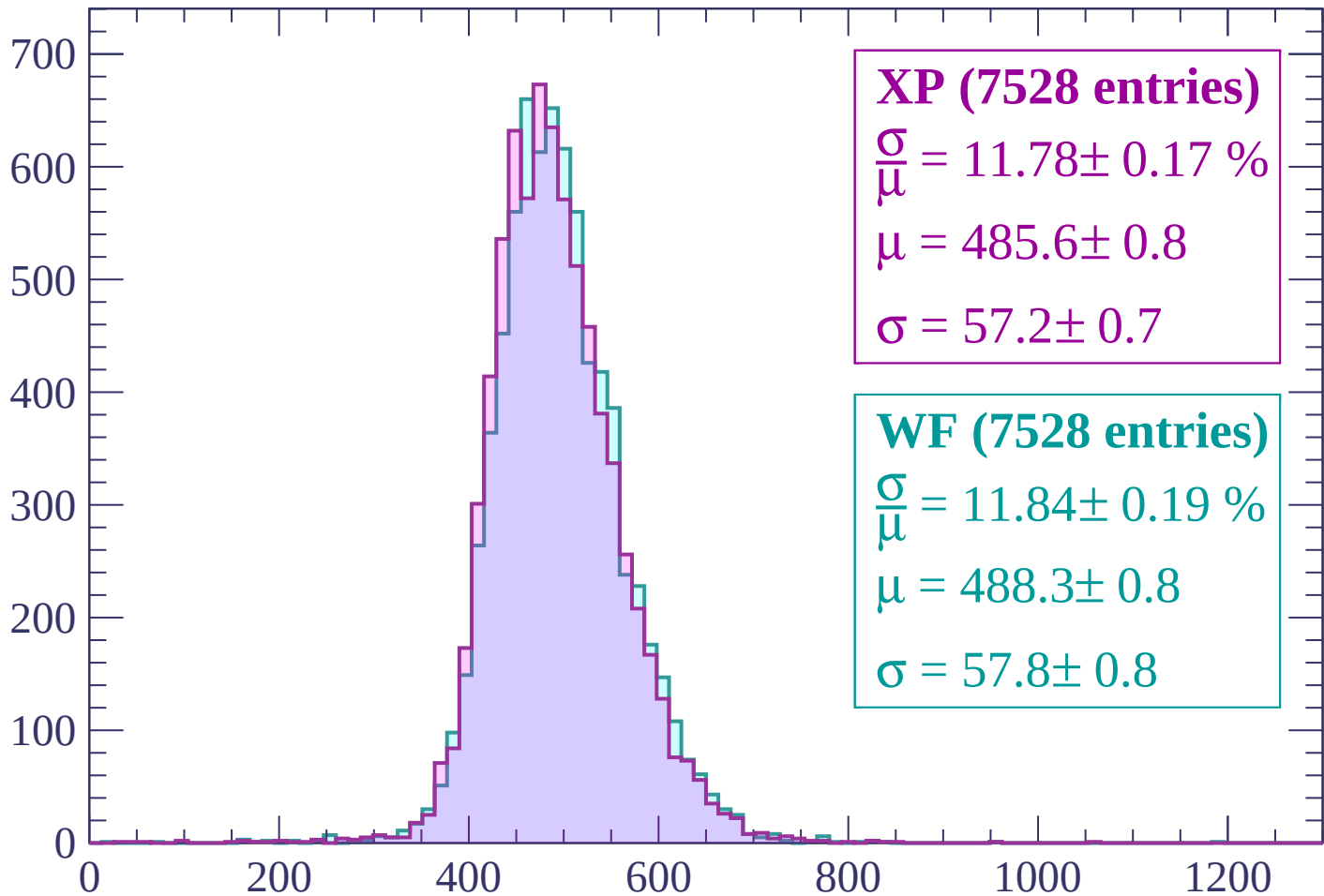
$$\mu = 486.5 \pm 0.8$$

$$\sigma = 56.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (7528 entries)

$\frac{\sigma}{\mu} = 11.78 \pm 0.17 \%$

$\mu = 485.6 \pm 0.8$

$\sigma = 57.2 \pm 0.7$

WF (7528 entries)

$\frac{\sigma}{\mu} = 11.84 \pm 0.19 \%$

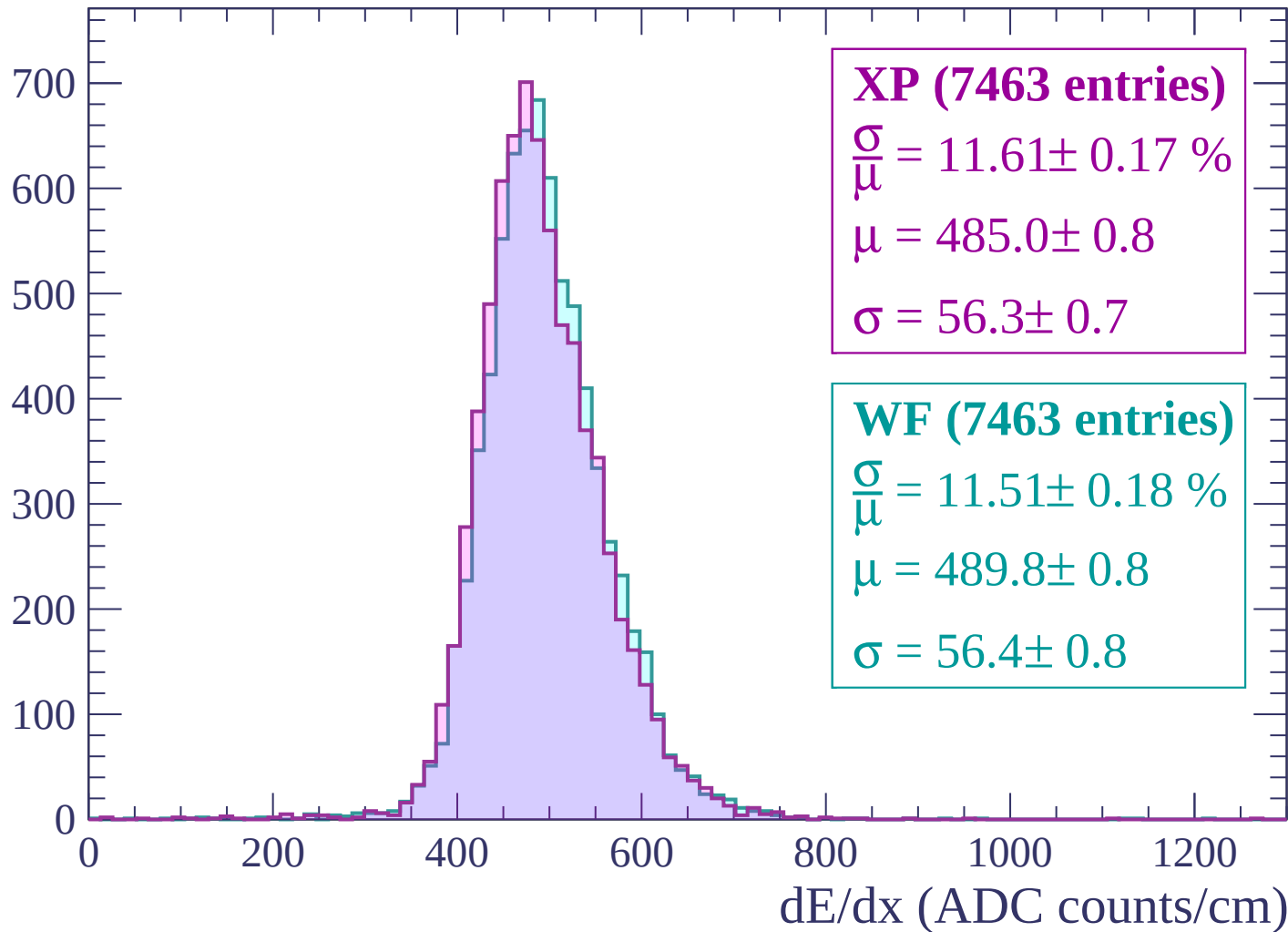
$\mu = 488.3 \pm 0.8$

$\sigma = 57.8 \pm 0.8$

dE/dx (ADC counts/cm)

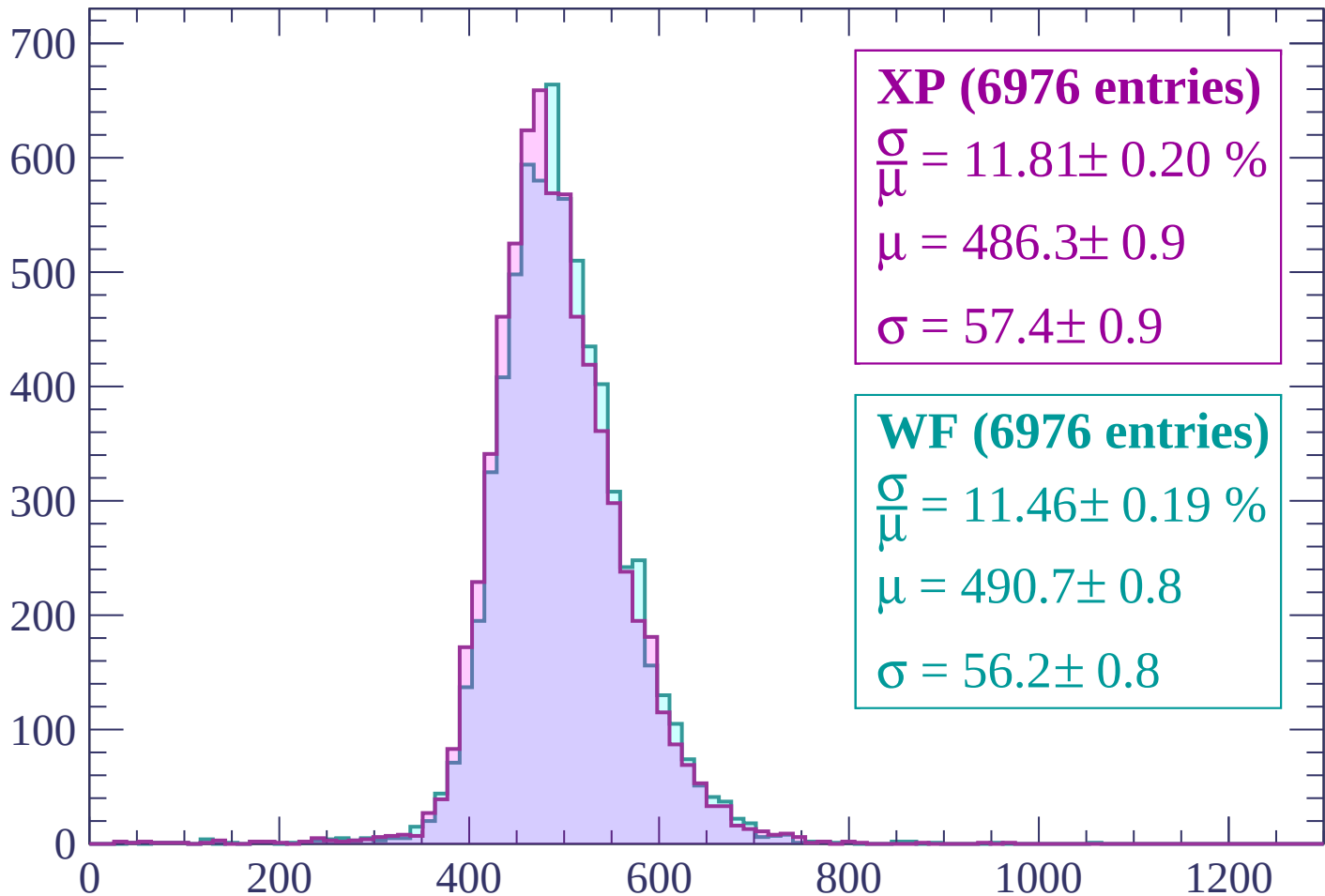
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (6976 entries)

$$\frac{\sigma}{\mu} = 11.81 \pm 0.20 \%$$

$$\mu = 486.3 \pm 0.9$$

$$\sigma = 57.4 \pm 0.9$$

WF (6976 entries)

$$\frac{\sigma}{\mu} = 11.46 \pm 0.19 \%$$

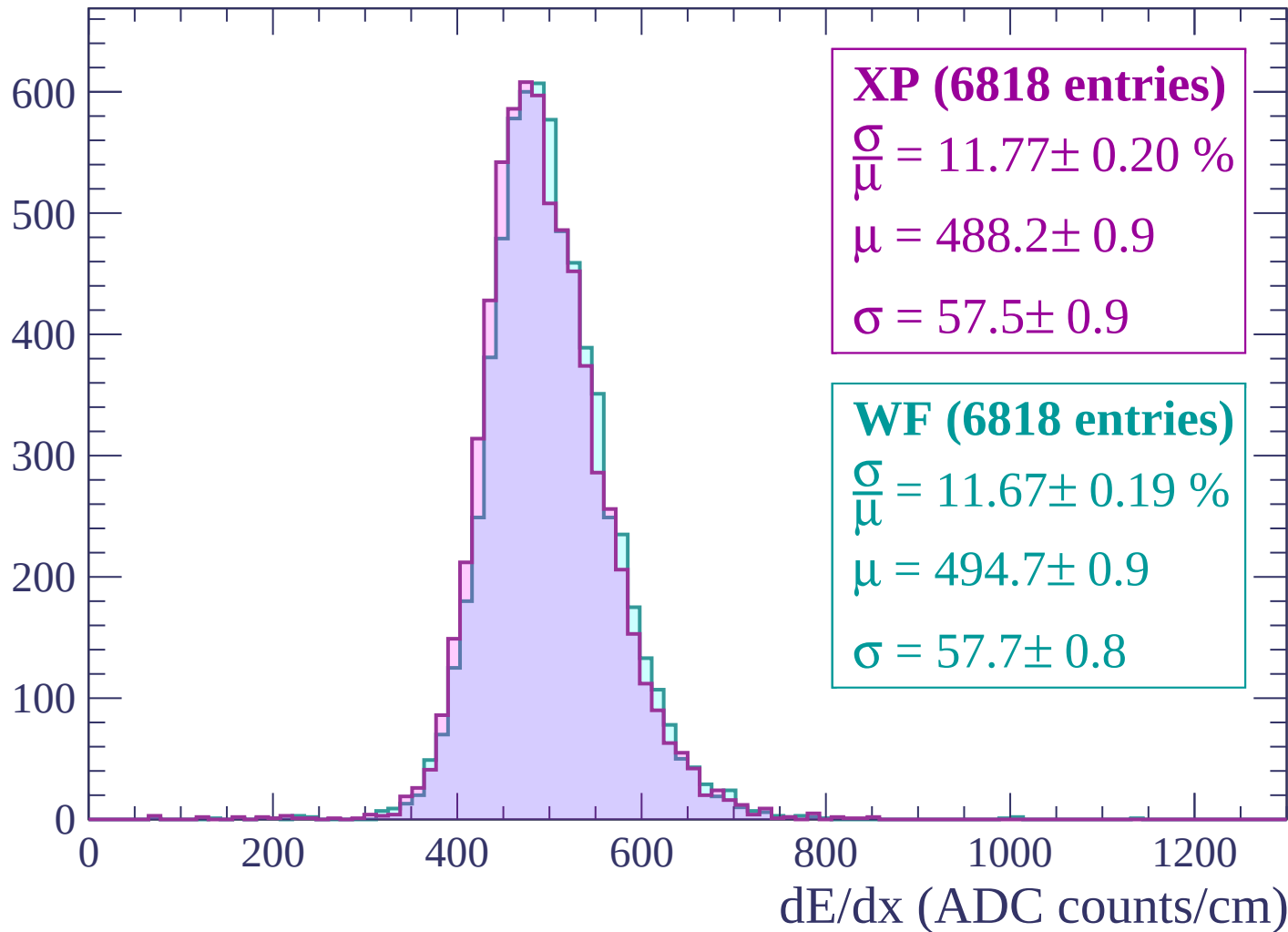
$$\mu = 490.7 \pm 0.8$$

$$\sigma = 56.2 \pm 0.8$$

dE/dx (ADC counts/cm)

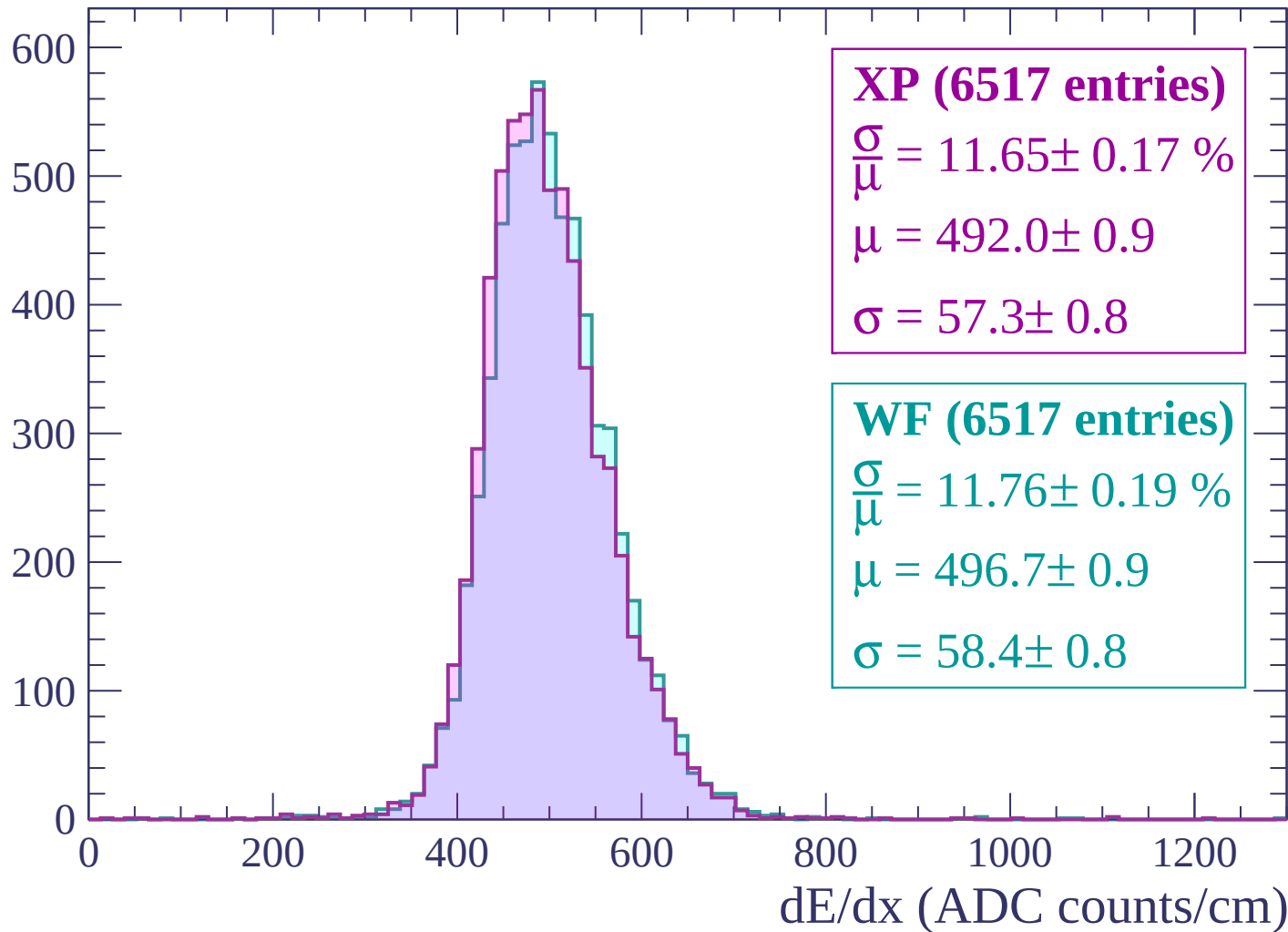
Energy loss | $1920 < p < 1980$

Count



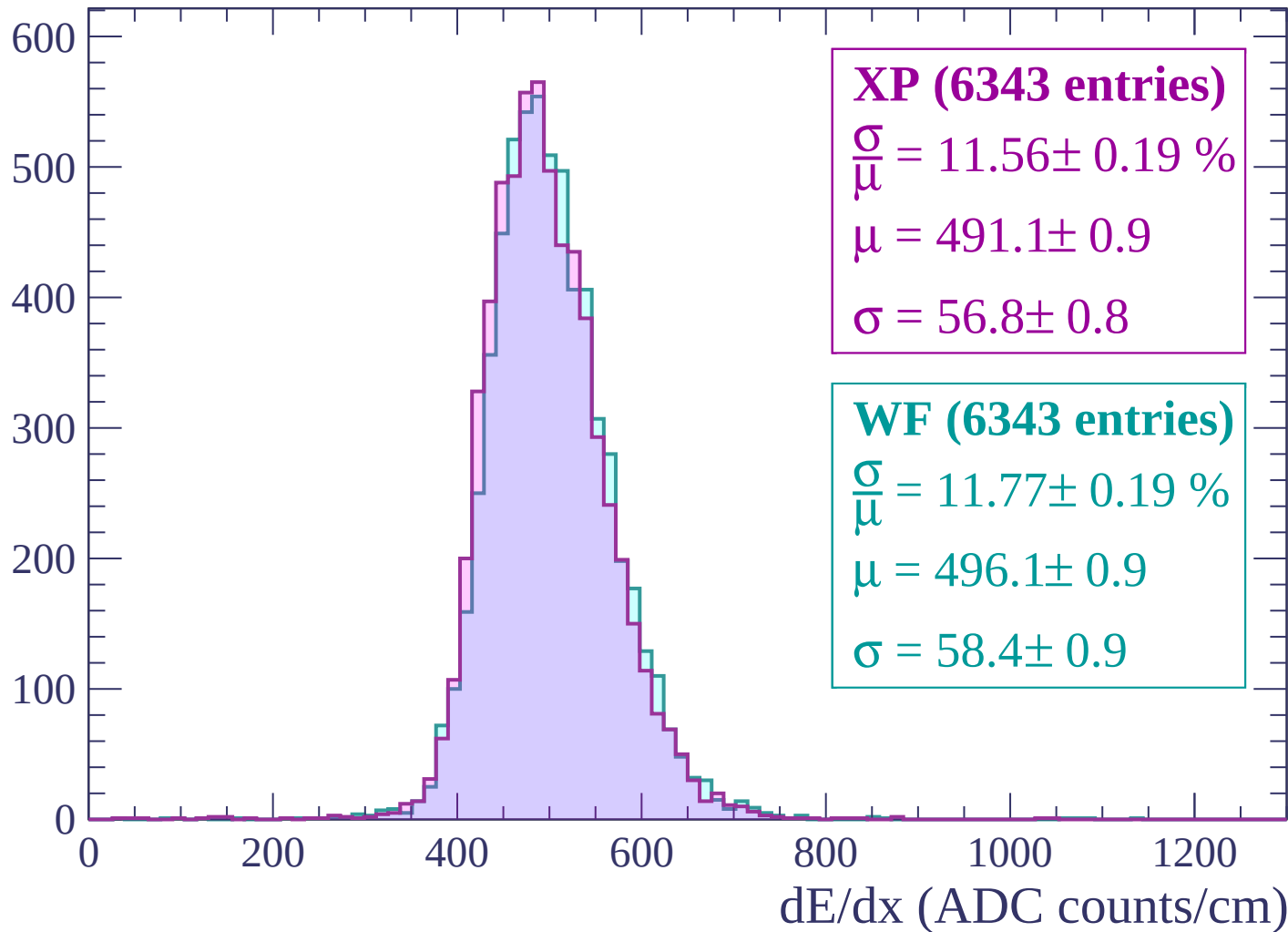
Energy loss | $1980 < p < 2040$

Count



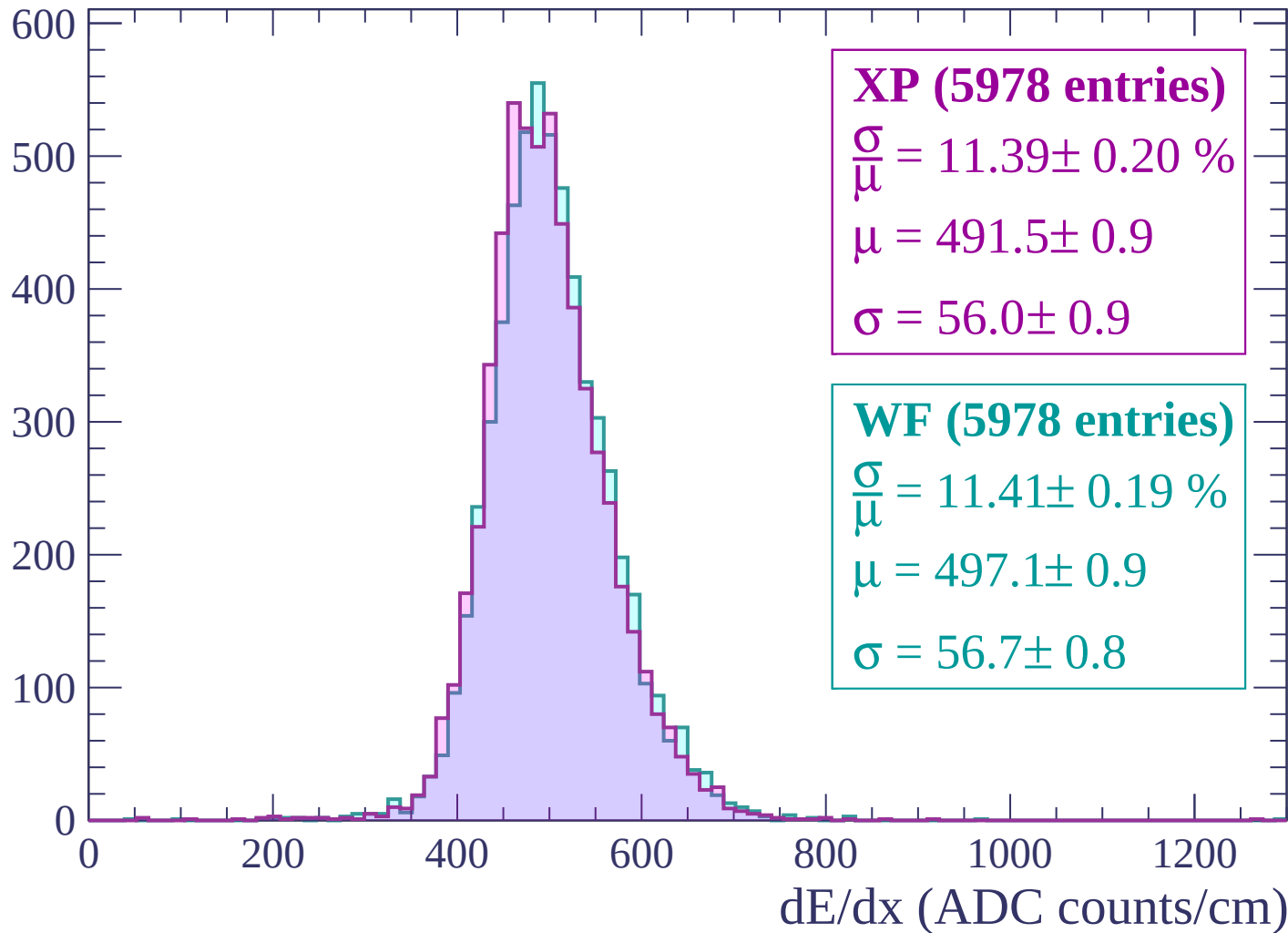
Energy loss | $2040 < p < 2100$

Count



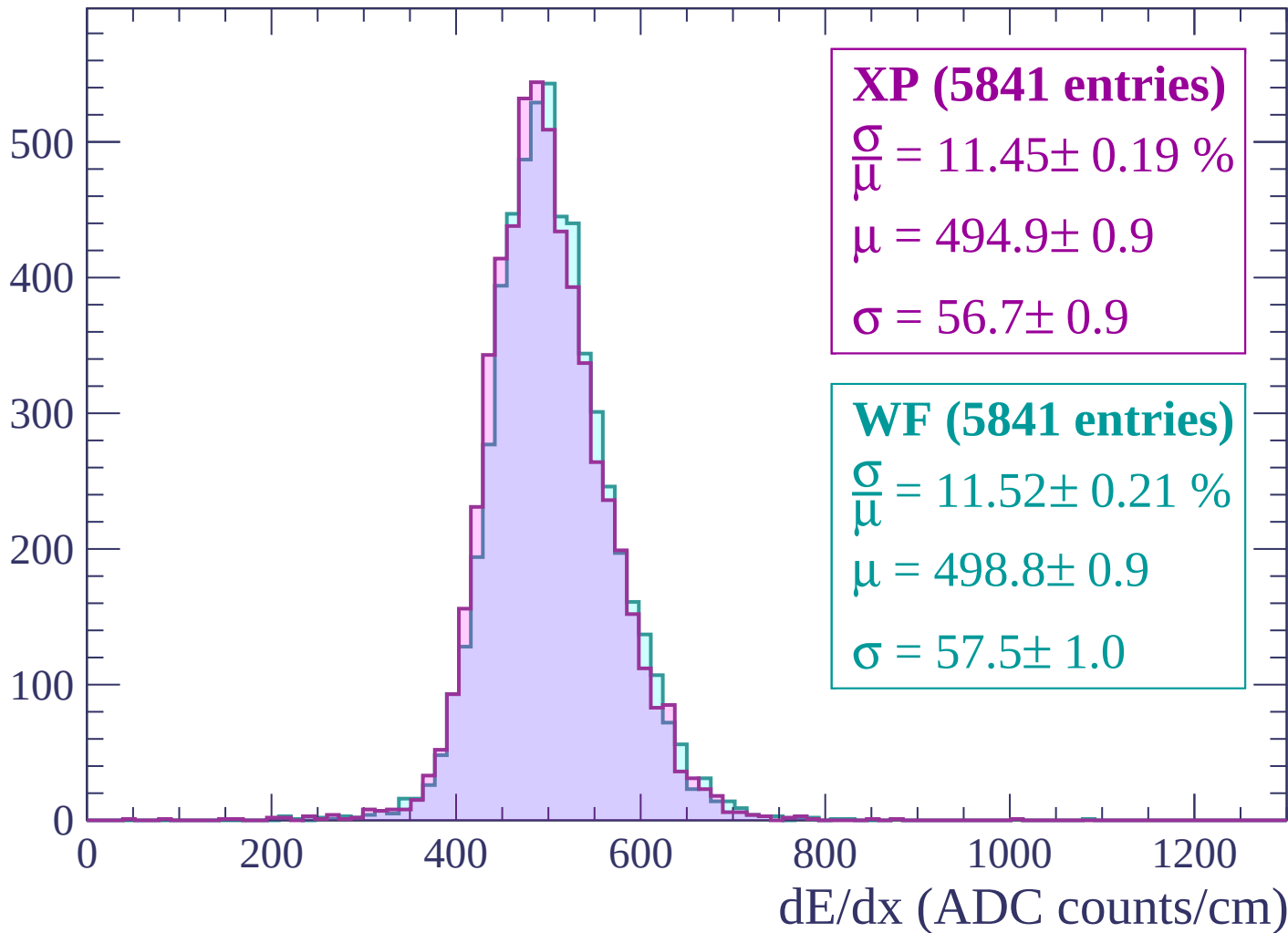
Energy loss | $2100 < p < 2160$

Count



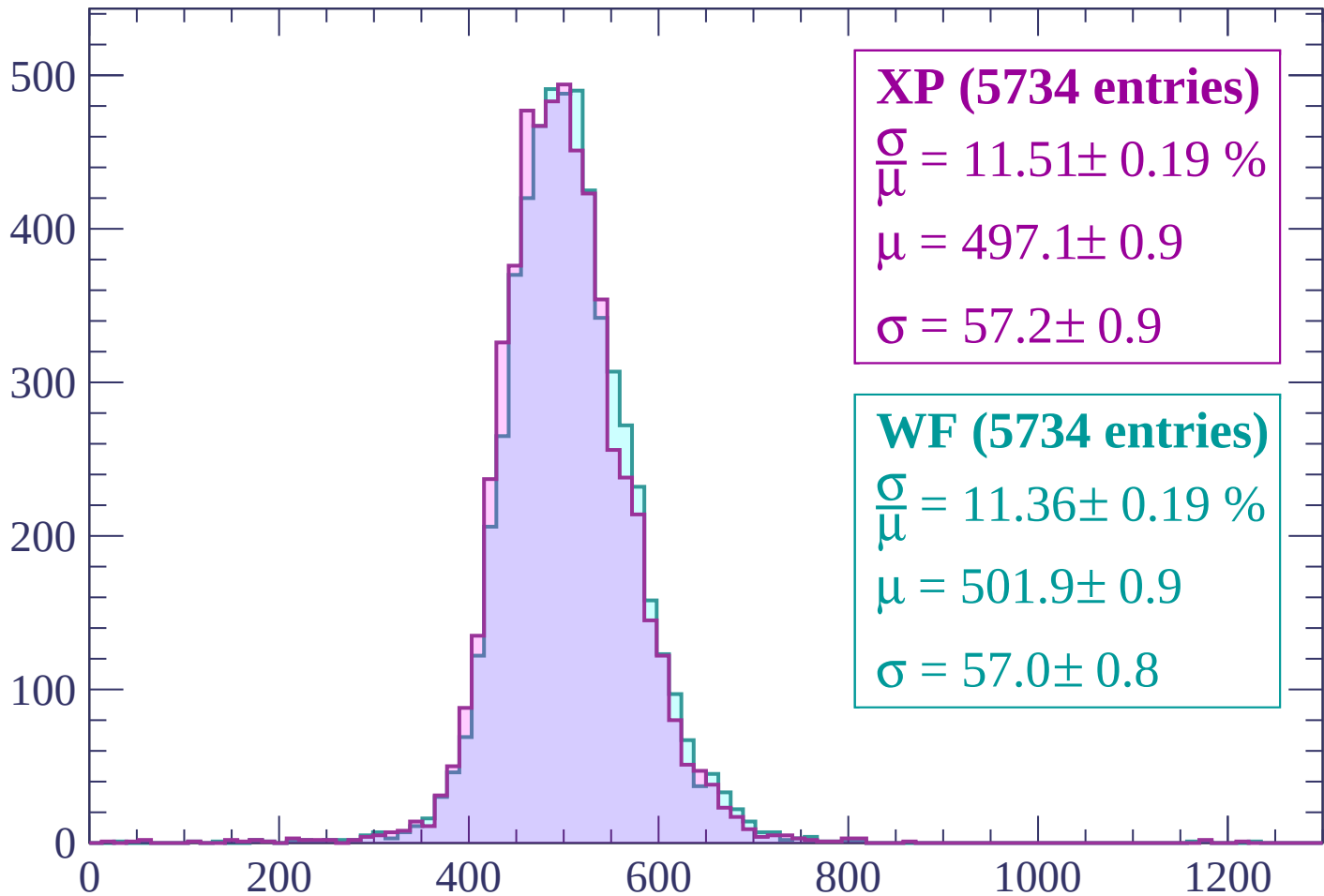
Energy loss | $2160 < p < 2220$

Count



Energy loss | $2220 < p < 2280$

Count



XP (5734 entries)

$\frac{\sigma}{\mu} = 11.51 \pm 0.19 \%$

$\mu = 497.1 \pm 0.9$

$\sigma = 57.2 \pm 0.9$

WF (5734 entries)

$\frac{\sigma}{\mu} = 11.36 \pm 0.19 \%$

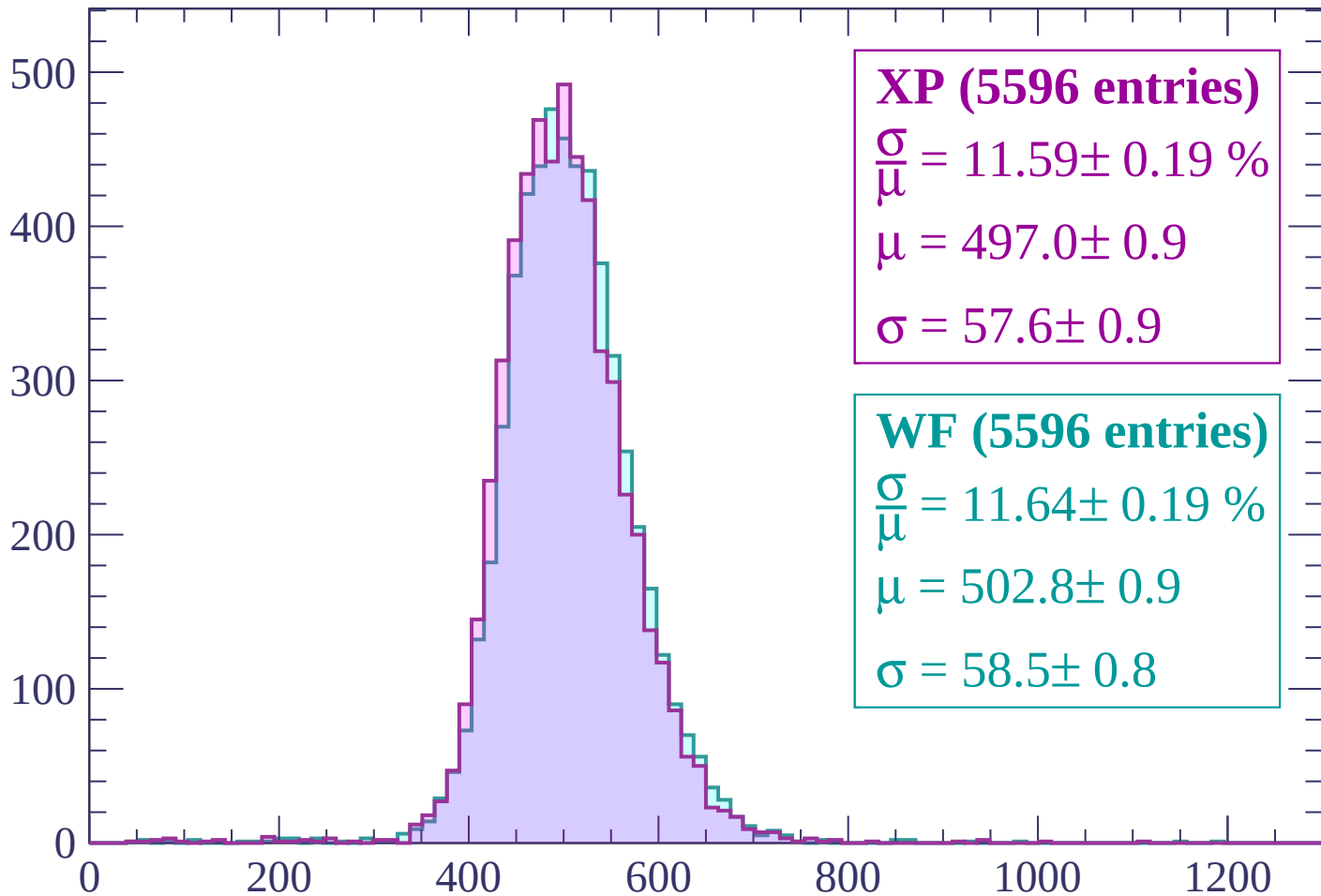
$\mu = 501.9 \pm 0.9$

$\sigma = 57.0 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count



XP (5596 entries)

$$\frac{\sigma}{\mu} = 11.59 \pm 0.19 \%$$

$$\mu = 497.0 \pm 0.9$$

$$\sigma = 57.6 \pm 0.9$$

WF (5596 entries)

$$\frac{\sigma}{\mu} = 11.64 \pm 0.19 \%$$

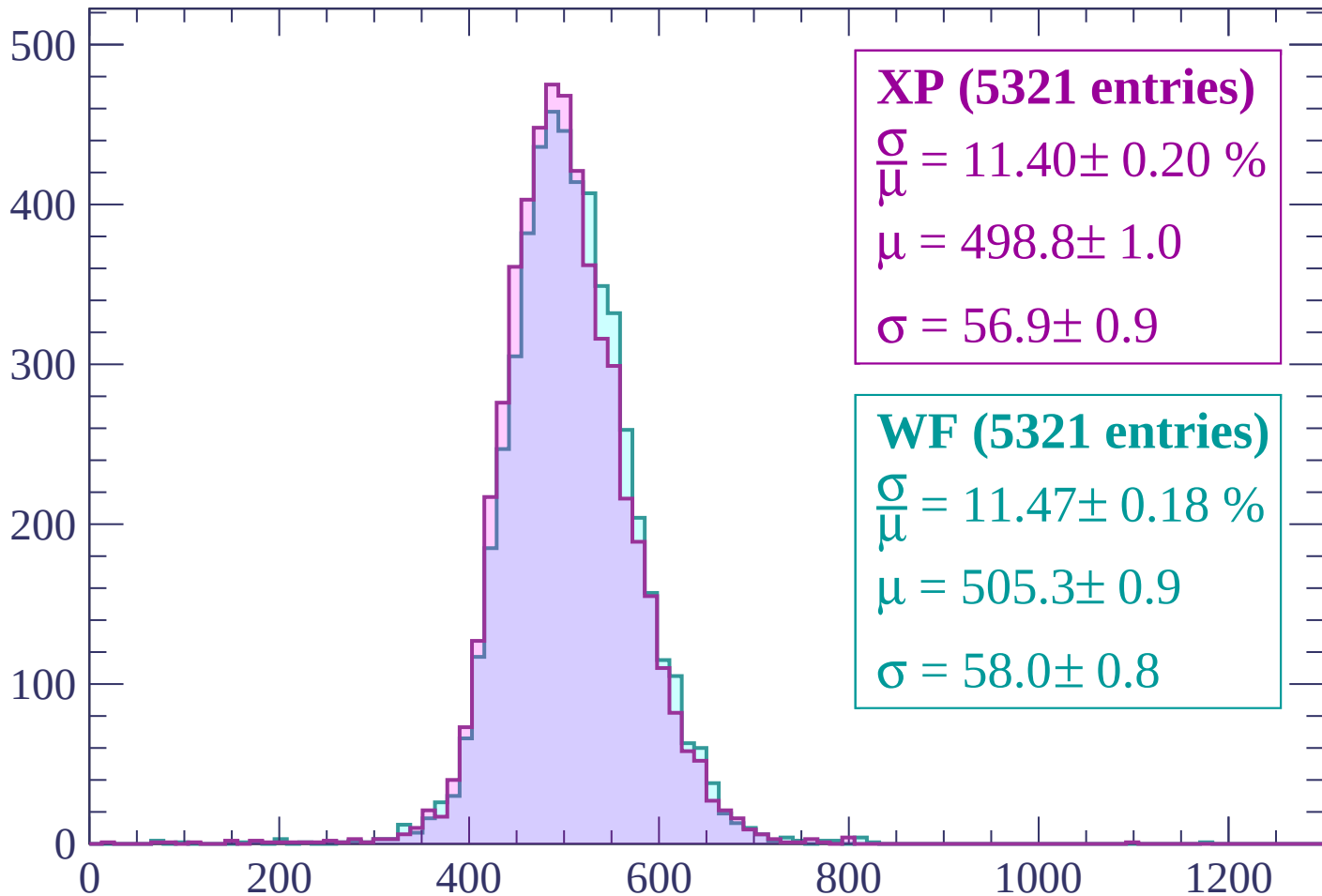
$$\mu = 502.8 \pm 0.9$$

$$\sigma = 58.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2340 < p < 2400$

Count



XP (5321 entries)

$\frac{\sigma}{\mu} = 11.40 \pm 0.20 \%$

$\mu = 498.8 \pm 1.0$

$\sigma = 56.9 \pm 0.9$

WF (5321 entries)

$\frac{\sigma}{\mu} = 11.47 \pm 0.18 \%$

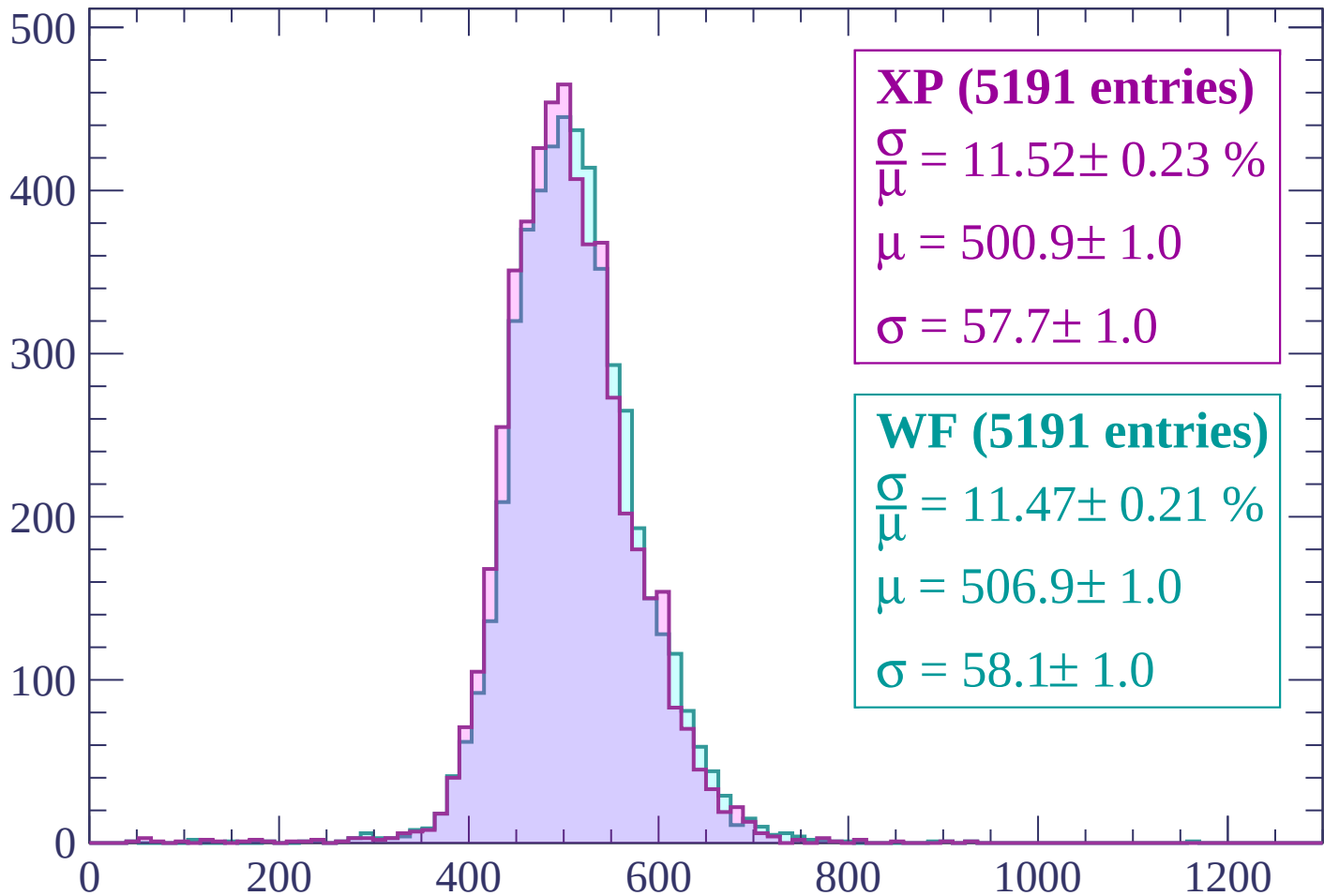
$\mu = 505.3 \pm 0.9$

$\sigma = 58.0 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count



XP (5191 entries)

$\frac{\sigma}{\mu} = 11.52 \pm 0.23 \%$

$\mu = 500.9 \pm 1.0$

$\sigma = 57.7 \pm 1.0$

WF (5191 entries)

$\frac{\sigma}{\mu} = 11.47 \pm 0.21 \%$

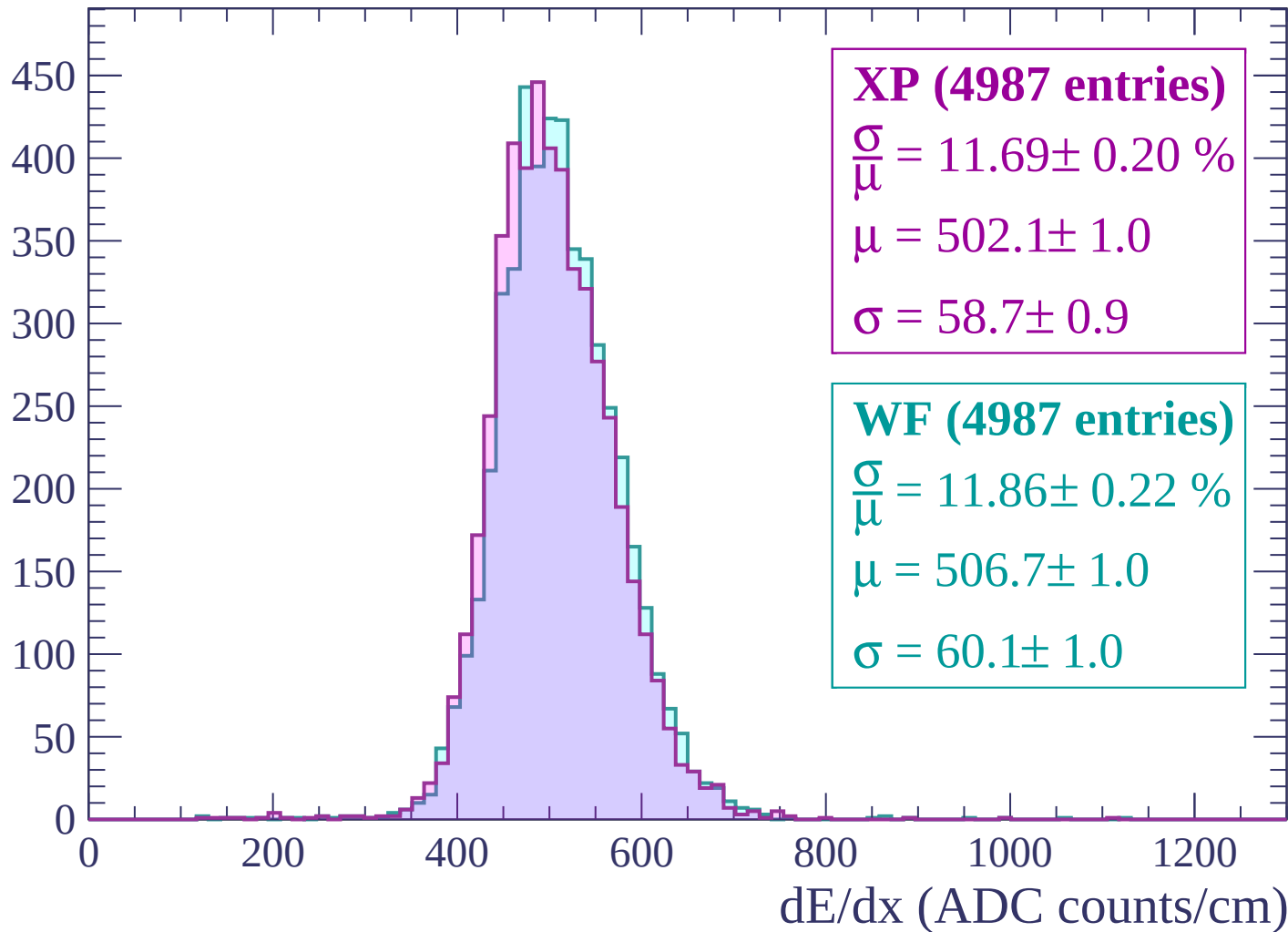
$\mu = 506.9 \pm 1.0$

$\sigma = 58.1 \pm 1.0$

dE/dx (ADC counts/cm)

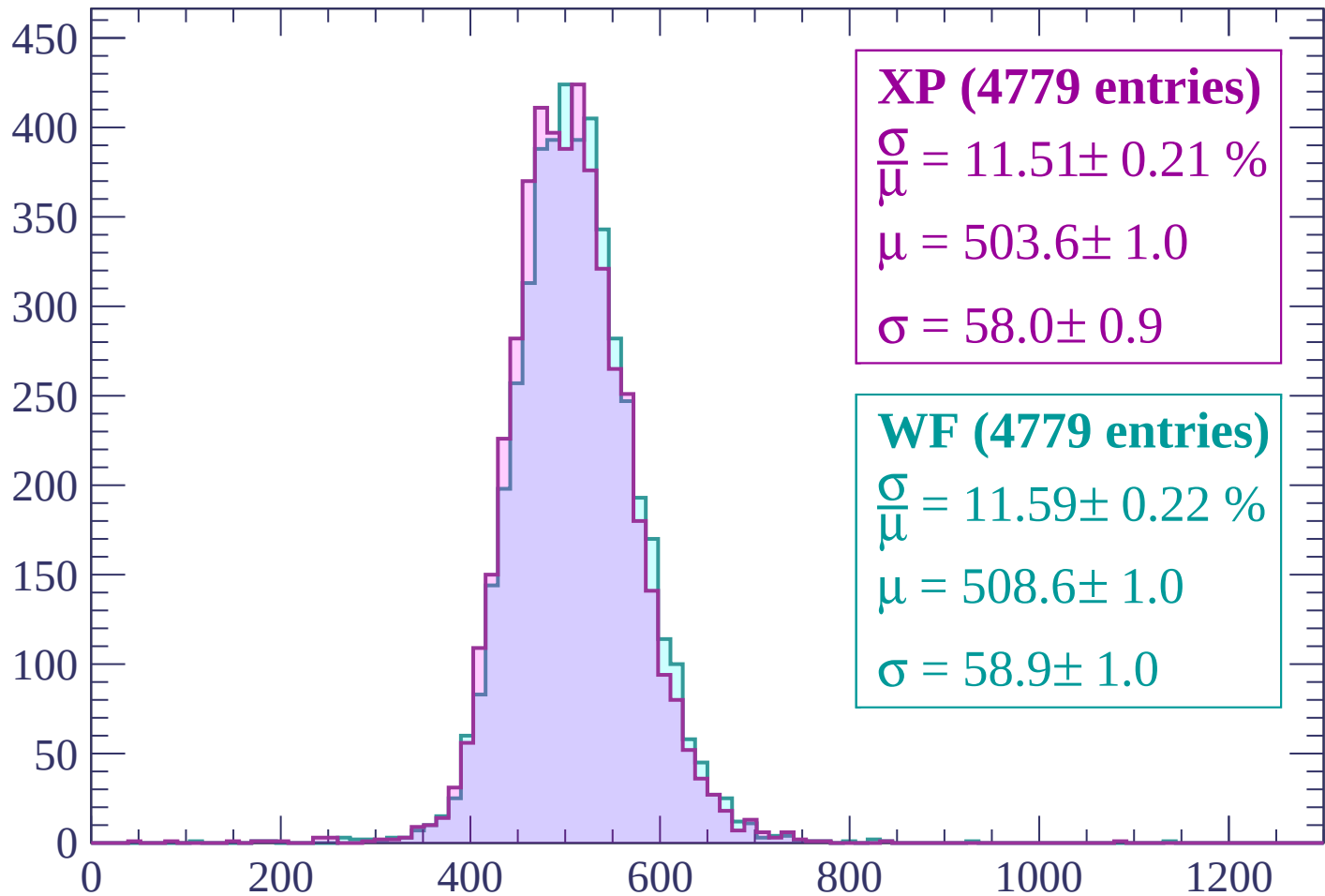
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (4779 entries)

$\frac{\sigma}{\mu} = 11.51 \pm 0.21 \%$

$\mu = 503.6 \pm 1.0$

$\sigma = 58.0 \pm 0.9$

WF (4779 entries)

$\frac{\sigma}{\mu} = 11.59 \pm 0.22 \%$

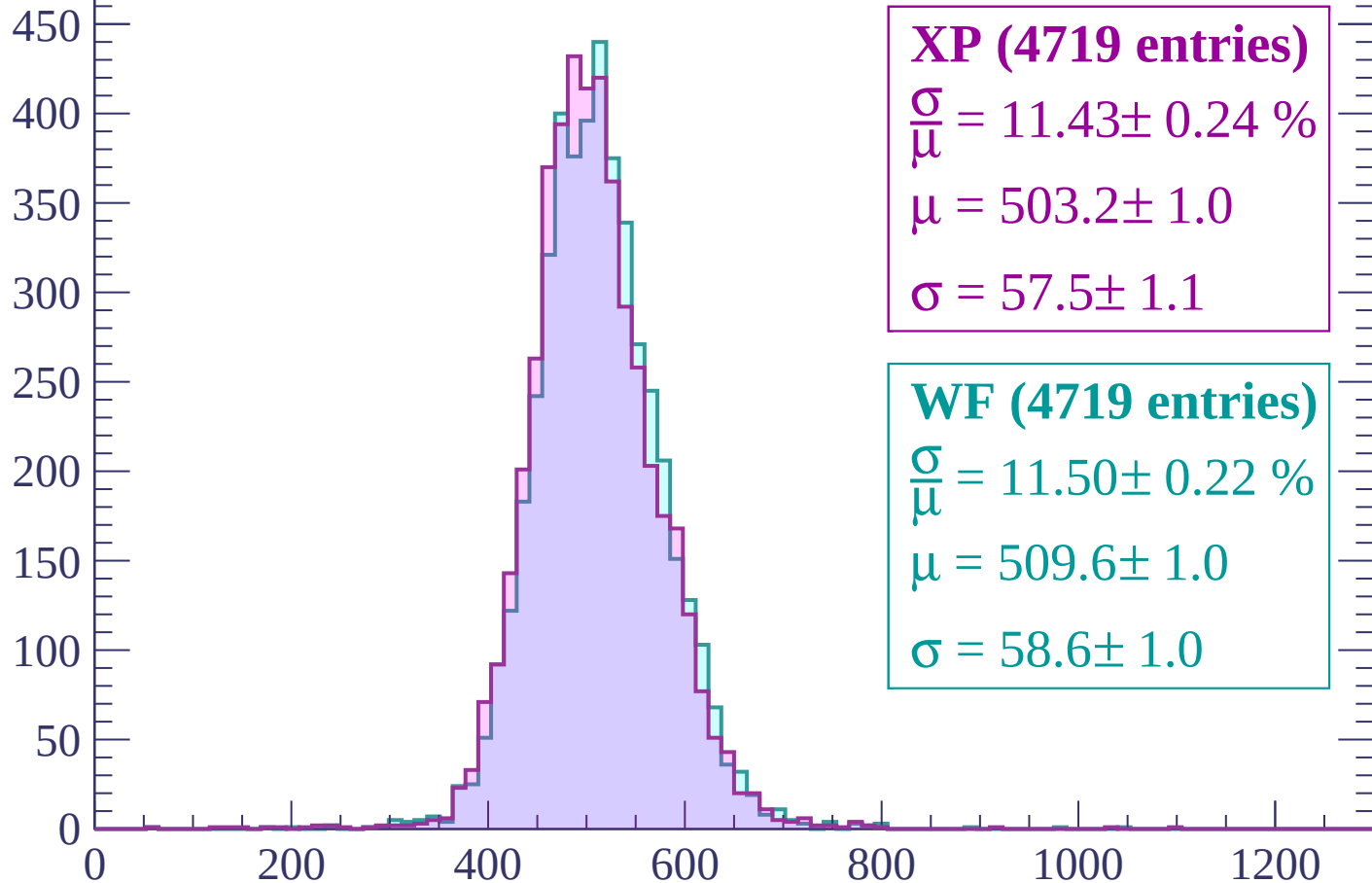
$\mu = 508.6 \pm 1.0$

$\sigma = 58.9 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (4719 entries)

$$\frac{\sigma}{\mu} = 11.43 \pm 0.24 \%$$

$$\mu = 503.2 \pm 1.0$$

$$\sigma = 57.5 \pm 1.1$$

WF (4719 entries)

$$\frac{\sigma}{\mu} = 11.50 \pm 0.22 \%$$

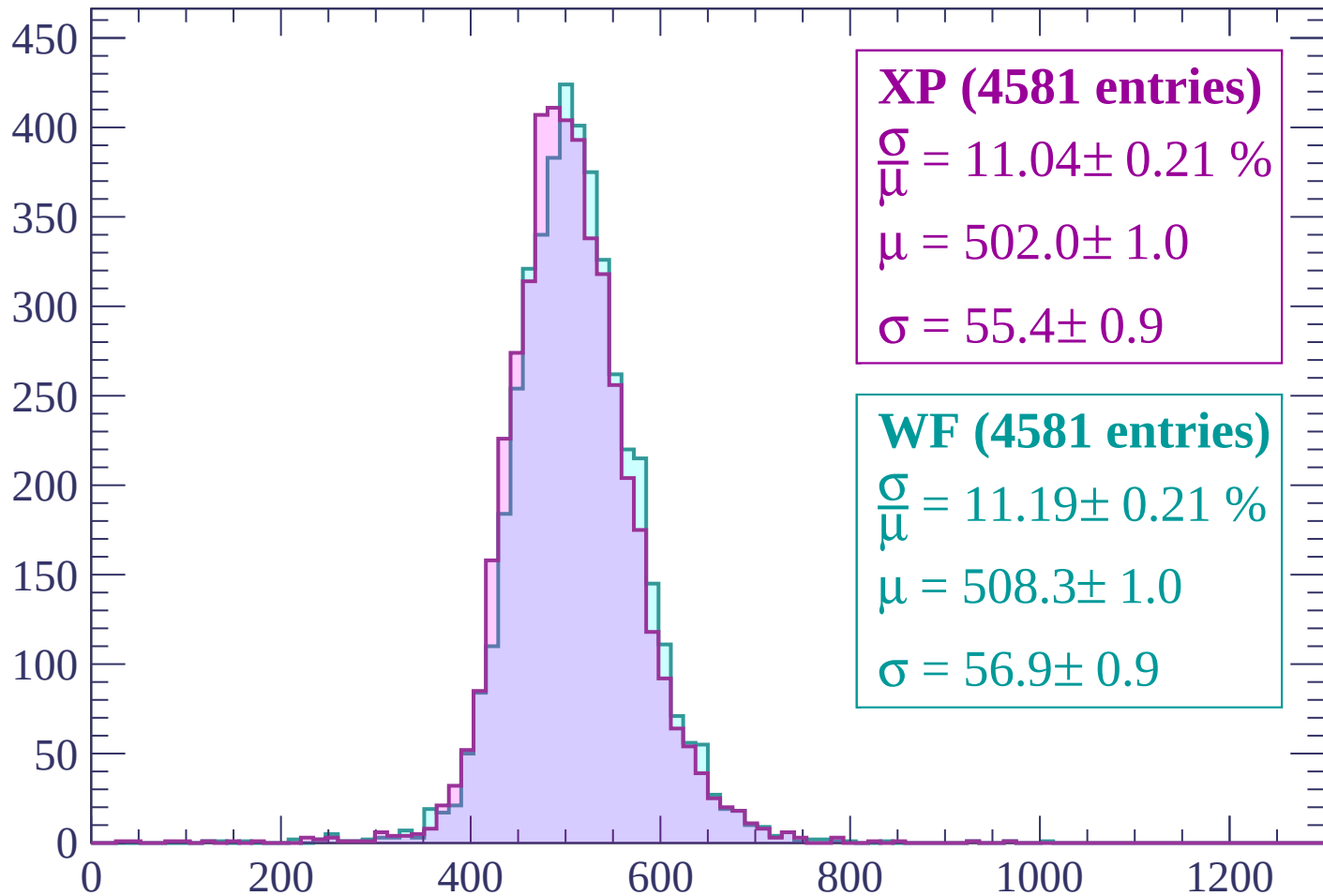
$$\mu = 509.6 \pm 1.0$$

$$\sigma = 58.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2640 < p < 2700$

Count



XP (4581 entries)

$$\frac{\sigma}{\mu} = 11.04 \pm 0.21 \%$$

$$\mu = 502.0 \pm 1.0$$

$$\sigma = 55.4 \pm 0.9$$

WF (4581 entries)

$$\frac{\sigma}{\mu} = 11.19 \pm 0.21 \%$$

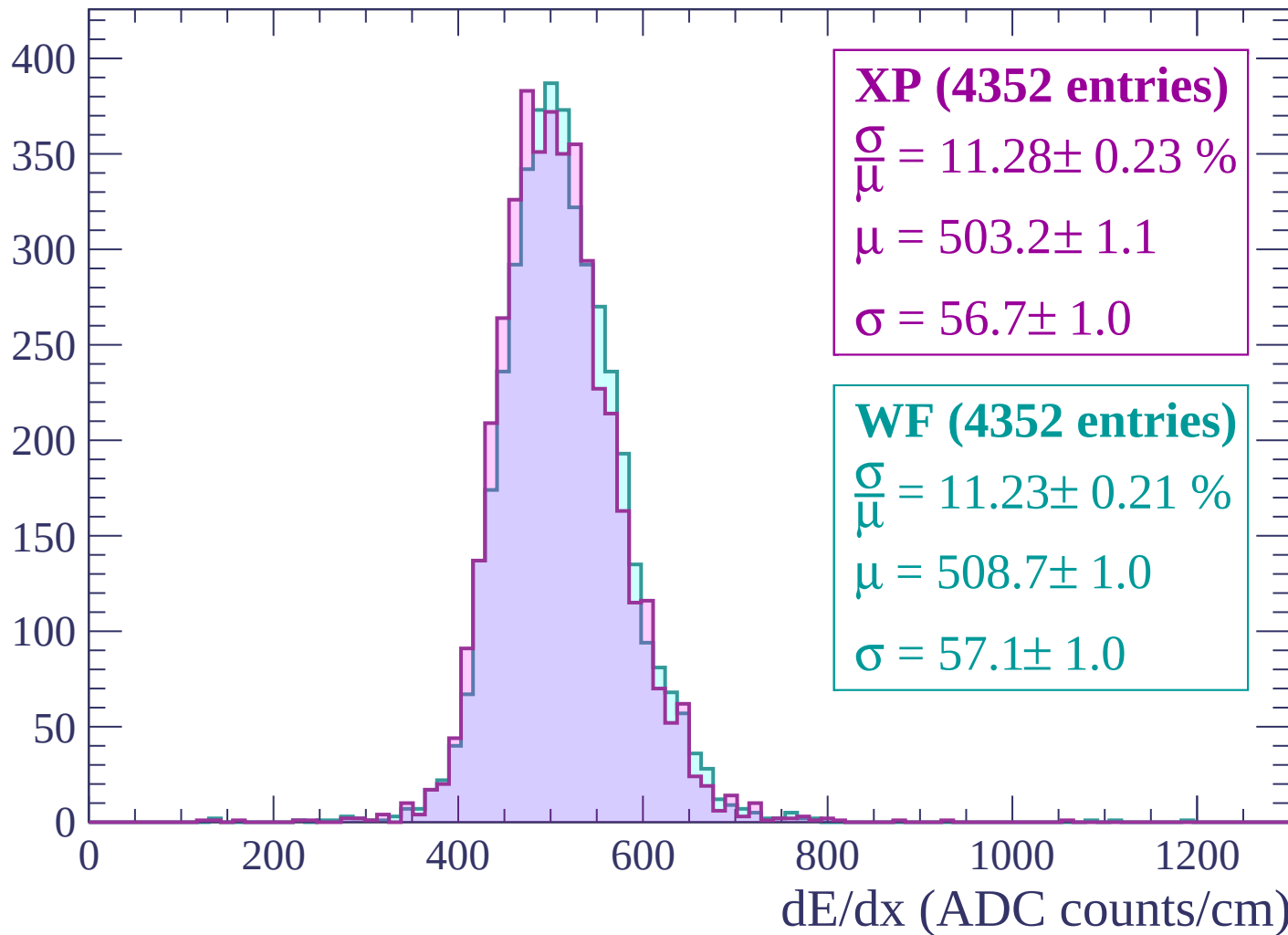
$$\mu = 508.3 \pm 1.0$$

$$\sigma = 56.9 \pm 0.9$$

dE/dx (ADC counts/cm)

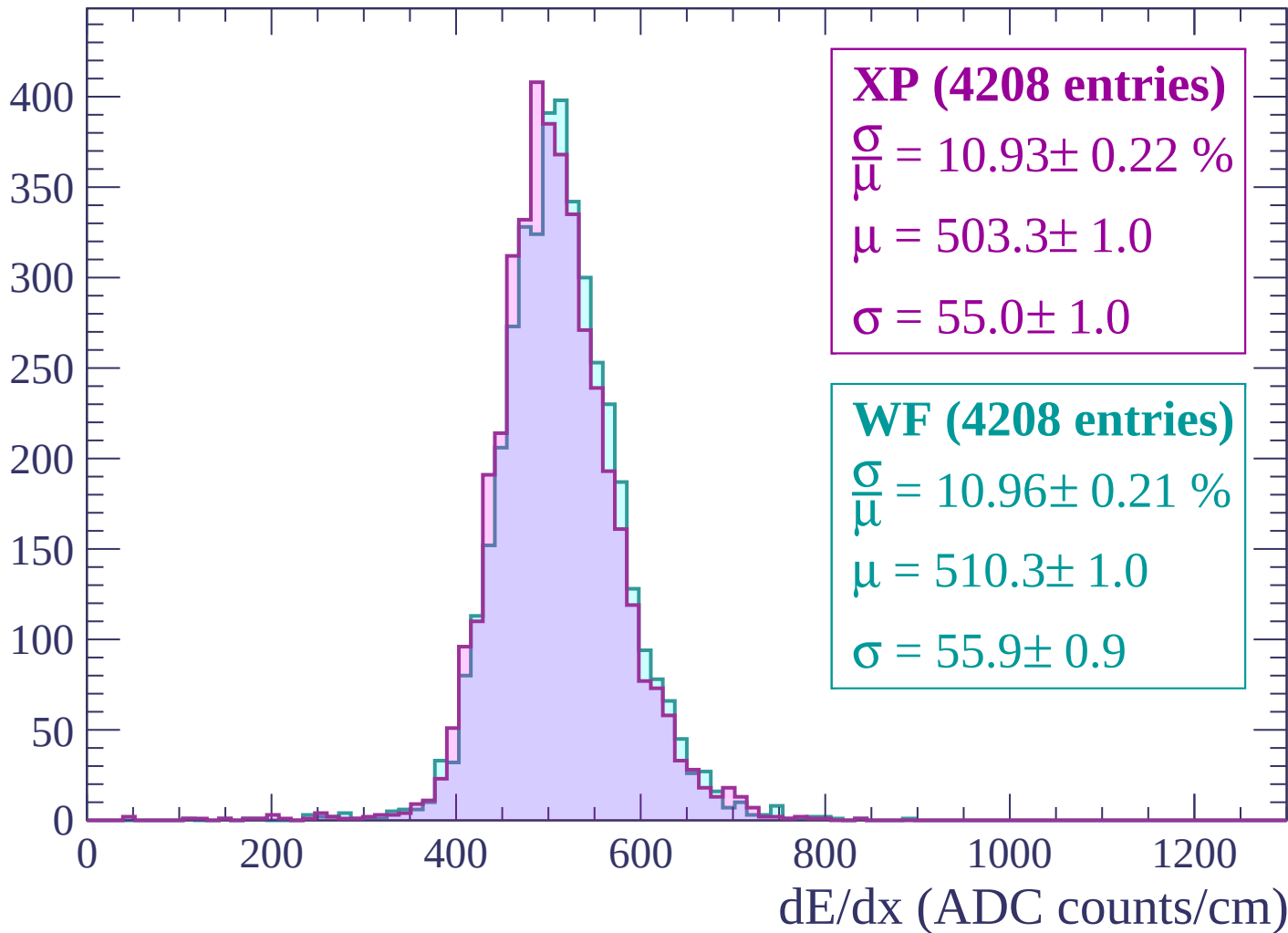
Energy loss | $2700 < p < 2760$

Count



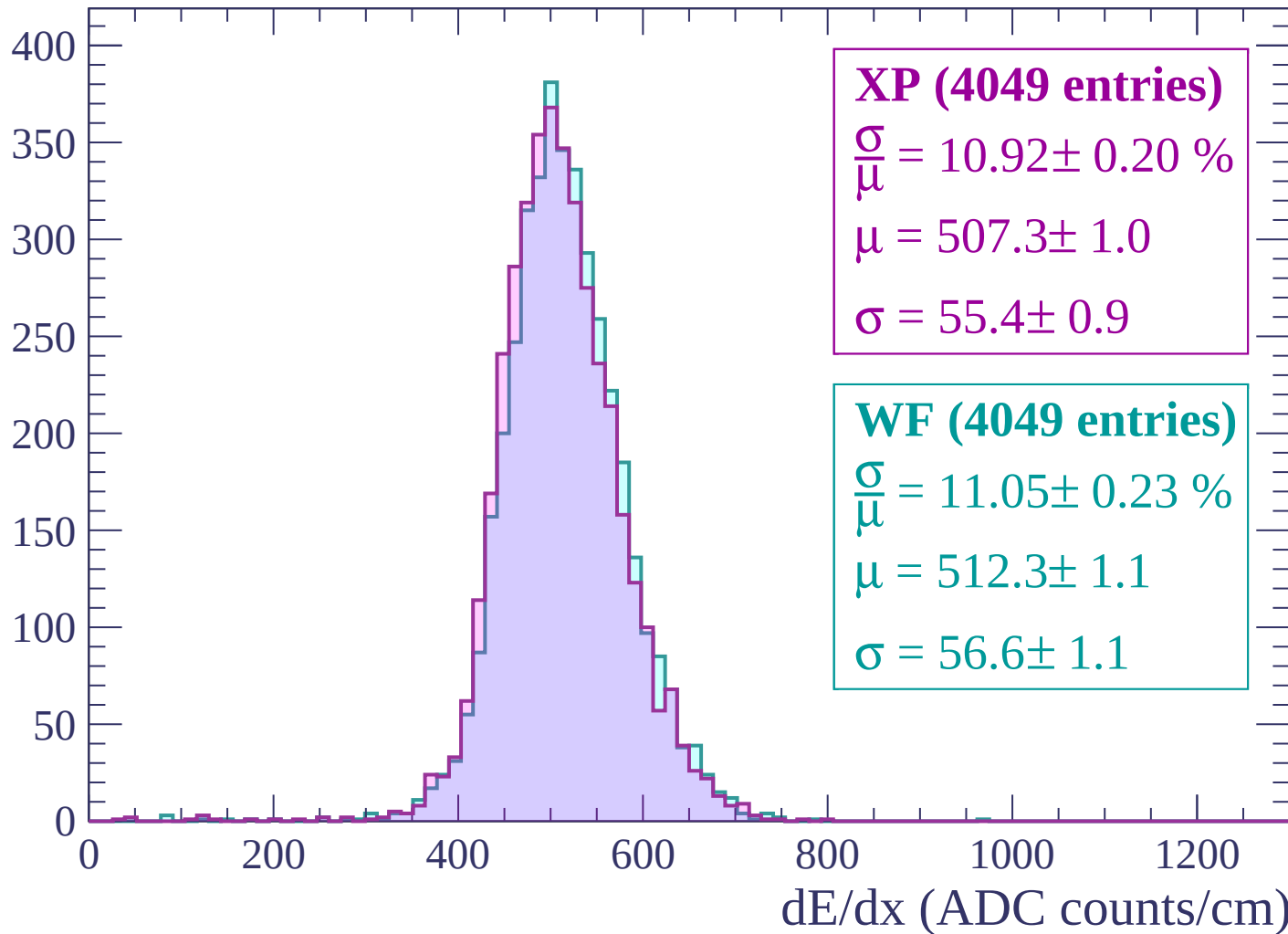
Energy loss | $2760 < p < 2820$

Count



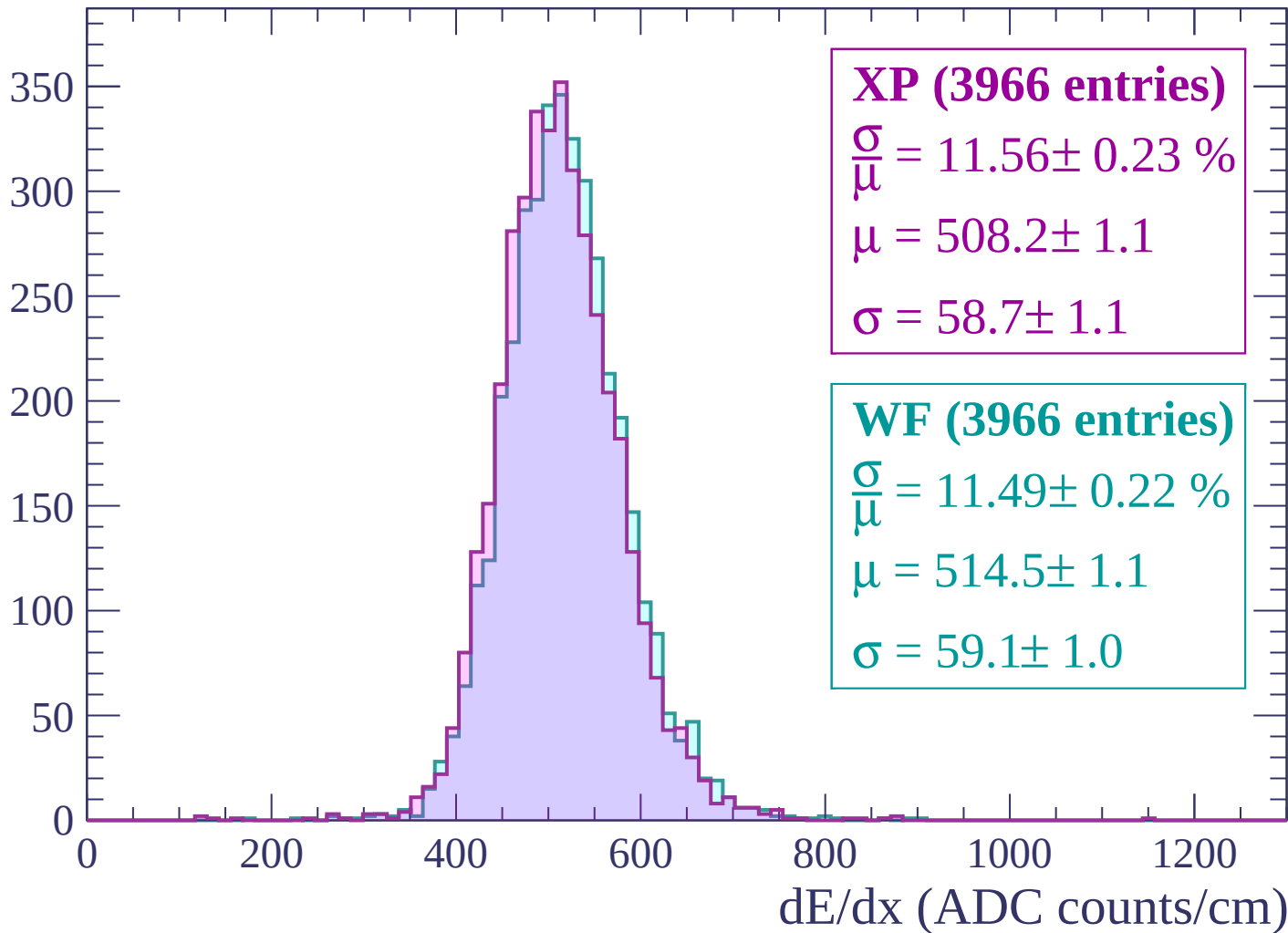
Energy loss | $2820 < p < 2880$

Count



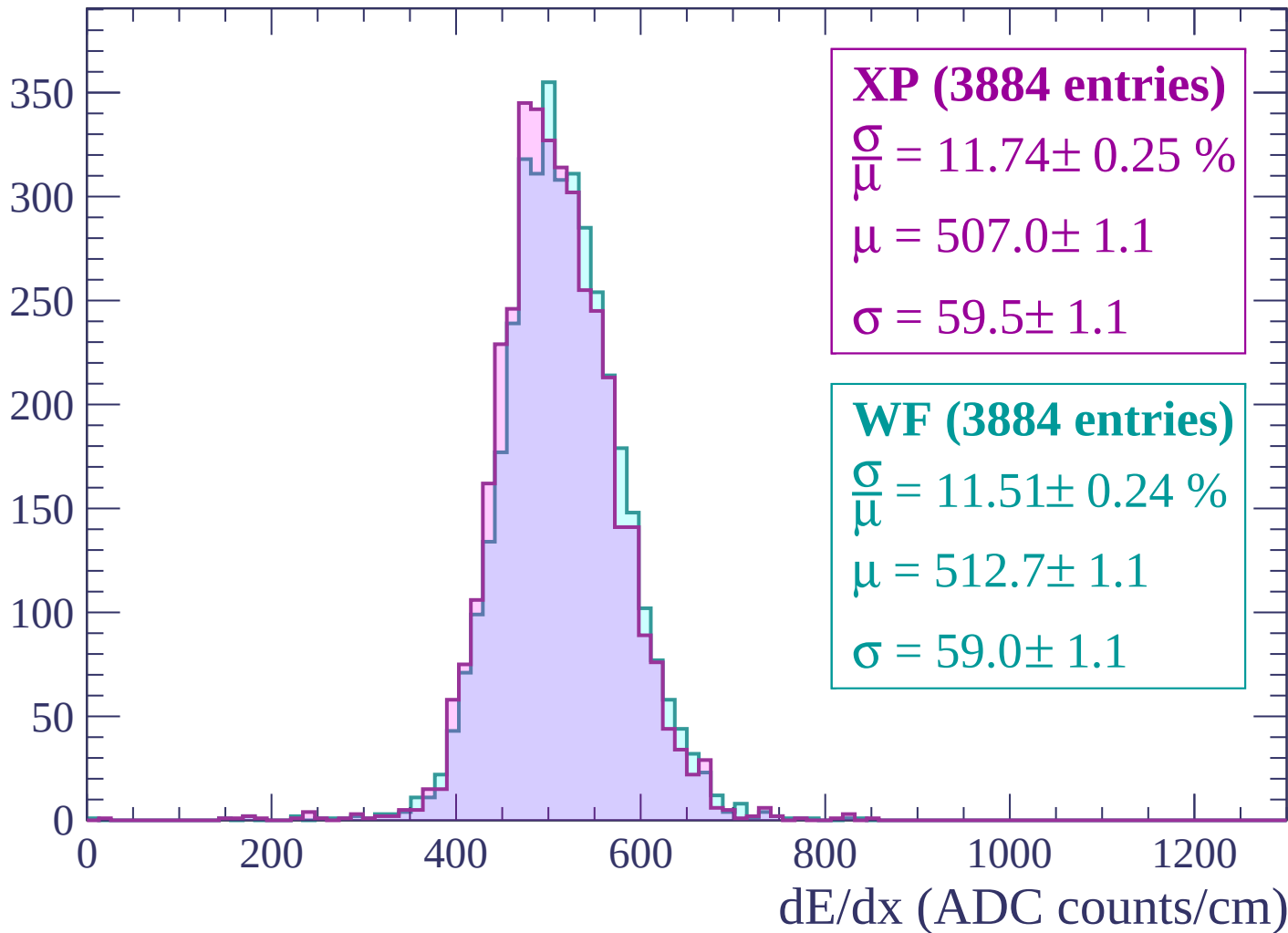
Energy loss | $2880 < p < 2940$

Count



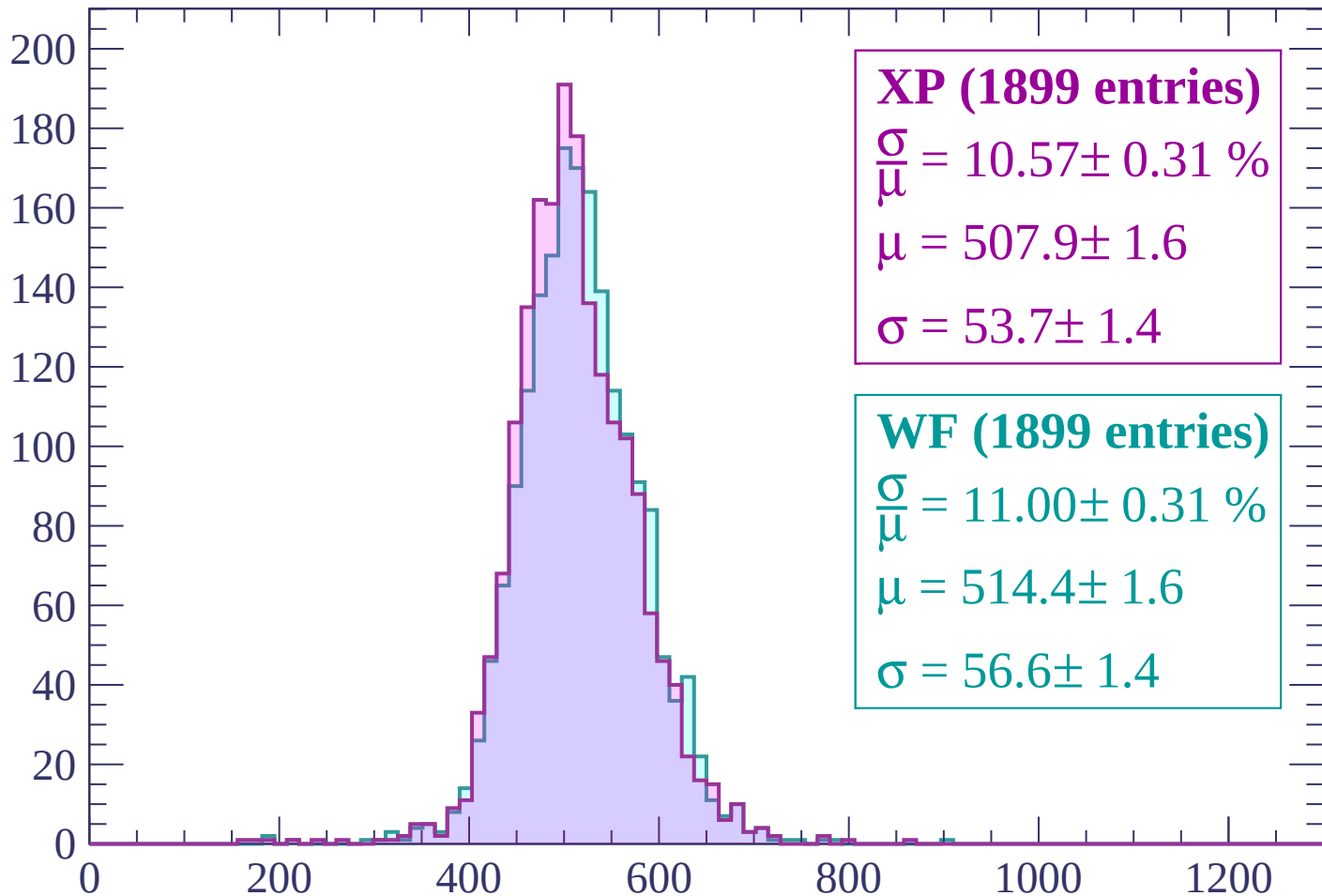
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



XP (1899 entries)

$$\frac{\sigma}{\mu} = 10.57 \pm 0.31 \%$$

$$\mu = 507.9 \pm 1.6$$

$$\sigma = 53.7 \pm 1.4$$

WF (1899 entries)

$$\frac{\sigma}{\mu} = 11.00 \pm 0.31 \%$$

$$\mu = 514.4 \pm 1.6$$

$$\sigma = 56.6 \pm 1.4$$

dE/dx (ADC counts/cm)

